Treatise on Plasma Physics

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Preface

This script is the result of my ceaseless, unwavering interest in plasma physics and was originally composed as a personal draft of introductory notes on the subject. Fortunately, the substantial effort invested in the craft of such a script has further motivated me to reformat it into a comprehensive review of plasma physics, which has, before long, become far more detailed than any introductory script in the relevant literature. Each chapter of the script initially addresses its topic at an introductory level, thereby motivating readers to continue following my theoretical development of the subject at an advanced level. The script thus presents a complete graduate-level review of plasma physics.

The first chapter is dedicated to the theoretical development of the underlying concepts in plasma physics, hence the title. We present the subject's main hypotheses and discuss their relevance in particular regimes. The topics covered include Debye shielding and quasi-neutrality, which together describe the response of plasmas to small-amplitude perturbations in the electrostatic potential, as well as transport phenomena at an introductory level. By creating the link between Debye shielding, quasi-neutrality, and collisional processes, we shall establish the fact that plasmas are weakly collisional.

The second chapter offers a complete review of charged particle motion in various configurations of fields, thus exploring the dynamics of plasma particles. The concept of drift velocity proves fundamental in the analytical approach to the kinematics of charged particles. This chapter also introduces readers to non-conservative phenomena and to the concepts of adiabatic invariants and symmetry breaking.

The third chapter elaborates on plasma dynamics from the statistical mechanical point of view by introducing the phase space and distribution functions within it. These considerations will serve as a foundation for the theoretical development of the three main models of plasma dynamics: the Vlasov model, the two-fluid model, and the magnetohydrodynamic model, each of which views plasma behaviour from a perspective relevant to its own valid regimes. Finally, we shall explore the conservation laws and working regimes relevant to each model.

The fourth chapter is fully consecrated to the study of waves and oscillations in plasmas. Our considerations will begin with the simplest model, the magnetohydrodynamic model, on which we lay the groundwork for examining oscillatory plasma behaviour. We will subsequently extend the work by applying the two-fluid and Vlasov models and comparing their predictions in the relevant regimes. The topics covered include Alfvén waves in the magnetohydrodynamic and the two-fluid description, cold and warm electrostatic plasma waves, as well as elementary cold and warm waves in magnetised plasmas in the two-fluid and the Vlasov description. The latter shall be divided into the studies of perpendicular and parallel propagation to an externally applied magnetic field. The criteria of wave stability and wave damping are also discussed by developing the Landau approach to electrostatic waves using the Vlasov-Poisson model. Our considerations of dynamics in a magnetised environment will later be expanded to treat the kinetics of plasma particles in magnetic fields.

The fifth chapter is dedicated to the study of the structure of magnetic fields in plasmas. We formally introduce the concepts of magnetic field-lines, magnetic surfaces, and the coordinate systems used to describe them. The main results of our study are magnetic flux coordinate systems, the magnetic-field-line Hamiltonian, and the quasi-symmetry of magnetic fields. These theoretical developments will later serve as a foundation for developing the theory of magnetic confinement.

The sixth chapter firstly formally introduces the reader to the Fokker-Planck collision theory. We revisit our considerations on collisions in plasmas from the first chapter and expand on them. The main theoretical result is the Fokker-Planck equation, which describes the temporal evolution of the plasma particle distribution functions due to collisional processes. Subsequently, we shall develop the theory of neoclassical transport.

Contents

Chapter 1

Preliminaries

A gaseous substance, ionised through a strong rise in temperature or by photo-ionization, whose constituent particles are predominantly electrons and ions of various species, yet whose density is sufficiently low for the constituent particles to follow the Maxwell distribution rather than that of Paul-Dirac or Bose-Einstein, is termed a *plasma*. The differential equations governing plasma behaviour generally lack simple exact models, because the dynamics of the constituent particles intertwine delicately with their trajectories at any given instant. Furthermore, the overabundance of particles renders the search for exact solutions in practice infeasible. However, due to the same overabundance, statistical tools prove particularly useful, as they provide a simplified approach to describing plasma phenomena.

1.1 Parameters and examples of plasma

The three fundamental parameters that characterize plasmas are:

1. The density n of the constituent particles, measured in particles per cubic meter,
2. The temperature T of each species, often measured in eV (where $1\,\mathrm{eV} = 11\,605\,\mathrm{K}$),
3. The magnetic field $\mathbf{B}$ in steady state, expressed in teslas.

Numerous additional parameters, such as the Debye length, the Larmor radius, and the thermal velocity of each species, among many others, are derived from these three main parameters. It is worth mentioning, as an exception, some remarkable peculiarities of partially ionized plasmas, where the dominant presence of neutral particles is of comparable relevance to that of charged particles.

1.2 Logical scheme of plasma physics

Plasmas exist within a wide range of conditions, spanning several orders of magnitude in each of the aforementioned quantities. Plasma physics cannot, in many cases, be considered an exact science; rather, it comprises numerous regimes and associated models grounded in well-justified conditions and approximations, each designed to describe a limited range of situations. In this first introductory chapter to plasma physics, we focus on expanding and deepening our understanding of plasma dynamics. Our study of plasma physics will adopt a selective focus on specific viewpoints in plasma dynamics, while preserving a global perspective that connects them into a coherent network of plasma phenomena. Plasma dynamics is primarily determined by the interaction between electromagnetic fields and the motion of constituent particles. From a pragmatic statistical perspective, the latter are considered to be present in very large numbers. Assuming time were not a constraint and measurement precision were absolute, the evolution of any form of plasma could be modelled according to the following procedure:

1. Given the trajectory $\mathbf{x}_i(t)$ and velocity $\mathbf{v}_i(t)$ of any particle, the electric field $\mathbf{E}(\mathbf{x}, t)$ and magnetic field $\mathbf{B}(\mathbf{x}, t)$ can be evaluated using Maxwell's equations.
2. Given the *instantaneous* electric and magnetic fields $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, the forces acting on any particle can be evaluated using the Lorentz equation to update their trajectories and velocities.

However, such procedures are not practicable given the very large number of particles and, to a lesser extent, the complexity of the electromagnetic field. To achieve a more practical understanding, we will abandon the precise evaluation of complex behaviour in its entirety and restrict our considerations to specific phenomena within a given regime. A *regime* is a situation in which certain approximations remain relevant and allow for a system's coherent description. The following categories of approximations allow us to establish numerous regimes:

1. Approximations concerning the electromagnetic field:
   (a) Assumption of the absence of a magnetic field (unmagnetised plasma);
   (b) Assumption of the absence of inductive fields (electrostatic approximation);
   (c) Neglect of the displacement current in Ampère's law (applicable to phenomena where certain characteristic speeds are much smaller than the speed of light);
   (d) Assumption that any magnetic field originates from an external conductor;
   (e) Assumptions concerning the geometry of the system (existence of symmetries, uniformity).
2. Approximations concerning particle properties:
   (a) Considerations on the Lorentz force acting on certain groups of particles:
      i. Using a distribution function $f_\sigma(\mathbf{x}, \mathbf{v}, t)$ of a species σ whose velocity is $\mathbf{v}$ at position $\mathbf{x}$ at time t (Vlasov theory).
      ii. By taking the average of the velocities of all particles of species σ located in the vicinity of $\mathbf{x}$, and characterizing, using the density $n_\sigma(\mathbf{x}, t)$, the mean velocity $\mathbf{u}_\sigma(\mathbf{x}, t)$ and the pressure of the species $P_\sigma(\mathbf{x}, t)$ (two-fluid theory).
      iii. By taking the average of the momenta of all particles of any species and characterizing the plasma by the centre-of-mass density $\rho(\mathbf{x}, t)$, the centre-of-mass velocity $\mathbf{U}(\mathbf{x}, t)$, and the pressure $P(\mathbf{x}, t)$ defined relative to the centre-of-mass velocity $\mathbf{U}(\mathbf{x}, t)$ (magnetohydrodynamic theory).
   (b) Assumptions concerning time (that the frequency of the considered phenomenon is lower or higher than certain characteristic frequencies of the plasma);
   (c) Assumptions concerning space and its dimensions (that the domain of influence of the considered phenomenon is larger or smaller than certain characteristic lengths of the plasma);
   (d) Assumptions concerning particle velocities (that the phenomenon spreads more or less quickly than the thermal speed of species σ).

If we consider the approximations listed above again, we will not necessarily see a structured hierarchy. Unfortunately, developing an intuition first requires admitting some of these approximations. This leaves us uncertain about how to approach the subject, as we must accept certain concepts on faith that their results will be valid, in the hope of later affirming their validity. Consequently, it will be necessary to consult the following scheme from time to time:

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Figure 1.1: Hierarchy of plasma models

Given the circular nature of the scheme from which a clear starting point cannot be isolated, it is thus left to the author's discretion to determine the appropriate first step. To make this introduction as accessible as possible, we first discuss Debye shielding, marking the beginning of our plasma physics studies. Debye shielding, Rutherford scattering, and elementary statistics will serve as a basis to justify that plasmas are *weakly collisional*. We shall then undertake the Vlasov equation, define the phase space, and the distribution functions within it. By averaging the Vlasov equation with respect to certain quantities, we obtain the two-fluid and magnetohydrodynamic equations. This approach will allow us to consolidate our recently acquired knowledge, as illustrated in Figure (1.1).

1.3 Debye Shielding

A careful analysis of the concept of Debye shielding, also known as *Debye screening*, is essential for a deep understanding of plasma dynamics. We will consider a finite-temperature plasma composed of a very large number of electrons and ions, with equal densities initially assumed to be uniform in space. Note that these two species need not be in thermal equilibrium, as we will see later. Let T_e and T_i be the temperatures of electrons and ions, respectively. Given that these temperatures are non-zero, the random thermal motion of the constituent particles generates tiny ephemeral fluctuations in the electrostatic potential, denoted ϕ. Following the circular scheme of figure (1.1), we admit the following assumptions:

1. The plasma is assumed to be weakly collisional. In other words, we neglect higher terms in the collision approximation between constituent particles and keep only the first-order terms.

2. We will consider species σ as a fluid of density n_σ, temperature T_σ, and pressure $P_\sigma = n_\sigma k_\mathrm{B} T_\sigma$ moving at a mean velocity $\mathbf{u}_\sigma$. Every particle is subject to the following forces: the Lorentz force produced in part by the perturbed potential ϕ, and the force due to fluid pressure. By taking the average over all particles, Newton's second law applied to this fluid of particles reads:

$$m_\sigma \frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} = q_\sigma \mathbf{E} - \frac{1}{n_\sigma} \nabla P_\sigma \tag{1.1}$$

   where m_σ is the mass of particles of species σ, q_σ is the particle charge, and $\mathbf{E}$ is the electric field.

Let us assume that the observed potential perturbation obeys a time dependence sufficiently slow for the following hypotheses to be considered valid:

1. The inertia term $\sim \mathrm{d}/\mathrm{d}t$ is negligible.

2. Inductive electric fields are negligible, so $\mathbf{E} \simeq -\nabla\phi$.

3. Any temperature gradient dissolves at a speed sufficient to ensure temperature homogeneity within each species.

4. During the perturbation, thermal equilibrium is assumed to be maintained for each species. Consequently, the temperature of each species is a well-defined quantity.

By adopting the approximations stated previously, equation (1.1) is simplified into a balance between the electrostatic force and the isothermal pressure gradient:

$$-q_\sigma \nabla\phi - \frac{k_\mathrm{B} T_\sigma}{n_\sigma} \nabla n_\sigma \simeq 0 \tag{1.2}$$

The solution to equation (1.2) is trivially determined, and it allows us to establish the *Boltzmann relation*:

$$n_\sigma = n_\sigma^{\square} \exp\left(-\frac{q_\sigma \phi}{k_\mathrm{B} T_\sigma}\right) \tag{1.3}$$

where $n_\sigma^{\square}$ is a simple constant. It is appropriate to revisit the initial assumptions to discuss them. The relevance of the Boltzmann relation depends on the speed at which the perturbation diffuses through the plasma. If this is too fast, induced electromagnetic fields, inertia effects, and even temperature gradients can generate behaviour that deviates significantly from the Boltzmann relation. In certain situations, the Boltzmann relation remains applicable, but only to electrons, whose mass is much lower than that of ions, thus having a much larger thermal velocity, allowing the electrons to thermalise among themselves. Let us now imagine that we introduce at position $\mathbf{r}_0$ a single particle of charge q_0 into the initially unperturbed plasma, uniform on the spatial scale and neutral ($n_i q_i = n_e e$). Before the introduction of the particle, the electrostatic potential was uniform due to the equality of densities Zn_i and n_e. We can exploit the initial conditions by setting the unperturbed potential to zero. Once the particle is introduced, a perturbation will occur in the vicinity of point $\mathbf{r}_0 = \mathbf{0}$. Plasma particles of the same polarity as q_0 will be slightly repelled, while those of opposite polarity will be attracted towards $\mathbf{r}_0$. These deviations in the perturbed distribution will result in the formation of a new potential in the plasma. By virtue of the superposition principle, this new potential is equivalent to the simple superposition of the potential created by the introduced charge q_0 and that created by the displaced plasma particles. The slight displacement of plasma particles is designated by the terms *shielding*, *masking*, or *screening* of the introduced charge, as it depletes the long-range influence of the introduced particle. The perturbation of the distribution of the plasma's constituent particles around the introduced charge is called a *shielding cloud*. It should be emphasized that the polarity of the shielding cloud is opposite to that of q_0. The potential generated by Debye shielding is described by Poisson's equation, whose potential sources are the particle of charge q_0 and its associated shielding cloud:

$$\nabla^2\phi = -\frac{1}{\varepsilon_0}\left(q_0\delta^3(\mathbf{r}-\mathbf{r}_0) + \sum_\sigma n_\sigma(\mathbf{r})q_\sigma\right) \tag{1.4}$$

Provided that the particle is introduced into the plasma with sufficient delicacy, the plasma's response will be determined by the Boltzmann relation, stated by equation (1.3). The perturbation being consequently assumed infinitesimal, we take advantage of this by establishing the following approximation:

$$|q_\sigma\phi| \ll k_\mathrm{B}T_\sigma \implies n_\sigma(\mathbf{r}) \simeq n_\sigma^\square\left(1-\frac{q_\sigma\phi}{k_\mathrm{B}T_\sigma}\right) \tag{1.5}$$

$$\nabla^2\phi = -\frac{1}{\varepsilon_0}\left(q_0\delta^3(\mathbf{r}-\mathbf{r_0}) + \sum_\sigma\left(1-\frac{q_\sigma\phi}{k_\mathrm{B}T_\sigma}\right)n_\sigma^\square q_\sigma\right) \tag{1.6}$$

The initial neutrality of the plasma is determined as follows, consequently simplifying equation (1.6):

$$\sum_\sigma n_\sigma^\square q_\sigma = 0 \implies \nabla^2\phi = -\frac{1}{\varepsilon_0}\left(q_0\delta^3(\mathbf{r}-\mathbf{r}_0) - \sum_\sigma\frac{n_\sigma^\square q_\sigma^2}{k_\mathrm{B}T_\sigma}\phi\right) \tag{1.7}$$

We define the Debye length of species σ and the effective Debye length as follows:

$$\lambda_\sigma^2 = \frac{\varepsilon_0 k_\mathrm{B}T_\sigma}{n_\sigma^\square q_\sigma^2} \implies \frac{1}{\lambda_\mathrm{d}^2} = \sum_\sigma\frac{1}{\lambda_\sigma^2} \tag{1.8}$$

Equation (1.7) simplifies greatly, and we obtain the following partial differential equation:

$$\nabla^2\phi - \frac{1}{\lambda_\mathrm{d}^2}\phi = -\frac{q_0}{\varepsilon_0}\delta^3(\mathbf{r}-\mathbf{r}_0) \tag{1.9}$$

It is possible to solve equation (1.9) by applying Gauss's law. However, a more general approach is to use the Fourier transform to illustrate more formal and powerful mathematical techniques. The first step consists of initiating the Fourier transform in three dimensions:

$$\widetilde{\phi}\,(\mathbf{k}) = \iiint_{\mathbb{R}^3}\phi(\mathbf{r})e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{r} \implies \phi(\mathbf{r}) = \frac{1}{(2\pi)^3}\iiint_{\mathbb{R}^3}\widetilde{\phi}\,(\mathbf{k})e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{k} \tag{1.10}$$

The transform of the Dirac distribution is simply given as follows:

$$\iiint_{\mathbb{R}^3}\delta^3(\mathbf{r}-\mathbf{r}_0)e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{r} = e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}_0} \implies \delta^3(\mathbf{r}-\mathbf{r}_0) = \frac{1}{(2\pi)^3}\iiint_{\mathbb{R}^3}e^{\mathrm{i}\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}_0)}\,\mathrm{d}^3\mathbf{k} \tag{1.11}$$

$$\iiint_{\mathbb{R}^3}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right)e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{r} = (i\mathbf{k})^2\iiint_{\mathbb{R}^3}\phi(\mathbf{r})e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\,\mathrm{d}^3\mathbf{k} \tag{1.12}$$

$$-\left(\mathbf{k}^2+\frac{1}{\lambda_\mathrm{d}^2}\right)\widetilde{\phi}\,(\mathbf{k}) = -\frac{q_0}{\varepsilon_0}e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r_0}} \implies \widetilde{\phi}\,(\mathbf{k}) = \frac{q_0/\varepsilon_0}{\mathbf{k}^2+\lambda_\mathrm{d}^{-2}}e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r_0}} \tag{1.13}$$

We choose the axis $\hat{\mathbf{z}}_k$ of the transformed coordinate system to be oriented in the same direction as $\hat{\mathbf{z}}$ of the original system, i.e., $\mathbf{k}\cdot\mathbf{r} = kr\cos(\vartheta)$ and $\mathbf{k}\cdot\mathbf{r_0} = kr_0$. Let us recall $\mathbf{r}_0 = \mathbf{0}$:

$$\phi(\mathbf{r}) = \frac{1}{(2\pi)^3}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty}\frac{q_0/\varepsilon_0}{k^2+\lambda_\mathrm{d}^{-2}}e^{ikr\cos(\vartheta)}k^2\sin(\vartheta)\,\mathrm{d}k\,\mathrm{d}\vartheta\,\mathrm{d}\varphi = \tag{1.14}$$

$$= \frac{q_0}{(2\pi)^3\varepsilon_0}\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\infty}\frac{k^2}{k^2+\lambda_\mathrm{d}^{-2}}\int_0^{\pi}e^{ikr\cos(\vartheta)}\sin(\vartheta)\,\mathrm{d}\vartheta\,\mathrm{d}k = \tag{1.15}$$

$$= \left\{\int_0^{\pi}e^{ikr\cos(\vartheta)}\sin(\vartheta)\,\mathrm{d}\vartheta = \int_{-1}^{1}e^{ikrt}\,\mathrm{d}t = \left.\frac{e^{ikrt}}{ikr}\right|_{-1}^{1} = \frac{e^{ikr}-e^{-ikr}}{ikr}\right\} = \tag{1.16}$$

$$= \{k\to -k\} = \frac{q_0}{(2\pi)^3\varepsilon_0}\times 2\pi\times\int_{-\infty}^{\infty}\frac{k^2}{k^2+\lambda_\mathrm{d}^{-2}}\frac{e^{ikr}}{ikr}\,\mathrm{d}k = \tag{1.17}$$

$$= \frac{q_0}{4\pi\varepsilon_0 r}\frac{1}{\pi \mathrm{i}}\int_{-\infty}^{\infty}\frac{k}{k^2+\lambda_\mathrm{d}^{-2}}e^{\mathrm{i}kr}\,\mathrm{d}k = \frac{q_0}{4\pi\varepsilon_0 r}\times I_{\lambda_\mathrm{d}}(r) \tag{1.18}$$

To evaluate the last integral, we shall resort to complex analysis, specifically the Cauchy residue theorem. The integrand has two poles, at $k = \mathrm{i}\lambda_\mathrm{d}^{-1}$ and $k = -\mathrm{i}\lambda_\mathrm{d}^{-1}$. The procedure followed is performed on the upper Jordan semicircle:

$$\int_{-\infty}^{\infty}\frac{k\times e^{\mathrm{i}kr}}{\left(k+\mathrm{i}\lambda_\mathrm{d}^{-1}\right)\left(k-\mathrm{i}\lambda_\mathrm{d}^{-1}\right)}\,\mathrm{d}k = 2\pi\mathrm{i}\lim_{k\to\mathrm{i}\lambda_\mathrm{d}^{-1}}\left((k-\mathrm{i}\lambda_\mathrm{d}^{-1})\times\frac{k\times e^{\mathrm{i}kr}}{(k-\mathrm{i}\lambda_\mathrm{d}^{-1})(k+\mathrm{i}\lambda_\mathrm{d}^{-1})}\right) = \tag{1.19}$$

$$= 2\pi\mathrm{i}\lim_{k\to\mathrm{i}\lambda_\mathrm{d}^{-1}}\frac{k\times e^{\mathrm{i}kr}}{k+\mathrm{i}\lambda_\mathrm{d}^{-1}} = 2\pi\mathrm{i}\times\frac{\mathrm{i}\lambda_\mathrm{d}^{-1}e^{-\lambda_\mathrm{d}^{-1}r}}{2\mathrm{i}\lambda_\mathrm{d}^{-1}} = \pi\mathrm{i}\exp\left(-\frac{r}{\lambda_\mathrm{d}}\right) \tag{1.20}$$

The form of the solution is commonly known by the term *Yukawa potential*:

$$\phi(\mathbf{r}) = \frac{q_0}{4\pi\varepsilon_0 r}\frac{1}{\pi\mathrm{i}}\times\pi\mathrm{i}\exp\left(-\frac{r}{\lambda_\mathrm{d}}\right) \implies \phi(r) = \frac{q_0}{4\pi\varepsilon_0 r}\exp\left(-\frac{r}{\lambda_\mathrm{d}}\right) \tag{1.21}$$

It is now appropriate to examine the obtained solution. In the immediate vicinity of the particle of charge q_0, when the inequality $r \ll \lambda_\mathrm{d}$ is well verified, the potential is equivalent to that which this particle exerts in a vacuum. As one moves farther from the introduced particle, the potential vanishes due to screening by the shielding cloud. The validity of Debye screening holds only when the plasma particle density is sufficiently high, that is, when $n_\sigma\lambda_\mathrm{d}^3 \gg 1$. It will be shown later that this condition precisely corresponds to the hypothesis of a weakly collisional plasma, one of the initial approximations in this section. Furthermore, the concept of Debye shielding is only relevant if the Debye length is considerably smaller than the spatial dimensions of the plasma. Otherwise, any plasma particle is necessarily inside the shielding cloud. It should be noted that the introduced particle could be any particle of the plasma. However, from a statistical point of view, such a particle, selected at random, moves at a defined velocity and is no longer characterized by the thermal speed of its species.

1.4 Quasi-neutrality

The study of Debye shielding relies on the hypothesis that the plasma was initially neutral, i.e., that electron and ion densities were equal and uniform at the initial instant. In this section, we demonstrate that the hypothesis we have formulated is excellent. More precisely, the result established in this section is that the plasma exhibits, remarkably, a state of neutrality when the Debye length is microscopic compared to the dimensions of the plasma. Indeed, conventional plasmas manifest a propensity to be quasi-neutral, because they generally do not possess sufficient internal energy to reach a non-neutral state over distances that considerably exceed the Debye length. We will demonstrate the argument by considering a plasma initially in a state of neutrality, with electrons at temperature T_e. Consider a sphere susceptible to spontaneously depleting all its electrons due to thermal fluctuations. We evaluate the maximum radius of such a sphere and denote it R_max. The depletion of electrons causes them to stop at the surface of the sphere. The energy required to create a space devoid of electrons is indeed the energy of the electrostatic field created by the ions remaining inside the sphere. Gauss's law allows us to determine the form of the electrostatic field under consideration, and consequently, the potential energy of the configuration:

$$\nabla\cdot\mathbf{E} \equiv \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2E_r\right) = \frac{\rho}{\varepsilon_0} = \frac{n_e e}{\varepsilon_0} \tag{1.22}$$

where we have omitted the azimuthal and polar terms due to the symmetry. The solution follows trivially:

$$\mathbf{E}(\mathbf{r}) = \frac{n_e e}{3\varepsilon_0}\mathbf{r} \tag{1.23}$$

$$W = \frac{\varepsilon_0}{2}\iiint_{B(\mathbf{0},R)}\mathbf{E}^2\,\mathrm{d}^3\mathbf{r} = 2\pi\varepsilon_0\times\frac{e^2n_e^2}{9\varepsilon_0^2}\int_0^{R_\mathrm{max}}r^4\,\mathrm{d}r = \frac{2\pi e^2n_e^2}{45\varepsilon_0}R_\mathrm{max}^5 \tag{1.24}$$

The potential energy is equal to the net initial kinetic energy of all evacuated electrons, hence:

$$K = \frac{3}{2}n_ek_\mathrm{B}T\times\frac{4}{3}\pi R_\mathrm{max}^3 = 2\pi k_\mathrm{B}Tn_eR_\mathrm{max}^3 \tag{1.25}$$

$$R_{\max}^2 = 45\frac{\varepsilon_0 k_{\mathrm{B}} T}{e^2 n_e} = 45\lambda_e^2 \tag{1.26}$$

The estimation $R_{\max} \lesssim 7\lambda_e$ follows. The maximum radius of a space completely devoid of electrons is, for its part, limited to a few Debye lengths. Considering the extremely low probability that the velocities of all these electrons are directed radially relative to a common centre, we deduce that plasmas are indeed quasi-neutral on a scale that largely exceeds their Debye length. It emerges from this analysis that the extent of the non-neutral region caused by a potential perturbation, henceforth termed *sheath*, does not exceed the order of the associated Debye length.

1.5 Collisions in plasma

Let us first analyse binary Coulomb collisions. Let there be two particles of possibly different species σ and ρ, masses m_σ and m_ρ, charges q_σ and q_ρ, as well as $\mathbf{r}_\sigma$ and $\mathbf{r}_\rho$ as their initial positions:

$$m_\sigma \frac{\mathrm{d}^2\mathbf{r}_\sigma}{\mathrm{d}t^2} = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0}\frac{\mathbf{r}_\sigma - \mathbf{r}_\rho}{\|\mathbf{r}_\sigma - \mathbf{r}_\rho\|^3} \qquad m_\rho \frac{\mathrm{d}^2\mathbf{r}_\rho}{\mathrm{d}t^2} = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0}\frac{\mathbf{r}_\rho - \mathbf{r}_\sigma}{\|\mathbf{r}_\rho - \mathbf{r}_\sigma\|} \tag{1.27}$$

Let $\mathbf{v}_\sigma$ be the velocity of the first particle in the frame of the other, when the two particles are infinitely far from each other. The relative position between the two particles verifies the following equations:

$$\mathbf{r} = \mathbf{r}_\sigma - \mathbf{r}_\rho \implies \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = \left(\frac{1}{m_\sigma} + \frac{1}{m_\rho}\right)\frac{q_\sigma q_\rho}{4\pi\varepsilon_0}\frac{\mathbf{r}}{\|\mathbf{r}\|^3} \qquad \mathbf{r}_\sigma = \mathbf{R}_{\mathrm{CM}} + \frac{m_\rho}{m_\sigma + m_\rho}\mathbf{r} \tag{1.28}$$

hence the definition of the reduced mass $\mu_{\sigma\rho} = m_\sigma m_\rho / (m_\sigma + m_\rho)$. The dynamics of the first particle is thus identical to that of a particle of mass $\mu_{\sigma\rho}$ and charge q_σ around a fixed source of charge q_ρ, which is equivalent to considering the problem in the centre-of-mass frame:

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Figure 1.2: Illustration of the binary Coulomb collision problem.

Let us define the axis $\hat{\mathbf{z}} = \mathbf{v}_\sigma / \|\mathbf{v}_\sigma\|$ and the axes $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$ as in figure (1.1). We will exploit the conservation of the Laplace-Runge-Lenz vector, valid for the electrostatic force, a central force:

$$\boldsymbol{\epsilon}_\sigma = -\frac{4\pi\varepsilon_0}{\mu_{\sigma\rho} q_\sigma q_\rho}\mathbf{p}_\sigma \times \mathcal{L}_\sigma - \hat{\mathbf{r}} = -\frac{4\pi\varepsilon_0}{\mu_{\sigma\rho} q_\sigma q_\rho}(\mu_{\sigma\rho} v_\sigma \hat{\mathbf{z}}) \times (-\mu_{\sigma\rho} v_\sigma b\, \hat{\mathbf{y}}) - (-\hat{\mathbf{z}}) = -\frac{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b}{q_\sigma q_\rho}\hat{\mathbf{x}} + \hat{\mathbf{z}} \tag{1.29}$$

The norm of the exiting velocity at infinity is equal to the incident, $\|\mathbf{v}_\sigma\|$, by conservation of energy. Note that the angular momentum is constant because the electrostatic force is central. The Laplace-Runge-Lenz vector when $t \to \infty$ verifies:

$$\boldsymbol{\epsilon}_\sigma = -\frac{4\pi\varepsilon_0}{\mu_{\sigma\rho} q_\sigma q_\rho}\left[\mu_{\sigma\rho}\left(v_\sigma \cos(\theta_c)\,\hat{\mathbf{z}} + v_\sigma \sin(\theta_c)\,\hat{\mathbf{x}}\right)\right] \times (-\mu_{\sigma\rho} v_\sigma b\,\hat{\mathbf{y}}) - \left[\cos(\theta_c)\,\hat{\mathbf{z}} + \sin(\theta_c)\,\hat{\mathbf{x}}\right] = \tag{1.30}$$

$$= \left[\sin(\theta_c) - \frac{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b}{q_\sigma q_\rho}\cos(\theta_c)\right]\hat{\mathbf{x}} + \left[\frac{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b}{q_\sigma q_\rho}\sin(\theta_c) - \cos(\theta_c)\right]\hat{\mathbf{z}} \tag{1.31}$$

By comparing equations (1.29) and (1.31), in particular the components along $\hat{\mathbf{z}}$, we can establish the expression for the deflection angle θ_c:

$$\sin(\theta_c) = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b}\left[1 + \cos(\theta_c)\right] \iff \tan\left(\frac{\theta_c}{2}\right) = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b} \tag{1.32}$$

Let us define by $b_{\pi/2}$ the impact parameter corresponding to a deflection of $\theta_c = \pi/2$, then:

$$b_{\pi/2} = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2} \tag{1.33}$$

The scattering angle in the laboratory frame, defined as the angle between the outgoing and incident velocities, is, however, different. We can write the outgoing velocities in both frames, and we apologize for any confusion due to the following notations:

$$\mathbf{v}'\big|_{\text{lab}} = \frac{\mathrm{d}\mathbf{R}_{\text{CM}}}{\mathrm{d}t} + \frac{m_\rho}{m_\sigma + m_\rho}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \left[v_{\text{CM}} + \frac{m_\rho v_\sigma}{m_\sigma + m_\rho}\cos(\theta_c)\right]\hat{\mathbf{z}} + \frac{m_\rho v_\sigma}{m_\sigma + m_\rho}\sin(\theta_c)\,\hat{\mathbf{x}} \tag{1.34}$$

from which we can determine exactly the deflection angle χ_c in the laboratory frame:

$$\cot(\chi_c) = \frac{v_{\text{CM}} + \frac{m_\rho v_\sigma}{m_\sigma+m_\rho}\cos(\theta_c)}{\frac{m_\rho v_\sigma}{m_\sigma+m_\rho}\sin(\theta_c)} = \frac{v_{\text{CM}}}{v_\sigma}\frac{m_\sigma + m_\rho}{m_\rho}\csc(\theta_c) + \cot(\theta_c) \tag{1.35}$$

$$v_{\text{CM}} = \frac{m_\sigma v_\sigma}{m_\sigma + m_\rho} \implies \cot(\chi_c) = \frac{m_\sigma}{m_\rho}\csc(\theta_c) + \cot(\theta_c) \tag{1.36}$$

1.5.1 Digression: Cross-sections

The cross-section associated with a particular collisional phenomenon is the probability that the phenomenon occurs. It is a surface area associated with each target particle of the phenomenon. The cross-section allows the probability of the phenomenon in question to be illustrated as the ratio of the interaction area to the density of target particles n along a segment $\mathrm{d}\ell$ of their path through a surface $\mathrm{d}S$. As an illustrative example, let us determine the mean free path of a stationary ideal gas, i.e., the average distance travelled by a particle between collisions. Let $\Phi(t)$ be the flux of particles entering through any surface. The flux of particles exiting the same surface after traversing a distance $\mathrm{d}\ell = v\,\mathrm{d}t$ is reduced by a multiplicative factor of $1 - n\sigma v\,\mathrm{d}t$:

$$\Phi(t + \mathrm{d}t) = (1 - n\sigma v\,\mathrm{d}t)\Phi(t) \implies \mathrm{d}\Phi = -\Phi n\sigma v\,\mathrm{d}t \tag{1.37}$$

We thus find the following expression, which allows us to define the mean time between two collisions τ:

$$\Phi(t) = \Phi(0)e^{-n\sigma v t} \iff \tau = \frac{1}{n\sigma v} \tag{1.38}$$

The mean free path is thus found by multiplying τ by the mean speed:

$$\lambda = \langle v\rangle\tau = \frac{\langle v\rangle}{n\sigma v} \tag{1.39}$$

It is now necessary to choose a value v to formulate a statistically significant quantity. Let us consider two groups of particles, the first of which moves at velocity $\mathbf{v}$, while the second moves at velocity $\mathbf{u}$. Focusing on the frame of the second group, the particles offer a cross-section given by $n\sigma f(\mathbf{u})\,\mathrm{d}^3\mathbf{u}$ where $f(\mathbf{u})$ simply represents the velocity distribution. The volume swept by the target particles after a time $\mathrm{d}t$ is given by $n\sigma\|\mathbf{v} - \mathbf{u}\| f(\mathbf{u})\,\mathrm{d}^3\mathbf{u}\,\mathrm{d}t$. The number of encounters is determined by multiplying the latter by the probability of finding a particle belonging to the first group in the swept volume:

$$n\sigma\|\mathbf{v} - \mathbf{u}\| f(\mathbf{u})\,\mathrm{d}^3\mathbf{u}\, f(\mathbf{v})\,\mathrm{d}^3\mathbf{v}\,\mathrm{d}t \tag{1.40}$$

The collision rate is found by integrating over all possible velocities $\mathbf{v}$ and $\mathbf{u}$:

$$\frac{\mathrm{d}N_{\text{coll}}}{\mathrm{d}t} = \iiint_{\mathbb{R}^3}\iiint_{\mathbb{R}^3} n\sigma\|\mathbf{v} - \mathbf{u}\| f(\mathbf{u}) f(\mathbf{v})\,\mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{u} \tag{1.41}$$

Using the following substitution $\mathbf{x} = (\mathbf{v} - \mathbf{u})/\sqrt{2}$ and $\mathbf{y} = (\mathbf{v} + \mathbf{u})/\sqrt{2}$, we establish:

$$\frac{\mathrm{d}N_{\text{coll}}}{\mathrm{d}t} = n\sigma\sqrt{2}\iiint_{\mathbb{R}^3}\|\mathbf{x}\| f(\mathbf{x})\,\mathrm{d}^3\mathbf{x}\iiint_{\mathbb{R}^3} f(\mathbf{y})\,\mathrm{d}^3\mathbf{y} = n\sigma\sqrt{2}\times\langle v\rangle\times 1 \tag{1.42}$$

$$\lambda = \langle v\rangle \bigg/ \frac{\mathrm{d}N_{\text{coll}}}{\mathrm{d}t} = \frac{1}{n\sigma\sqrt{2}} \tag{1.43}$$

We remain in the frame of the particle of species ρ. Let us specifically examine an annular surface area $2\pi b\,\mathrm{d}b$ through which incident particles propagate, undergoing an angular deflection between θ_c and $\theta_c + \mathrm{d}\theta_c$. The deflected particles then pierce the solid angle $\mathrm{d}\Omega = 2\pi\sin(\theta_c)\mathrm{d}\theta_c$. By conservation of the net number of particles, the following relation is established:

$$\frac{\mathrm{d}\sigma(\theta_c)}{\mathrm{d}\Omega}\mathrm{d}\Omega = 2\pi b\,\mathrm{d}b \implies \frac{\mathrm{d}\sigma(\theta_c)}{\mathrm{d}\Omega} = \frac{b}{\sin(\theta_c)}\frac{\mathrm{d}b}{\mathrm{d}\theta_c} \tag{1.44}$$

The sought derivative is found via eq. (1.32), giving the *Rutherford cross-section* relation:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}(\theta_c) = \frac{1}{4}\left(\frac{q_1 q_2}{4\pi\varepsilon_0 \mu v_0^2}\csc^2\left(\frac{\theta_c}{2}\right)\right)^2 = \frac{1}{4}\left(\frac{b_{\pi/2}}{\sin^2(\theta_c/2)}\right)^2 \tag{1.45}$$

1.5.2 Collisions with small and large angles of deflection

At the core of this section, we examine the validity of the approximation that plasmas are weakly collisional, i.e., that collisions resulting in small deflections, whose cumulative effect is equivalent to a collision resulting in a large deflection, are vastly more frequent than those resulting in a large deflection. Let us first analyse the statistics of collisions that cause small deflections, then proceed to the proof. For weak deflections, we take advantage by making the small-angle approximation and defining the deviation of each collision $\Delta\theta$:

$$\tan\left(\frac{\Delta\theta}{2}\right) \simeq \frac{\Delta\theta}{2} \Longleftrightarrow \Delta\theta = 2\frac{b_{\pi/2}}{b} = \frac{q_\sigma q_\rho}{2\pi\varepsilon_0 \mu_{\sigma\rho} v_\sigma^2 b} \tag{1.46}$$

The average of $\Delta\theta$, following the random walk statistics, is zero, while that of $\Delta(\theta^2)$ is:

$$\langle\Delta(\theta^2)\rangle = \frac{1}{\pi b_{\max}^2}\int_{b_{\min}}^{b_{\max}} \Delta(\theta^2)\, 2\pi b\,\mathrm{d}b = \frac{8\pi b_{\pi/2}^2}{\pi b_{\max}^2}\int_{b_{\min}}^{b_{\max}} \frac{\mathrm{d}b}{b} \tag{1.47}$$

The question of choosing $b_{\max}$ arises. We find inspiration in the context of Debye shielding. Observe why. Consequently, we choose $b_{\max} = \lambda_{\mathrm{d}}$, as well as $b_{\min} = b_{\pi/2}$:

$$\langle\Delta(\theta^2)\rangle = \frac{8 b_{\pi/2}^2}{b_{\max}^2}\ln(\Lambda) \quad \text{where} \quad \Lambda = \frac{\lambda_{\mathrm{d}}}{b_{\pi/2}} \tag{1.48}$$

It should, however, be noted that $b_{\max}$ and $b_{\frac{\pi}{2}}$ depend on the particles constituting the considered plasma. At time t, an electron undergoes $n_c \pi b_{\max}^2 vt$ collisions with the target ions of the plasma. Since the deflection and collision frequency processes are independent, we derive the following expression:

$$\langle\Delta_{\mathrm{net}}(\theta^2)\rangle(t) = \langle\Delta(\theta)^2\rangle n_i \pi b_{\max}^2 vt = 8\pi n_i b_{\frac{\pi}{2}}^2 vt \tag{1.49}$$

Using the last relation, we define the mean time necessary to have a net angular deflection equal to 90° due to collisions resulting in small deflections:

$$\left(\frac{\pi}{2}\right)^2 = 8\pi n_i b_{\frac{\pi}{2}}^2 \langle v\rangle \tau_{\frac{\pi}{2}} \implies \tau_{\frac{\pi}{2}} = \frac{\pi}{32 n_i b_0^2 \ln(\Lambda)\langle v\rangle} \tag{1.50}$$

Let us now consider the net number of collisions N necessary to produce a deflection verifying $N\langle\Delta(\theta^2)\rangle \simeq 1$. We achieve this by considering an arbitrary particle and observing the incident particle flux density, $\mathbf{\Phi} = n_\sigma(\mathbf{v} - \mathbf{u})$, where $\mathbf{u}$ represents the velocity of the chosen particle in the laboratory frame. The number of collisions resulting in small deflections is simply:

$$N\langle\Delta\theta\rangle = \int_0^t \int_{b_{\pi/2}}^{b_{\max}} \iiint_{\mathbb{R}^3}\iiint_{\mathbb{R}^3} 2\pi n_\sigma b\,[\Delta\theta(b)]^2\, \|\mathbf{v} - \mathbf{u}\| f(\mathbf{v}) f(\mathbf{u})\,\mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{u}\,\mathrm{d}b\,\mathrm{d}t \tag{1.51}$$

By a previous result, specifically, equation (1.42), we find:

$$N\langle\Delta(\theta^2)\rangle = n_\sigma\langle v\rangle t\sqrt{2}\int_{b_{\pi/2}}^{b_{\max}} 2\pi b\,\Delta(\theta)^2(b)\,\mathrm{d}b \overset{\text{def}}{=} n_\sigma \sigma_{\mathrm{effc}}\langle v\rangle t\sqrt{2} \tag{1.52}$$

hence the expression of the effective area, which corresponds to the cumulative effect of collisions resulting in small deflections:

$$\sigma_{\mathrm{eff}} = 2\pi\int_{b_{\pi/2}}^{\lambda_{\mathrm{d}}} b\left(\frac{q_\sigma q_\rho}{2\pi\varepsilon_0\mu_{\sigma\rho}v_\sigma^2 b}\right)\mathrm{d}b = 8\pi b_{\frac{\pi}{2}}^2 \ln\left(\frac{\lambda_{\mathrm{d}}}{b_{\pi/2}}\right) = \frac{1}{2\pi}\left(\frac{q_\sigma q_\rho}{\varepsilon_0 \mu v_0^2}\right)^2 \ln\left(\frac{\lambda_{\mathrm{d}}}{b_{\pi/2}}\right) \tag{1.53}$$

The cross-section of collisions resulting in the cumulative effect of small deflections will largely exceed that of a collision resulting once in a large deflection, provided that:

$$\lambda_\mathrm{d} \gg b_{\frac{\pi}{2}} \implies \left\{ b_{\frac{\pi}{2}} \simeq \frac{1}{2n\lambda_\mathrm{d}^2} \right\} \implies n\lambda_\mathrm{d}^3 \gg 1 \tag{1.54}$$

which corresponds precisely to the criterion of having a large number of particles in a sphere of radius λ_d, commonly called the *Debye sphere*. It should be noted that the cumulative effective cross-section decreases approximately as v^{-4}. In high-temperature plasmas where velocities reach large values, the cumulative effective cross-section proves negligible, and Coulomb collisions thus become far less significant than other plasma phenomena.

1.5.3 Slowing down of a mono-energetic beam

Nota bene: It is at the reader's discretion to consult section 9.1 to understand the following sections.

We consider a mono-energetic beam composed of particles of species σ in a quasi-neutral plasma, consisting of ions and electrons at different temperatures, each following a Maxwell distribution. The beam is defined by the following distribution function:

$$f_\sigma(\mathbf{v}_\sigma) = n_\sigma \delta(\mathbf{v}_\sigma - \mathbf{u}_\sigma) \quad \text{and} \quad f_\rho = n_\rho \left(\frac{m_\rho}{2\pi k_\mathrm{B} T_\rho} \right)^{3/2} \exp\left[-\frac{m_\rho \mathbf{v}_\rho^2}{2k_\mathrm{B} T_\rho} \right] \quad \text{where } \rho = \mathrm{i}, \mathrm{e} \tag{1.55}$$

Note that we shall employ here results from chapter 9.1 concerning the Fokker-Planck collision theory. The Rosenbluth h-potential of species ρ reads:

$$h_\rho(\mathbf{v}_\sigma) = \frac{n_\rho m_\sigma}{\mu_{\sigma\rho} \|\mathbf{v}_\sigma\|} \mathrm{erf}\left(\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}} \right) \tag{1.56}$$

It is appropriate to introduce the following variable, which aims to express the ratio between the velocity of incident particles and the characteristic velocity of target particles:

$$\zeta_\rho = \|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}} \implies \left. \frac{\partial h_\rho}{\partial \mathbf{v}_\sigma} \right|_{\mathbf{u}_\sigma} = \frac{n_\rho m_\sigma}{\mu_{\sigma\rho}} \frac{m_\rho}{2k_\mathrm{B} T_\rho} \frac{\partial}{\partial \zeta_\rho} \left. \left(\frac{\mathrm{erf}\ (\zeta_\rho)}{\zeta_\rho} \right) \right|_{\mathbf{u}_\sigma} \tag{1.57}$$

The following two limiting cases are possible, considering the ratio between these two velocities:

1. $\zeta_\rho \ll 1 \implies$ The velocity of the beam particles of species σ is much lower than the characteristic thermal velocity of particles of species ρ.
2. $\zeta_\rho \gg 1 \implies$ The velocity of the beam particles of species σ is much greater than the characteristic thermal velocity of particles of species ρ.

It is then convenient to consider the limiting cases of the error function:

$$\lim_{x \ll 1} \mathrm{erf}\ (x) \simeq \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3} \right) \quad \text{and} \quad \lim_{x \gg 1} \mathrm{erf}\ (x) \simeq 1 \tag{1.58}$$

$$\lim_{x \ll 1} \left[-\frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\mathrm{erf}\ (x)}{x} \right) \right] \simeq \frac{4x}{3\sqrt{\pi}} \quad \text{and} \quad \lim_{x \gg 1} \left[-\frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{\mathrm{erf}\ (x)}{x} \right) \right] \simeq \frac{1}{x^2} \tag{1.59}$$

Applying everything to our consideration of a plasma composed of ions and electrons, it follows:

$$\begin{aligned} \frac{\partial u_\sigma}{\partial t} = &-\frac{n_\mathrm{i} q_\sigma^2 q_\mathrm{i}^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_\sigma^2} \left(1 + \frac{m_\sigma}{m_\mathrm{i}} \right) \left[\frac{\mathrm{d}}{\mathrm{d}u_\sigma} \left(\frac{1}{u_\sigma} \mathrm{erf}\ \left(u_\sigma \sqrt{\frac{m_\mathrm{i}}{2k_\mathrm{B} T_\mathrm{i}}} \right) \right) \right] + \\ &-\frac{n_\mathrm{e} q_\sigma^2 e^2 \ln \Lambda}{4\pi \varepsilon_0^2 m_\sigma^2} \left(1 + \frac{m_\sigma}{m_\mathrm{e}} \right) \left[\frac{\mathrm{d}}{\mathrm{d}u_\sigma} \left(\frac{1}{u_\sigma} \mathrm{erf}\ \left(u_\sigma \sqrt{\frac{m_\mathrm{e}}{2k_\mathrm{B} T_\mathrm{e}}} \right) \right) \right] \end{aligned} \tag{1.60}$$

Due to the quasi-neutrality of the plasma under consideration, $n_\mathrm{i} q_\mathrm{i} = n_\mathrm{i} Z e \simeq n_\mathrm{e} e$, hence the following:

$$\frac{\partial u_\sigma}{\partial t} = -\frac{n_\mathrm{e} e^2 \ln \Lambda}{4\pi \varepsilon_0^2} \frac{q_\sigma^2}{m_\sigma^2} \left[Z \left(1 + \frac{m_\sigma}{m_\mathrm{i}} \right) \frac{m_\mathrm{i}}{2k_\mathrm{B} T_\mathrm{i}} \frac{\mathrm{d}}{\mathrm{d}\xi_\mathrm{i}} \left(\frac{\mathrm{erf}\ (\xi_\mathrm{i})}{\xi_\mathrm{i}} \right) + \left(1 + \frac{m_\sigma}{m_\mathrm{e}} \right) \frac{m_\mathrm{e}}{2k_\mathrm{B} T_\mathrm{e}} \frac{\mathrm{d}}{\mathrm{d}\xi_\mathrm{e}} \left(\frac{\mathrm{erf}\ (\xi_\mathrm{e})}{\xi_\mathrm{e}} \right) \right] \tag{1.61}$$

A mono-energetic beam of very low energy particles corresponds to the first limiting case, i.e., $\zeta_\mathrm{i} \ll 1$ and $\zeta_\mathrm{e} \ll 1$. The slowing strongly depends on the temperatures of the ions and electrons, and the collision frequency does not depend on the velocities of the beam particles. This is explained by the fact that incident particles, having a negligible velocity compared to the characteristic velocities of the plasma's constituent particles, encounter a number of target particles that weakly depends on their velocity:

$$\frac{\partial u_\sigma}{\partial t} \simeq -\frac{n_\mathrm{e} e^2 u_\sigma \ln\Lambda}{3\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\left[Z\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)\left(\frac{m_\mathrm{i}}{2\pi k_\mathrm{B} T_\mathrm{i}}\right)^{3/2}+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\left(\frac{m_\mathrm{e}}{2\pi k_\mathrm{B} T_\mathrm{e}}\right)^{3/2}\right] \tag{1.62}$$

$$\nu_{\sigma\to\mathrm{i,e}} \simeq \frac{n_\mathrm{e} e^2 \ln\Lambda}{3\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\left[Z\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)\left(\frac{m_\mathrm{i}}{2\pi k_\mathrm{B} T_\mathrm{i}}\right)^{3/2}+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\left(\frac{m_\mathrm{e}}{2\pi k_\mathrm{B} T_\mathrm{e}}\right)^{3/2}\right] \tag{1.63}$$

Conversely, the second case, where ζ_i and $\zeta_\mathrm{e} \gg 1$, corresponds to a mono-energetic beam of high-energy particles. The slowing down depends on ion and electron temperatures only through the logarithmic term $\ln(\lambda_\mathrm{d}/b_{\pi/2})$, which grows slowly with these temperatures. The collision frequency, however, depends on the cube of the velocity of the beam particles, as indicated in chapter 9.1:

$$\frac{\partial u_\sigma}{\partial t} \simeq -\frac{n_\mathrm{e} e^2 \ln\Lambda}{4\pi\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\frac{1}{u_\sigma^2}\left[Z\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\right] \tag{1.64}$$

$$\nu_{\sigma\to\mathrm{i,e}} \simeq \frac{n_\mathrm{e} e^2 \ln\Lambda}{4\pi\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\frac{1}{u_\sigma^3}\left[Z\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\right] \tag{1.65}$$

It is also appropriate to consider the case where the velocity of the beam particles exceeds that of the ions, but remains negligible compared to that of the electrons, which is motivated by the large mass difference between ions and electrons, $m_\mathrm{i} \gg m_\mathrm{e}$, translating to the following limits: $\zeta_\mathrm{e} \ll 1$ and $\zeta_\mathrm{i} \gg 1$.

$$\frac{\partial u_\sigma}{\partial t} \simeq -\frac{n_\mathrm{e} e^2 \ln\Lambda}{4\pi\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\left[\frac{4Zu_\sigma}{3\sqrt{\pi}}\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)\left(\frac{m_\mathrm{i}}{2k_\mathrm{B} T_\mathrm{i}}\right)^{3/2}+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\frac{1}{u_\sigma^2}\right] \tag{1.66}$$

$$\nu_{\sigma\to\mathrm{i,e}} \simeq \frac{n_\mathrm{e} e^2 \ln\Lambda}{4\pi\varepsilon_0^2}\frac{q_\sigma^2}{m_\sigma^2}\left[\frac{4Z}{3\sqrt{\pi}}\left(1+\frac{m_\sigma}{m_\mathrm{i}}\right)\left(\frac{m_\mathrm{i}}{2k_\mathrm{B} T_\mathrm{i}}\right)^{3/2}+\left(1+\frac{m_\sigma}{m_\mathrm{e}}\right)\frac{1}{u_\sigma^3}\right] \tag{1.67}$$

1.5.4 Types and frequencies of momentum dissipation

We refer to momentum dissipation as the process of the slowing down of a flow of some plasma species. Among the fundamental physical constants that significantly influence plasma behaviour, one is the ratio of the masses of the constituent particles, particularly the electron mass and the masses of various ions. This ratio frequently attains high values because ions have at least one proton, leading to distinct behaviours attributed to the different species of constituent particles. It must be recognized that, in certain circumstances, plasmas can be clearly characterized by a single species, while other species exert negligible or even zero influence. The mass ratio affects the following two relaxation processes relating to momentum dissipation:

1. momentum dissipation of an incident particle due to collisions with particles of the same species, for example, electron-electron or ion-ion;
2. momentum dissipation of an incident particle due to collisions with particles of different species, for illustration, electron-ion, which underpins electron scattering on ions, as well as ion-electron, which underpins the reciprocal.

Momentum dissipation of plasma species σ is here considered as the decay rate of the macroscopic velocity of said species, thus characterised by the collision frequency in the expression (9.80). The different collision frequencies according to the type of momentum dissipation in a typical plasma consisting of an ion and an electron species are denoted as follows: $\nu_{\mathrm{e}\to\mathrm{e}}$, $\nu_{\mathrm{e}\to\mathrm{i}}$, $\nu_{\mathrm{i}\to\mathrm{i}}$ and $\nu_{\mathrm{i}\to\mathrm{e}}$.

1.5.4.1 Electron-electron collisions

We shall now employ our considerations from the Fokker-Planck theory in chapter 9.1. In the case of electron-electron collisions, we simply impose $T_\sigma = T_\rho = T_\mathrm{e}$ and $m_\sigma = m_\rho = m_\mathrm{e}$, giving the following result for the momentum dissipation frequency among electrons:

$$\nu_{\mathrm{e}\to\mathrm{e}} = \frac{n_\mathrm{e} e^4 \ln\Lambda}{12\varepsilon_0^2\sqrt{\pi^3 m_\mathrm{e} k_\mathrm{B}^3 T_\mathrm{e}^3}} \tag{1.68}$$

1.5.4.2 Ion-ion collisions

In the case of ion-ion collisions, we impose $T_\sigma = T_\rho = T_\mathrm{i}$ and $m_\sigma = m_\rho = m_\mathrm{i}$, along with $q_\mathrm{i} = Ze$ so that quasi-neutrality gives:

$$\nu_{\mathrm{i}\to\mathrm{i}} = \frac{Z^4 n_\mathrm{i} e^4 \ln\Lambda}{12\varepsilon_0^2\sqrt{\pi^3 m_\mathrm{i} k_\mathrm{B}^3 T_\mathrm{i}^3}} = \frac{Z^3 n_\mathrm{e} e^4 \ln\Lambda}{12\varepsilon_0^2\sqrt{\pi^3 m_\mathrm{i} k_\mathrm{B}^3 T_\mathrm{i}^3}} \tag{1.69}$$

1.5.4.3 Electron-ion collisions

In the case of electron-ion collisions, we shall impose $T_\sigma = T_\mathrm{e}$, $T_\rho = T_\mathrm{i}$, $m_\sigma = m_\mathrm{e}$ and $m_\rho = m_\mathrm{i}$. Quasi-neutrality also gives $Zn_\mathrm{i} \simeq n_\mathrm{e}$. We shall also employ the fact that $m_\mathrm{e} \ll m_\mathrm{i}$ and generally $T_\mathrm{i} \lesssim T_\mathrm{e}$:

$$\nu_{\mathrm{e}\to\mathrm{i}} \simeq \frac{Z n_\mathrm{e} e^4 \ln\Lambda}{6\varepsilon_0^2\sqrt{2\pi^3 m_\mathrm{e} k_\mathrm{B}^3 T_\mathrm{e}^3}} \tag{1.70}$$

1.5.4.4 Ion-electron collisions

In the case of ion-electron collisions, we may impose the reciprocal of the previous, or simply use momentum conservation as shown in equation (9.83):

$$\nu_{\mathrm{i}\to\mathrm{e}} \simeq \frac{Zm_\mathrm{e}}{m_\mathrm{i}}\frac{Z n_\mathrm{e} e^4 \ln\Lambda}{6\varepsilon_0^2\sqrt{2\pi^3 m_\mathrm{e} k_\mathrm{B}^3 T_\mathrm{e}^3}} = \frac{Z^2 n_\mathrm{e} e^4 \sqrt{m_\mathrm{e}}\ln\Lambda}{6\varepsilon_0^2 m_\mathrm{i}\sqrt{2\pi^3 k_\mathrm{B}^3 T_\mathrm{e}^3}} \tag{1.71}$$

1.5.4.5 Comparison of momentum dissipation frequencies

We conclude the study of momentum dissipation with the following comparisons:

$$\nu_{\mathrm{i}\to\mathrm{i}} = Z^3\sqrt{\frac{m_\mathrm{e} T_\mathrm{e}^3}{m_\mathrm{i} T_\mathrm{i}^3}}\nu_{\mathrm{e}\to\mathrm{e}} \qquad \nu_{\mathrm{e}\to\mathrm{i}} \simeq Z\sqrt{2}\nu_{\mathrm{e}\to\mathrm{e}} \qquad \nu_{\mathrm{i}\to\mathrm{e}} = \frac{Zm_\mathrm{e}}{m_\mathrm{i}}\nu_{\mathrm{e}\to\mathrm{i}} \simeq \frac{Z^2 m_\mathrm{e}\sqrt{2}}{m_\mathrm{i}}\nu_{\mathrm{e}\to\mathrm{e}} \tag{1.72}$$

Notice that these relations show that specific collision frequencies reside at particular time scales. Electron-electron and electron-ion collision frequencies are thus almost equally fast, then at a time scale $\sqrt{m_\mathrm{i}/m_\mathrm{e}}$ larger come ion-ion frequencies, and lastly, the ion-electron frequency at a time scale of order $m_\mathrm{i}/m_\mathrm{e}$ longer than electron-electron frequency. Hence, we conclude that electrons diffuse momentum amongst themselves equally fast as they diffuse momentum to ions, then after a time scale $\sqrt{m_\mathrm{i}/m_\mathrm{e}}$ larger, ions diffuse momentum amongst themselves, and only at a time scale $m_\mathrm{i}/m_\mathrm{e}$ larger than time scales corresponding to electron-electron collisions will ions diffuse momentum to electrons.

In weakly ionized plasmas, it is also necessary to account for collisions involving neutral particles. Since these collisions are short-ranged, they can be modelled by the rigid sphere potential. It is nevertheless necessary to recognize the possibility of inelastic collisions, linked to the excitation and ionization of neutral particles, which are likely to occur. In this case, we speak of inelastic diffusion of kinetic energy. Another plausible phenomenon is charge exchange, which occurs when an incident ion collides with a neutral particle, ionizing the neutral particle and transferring charge to the ion. Charge exchange is a widely used technique for producing energetic neutral particle beams. Ions and neutral particles differ very little in mass, so these two species tend to reach a mutual thermal equilibrium when the plasma is weakly ionized. Using the fact that neutral particles efficiently reach thermal equilibrium with the container walls, ions in plasmas are generally cold.

1.6 Simple transport phenomena

The discussion of inter-particle collisions enables us to study transport phenomena in a plasma, particularly the coefficients that characterize them. The analysis of particle diffusion will be addressed with particular attention once we have examined the dynamics of these particles in various forms of electromagnetic fields.

1.6.1 Electrical resistivity

A uniform electric field in a plasma accelerates ions and electrons. The accelerated particles collide, generating a friction force between the two species. A steady state may be established where the friction force compensates for the electrostatic force. Due to quasi-neutrality, the net density of the electrostatic force within the plasma is zero. The centre of mass velocity is therefore conserved. Furthermore, electrons can exchange momentum only with ions, and vice versa, implying that electrical resistivity depends solely on interspecies collisions. We postulate that the electric field leads to a simple modification of the Maxwellian distribution of the two species:

$$f_{\mathrm{i}}(\mathbf{v}_{\mathrm{i}}) = n_{\mathrm{i}} \left(\frac{m_{\mathrm{i}}}{2\pi k_{\mathrm{B}} T_{\mathrm{i}}}\right)^{3/2} \exp\left[-\frac{m_{\mathrm{i}}(\mathbf{v}_{\mathrm{i}} - \mathbf{u}_{\mathrm{i}})^2}{2k_{\mathrm{B}} T_{\mathrm{i}}}\right] \tag{1.73}$$

$$f_{\mathrm{e}}(\mathbf{v}_{\mathrm{e}}) = n_{\mathrm{e}} \left(\frac{m_{\mathrm{e}}}{2\pi k_{\mathrm{B}} T_{\mathrm{e}}}\right)^{3/2} \exp\left[-\frac{m_{\mathrm{e}}(\mathbf{v}_{\mathrm{e}} - \mathbf{u}_{\mathrm{e}})^2}{2k_{\mathrm{B}} T_{\mathrm{e}}}\right] \tag{1.74}$$

where $\mathbf{u}_{\mathrm{i}}$ and $\mathbf{u}_{\mathrm{e}}$ are the drift velocities of ions and electrons, respectively, and, by assumption, the mean velocities of the two species. The current density is, by definition, given by:

$$\mathbf{J} = n_{\mathrm{i}} q_{\mathrm{i}} \mathbf{u}_{\mathrm{i}} + n_{\mathrm{e}} q_{\mathrm{e}} \mathbf{u}_{\mathrm{e}} \tag{1.75}$$

It is appropriate to consider the motions of both species in the electron frame where their mean velocity is zero, while the ion velocity is $\mathbf{u}_{\mathrm{rel}} = \mathbf{u}_{\mathrm{i}} - \mathbf{u}_{\mathrm{e}}$. Due to quasi-neutrality, the current density is the same in this frame. The distribution functions will be transformed as follows:

$$f_{\mathrm{i}}(\mathbf{v}_{\mathrm{i}}) = n_{\mathrm{i}} \left(\frac{m_{\mathrm{i}}}{2\pi k_{\mathrm{B}} T_{\mathrm{i}}}\right)^{3/2} \exp\left[-\frac{m_{\mathrm{i}}\left(\mathbf{v}_{\mathrm{i}} - \mathbf{u}_{\mathrm{rel}}\right)^2}{2k_{\mathrm{B}} T_{\mathrm{i}}}\right] \quad \text{and} \quad f_{\mathrm{e}} = n_{\mathrm{e}} \left(\frac{m_{\mathrm{e}}}{2\pi k_{\mathrm{B}} T_{\mathrm{e}}}\right) \exp\left[-\frac{m_{\mathrm{e}} \mathbf{v}_{\mathrm{e}}^2}{2k_{\mathrm{B}} T_{\mathrm{e}}}\right] \tag{1.76}$$

Since ions are much more massive than electrons, their distribution function is much thinner than that of electrons. This implies the reasonable approximation of considering ion motion as a quasi-mono-energetic beam. Applying equation (1.60) for $\rho = \mathrm{i}$ and $\sigma = \mathrm{e}$, we recover the equation of ion motion:

$$m_{\mathrm{i}} \frac{\partial \mathbf{u}_{\mathrm{rel}}}{\partial t} \simeq \frac{n_{\mathrm{e}} q_{\mathrm{i}}^2 q_{\mathrm{e}}^2 \ln \Lambda}{4\pi \varepsilon_0^2 \mu_{\mathrm{ie}}} \left[\frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{\|\mathbf{v}\|} \mathrm{erf}\left(\|\mathbf{v}\| \sqrt{\frac{m_{\mathrm{e}}}{2k_{\mathrm{B}} T_{\mathrm{e}}}}\right)\right)\right]\Bigg|_{\mathbf{u}_{\mathrm{rel}}} + q_{\mathrm{i}} \mathbf{E} \tag{1.77}$$

For a steady state, the two terms on the right side of the previous equation compensate each other, hence the following expression for the electric field in the plasma:

$$\mathbf{E} \simeq -\frac{n_{\mathrm{e}} q_{\mathrm{i}} e^2 \ln \Lambda}{4\pi \varepsilon_0^2 \mu_{\mathrm{ie}}} \left[\frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{\|\mathbf{v}\|} \mathrm{erf}\left(\|\mathbf{v}\| \sqrt{\frac{m_{\mathrm{e}}}{2k_{\mathrm{B}} T_{\mathrm{e}}}}\right)\right)\right]\Bigg|_{\mathbf{u}_{\mathrm{rel}}} \tag{1.78}$$

The large mass difference between ions and electrons further leads to the assumption that the relative velocity of ions is much lower than the thermal velocity of electrons, which corresponds to the first case where $\zeta_{\mathrm{e}} \ll 1$, thus allowing the application of the derivative approximation:

$$\mathbf{E} \simeq \frac{n_{\mathrm{e}} q_{\mathrm{i}} e^2 \ln \Lambda}{3\varepsilon_0^2 \mu_{\mathrm{ie}}} \left(\frac{m_{\mathrm{e}}}{2\pi k_{\mathrm{B}} T_{\mathrm{e}}}\right)^{3/2} \mathbf{u}_{\mathrm{rel}} \simeq \frac{Z e^2 \ln \Lambda}{3\varepsilon_0^2 m_{\mathrm{e}}} \left(\frac{m_{\mathrm{e}}}{2\pi k_{\mathrm{B}} T_{\mathrm{e}}}\right)^{3/2} \mathbf{J} \tag{1.79}$$

The previous result precisely defines the electrical resistivity of a plasma:

$$\eta = \frac{Z e^2 \sqrt{m_{\mathrm{e}}} \ln \Lambda}{6\varepsilon_0^2 \sqrt{2\pi^3 k_{\mathrm{B}}^3 T_{\mathrm{e}}^3}} \tag{1.80}$$

1.6.1.1 Dreicer electric field

The evaluation of the derivative in equation (1.77) implied the assumption that the relative velocity, $\mathbf{u}_{\mathrm{rel}}$, is much lower than the thermal velocities. This corresponds to the fact that ζ_{i} is in the vicinity of zero. Nevertheless, by plotting the curve $(x^{-1}\mathrm{erf}\,(x))'$, we observe a maximum value of $\max \mathrm{erf}\,(x) \simeq 0.427998$ reached for $x \simeq 0.967857$. This implies that the steady state assumption is no longer valid provided that the electric field exceeds this maximum value:

$$E > \max_{v \geq 0} \left\{ -\frac{n_e q_i e^2 \ln \Lambda}{4\pi\varepsilon_0^2 \mu_{ie}} \left[\frac{\partial}{\partial v} \left(\frac{1}{v} \mathrm{erf} \left(v \sqrt{\frac{m_e}{2k_B T_e}} \right) \right) \right] \right\} \simeq 0.427998 \frac{n_e Z e^3}{8\pi\varepsilon_0^2 k_B T_e} \ln \frac{\lambda_d}{b_{\pi/2}} \tag{1.81}$$

When the electric field satisfies the preceding inequality, the concept of electrical resistivity collapses, because the steady state assumption is no longer valid. The lower bound of the electric field is called the *Dreicer electric field*, and when the magnitude of the electric field exceeds this value, the friction force necessary to compensate for the electric force cannot be established. Due to the decrease of the derivative in equation (1.77) for any value of ζ_i greater than the maximum, the acceleration due to the electric field leads to an increased rise in the velocity of incident particles. These particles, moving faster, experience less friction from interspecies collisions, increasing the mean free path and allowing them to accelerate further. The particle velocity will therefore increase without limit if the system is infinite and uniform. In reality, runaway particles can either escape the system or radiate excess energy.

1.6.2 Temperature equilibration

When $T_e \neq T_i$, interspecific collisions result, on the macroscopic scale, in a net energy exchange, thus leading to the equalization of the temperatures of the two species. The thermal exchange is indeed, at the microscopic level, driven by energy dissipation, mathematically treated in chapter 9.1, namely result (9.130). In the particular case of a quasi-neutral plasma consisting of stationary ion and electron species:

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_e k_B T_e \right) = \frac{Z n_e^2 e^4 \ln \Lambda}{2\pi^{3/2} \varepsilon_0^2} \frac{2k_B (T_i - T_e)}{m_e m_i} \bigg/ \left(\frac{2k_B T_e}{m_e} + \frac{2k_B T_i}{m_i} \right)^{3/2} \tag{1.82}$$

Since typical ion species have temperature $T_i \lesssim T_e$ and $m_e \ll m_i$, we may approximate:

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_e k_B T_e \right) \simeq \frac{Z n_e^2 e^4 \ln \Lambda}{(2\pi)^{3/2} \varepsilon_0^2 m_i} \left(\frac{T_i}{T_e} - 1 \right) \sqrt{\frac{m_e}{k_B T_e}} \tag{1.83}$$

Since the species are assumed stationary, continuity equation gives $\partial n_e / \partial t = 0$, so that:

$$\frac{\partial T_e}{\partial t} \simeq \frac{Z n_e e^4 \ln \Lambda}{3\pi^{3/2} \varepsilon_0^2 k_B m_i} \left(\frac{T_i}{T_e} - 1 \right) \sqrt{\frac{m_e}{2k_B T_e}} \tag{1.84}$$

$$\frac{\partial T_i}{\partial t} \simeq \frac{n_e e^4 \ln \Lambda}{3\pi^{3/2} \varepsilon_0^2 k_B m_i} \left(1 - \frac{T_i}{T_e} \right) \sqrt{\frac{m_e}{2k_B T_e}} \tag{1.85}$$

Chapter 2

Charge motion in force fields

In the study of plasmas, which consist of charged particles, it is essential to know their motion in various force field configurations, whether electric or magnetic, static or oscillatory. It should be noted that charged particles, for their part, create currents that modify these electromagnetic fields. In our considerations, we will ignore the second part of the problem. In other words, we will consider force fields imposed on an arbitrary plasma. The motion of said particles in electromagnetic fields is described by the Lorentz equation:

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = q\left[\mathbf{E}(\mathbf{x},t) + \mathbf{v}\times\mathbf{B}(\mathbf{x},t)\right] = q\left[\mathbf{E}(\mathbf{x},t) + \frac{\mathbf{p}}{\gamma m}\times\mathbf{B}(\mathbf{x},t)\right] \tag{2.1}$$

where m is the rest mass of the particle. In certain cases, we will ignore relativistic effects imposed by special or general relativity.

2.1 Constant uniform electric field

The equation of motion is, in this case, trivial to solve:

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = q\mathbf{E}_0 \implies \mathbf{p}(t) = \mathbf{p}_0 + q\mathbf{E}_0 t \tag{2.2}$$

$$\mathbf{v}(t) = \frac{\mathbf{p}(t)}{\gamma(t)m} = \frac{\mathbf{p}_0 + q\mathbf{E}_0 t}{\sqrt{1+\frac{(\mathbf{p}_0+q\mathbf{E}_0 t)^2}{m^2c^2}}} \qquad \lim_{t\to\infty}\|\mathbf{v}(t)\| = c \tag{2.3}$$

2.2 Constant uniform magnetic field

The equation of motion is also trivial to solve, as in the previous case:

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = q\mathbf{v}\times\mathbf{B}_0 \implies \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{v}\times\frac{q\mathbf{B_0}}{\gamma m} \equiv \mathbf{v}\times\mathbf{\Omega_c} \qquad \mathbf{\Omega}_c = \frac{q\mathbf{B}_0}{\gamma m} \tag{2.4}$$

where Ω_c designates the relativistic angular velocity of the particle considered. The norm of the charge's velocity $\|\mathbf{v}\|$ is conserved:

$$\mathbf{v}\cdot\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{v}^2\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\|\mathbf{v}\|\right) = \mathbf{v}\cdot\left(\mathbf{v}\times\Omega_c\right) = 0 \implies \frac{\mathrm{d}\gamma}{\mathrm{d}t} = 0 \tag{2.5}$$

The motion along the magnetic field is uniform. Let us choose $\mathbf{B}_0$ along the $\hat{\mathbf{z}}$ axis, then:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{v}\cdot\hat{\mathbf{z}}\right) = 0 \qquad \mathbf{v} - \left(\mathbf{v}\cdot\hat{\mathbf{z}}\right)\hat{\mathbf{z}} = v_{\mathrm{x}}\,\hat{\mathbf{x}} + v_{\mathrm{y}}\,\hat{\mathbf{y}} \implies v_\perp = \sqrt{v_{\mathrm{x}}^2 + v_{\mathrm{y}}^2} \tag{2.6}$$

$$\frac{\mathrm{d}v_{\mathrm{x}}}{\mathrm{d}t} = \frac{qB_0}{\gamma m}v_{\mathrm{y}} \qquad \frac{\mathrm{d}v_{\mathrm{y}}}{\mathrm{d}t} = -\frac{qB_0}{\gamma m}v_{\mathrm{x}} \tag{2.7}$$

$$\frac{\mathrm{d}^2 v_{\mathrm{x}}}{\mathrm{d}t^2} = \frac{qB_0}{\gamma m}\frac{\mathrm{d}v_{\mathrm{y}}}{\mathrm{d}t} = -\left(\frac{qB_0}{\gamma m}\right)^2 v_{\mathrm{x}} \qquad \frac{\mathrm{d}^2 v_{\mathrm{y}}}{\mathrm{d}t^2} = -\frac{qB_0}{\gamma m}\frac{\mathrm{d}v_{\mathrm{x}}}{\mathrm{d}t} = -\left(\frac{qB_0}{\gamma m}\right)^2 v_{\mathrm{y}} \tag{2.8}$$

$$v_x(t) = v_\perp \cos(\Omega_c t + \varphi) \qquad v_y(t) = v_\perp \sin(\Omega_c t + \varphi) \tag{2.9}$$

$$x(t) = \int v_x(t)\,\mathrm{d}t = x_0 + \frac{v_\perp}{\Omega_c}\sin(\varphi) + \frac{v_\perp}{\Omega_c}\sin(\Omega_c t + \varphi) \tag{2.10}$$

$$y(t) = \int v_y(t)\,\mathrm{d}t = y_0 - \frac{v_\perp}{\Omega_c}\cos(\varphi) - \frac{v_\perp}{\Omega_c}\cos(\Omega_c t + \varphi) \tag{2.11}$$

We will take further advantage of this by defining the angle $\alpha = \varphi - \frac{\pi}{2}$:

$$x(t) = x_g + r_L \cos(\Omega_c t + \alpha) \qquad y(t) = y_g + r_L \sin(\Omega_c t + \alpha) \tag{2.12}$$

$$x_g = x_0 - r_L \cos(\alpha) \qquad y_g = y_0 - r_L \sin(\alpha) \qquad r_L = \frac{\gamma m v_\perp}{\|q\mathbf{B_0}\|} = \frac{\|\mathbf{p}_\perp\|}{\|q\mathbf{B}_0\|} \tag{2.13}$$

The centre of the particle rotation orbits is termed the *guiding centre* or *gyro-centre*. It is defined by the vector $\mathbf{r}_g(t) = x_g\,\hat{\mathbf{x}} + y_g\,\hat{\mathbf{y}} + z(t)\,\hat{\mathbf{z}}$. The quantity r_L is known as the *Larmor radius* or *gyro radius*. It should be noted that the direction of rotation for positively charged particles is clockwise (left-handed), whereas the direction of rotation for negatively charged particles is counterclockwise (right-handed). This results in the magnetic dipole created by the motion of the concerned particle being oriented in the opposite direction to the external magnetic field. In other words, the particles tend to reduce the magnetic field imposed on them. This is precisely the property of magnetic materials termed *diamagnetism*. Plasma is therefore a diamagnetic medium.

2.3 Constant uniform magnetic and electric fields

The non-relativistic Lorentz equation appears in the following form:

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{2.14}$$

We will decompose the velocity $\mathbf{v}$ into three components: the velocity $\mathbf{v}_\parallel$ parallel to the magnetic field $\mathbf{B}$, the gyration velocity $\mathbf{v}_L$, and another velocity, introduced under the term *drift velocity*, $\mathbf{v}_d$:

$$m\,\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{v}_\parallel + \mathbf{v}_d + \mathbf{v}_L\right) = q\left[\mathbf{E} + \left(\mathbf{v}_\parallel + \mathbf{v}_d + \mathbf{v}_L\right) \times \mathbf{B}\right] \tag{2.15}$$

It is easy to see that one must define the drift velocity such that:

$$\mathbf{E} + \mathbf{v}_d \times \mathbf{B} = \mathbf{0} \tag{2.16}$$

so that we have a simple cyclotron motion satisfied by $\mathbf{v}_L$ as in section (2.2). By taking the cross product of equation (2.16) with $\mathbf{B}$, it follows:

$$\mathbf{E} \times \mathbf{B} + (\mathbf{v}_d \times \mathbf{B}) \times \mathbf{B} = \mathbf{E} \times \mathbf{B} + \underbrace{(\mathbf{v}_d \cdot \mathbf{B})}_{\mathbf{0}}\mathbf{B} - \mathbf{B}^2\mathbf{v}_d = \mathbf{0} \tag{2.17}$$

The expression for the drift velocity is thus easily determined:

$$\mathbf{v}_d = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2} \tag{2.18}$$

It should be noted that the drift velocity $\mathbf{v}_d$ is independent of the particle's properties; thus, it points in the same direction for positive and negative charges. The physical intuition behind the drift velocity is that, in the frame moving at this velocity, the electric field is transformed so that it is zero. When introducing relativistic effects, one must distinguish between the following two cases: $\|\mathbf{E}\| < c\|\mathbf{B}\|$ and $\|\mathbf{E}\| > c\|\mathbf{B}\|$. Otherwise, we would encounter contradictions. Without loss of generality, let $\mathbf{E} \perp \mathbf{B}$, then:

$$\|\mathbf{E}\| > c\|\mathbf{B}\| \implies \|\mathbf{v}_d\| = \left\|\frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2}\right\| > c \quad \perp \tag{2.19}$$

Let us consider the frame moving, relative to the laboratory frame where electric and magnetic fields are defined such that $\|\mathbf{E}\| < c\|\mathbf{B}\|$, at the following velocity:

$$\mathbf{v}_{\mathrm{d}} = \frac{\mathbf{E}\times\mathbf{B}}{\|\mathbf{B}\|^2} \implies \|\mathbf{v}_{\mathrm{d}}\| < c \tag{2.20}$$

Using Lorentz transformations, we recover the following fields in the new frame:

$$\mathbf{E}' = \gamma_{\mathrm{d}}\left(\mathbf{E}+\mathbf{v}_{\mathrm{d}}\times\mathbf{B}\right) - (\gamma_{\mathrm{d}}-1)\underbrace{\left(\mathbf{E}\cdot\hat{\mathbf{v}}_{\mathrm{d}}\right)}_{0}\hat{\mathbf{v}}_{\mathrm{d}} = \gamma_{\mathrm{d}}\left[\mathbf{E}+\frac{(\mathbf{E}\times\mathbf{B})}{\|\mathbf{B}\|^2}\times\mathbf{B}\right] = \gamma_{\mathrm{d}}\frac{\mathbf{E}\cdot\mathbf{B}}{\|\mathbf{B}\|^2}\mathbf{B} \tag{2.21}$$

$$\mathbf{B}' = \gamma_{\mathrm{d}}\left(\mathbf{B}-\frac{\mathbf{v}_{\mathrm{d}}\times\mathbf{E}}{c^2}\right) - (\gamma_{\mathrm{d}}-1)\underbrace{(\mathbf{B}\cdot\hat{\mathbf{v}}_{\mathbf{d}})}_{0}\hat{\mathbf{v}}_{d} = \gamma_{\mathrm{d}}\left[\mathbf{B}+\frac{(\mathbf{E}\cdot\mathbf{B})\mathbf{E}-\|\mathbf{E}\|^2\mathbf{B}}{c^2\|\mathbf{B}\|^2}\right] \tag{2.22}$$

In the specific case where $\mathbf{E}\perp\mathbf{B}$, we observe a significant simplification of the preceding expressions:

$$\mathbf{E}' = 0 \quad \text{et} \quad \mathbf{B}' = \gamma_{\mathrm{d}}\left(1-\frac{\mathbf{E}^2}{c^2\mathbf{B}^2}\right)\mathbf{B} = \frac{\mathbf{B}}{\gamma_{\mathrm{d}}} \tag{2.23}$$

Similarly, in the case where $\|\mathbf{E}\| > c\|\mathbf{B}\|$, and furthermore $\mathbf{E}\perp\mathbf{B}$, we simply admit that the expression for the drift velocity is of the following form:

$$\mathbf{v}_{\mathrm{d}} = c^2\frac{\mathbf{E}\times\mathbf{B}}{\mathbf{E}^2} \implies \mathbf{E}' = \frac{\mathbf{E}}{\gamma_{\mathrm{d}}} \quad \text{et} \quad \mathbf{B}' = \mathbf{0} \tag{2.24}$$

2.4 Constant uniform magnetic field and force field

The equation of motion in the case where the considered particle undergoes a force $\mathbf{F}$ reads:

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{F}+q\mathbf{v}\times\mathbf{B} = q\left(\frac{\mathbf{F}}{q}+\mathbf{v}\times\mathbf{B}\right) \tag{2.25}$$

Identically to the previous section, we can prove that the drift velocity appears in the following form, the derivation of which is omitted due to redundancy:

$$\mathbf{v}_{\mathrm{d}} = \frac{1}{q}\frac{\mathbf{F}\times\mathbf{B}}{\mathbf{B}^2} \tag{2.26}$$

2.5 Constant non-uniform magnetic field

2.5.1 Direction of variation perpendicular to the field

We examine the motion of a particle of charge q and mass m subjected to a magnetic field $\mathbf{B}$ characterized by a non-zero dyadic tensor $\nabla\mathbf{B}$. For the sake of simplicity and physical intuition, we will adopt the following assumptions:

1. Any velocity considered is much smaller than the speed of light. This allows us to neglect relativistic effects, which are indeed, for the most part, negligible.

2. The magnetic field undergoes minuscule variations on the scale of the Larmor radius of the particle considered, i.e. $r_{\mathrm{L}}\|\nabla\mathbf{B}\| \ll \|\mathbf{B}\|$, the approximation $\mathbf{B}\simeq\mathbf{B}_0+(\mathbf{r}\cdot\nabla)\mathbf{B}$ holds very well.

3. The dyadic tensor is a tensor with a single non-zero component, $\hat{\mathbf{x}}\hat{\mathbf{B}}$, where $\hat{\mathbf{x}}\perp\mathbf{B}$.

4. The drift velocity is much smaller than the Larmor velocity of the particle considered, i.e. $\|\mathbf{v}_{\mathrm{d}}\| \ll \|\mathbf{v}_{\mathrm{L}}\|$ holds very well. The validity of this assumption will be examined shortly.

5. The drift velocity indeed undergoes variations, but these variations are perfectly negligible. The validity of this assumption will be established at the close of the subsection.

By virtue of the preceding analyses, we impose the velocity decomposition $\mathbf{v} = \mathbf{v}_\mathrm{d} + \mathbf{v}_\mathrm{L}$ such that the Lorentz equation applied to the particle considered reads:

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \equiv m\frac{\mathrm{d}\mathbf{v}_\mathrm{L}}{\mathrm{d}t} = q\left(\mathbf{v}_\mathrm{d} + \mathbf{v}_\mathrm{L}\right) \times \mathbf{B} \tag{2.27}$$

Using assumption (2), we expand the cross product as follows:

$$m\frac{\mathrm{d}\mathbf{v}_\mathrm{L}}{\mathrm{d}t} = q\left[\mathbf{v}_\mathrm{L} \times \mathbf{B}_0 + \underbrace{\mathbf{v}_\mathrm{d} \times \mathbf{B}_0 + (\mathbf{v}_\mathrm{d} + \mathbf{v}_\mathrm{L}) \times (\mathbf{r}\cdot\nabla)\mathbf{B}}_{\mathbf{0}}\right] \tag{2.28}$$

We impose the drift-velocity condition, such that the underlined sum results in zero:

$$\mathbf{v}_\mathrm{d} \times \mathbf{B}_0 + \mathbf{v}_\mathrm{d} \times (\mathbf{r}\cdot\nabla)\mathbf{B} + \underbrace{\mathbf{v}_\mathrm{L} \times (\mathbf{r}\cdot\nabla)\mathbf{B}}_{\langle\cdots\rangle=\mathbf{0}} = \mathbf{0} \tag{2.29}$$

We identify the second term as the most negligible thanks to assumptions (2) and (4). Before beginning the resolution of the condition given by equation (2.29), we impose the decomposition of the position vector $\mathbf{r} = \mathbf{r}_\mathrm{g} + \mathbf{r}_\mathrm{L}$, in other words, into guiding centre and Larmor radius:

$$\mathbf{v}_\mathrm{d} \times \mathbf{B} + \mathbf{v}_\mathrm{L} \times (\mathbf{r}_\mathrm{L}\cdot\nabla)\mathbf{B} + \mathbf{v}_\mathrm{L} \times (\mathbf{r}_\mathrm{g}\cdot\nabla)\mathbf{B} = \mathbf{0} \tag{2.30}$$

We will proceed by taking the average of equation (2.30) over a rotation period. It is easy to see that the last term of the previous equation vanishes when its average is taken. Assumption (3) gives:

$$\mathbf{r}_\mathrm{L} = \frac{v_{\perp\mathbf{B}}}{\Omega_\mathrm{c}}\begin{pmatrix}\cos(\Omega_\mathrm{c}t + \varphi_0)\\ \sin(\Omega_\mathrm{c}t + \varphi_0)\end{pmatrix} \implies \mathbf{v}_\mathrm{L} = v_{\perp\mathbf{B}}\begin{pmatrix}-\sin(\Omega_\mathrm{c}t + \varphi_0)\\ \cos(\Omega_\mathrm{c}t + \varphi_0)\end{pmatrix} \quad \text{et} \quad (\mathbf{r}\cdot\nabla)\mathbf{B} = x\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}x} \tag{2.31}$$

We examine the second term of equation (2.30) more closely by decomposing it along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}} \perp \mathbf{B}$:

$$\left[\mathbf{v}_\mathrm{L} \times \left(x\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}x}\right)\right]\cdot\hat{\mathbf{x}} = v_\mathrm{Ly}x\frac{\mathrm{d}B}{\mathrm{d}x} \quad \text{et} \quad \left[\mathbf{v}_\mathrm{L} \times \left(x\frac{\mathrm{d}\mathbf{B}}{\mathrm{d}x}\right)\right]\cdot\hat{\mathbf{y}} = -v_\mathrm{Lx}x\frac{\mathrm{d}B}{\mathrm{d}x} \tag{2.32}$$

$$\langle v_\mathrm{Ly}x\rangle = \frac{v_{\perp\mathbf{B}}^2}{\Omega_c}\langle\cos(\Omega_c t + \varphi_0)\cos(\Omega_c t + \varphi_0)\rangle = \frac{v_{\perp\mathbf{B}}^2}{2\Omega_c} \tag{2.33}$$

$$\langle v_\mathrm{Lx}x\rangle = \frac{v_{\perp\mathbf{B}}}{\Omega_c}\langle -\sin(\Omega_c t + \varphi_0)\cos(\Omega_c t + \varphi_0)\rangle = 0 \tag{2.34}$$

Returning to equation (2.30), we obtain the expression for the drift velocity in an analogous manner as in the previous sections, hence:

$$\mathbf{v}_{\nabla\mathbf{B}} = \frac{1}{q}\frac{mv_\perp^2}{2}\frac{\mathbf{B}\times(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mathbf{B}^4} \tag{2.35}$$

This drift velocity is commonly referred to as *grad B-drift*. It is worth confirming the consistency between the assumptions established at the beginning of the discussion and the result obtained. We first test assumption (4) and find that it is well verified. Let us note assumption (3):

$$\|\mathbf{v}_{\nabla\mathbf{B}}\| = \left\|\frac{1}{q}\frac{mv_\perp^2}{2}\frac{\mathbf{B}\times(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mathbf{B}^4}\right\| = \frac{mv_\perp^2}{2|q|r_\mathrm{L}}\frac{r_\mathrm{L}\|\nabla\mathbf{B}\|}{\mathbf{B}^2} = \frac{mv_\perp^2}{2|q|}\frac{\|q\mathbf{B}\|}{mv_\perp}\frac{r_\mathrm{L}\|\nabla\mathbf{B}\|}{\mathbf{B}^2} \ll \frac{v_\perp}{2} = \frac{\|\mathbf{v}_\mathrm{L}\|}{2} \tag{2.36}$$

The variation of the drift velocity is given by the following:

$$\delta\|\mathbf{v}_{\nabla\mathbf{B}}\| = \frac{mv_\perp^2}{2|q|}\delta\left(\frac{\|\nabla\mathbf{B}\|}{\mathbf{B}^2}\right) \tag{2.37}$$

The order of the variation of the perpendicular velocity is the same as that of the magnetic field. Thus, assumption (2) ensures the minimality of the variation of $\mathbf{v}_{\nabla\mathbf{B}}$.

2.5.2 Direction of variation parallel to the field

When a magnetic field experiences intensification due to the dyadic tensor $\nabla\mathbf{B}$, whose component $\hat{\mathbf{z}}\hat{\mathbf{B}}$ is non-zero, this has the effect of compiling the field lines alongside each other. In other words, the magnetic field lines locally form a section cut by a cone, and the radial component is no longer zero, i.e. $\mathbf{B}\cdot\hat{\mathbf{r}} \neq 0$, however, such that the inequality $\mathbf{B}\cdot\hat{\mathbf{r}} \ll \|\mathbf{B}\|$ is satisfied. This strong inequality is analogous to assumption (2) of subsection (2.5.1). This entails that the magnetic field varies very little along the $\hat{\mathbf{z}}$ axis. By taking the average of the Lorentz force over a rotation period, there exists a non-zero component of this force parallel to the direction of magnetic field increase $\hat{\mathbf{z}}$:

$$\langle \mathbf{F}\cdot\hat{\mathbf{z}}\rangle = \langle q(\mathbf{v}\times\mathbf{B})\cdot\mathbf{z}\rangle = q\langle \mathbf{v}\cdot(\mathbf{B}\times\hat{\mathbf{z}})\rangle = qv_\perp \mathbf{B}\cdot\hat{\mathbf{r}} = -qv_\perp\|\mathbf{B}\|\sin(\theta) \tag{2.38}$$

where θ is the angle between the magnetic field $\mathbf{B}$ and the $\hat{\mathbf{z}}$ axis in the trigonometric sense. Using a Maxwell equation, we can relate the components of the magnetic field along the axes $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$:

$$\nabla\cdot\mathbf{B} = 0 \overset{B_\varphi=0}{\Longrightarrow} \frac{1}{r}\frac{\partial}{\partial r}(rB_\mathrm{r}) + \frac{\partial B_\mathrm{z}}{\partial z} = 0 \implies rB_\mathrm{r} = -\int r\frac{\partial B_\mathrm{z}}{\partial z}\,\mathrm{d}r \tag{2.39}$$

Under the assumption that the Larmor radius is sufficiently small for the variation $\partial_z B_\mathrm{z}$ to be approximately constant, we obtain:

$$rB_r\Big|_0^{r_\mathrm{L}} \simeq -\frac{\partial B_\mathrm{z}}{\partial z}\int_0^{r_\mathrm{L}} r\,\mathrm{d}r = -\frac{1}{2}r_\mathrm{L}^2\frac{\partial B_\mathrm{z}}{\partial z} \implies B_\mathrm{r} \simeq -\frac{r_\mathrm{L}}{2}\frac{\partial B_\mathrm{z}}{\partial z} \implies \sin(\theta) = \frac{r_\mathrm{L}}{2}\frac{1}{\|\mathbf{B}\|}\frac{\partial B_\mathrm{z}}{\partial z} \tag{2.40}$$

Consequently, the particle undergoes a recoil force given by the following expression:

$$\langle \mathbf{F}\cdot\hat{\mathbf{z}}\rangle = -\frac{qv_\perp r_\mathrm{L}}{2}\frac{\partial B_\mathrm{z}}{\partial z} = -\frac{mv_\perp^2}{2\|\mathbf{B}\|}\frac{\partial B_\mathrm{z}}{\partial z} \tag{2.41}$$

2.6 Magnetic field presenting a radius of curvature

We will next analyse the motion of a particle of charge q and mass m in a uniform and constant magnetic field of magnitude $\|\mathbf{B}\|$ and radius of curvature $\mathbf{R}$. By entering the rotating frame in which the particle follows a simple cyclotron motion, it undergoes inertial forces, in particular the centrifugal force and the Coriolis force. For simplicity, we admit straight away that the Coriolis effect does not influence the particle's motion significantly at all. The proof of this assertion relies on the fact that the Coriolis effect vanishes when taking its average over a rotation period. Let $\mathbf{v}_\parallel = (\mathbf{v}\cdot\hat{\mathbf{B}})\,\hat{\mathbf{B}}$ be the component of the particle's velocity parallel to the magnetic field, assumed constant. Then, the centrifugal force reads:

$$\mathbf{F}_\mathrm{cf} = mv_\parallel^2\frac{\mathbf{R}}{\mathbf{R}^2} \tag{2.42}$$

Using equation (2.26), the drift velocity induced by the curvature of the magnetic field is:

$$\mathbf{v}_\mathbf{R} = \frac{mv_\parallel^2}{q}\frac{\mathbf{R}\times\mathbf{B}}{\mathbf{R}^2\mathbf{B}^2} \tag{2.43}$$

The fact that the magnitude of the magnetic field is constant does not prevent its gradient from being non-zero. It is therefore appropriate to relate the dyadic tensor $\nabla\mathbf{B}$ to the radius of curvature. By introducing a cylindrical coordinate system such that $\mathbf{B}\cdot\hat{\mathbf{r}} = 0 = \mathbf{B}\cdot\hat{\mathbf{z}}$, and by taking advantage of a Maxwell equation, we will achieve this elegantly:

$$\nabla\times\mathbf{B} = \mathbf{0} \implies (\nabla\times\mathbf{B})\cdot\hat{\mathbf{z}} = \frac{1}{r}\frac{\partial}{\partial r}(rB_\varphi) - \frac{1}{r}\underbrace{\frac{\partial B_r}{\partial\varphi}}_{0} = 0 \tag{2.44}$$

$$\frac{\partial B_\varphi}{\partial r} = -\frac{B_\varphi}{r} \implies \nabla\mathbf{B} = -\frac{\mathbf{RB}}{\mathbf{R}^2} \implies (\mathbf{B}\cdot\nabla)\,\mathbf{B}\perp\mathbf{B} \tag{2.45}$$

Using the result (2.35), the drift velocity induced by the magnetic field gradient, in this specific case, reads:

$$\mathbf{v}_{\nabla\mathbf{B}} = \frac{1}{q}\frac{mv_\perp^2}{2}\frac{\mathbf{B}}{\|\mathbf{B}\|^3}\times\left(-\frac{\mathbf{R}}{\mathbf{R}^2}\right)\|\mathbf{B}\| = \frac{1}{q}\frac{mv_\perp^2}{2}\frac{\mathbf{R}\times\mathbf{B}}{\mathbf{R}^2\mathbf{B}^2} \tag{2.46}$$

The total drift is the simple sum of the two drift velocities obtained:

$$\mathbf{v}_\mathrm{d} = \mathbf{v}_{\nabla\mathbf{B}} + \mathbf{v}_\mathbf{R} = \frac{1}{q}\left(mv_\parallel^2 + \frac{1}{2}mv_\perp^2\right)\frac{\mathbf{R}\times\mathbf{B}}{\mathbf{R}^2\mathbf{B}^2} \tag{2.47}$$

It should be noted that the average of the preceding drift velocity can be evaluated quickly thanks to the equipartition theorem:

$$\left\langle\frac{1}{2}mv_\parallel^2\right\rangle = \frac{1}{2}k_\mathrm{B}T \quad \text{et} \quad \left\langle\frac{1}{2}mv_\perp^2\right\rangle = 2\times\frac{1}{2}k_\mathrm{B}T \implies \langle\mathbf{v}_\mathrm{d}\rangle = \frac{2k_\mathrm{B}T}{q}\frac{\mathbf{R}\times\mathbf{B}}{\mathbf{R}^2\mathbf{B}^2} \tag{2.48}$$

2.7 Variable electric and constant magnetic field

The drift velocity undergoes a temporal variation because the electric field, for its part, depends on time:

$$\mathbf{v}_\mathrm{d}(t) = \frac{\mathbf{E}(t)\times\mathbf{B}}{\mathbf{B}^2} \implies \mathbf{a}_\mathrm{d} = \frac{\mathrm{d}\mathbf{v}_\mathrm{d}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathbf{E}\times\mathbf{B}}{\mathbf{B}^2}\right) \tag{2.49}$$

The guiding centre undergoes acceleration, as is also the case for the particle considered. We will examine the specific case of an oscillatory electric field perpendicular to the magnetic field:

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = q\left(\mathbf{E}e^{i\omega t} + \mathbf{v}\times\mathbf{B}\right) \tag{2.50}$$

Complex notation implies only the real component of the number $e^{\mathrm{i}\omega t}$. Multiplication by $i\omega$ is thus equivalent to the derivative $\mathrm{d}/\mathrm{d}t$. We impose the following decomposition:

$$\mathbf{v} = \mathbf{v}_\mathrm{d}e^{i\omega t} + \mathbf{v}_\mathrm{L} \quad \text{où} \quad m\frac{\mathrm{d}\mathbf{v}_\mathrm{L}}{\mathrm{d}t} = q\mathbf{v}_\mathrm{L}\times\mathbf{B} \tag{2.51}$$

$$im\omega\,\mathbf{v}_\mathrm{d} = q\left(\mathbf{E} + \mathbf{v}_\mathrm{d}\times\mathbf{B}\right)e^{i\omega t} \implies im\omega\,\mathbf{v}_\mathrm{d}\times\mathbf{B} = q\left[\mathbf{E}\times\mathbf{B} + (\mathbf{B}\cdot\mathbf{v}_\mathrm{d})\mathbf{B} - \mathbf{B}^2\mathbf{v}_\mathrm{d}\right]e^{i\omega t} \tag{2.52}$$

To eliminate $\mathbf{v}_\mathrm{d}\times\mathbf{B}$, we multiply the first equation of (2.52) by $-im\omega$ and the second by q:

$$m^2\omega^2\mathbf{v}_\mathrm{d} = -\mathrm{i}mq\omega\left(\mathbf{E} + \mathbf{v}_\mathrm{d}\times\mathbf{B}\right)e^{\mathrm{i}\omega t} \quad \text{et} \quad \mathrm{i}mq\omega\,\mathbf{v}_\mathrm{d}\times\mathbf{B} = q^2\left[\mathbf{E}\times\mathbf{B} + (\mathbf{B}\cdot\mathbf{v}_\mathrm{d})\mathbf{B} - \mathbf{B}^2\mathbf{v}_\mathrm{d}\right]e^{\mathrm{i}\omega t} \tag{2.53}$$

$$m^2\omega^2\mathbf{v}_\mathrm{d} + q^2\left(\mathbf{E}\times\mathbf{B} - \mathbf{B}^2\mathbf{v}_\mathrm{d}\right) = \mathrm{i}mq\omega\,\mathbf{E} \equiv mq\frac{\mathrm{d}\mathbf{E}}{\mathrm{d}t} \tag{2.54}$$

This drift velocity is also known as *polarization drift*:

$$\mathbf{v}_\mathrm{d} = \left(\frac{\mathbf{E}\times\mathbf{B}}{\mathbf{B}^2} + \frac{m}{q\mathbf{B}^2}\frac{\mathrm{d}\mathbf{E}}{\mathrm{d}t}\right)\Bigg/\left(1 - \frac{m^2\omega^2}{q^2\mathbf{B}^2}\right) \tag{2.55}$$

We are obliged to follow a different procedure when these temporal variations of the electric field do not present oscillations.

2.8 Non-uniform electric and uniform magnetic field

The electric field is assumed to be slightly non-uniform, analogously to the magnetic field, in section (2.5). In other words, we adopt the following assumption:

$$\mathbf{E} \simeq \mathbf{E}_0 + (\mathbf{r}\cdot\nabla)\mathbf{E} \quad \text{where } \|\mathbf{r}\| \text{ is of the same order as the Larmor radius} \tag{2.56}$$

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = q\left[\mathbf{E}(r) + \mathbf{v}\times\mathbf{B}\right] \tag{2.57}$$

We seek a solution for $\mathbf{v}_\mathrm{d}$ such that the following equation is satisfied:

$$m\left\langle\frac{\mathrm{d}\mathbf{v}_\mathrm{d}}{\mathrm{d}t}\right\rangle = q\left\langle\mathbf{E}(r) + \mathbf{v}_\mathrm{d}\times\mathbf{B}\right\rangle = q\left[\langle\mathbf{E}\rangle + \mathbf{v}_\mathrm{d}\times\mathbf{B}\right] = \mathbf{0} \tag{2.58}$$

We know the solution very well, so we only need to determine the average magnetic field over a rotation period. The averages of the mixed terms are all zero thanks to orthogonality, so we are left with:

$$\langle \mathbf{E} \rangle = \mathbf{E}_0 + (\langle \mathbf{r}_\mathrm{L} \rangle \cdot \nabla)\mathbf{E} + \frac{1}{2}\langle (\mathbf{r}_\mathrm{L} \cdot \hat{\mathbf{x}})^2 \rangle \frac{\partial^2 \mathbf{E}}{\partial x^2} + \frac{1}{2}\langle (\mathbf{r}_\mathrm{L} \cdot \hat{\mathbf{y}})^2 \rangle \frac{\partial^2 \mathbf{E}}{\partial y^2} \tag{2.59}$$

We easily see $\langle \mathbf{r}_\mathrm{L} \rangle = \mathbf{0}$, $\langle (\mathbf{r}_\mathrm{L} \cdot \hat{\mathbf{x}})^2 \rangle = r_\mathrm{L}^2/2$ and likewise for the component along $\hat{\mathbf{y}}$:

$$\mathbf{v}_\mathrm{d} = \mathbf{v}_\mathbf{E} + \mathbf{v}_{\nabla\mathbf{E}} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2} + \frac{r_\mathrm{L}^2}{4}\nabla^2 \mathbf{E} = \left(1 + \frac{1}{4}\frac{m^2 v_\perp^2}{q^2 \mathbf{B}^2}\nabla^2\right)\frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2} \tag{2.60}$$

2.9 Adiabatic invariants

Most periodic physical systems are characterized by a physical quantity which, despite possible *slow* deformations of certain parameters, remains constant. This quantity is commonly designated by the term *adiabatic invariant*. More precisely, adiabatic invariants are first-order approximations of *Poincaré invariants*. We will adopt an analytical mechanics approach in order to study and theoretically develop the basic notions. Subsequently, we will draw on the preceding sections to illustrate several examples of adiabatic invariants. A Poincaré invariant takes the following form:

$$\mathcal{I} = \oint_{\mathcal{C}(t)} \mathbf{p} \cdot \mathrm{d}\mathbf{q} \tag{2.61}$$

where all points belonging to the closed curve $\mathcal{C}(t)$ in phase space move according to the system's equations of motion. We introduce a periodic variable $s \in \mathcal{S}$, which parametrizes the points of the curve $\mathcal{C}(t)$. The general coordinates of a point on this curve are then written in the form $q_i = q_i(s,t)$ and $p_i = p_i(s,t)$. The time derivative then reads:

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \oint_{\mathcal{S}} \left(p_i \frac{\partial^2 q_i}{\partial t \partial s} + \frac{\partial p_i}{\partial t}\frac{\partial q_i}{\partial s} \right) \mathrm{d}s \tag{2.62}$$

We integrate the first term by parts and then substitute Hamilton's equations:

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \oint_{\mathcal{S}} \left(-\frac{\partial q_i}{\partial t}\frac{\partial p_i}{\partial s} + \frac{\partial p_i}{\partial t}\frac{\partial q_i}{\partial s} \right) \mathrm{d}s = -\oint_{\mathcal{S}} \left(\frac{\partial \mathcal{H}}{\partial p_i}\frac{\partial p_i}{\partial s} + \frac{\partial \mathcal{H}}{\partial q_i}\frac{\partial q_i}{\partial s} \right) \mathrm{d}s \tag{2.63}$$

where $\mathcal{H}(\mathbf{q}, \mathbf{p}, t)$ represents the Hamiltonian of the system. If the system is conservative, i.e., if the Hamiltonian doesn't explicitly depend on time, the integrand is the total derivative of the Hamiltonian with respect to the variable s along the curve $\mathcal{C}$. The time-independent Hamiltonian being a function of state, the integral evaluates to zero, thus demonstrating that $\mathcal{I}$ is indeed invariant:

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = -\oint_{\mathcal{S}} \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}s}\,\mathrm{d}s = 0 \tag{2.64}$$

2.9.1 First adiabatic invariant

Poincaré invariants are generally of little interest in practice unless the curve $\mathcal{C}$ corresponds closely to the actual trajectories of particles. As in previous studies, we consider the case where the curve $\mathcal{C}$ represents a gyration period in phase space. The Poincaré invariant is thus approximated by the associated adiabatic invariant:

$$\mathcal{I} \simeq \mathcal{J} = \oint_{\mathcal{C}} \boldsymbol{\mathcal{P}} \cdot \frac{\partial \mathbf{q}}{\partial \varphi}\,\mathrm{d}\varphi \tag{2.65}$$

where φ corresponds indeed to the angular parameter of the position $\mathbf{r}_\mathrm{L}(t)$, i.e. $\varphi = \Omega_c t + \varphi_0$. The canonical momentum of a particle of mass m of charge q appears in the following form:

$$\boldsymbol{\mathcal{P}} = m\mathbf{v} + q\mathbf{A} \tag{2.66}$$

where $\mathbf{A}$ designates the vector potential which satisfies $\nabla \times \mathbf{A} = \mathbf{B}$. The expression of $\mathbf{A}$ in the vicinity of the guiding centre can be written by a Taylor expansion:

$$\mathbf{A}(\mathbf{r}) = \mathbf{A}_0 + (\mathbf{r} \cdot \nabla)\mathbf{A} + \mathcal{O}(\mathbf{r}^2) \tag{2.67}$$

It is clear that the next step is to account for a slightly variable magnetic field. In other words, during a gyration period, the magnitude $\|\mathbf{B}\|$ remains approximately constant. The angular parameter φ satisfies the following relation:

$$\mathbf{v}_\mathrm{L} = \Omega_c \frac{\mathrm{d}\mathbf{r}_\mathrm{L}}{\mathrm{d}\varphi} \implies \mathrm{d}\boldsymbol{\ell} = \frac{\mathbf{v}_\mathrm{L}}{\Omega_c}\,\mathrm{d}\varphi \tag{2.68}$$

where $\mathrm{d}\boldsymbol{\ell}$ represents a length element belonging to the curve $\mathcal{C} \equiv [0, 2\pi)$:

$$\mathcal{J} = \oint_\mathcal{C} \Big[m(\mathbf{v}_\mathrm{d} + \mathbf{v}_\mathrm{L}) + q(\mathbf{A}_0 + (\mathbf{r}_\mathrm{L} \cdot \nabla)\mathbf{A}) \Big] \cdot \frac{m\mathbf{v}_\mathrm{L}}{q\|\mathbf{B}\|}\,\mathrm{d}\varphi \tag{2.69}$$

It is easy to see that the last expression is equivalent to evaluating the mean quantities:

$$\mathcal{J} = \oint_\mathcal{C} \left[\frac{m^2\mathbf{v}_\mathrm{d}\cdot\mathbf{v}_\mathrm{L}}{q\|\mathbf{B}\|} + \frac{m^2\mathbf{v}_\mathrm{L}^2}{q\|\mathbf{B}\|} + \frac{m\mathbf{A_0}\cdot\mathbf{v}_\mathrm{L}}{\|\mathbf{B}\|} + \frac{m\mathbf{v}_\mathrm{L}}{\|\mathbf{B}\|}\cdot(\mathbf{r}_\mathrm{L}\cdot\nabla)\mathbf{A} \right] \mathrm{d}\varphi \tag{2.70}$$

The first and third terms are zero by orthogonality. The second term is assumed constant, so it comes out of the integral without fail. It remains for us to evaluate the last term:

$$\mathcal{J} = \frac{2\pi m\mathbf{v}_\mathrm{L}^2}{q\|\mathbf{B}\|} + \frac{2\pi m}{\|\mathbf{B}\|}\,\langle \mathbf{v}_\mathrm{L}\cdot(\mathbf{r}_\mathrm{L}\cdot\nabla)\mathbf{A}\rangle \tag{2.71}$$

$$\langle \mathbf{v}_\mathrm{L}\cdot(\mathbf{r}_\mathrm{L}\cdot\nabla)\mathbf{A}\rangle \equiv \langle v^j x^i \partial_i A_j\rangle \simeq \langle v^i x^j\rangle \partial_i A_j = -\frac{m\mathbf{v}_\mathrm{L}^2}{2q\|\mathbf{B}\|}\,(\nabla\times\mathbf{A})\cdot\hat{\mathbf{B}} = -\frac{m\mathbf{v}_\mathrm{L}^2}{2q} \tag{2.72}$$

$$\mathcal{J} = \frac{2\pi m^2\mathbf{v}_\mathrm{L}^2}{q\|\mathbf{B}\|} - \frac{2\pi m}{\|\mathbf{B}\|}\frac{m\mathbf{v}_\mathrm{L}^2}{2q} = \frac{\pi m^2\mathbf{v}_\mathrm{L}^2}{q\|\mathbf{B}\|} = 2\pi\times\frac{m}{q}\times\frac{m\mathbf{v}_\mathrm{L}^2}{2\|\mathbf{B}\|} \tag{2.73}$$

The reason why we separated the term $m\mathbf{v}_\mathrm{L}^2/2\|\mathbf{B}\|$ is that it represents precisely the magnetic moment of the particle considered. The magnetic moment of any current localization on $\Omega \subset \mathbb{R}^3$ is defined by the following integral. We apply it directly to the particle:

$$\boldsymbol{\mu} = \frac{1}{2}\iiint_\Omega \mathbf{r}\times\mathbf{J}\,\mathrm{d}V = \frac{1}{2}\mathbf{r}_\mathrm{L}\times q\mathbf{v}_\mathrm{L} = -\frac{q}{2}\frac{m\mathbf{v}_\mathrm{L}^2}{q\|\mathbf{B}\|}\,\hat{\mathbf{B}} = -\frac{m\mathbf{v}_\mathrm{L}^2}{2\mathbf{B}^2}\mathbf{B} \tag{2.74}$$

It then follows that, to the lowest order, the adiabatic invariant is proportional to the magnetic moment $\|\boldsymbol{\mu}\|$. The magnetic moment of a charged particle in a magnetic field is commonly designated by the term *first adiabatic invariant.*

2.9.2 Second adiabatic invariant

The second adiabatic invariant concerns the motion of particles in a magnetic field, as examined in subsection (2.5.2), subject to the imposed assumptions. We consider a particle of mass m and charge q whose velocity component parallel to the field is $\mathbf{v}_\| = (\mathbf{v}\cdot\hat{\mathbf{B}})\,\hat{\mathbf{B}}$. In addition, we will consider the case in which the magnetic field lines are tightly compressed at a point, for example, by a Helmholtz coil. Certain particles, and we will determine which ones, will bounce back in the opposite direction, resulting in a round trip. The bouncing is precisely due to result (2.41) of subsection (2.5.2):

$$\langle \mathbf{F}\cdot\hat{\mathbf{z}}\rangle = -\frac{m\mathbf{v}_\mathrm{L}^2}{2\|\mathbf{B}\|}\frac{\partial B_\mathrm{z}}{\partial z} = -\|\boldsymbol{\mu}\|\frac{\partial B_\mathrm{z}}{\partial z} = (\boldsymbol{\mu}\cdot\nabla)\,\mathbf{B}\cdot\hat{\mathbf{z}} \tag{2.75}$$

Conservation of energy and the first adiabatic invariant allow us to write:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[K = \frac{1}{2}m\left(\mathbf{v}_\perp^2 + \mathbf{v}_\|^2\right)\right] = 0 \quad \text{et} \quad \frac{\mathrm{d}\boldsymbol{\mu}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m\mathbf{v}_\perp^2}{2\mathbf{B}^2}\mathbf{B}\right) = \mathbf{0} \tag{2.76}$$

Consider a confined particle. The component of its velocity parallel to the magnetic field vanishes at the squeezing location. Let $v_\perp$ and $v_\|$ be the velocity components when the particle is in the middle between the two boundaries, and let $u_\perp$ be the component perpendicular to the magnetic field $\mathbf{B}_\mathrm{m}$ when it reaches the turnaround point of its trajectory:

$$\frac{1}{2}m(v_\|^2 + v_\perp^2) \leq \frac{1}{2}mu_\perp^2 \quad \text{et} \quad \frac{mv_\perp^2}{2\|\mathbf{B}_0\|} = \frac{mu_\perp^2}{2\|\mathbf{B}_\mathrm{m}\|} \implies \frac{\|\mathbf{B}_0\|}{\|\mathbf{B}_\mathrm{m}\|} \geq \frac{v_\perp^2}{v_\perp^2 + v_\|^2} \tag{2.77}$$

It is appropriate to introduce the angle θ between the initial velocity $\mathbf{v}_0$ and the magnetic field $\mathbf{B}_0$, then:

$$\cos(\theta) = \hat{\mathbf{v}}_0 \cdot \hat{\mathbf{B}}_\mathbf{0} \implies \tan(\theta) = \frac{v_\perp}{v_\parallel} \implies \frac{\|\mathbf{B}_0\|}{\|\mathbf{B}_\mathrm{m}\|} \geq \sin^2(\theta) \implies \theta_c = \sqrt{\arcsin\left(\frac{\|\mathbf{B}_0\|}{\|\mathbf{B}_\mathrm{m}\|}\right)} \tag{2.78}$$

It is easy to see that the condition under which a particle is reflected by the magnetic field depends solely on the angle θ, that is to say, on the ratio between the components perpendicular and parallel to the magnetic field. The critical value is then given by the equality in the last inequality (2.78). It is appropriate to represent these components using a *loss cone*. It is so named because, if we place the velocity vector of a particle and it is contained within the cone, the particle will not be reflected:

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 2.1: Loss cone

We will next consider a magnetic field *squeezed* at two locations, by two currents, such that the field lines form a solid of revolution. Then, there exists an axial symmetry of the field lines. Particles satisfying (2.78) will be trapped between the two boundaries. Let $\mathcal{C}$ be the curve tracing the trajectory of the guiding centre. We will tolerate plausible deformations of the curve $\mathcal{C}$. Let $\mathbf{u}$ be the velocity of any point on the curve $\mathcal{C}$. We will impose the following assumption:

1. The curve $\mathcal{C}$ undergoes *slow* and *weak* deformations, or none at all, such that the field $\mathbf{u}$ satisfies $\|\mathbf{u}\| \ll r_\mathrm{L}/\tau$, where r_L is the Larmor radius of a trapped particle and τ represents the time necessary for the particle to make a complete round trip. This allows us to impose a following bound, regarding the divergence of the field $\mathbf{u}$, which we will need at the end:

$$\nabla \cdot \mathbf{u} \simeq \|\mathbf{u}\|/\mathbf{r}_L \ll 1/\tau \tag{2.79}$$

The second invariant will be represented by an integral whose derivative with respect to time we will prove to be *approximately* zero. Let $\mathcal{S}$ be a surface where $\mathcal{C} = \partial\mathcal{S}$. Let us immediately apply Stokes' theorem:

$$\mathcal{J} = \oint_\mathcal{C} \mathbf{p} \cdot \mathrm{d}\boldsymbol{\ell} = \iint_\mathcal{S} (\nabla \times \mathbf{p}) \cdot \mathrm{d}\mathbf{S} \quad \text{où} \quad \mathrm{d}\boldsymbol{\ell} \parallel \mathbf{B} \tag{2.80}$$

We evaluate the time derivative of expression (2.80). Note that $\mathbf{F} = \partial_t \mathbf{p}$:

$$\frac{\mathrm{d}\mathcal{J}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \iint_\mathcal{S} (\nabla \times \mathbf{p}) \cdot \mathrm{d}\mathbf{S} = \iint_\mathcal{S} [\nabla \times \mathbf{F} + \mathbf{u}\,(\nabla \cdot (\nabla \times \mathbf{p})) + \nabla \times ((\nabla \times \mathbf{p}) \times \mathbf{u})] \cdot \mathrm{d}\mathbf{S} \tag{2.81}$$

Note that the divergence of a curl is always zero, that is, $\nabla \cdot (\nabla \times \mathbf{p}) = 0$. Since the field lines form a solid of revolution, the surface $\mathcal{S}$ is perpendicular to the magnetic field, so the integration of the first term leads to its vanishing:

$$\nabla \times (q\mathbf{v} \times \mathbf{B}) = q(\underbrace{\nabla \cdot \mathbf{B}}_{0})\mathbf{v} + q\mathbf{B}(\nabla \cdot \mathbf{v}) = q(\nabla \cdot \mathbf{v})\mathbf{B} \perp \mathrm{d}\mathbf{S} \tag{2.82}$$

$$\nabla \times [(\nabla \times \mathbf{p}) \times \mathbf{u}] = \nabla \times [(\nabla \cdot \mathbf{u})\mathbf{p} - \nabla(\mathbf{p} \cdot \mathbf{u})] = \nabla \times (\nabla \cdot \mathbf{u})\mathbf{p} \tag{2.83}$$

$$\frac{\mathrm{d}\mathcal{J}}{\mathrm{d}t} = \iint_\mathcal{S} q(\nabla \cdot \mathbf{v})\underbrace{\mathbf{B} \cdot \mathrm{d}\mathbf{S}}_{0} + \oint_\mathcal{C} (\nabla \cdot \mathbf{u})\mathbf{p} \cdot \mathrm{d}\boldsymbol{\ell} \tag{2.84}$$

We apply the previously stated hypothesis. The adiabatic invariant is thus quasi-invariant unless the curve $\mathcal{C}$ undergoes uniform deformations, in which case $\nabla \cdot \mathbf{u} = 0$, thus maintaining the constancy of the invariant. The second adiabatic invariant is given by the following:

$$\frac{\mathrm{d}\mathcal{J}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \oint_\mathcal{C} \mathbf{p} \cdot \mathrm{d}\boldsymbol{\ell} = \oint_\mathcal{C} (\nabla \cdot \mathbf{u})\mathbf{p} \cdot \mathrm{d}\mathbf{S} \simeq 0 \tag{2.85}$$

2.9.3 Third adiabatic invariant

The third adiabatic invariant concerns the temporal variation of the magnetic field, which was omitted in the previous discussions, i.e., in the analyses of the first and second adiabatic invariants. In this subsection, we will explicitly examine a magnetic field subjected to a temporal variation under certain limiting conditions.

Consider a configuration of a trapping magnetic field. It is appropriate to examine the static effects separately, when the field does not vary, and those that the temporal variation of the field exerts on the dynamics of particles. When we assume that the field does not vary temporally, we refer to this configuration as *static conditions*. A precise description of static conditions is given by the set of following assumptions:

1. Let τ be the time necessary for a particle to traverse the curve $\mathcal{C}$ in the preceding considerations. It is more precisely given by the following expression:

$$\tau = \oint_{\mathcal{C}} \frac{\mathrm{d}\ell}{\|\mathbf{v}_\mathrm{d}\|} \tag{2.86}$$

   Once temporal variation is introduced, the magnetic field subjected to an *adiabatic* variation satisfies the following condition:

$$\tau \ll \inf \|\mathbf{B}\| \Big/ \sup \left\| \frac{\partial \mathbf{B}}{\partial t} \right\| \tag{2.87}$$

   where the infimum and supremum are taken over the surface $\mathcal{S}$ bounded by the particle's trajectory, in other words by the curve $\mathcal{C} = \partial\mathcal{S}$.

2. The uniformity of the magnetic field in space is very stable. This indeed presents assumption (2) in subsection (2.5.1).

Consider the magnetic flux through the surface $\mathcal{S}$ whose boundary is the curve $\mathcal{C}$:

$$\Phi_\mathbf{B} = \iint_{\mathcal{S}} \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \oint_{\mathcal{C}} \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} \tag{2.88}$$

$$\frac{\mathrm{d}\Phi_\mathbf{B}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \oint_{\mathcal{C}} \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} = \oint_{\mathcal{C}} \left[\frac{\partial \mathbf{A}}{\partial t} + (\nabla \cdot \mathbf{u})\mathbf{A} \right] \cdot \mathrm{d}\boldsymbol{\ell} \tag{2.89}$$

where $\mathbf{u}$ represents the same velocity field regarding possible deformations of the curve $\mathcal{C}$ as in the previous subsection. This velocity is precisely determined by the variation of the magnetic field over time. Note the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$:

$$\nabla \cdot \mathbf{u} = \nabla \cdot \left(\frac{\mathbf{E}_\mathrm{ind} \times \mathbf{B}}{\mathbf{B}^2} \right) = \nabla \cdot \left(-\frac{\partial \mathbf{A}}{\partial t} \times \frac{\mathbf{B}}{\mathbf{B}^2} \right) = \frac{\partial}{\partial t} \left[\nabla \cdot \left(-\frac{\nabla \mathbf{A}^2}{\mathbf{B}^2} \right) \right] \tag{2.90}$$

Assumption (2) of section (2.5.1) allows us to conclude that this effect of modification of the curve $\mathcal{C}$ is largely negligible compared to the purely temporal variation of the field:

$$\frac{\mathrm{d}\Phi_\mathbf{B}}{\mathrm{d}t} \simeq \oint_{\mathcal{C}} \frac{\partial \mathbf{A}}{\partial t} \cdot \mathrm{d}\boldsymbol{\ell} = \iint_{\mathcal{S}} \frac{\partial \mathbf{B}}{\partial t} \cdot \mathrm{d}\mathbf{S} \leq \sup_{\mathbf{B}|_\mathcal{S}} \frac{\partial \|\mathbf{B}\|}{\partial t} S \ll \frac{\inf \|\mathbf{B}\|}{\tau} S \leq \frac{\Phi_\mathbf{B}}{\tau} \tag{2.91}$$

This precisely defines the third adiabatic invariant as the magnetic flux enclosed by the particle's trajectory. It should be noted that, in the purely static case, the magnetic flux is effectively constant.

2.10 Violation of adiabatic invariants

Adiabatic invariants are not entirely conserved quantities, as they are approximations of Poincaré invariants. However, their stability is carefully verified under the aforementioned assumptions when considering a single particle. It is essential to note that the stability of adiabatic invariants in a plasma is considerably lower than that observed for a single particle. Indeed, collisional effects lead to violations of adiabatic invariants. It should be emphasized that the low collisionality of plasmas constitutes a significant advantage, ensuring that our analyses are not invalidated by uncertainty related to collisional effects. Indeed, an interparticle collision significantly modifies the velocities of the two particles involved. Consequently, during a collision, no adiabatic invariant is approximately conserved and undergoes a modification of order of magnitude that is entirely dependent on the collisional conditions. Estimates of these violations, with respect to the first and third adiabatic invariants, cannot be obtained without loss of generality. However, we will provide illustrative examples of various violations of adiabatic invariants.

2.10.1 Violation of the second adiabatic invariant

Consider a collection of charged particles trapped in a magnetic field configuration like that examined in subsection (2.9.2). Particles belonging to the loss cone illustrated in Figure (2.1) will escape confinement. This particle escape will cause non-uniformity in particle density in phase space. It will be shown later that this configuration in phase space is unstable, as residual particles will collide. Indeed, collisions serve as carriers of particle diffusion in phase space, thereby leveling and balancing density in this space. This dynamics leads to the entry of particles initially located outside the loss cone, allowing their escape from confinement.

2.10.2 Non-adiabatic motion in symmetric geometry

As we have demonstrated in previous sections, motion characterized by an adiabatic invariant occurs when variations, whether temporal or spatial, in electromagnetic fields during gyration orbits are sufficiently gradual to be treated as analytic. As mentioned previously, when non-adiabatic motion occurs, we do not have the necessary knowledge to develop dynamic models that effectively describe violations of adiabatic invariants. This lack of adequate tools is due to the non-analytic nature of electromagnetic fields in these conditions. Nevertheless, certain situations can be the subject of intuition-rich analysis in non-adiabatic motion. By way of illustration, we will consider a situation where the electromagnetic field is perfectly symmetric with respect to the coordinate q_i. It is precisely symmetry that will allow us to develop analytical descriptions of non-adiabatic motion. Symmetry along coordinate q_i ensures that the generalized momentum, denoted $\mathcal{P}_i$, is an exact constant of motion, by virtue of Noether's theorem. In other words, symmetry determines analyticity. The constancy of a motion variable allows the production of a partial, or even complete, solution to the considered non-adiabatic motion problem. Solutions to symmetric problems will help develop intuition that proves valuable when transitioning to non-analytical asymmetric problems.

2.10.3 Digression: Stream functions

When a two-dimensional flow satisfies the continuity equation, $\nabla \cdot \mathbf{u} = 0$, it is appropriate to introduce a scalar field, called the *stream function*. A zero-divergence field can always be represented by the curl of a vector potential $\mathbf{A}$ whose stream function is part of the coordinates. Depending on the geometry of the problem, it is possible to distinguish two types of stream functions: Lagrange functions and Stokes functions. First, we study Lagrange stream functions.

2.10.3.1 Lagrange Stream Function

We consider here an irrotational, purely two-dimensional flow in the xy plane. Let $\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}$ be the matrix corresponding to a rotation of $90°$ around the $\hat{\mathbf{x}}$ axis. It is undoubted that a two-dimensional flow $\mathbf{u}$, whether compressible or not, satisfies the following equation:

$$(\nabla \cdot \mathbf{u})\, \hat{\mathbf{z}} = -\nabla \times (\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u}) \tag{2.92}$$

In case of incompressibility, it follows that the curl of the field $\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u}$ is exactly zero. By virtue of Stokes' theorem, the integral of the field $\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u}$ vanishes on any closed curve:

$$\nabla \times \left(\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u}\right) = \mathbf{0} \implies \oint_{\partial\mathcal{C}} (\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u}) \cdot \mathrm{d}\boldsymbol{\ell} = 0 \tag{2.93}$$

The integral being path-independent, it follows, by virtue of the converse of the gradient theorem, that there exists a scalar field satisfying:

$$\mathcal{R}^{\hat{\mathbf{x}}}_{\pi/2}\mathbf{u} = \nabla\psi \tag{2.94}$$

Having demonstrated the existence of a stream function for an incompressible flow, we will write equation (2.94) in the following form:

$$\mathbf{u} = \nabla \times \mathbf{A} = \nabla \times (\psi\,\hat{\mathbf{z}}) = \nabla \times (\psi\,\nabla z) = \psi\,\underbrace{\nabla \times \hat{\mathbf{z}}}_{\mathbf{0}} + \nabla\psi \times \hat{\mathbf{z}} = \nabla\psi \times \nabla z \tag{2.95}$$

It follows that the flux through any curve is given by the following:

$$\Phi_\Gamma = \int_\Gamma \mathbf{u} \cdot \mathrm{d}\boldsymbol{\ell} = \int_\Gamma (\nabla\psi \times \nabla z) \cdot \mathrm{d}\boldsymbol{\ell} = \Delta\psi \tag{2.96}$$

Furthermore, the velocity field and the gradient of the stream function are orthogonal. This observation allows defining a streamline implicitly:

$$\mathbf{u} \cdot \nabla\psi = 0 \implies \psi(x, y, t) = \text{const. along a streamline} \tag{2.97}$$

2.10.3.2 Stokes Stream Function

An axially symmetric flow in cylindrical geometry is certainly not two-dimensional. However, the following symmetry allows us to define another form of stream function:

$$\frac{\partial \mathbf{u}}{\partial \varphi} = \mathbf{0} \implies \mathbf{u} \cdot \hat{\boldsymbol{\varphi}} = \text{const.} = 0 \text{ according to our needs} \tag{2.98}$$

Like definition (2.97), the implicit equation of a streamline is given by:

$$\psi(r, z, t) = \text{const. along a streamline} \tag{2.99}$$

Since the flow is symmetric under rotation by an angle φ, we determine that the Stokes stream function satisfies the following expression. Nota bene: $\nabla\varphi = \hat{\boldsymbol{\varphi}}/r$:

$$\mathbf{u} = \frac{1}{2\pi}\nabla\psi \times \nabla\varphi \tag{2.100}$$

A flow described by expressions (2.98) and (2.100) is called *poloidal*. Like equation (2.96), the flow flux through a circle of radius r centred at z in the radial plane is given by the following:

$$\Phi_{\mathcal{C}(r,z)}(t) = \iint_{\mathcal{C}(r,z)} \mathbf{u} \cdot \mathrm{d}\mathbf{S} = \frac{1}{2\pi}\iint_{\mathcal{C}(r,z)} (\nabla\psi \times \nabla\varphi) \cdot \mathrm{d}\mathbf{S} = \psi\Big|_{(0,z,t)}^{(r,z,t)} \tag{2.101}$$

We isolate the radial component of the field $\mathbf{u}$. Since this field is divergence-free, the radial component at $r = 0$ must vanish, so:

$$u_\mathrm{r} = -\frac{1}{2\pi r}\frac{\partial\psi}{\partial z} = 0 \implies \left.\frac{\partial\psi}{\partial z}\right|_{r=0} = 0 \implies \psi(0, z, t) = \text{const.} = 0 \text{ for convenience} \tag{2.102}$$

Thus, the flux through a circle of radius r centred at z in the radial plane is equal to the stream function evaluated on the edge of the circle, in other words, at (r, z):

$$\Phi_{\mathcal{C}(r,z)}(t) = \psi(r, z, t) \tag{2.103}$$

We will examine two illustrative examples of non-adiabatic motion. These examples concern (i) a sudden temporal reversal and (ii) a sudden spatial reversal of the polarity of an axially symmetric magnetic field with zero azimuthal component. Indeed, the magnetic field being a divergence-free field, it can be described using a Stokes stream function:

$$\mathbf{B}(r, z, t) = \frac{1}{2\pi}\nabla\psi(r, z, t) \times \nabla\varphi \implies \Phi_{\mathcal{C}(r,z)} = \iint_{\mathcal{C}(r,z)} \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \psi(r, z, t) \tag{2.104}$$

Like equation (2.95), the vector potential is given by the following:

$$\mathbf{A} = \frac{1}{2\pi r}\psi(r, z, t)\,\hat{\boldsymbol{\varphi}} \tag{2.105}$$

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 2.2: Poloidal $\mathbf{B}$. The flux through $\mathcal{C}(r,z)$ is $\psi(r,z,t)$.

Using a Maxwell equation, in the static case, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, we determine the electric current giving rise to the considered magnetic field. The current is purely azimuthal, quickly verified:

$$\mu_0 \mathbf{J} \cdot \hat{\boldsymbol{\varphi}} = r\nabla\varphi \cdot (\nabla \times \mathbf{B}) = r\nabla \cdot (\mathbf{B} \times \nabla\varphi) = -\frac{r}{2\pi}\nabla \cdot \left(\frac{1}{r^2}\nabla\psi\right) \tag{2.106}$$

$$J_\varphi = -\frac{r}{2\pi\mu_0}\left[\frac{\partial}{\partial r}\left(\frac{1}{r^2}\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial\psi}{\partial z^2}\right] \tag{2.107}$$

The current distribution giving rise to the magnetic field consists, in the most general case, of loops of finite radii r and vertical positions z. The axial component of the magnetic field is written:

$$B_z = \frac{1}{2\pi r}\frac{\partial\psi}{\partial r} \tag{2.108}$$

The stream function can be expanded in a Taylor series in the vicinity of $r = 0$:

$$\psi(r,z,t) = 0 + r\left.\frac{\partial\psi}{\partial r}\right|_{(0,z,t)} + \frac{1}{2}r^2\left.\frac{\partial^2\psi}{\partial r^2}\right|_{(0,z,t)} + \mathcal{O}(r^3) \tag{2.109}$$

Suppose the term $\partial_r\psi\,(0,z,t)$ is non-zero. Then, in the vicinity of $r = 0$, $\psi \sim r$. However, in this case, the right-hand side of equation (2.107) becomes infinite. This result is obviously unphysical, so we require that the first non-zero term of the Taylor expansion of ψ be the quadratic term. Furthermore, it is evident that the stream function $\psi(r,z,t)$ is positive semi-definite according to our conventions and reaches a maximum. Let $r_{\max}(z) \equiv r_{\max}$ such that:

$$\psi_{\max} = \sup_{r\in\mathbb{R}_+}\psi(r,z) = \psi(r_{\max},z) \tag{2.110}$$

The supremum exists and is reached because the stream function is analytic. The Lagrangian of a particle of mass m and charge q in this cylindrical coordinate system appears in the form:

$$\mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\varphi}^2 + \dot{z}^2\right) + qr\dot{\varphi}A_\varphi - q\phi(r,z,t) \tag{2.111}$$

The generalized momentum is then an invariant of motion:

$$\mathcal{P}_\varphi = \frac{\partial\mathcal{L}}{\partial\dot{\varphi}} = mr^2\dot{\varphi} + qrA_\varphi = mr^2\dot{\varphi} + \frac{q}{2\pi}\psi(r,z,t) = \text{const.} \tag{2.112}$$

The Hamiltonian is then given by the following, where we define $\chi(r,z,t)$ as the effective potential:

$$\mathcal{H} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\varphi}^2 + \dot{z}^2\right) + \phi(r,z,t) = \frac{1}{2}m(\dot{r}^2 + \dot{z}^2) + \frac{(\mathcal{P}_\varphi - q\psi(r,z,t)/2\pi)^2}{2mr^2} + \phi(r,z,t) \tag{2.113}$$

$$\chi(r,z,t) = \frac{1}{2m}\left[\frac{\mathcal{P}_\varphi - q\psi(r,z,t)/2\pi}{r}\right]^2 \tag{2.114}$$

For simplicity, it is appropriate to normalize the effective potential to an arbitrary value. Furthermore, the electrostatic potential will be fixed at zero, due to intuition already developed regarding particle motion in electrostatic fields:

$$\chi_0 = \frac{q\psi_0^2}{8\pi^2 mL^2} \implies \frac{\chi(r,z,t)}{\chi_0} = \left[\frac{(2\pi\mathcal{P}_\varphi/q) - \psi(r,z,t)}{(r/L)\psi_0}\right]^2 \tag{2.115}$$

The current distribution is assumed static when $t < 0$. Since the Lagrangian is time-independent, the energy $\mathcal{H}$ is a constant of motion, associated with $\mathcal{P}_\varphi$ by symmetry. We consider a particle initially located in the plane $z = z_0$ and radius $r_0 < r_{\max}$. Its motion depends on the sign of $q\psi/\mathcal{P}_\varphi$. We thus distinguish the following two cases:

1. $q\psi/\mathcal{P}_\varphi$ is positive. If $2\pi|\mathcal{P}_\varphi| < |q\psi_{\max}|$, there exist two loci at $r_0 < r_{\max}$ and $r_0' > r_{\max}$ respectively, where the effective potential χ is zero:

$$\mathcal{P}_\varphi = \frac{q}{2\pi}\psi(r_0, z_0) \tag{2.116}$$

The effective potential χ being positive semi-definite, these two loci are also its two global minima, and there exists a local maximum $\chi_{\max}$ between them. If the energy $\mathcal{H} < \chi_{\max}$, the particle will be confined in an effective potential well centred around the edge of a surface through which the flux is conditioned by equation (2.116). The angular velocity $\dot{\varphi}$ changes sign periodically by virtue of equation (2.112):

$$\frac{\mathrm{d}\varphi}{\mathrm{d}t}(r, z_0) = \frac{1}{mr^2}\left(\mathcal{P}_\varphi - \frac{q\psi(r, z_0)}{2\pi}\right) \tag{2.117}$$

This corresponds precisely to a localized gyration. We will refer to this type of motion by the term *non-encircling* of the $\hat{\mathbf{z}}$ axis.

2. $q\psi/\mathcal{P}_\varphi$ is negative. In this case, χ reaches zero nowhere, but it reaches a local minimum:

$$\left.\left[\left(\mathcal{P}_\varphi - \frac{q}{2\pi}\psi(r, z)\right)\frac{\partial}{\partial r}\left(\frac{\mathcal{P}_\varphi - q\psi(r, z)/2\pi}{r}\right)\right]\right|_{(r_{\min}, z_0)} = 0 \tag{2.118}$$

This equation is satisfied only if the partial derivative is zero:

$$\left.\frac{\partial}{\partial r}\left(\frac{\mathcal{P}_\varphi - q\psi(r, z)/2\pi}{r}\right)\right|_{(r_{\min}, z_0)} = 0 \implies \mathcal{P}_\varphi = -\left.\left[\frac{qr^2}{2\pi}\frac{\partial}{\partial r}\left(\frac{\psi(r, z)}{r}\right)\right]\right|_{(r_{\min}, z_0))} \tag{2.119}$$

Since the stream function ψ is positive semi-definite, $\psi \sim r^2$ as $r \to 0$ and $\psi \to 0$ as $r \to \infty$, we determine that $\psi/r \sim r \to 0$ as $r \to 0$ and $\psi/r \to 0$ as $r \to \infty$. Since the function ψ/r is also positive semi-definite, it reaches a maximum in a locus satisfying $r < r_{\max}$. Let $\sup_{r \leq r_{\max}}(\psi/r) = \psi(\rho)/\rho$. It should be noted that equation (2.119) is satisfied only on the condition that $|\mathcal{P}_\varphi|$ is not too large, because the right-hand side of equation (2.119) reaches a finite maximum. Furthermore, this equation is only valid for points surrounding the maximum of ψ/r. Since the function ψ/r increases on the interval $[0, \rho)$, then $\dot{\varphi}$ does not change sign, meaning the particle exerts a purely rotational motion around the $\hat{\mathbf{z}}$ axis. When r is very close to zero, we could approximate equation (2.119) by the following, clearly similar to equation (2.116):

$$\mathcal{P}_\varphi = -\frac{q}{2\pi}\psi(r_{\min}, z_0) \tag{2.120}$$

Nota bene! This motion is susceptible to undergoing subtle radial variations. Nevertheless, $\dot{\varphi}$ always retains the same sign. We will refer to motions during which $\dot{\varphi}$ retains its sign by the term *encircling* the $\hat{\mathbf{z}}$ axis.

2.10.4 Sudden temporal reversal of the magnetic field

We will now examine a highly non-adiabatic situation in which the current distribution, as considered thus far, undergoes a rapid sign change. In this configuration, the set of fields and fluxes reverses its sign. The energy of the particle cannot be considered a constant of motion, because the Lagrangian explicitly depends on time. However, the symmetry of the problem guarantees the constancy of the azimuthal component of the generalized momentum, $\mathcal{P}_\varphi$. The sign reversal of ψ, if performed, can lead to the presence of a minimum of the effective potential χ. A particle not encircling the $\hat{\mathbf{z}}$ axis at a position determined by equation (2.116) will become encircling. Under the assumption of a particle located near the $\hat{\mathbf{z}}$ axis, by comparison of equations (2.116) and (2.120), the radial position of the particle will be conserved. The particle, now encircling the $\hat{\mathbf{z}}$ axis compared to the contrary case before the sign reversal, allows us to observe an increase in its kinetic energy equivalent to the minimum value of χ of the encircling case.

It would also be appropriate to illustrate the process in question from the perspective of particle drifts. Initially, the particle was frozen by the magnetic field lines, so as to follow the surface characterized by

$\psi(r, z) = \text{const}$. Once the current distribution begins to decrease in intensity, the maximum flux $\psi_{\max}$ decreases accordingly. The contours of constant ψ inside the circle of radius $r_{\max}$ move towards this radius where $\psi = \psi_{\max}$. It is the same for contours of constant ψ outside the aforementioned circle. The annihilation of contours reaching radius $r_{\max}$ follows. The effect of the $\mathbf{E} \times \mathbf{B}$ drift maintains the motion of particles fixed to a surface of constant flux. Considering a fixed circle of radius r_0 at $z = z_0$, the Maxwell-Faraday equation is written as follows:

$$\oint_{\partial\mathcal{C}(r_0,z_0)} \mathbf{E} \cdot \mathrm{d}\boldsymbol{\ell} = 2\pi r_0 E_\varphi = -\left.\frac{\mathrm{d}\Phi_{\mathbf{B}}}{\mathrm{d}t}\right|_{\mathcal{C}(r_0,z_0)} = -\frac{\mathrm{d}}{\mathrm{d}t}\left[\psi(r_0, z_0, t)\right] = -\left.\frac{\partial \psi}{\partial t}\right|_{(r_0,z_0)} \tag{2.121}$$

The radial and axial components of the drift velocity give us:

$$E_\varphi + v_{\mathrm{z}} B_{\mathrm{r}} - v_{\mathrm{r}} B_{\mathrm{z}} = 0 \quad \text{with} \quad B_{\mathrm{r}} = -\frac{1}{2\pi r}\frac{\partial \psi}{\partial z} \quad \text{and} \quad B_{\mathrm{z}} = \frac{1}{2\pi r}\frac{\partial \psi}{\partial r} \tag{2.122}$$

$$\frac{\partial \psi}{\partial t} + v_{\mathrm{r}}\frac{\partial \psi}{\partial r} + v_{\mathrm{z}}\frac{\partial \psi}{\partial z} \equiv \frac{\partial \psi}{\partial t} + (\mathbf{v} \cdot \nabla)\psi = \frac{\mathrm{d}\psi}{\mathrm{d}t} = 0 \tag{2.123}$$

The convective derivative being zero, the particle's motion contours a surface of constant ψ. It should be noted that the concept of $\mathbf{E} \times \mathbf{B}$ drift collapses completely at the precise moment when $\mathbf{B} = \mathbf{0}$. This presents a failure of the adiabatic approximation. It should be noted that the particle will encircle the $\hat{\mathbf{z}}$ axis following the magnetic field reversal, provided that the stream function undergoes a sign reversal before the particle reaches the surface corresponding to the maximum of the stream function $\psi_{\max}$. The kinetic energy of the particle increases at the instant when the azimuthal electric field exerts work, when $\mathbf{B} = \mathbf{0}$. Once the sign reversal of ψ is established, the adiabatic approximation is valid again. The polarity of ψ being reversed, the increase in the magnitude of ψ is analogous to an increasing potential well. The particle thus moves away from the surface corresponding to $\psi_{\min} = -\psi_{\max}$. Once the reversal is completed, the particles will encircle the $\hat{\mathbf{z}}$ axis and acquire additional kinetic energy, acquired during the instant of sign reversal.

2.10.5 Sudden spatial reversal of the magnetic field

We consider an arrangement of two solenoids carrying constant currents, with their magnetic fields opposing each other. Since the currents are constant, the Lagrangian of a particle located inside one of the solenoids does not depend explicitly on time. The energy of the particle is thus a constant of its motion. The given geometry allows us to conclude that the stream function is antisymmetric in z, where $z = 0$ corresponds to the plane of symmetry between the two solenoids, as illustrated by Figure 2.3.

Consider a particle launched at initial velocity $\mathbf{v} = v_0\,\hat{\mathbf{z}}$ at $z = -L$ and radius $r = a$. Since the velocity component is perpendicular to the $\hat{\mathbf{z}}$ axis, the particle will simply follow a magnetic field line. Once the particle reaches the cusp region, the magnetic field lines begin to curve, as illustrated in Figure (2.3). Consequently, the particle develops curvature and drifts along the magnetic field gradient. When the particle approaches the plane $z = 0$, the magnetic field tends towards zero, which is followed by the collapse of the concept of drift. Despite the complexity of the particle's trajectory in the cusp region, it is possible to determine the condition for a particle to pass through this cusp. The constants of motion are precisely the energy $\mathcal{H}$ and the azimuthal component of the generalized momentum.

Figure 2.3: Lines of **B** and trajectory of a particle crossing the cusp in blue.

[[IMAGE DISCARDED DUE TO '/tikz/external/mode=list and make']]

$$\mathcal{P}_\varphi = mr^2\dot{\varphi} + \frac{q}{2\pi}\psi(r, z, t) = \frac{q}{2\pi}\psi(a, -L) = \text{const.} \tag{2.124}$$

$$\mathcal{H} = \frac{\mathbf{p}^2}{2m} = \frac{1}{2}mv_0^2 = \text{const.} \tag{2.125}$$

The minimum energy to overcome the cusp corresponds to motion encircling the $\hat{\mathbf{z}}$ axis and zero radial and axial components. Equations (2.124) and (2.125) give us the sought condition:

$$v_0^2 \geq r^2\dot{\varphi}^2 = \left(\frac{q}{2\pi m r}\right)^2\left[\psi(a, -L) - \psi(r, z)\right]^2 \tag{2.126}$$

Depending on the value of v_0, equation (2.126) will give us several solutions for r. Imaginary and negative solutions are unphysical. This corresponds to the particle's inability to penetrate the cusp. It will return towards the interior of the first solenoid. However, positive real solutions correspond to a genuine passage through the cusp, illustrated in figure (2.3).

Chapter 3

Models of plasma dynamics

We will devote the present chapter to the in-depth examination of the three main models underlying plasma dynamics. These models, which will be detailed in the sections to come, include the Vlasov theory, the two-fluid theory, and magnetohydrodynamics. Although the three aforementioned models differ in perspective, they share a common concept: the conservation of particles in phase space. The choice of a model when examining a given situation is, above all, a question of selecting the tool best suited to the specific problem. It is often judicious to alternate among these models when confronting a particular problem.

3.1 Distribution function and the Vlasov Equation

Consider a volume $\mathrm{d}^3\mathbf{p}\,\mathrm{d}^3\mathbf{x}$ occupied by plasma particles of species σ in phase space. When following the particles within the aforementioned volume, it is evident that their total number is conserved. This translates to the cancellation of the *convective* drift of the distribution function in phase space. Let us note the nuance concerning the convective derivative in phase space:

$$\frac{\mathrm{d}f_\sigma(\mathbf{x}_\sigma, \mathbf{p}_\sigma, t)}{\mathrm{d}t} = 0 \tag{3.1}$$

$$\frac{\mathrm{d}f_\sigma}{\mathrm{d}t} = \frac{\partial f_\sigma}{\partial t} + \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot (f_\sigma \mathbf{v}_\sigma) + \frac{\partial}{\partial \mathbf{p}_\sigma} \cdot (f_\sigma \mathbf{F}_\sigma) \tag{3.2}$$

In the non-relativistic case, we observe the cancellation of the mass of the considered particles. Note that we may interchange the space gradient with the velocity coordinate, since both are coordinates and thus commute. However, the same cannot, in general, be applied to the force on a particle and the momentum gradient. The equation (3.2) simplifies as follows, whose form shall be referred to as the *Vlasov equation*:

$$\frac{\partial f_\sigma(\mathbf{x}_\sigma, \mathbf{v}_\sigma, t)}{\partial t} + \mathbf{v}_\sigma \cdot \frac{\partial f(\mathbf{x}_\sigma, \mathbf{v}_\sigma, t)}{\partial \mathbf{x}_\sigma} + \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot (f(\mathbf{x}_\sigma, \mathbf{v}_\sigma, t)\mathbf{a}_\sigma) = 0 \tag{3.3}$$

The solutions of the Vlasov equation can take various forms depending on the problem at hand. Furthermore, it is possible to characterize the different types of distribution functions by one or more constants of motion. For example, the latter may be the conservation of energy or momentum. However, it is important to emphasize that, in certain circumstances, neither energy nor momentum is conserved. Nevertheless, it should be noted that for any distribution function, there certainly exists at least one constant of motion. It is trivial to notice that the initial conditions of the motion are invariant along the particle trajectories. Although this observation may seem superfluous, its implications can lead to resolutions of great sophistication. Furthermore, the equation (3.1) allows us to express the distribution function in terms of any quantity conserved in this reference frame.

It is important to note that the force and momentum terms in the equation (3.2) also underlie the possible interparticle collisions occurring during the considered motion. However, these latter lead to a very irregular functional behaviour. When two arbitrary particles collide, the interaction force can take large values, even $\|\mathbf{F}\| \to \infty$. Furthermore, during such collisions, the distribution function is not necessarily differentiable in $\mathbf{p}$. It is then appropriate to introduce the concept of annihilation and creation of particles, in such a way that it is entirely equivalent to interparticle collisions. When a particle in the considered volume undergoes a collision, it moves almost instantaneously in phase space only along the coordinate $\mathbf{p}$. Within the considered volume, it is annihilated, whereas in the new volume it appears as if it were created. We will accordingly modify the equation (3.3) to isolate the possible collisional terms:

$$\frac{\partial f_\sigma}{\partial t} + \mathbf{v}_\sigma \cdot \frac{\partial f_\sigma}{\partial \mathbf{x}_\sigma} + \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot (f_\sigma \mathbf{a}_\sigma) = \sum_\rho C_{\sigma\rho}(f_\sigma) \tag{3.4}$$

The collision operator $C_{\sigma\rho}$, which represents the rate of variation of the distribution function f_σ due to collisions between particles of species σ and ρ, necessarily satisfies the following conditions:

1. Conservation of particles – collisions do not modify the total number of particles at a particular position:

$$\iiint_{\mathbb{R}^3} C_{\sigma\rho}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma = 0 \tag{3.5}$$

2. Conservation of momentum – collisions between particles of the same species do not modify the total momentum of the species:

$$\iiint_{\mathbb{R}^3} m_\sigma \mathbf{v}_\sigma C_{\sigma\sigma}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma = \mathbf{0} \tag{3.6}$$

Interspecific collisions must, however, guarantee the conservation of momentum:

$$\iiint_{\mathbb{R}^3} m_\sigma \mathbf{v}_\sigma C_{\sigma\rho}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma + \iiint_{\mathbb{R}^3} m_\rho \mathbf{v}_\rho C_{\rho\sigma}(f_\rho)\,\mathrm{d}^3\mathbf{v}_\rho = \mathbf{0} \tag{3.7}$$

3. Conservation of energy – collisions between particles of the same species do not modify the total energy of the species:

$$\iiint_{\mathbb{R}^3} m_\sigma \mathbf{v}_\sigma^2 C_{\sigma\sigma}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma = 0 \tag{3.8}$$

However, interspecific collisions must guarantee the conservation of energy:

$$\iiint_{\mathbb{R}^3} m_\sigma \mathbf{v}_\sigma^2 C_{\sigma\rho}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma + \iiint_{\mathbb{R}^3} m_\rho \mathbf{v}_\rho^2 C_{\rho\sigma}(f_\rho)\,\mathrm{d}^3\mathbf{v}_\rho = 0 \tag{3.9}$$

3.2 The two-fluid equations

We will integrate the Vlasov equation, in particular the form given by equation (3.4), to obtain a set of partial differential equations. This latter precisely presents the two-fluid equations for ions and electrons. We integrate the equation (3.4) over velocity space:

$$\iiint_{\mathbb{R}^3} \left[\frac{\partial f_\sigma}{\partial t} + \mathbf{v}_\sigma \cdot \frac{\partial f_\sigma}{\partial \mathbf{x}_\sigma} + \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot (f_\sigma \mathbf{a}_\sigma)\right] \mathrm{d}^3\mathbf{v}_\sigma = \sum_\rho \underbrace{\iiint_{\mathbb{R}^3} C_{\sigma\rho}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma}_{0} \tag{3.10}$$

$$\iiint_{\mathbb{R}^3} \left[\frac{\partial f_\sigma}{\partial t} + \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot (f_\sigma \mathbf{v}_\sigma) \cdot\right] \mathrm{d}^3\mathbf{v}_\sigma = \frac{\partial}{\partial t} \iiint_{\mathbb{R}^3} f_\sigma\,\mathrm{d}^3\mathbf{v}_\sigma + \nabla \cdot \iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma\,\mathrm{d}^3\mathbf{v}_\sigma = \frac{\partial n_\sigma}{\partial t} + \nabla \cdot (n_\sigma \mathbf{u}_\sigma) \tag{3.11}$$

$$\iiint_{\mathbb{R}^3} \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot (f_\sigma \mathbf{a}_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma = \oiint_{\partial\mathbb{R}^3} f_\sigma \mathbf{a}_\sigma \cdot \mathrm{d}^2\mathbf{\Sigma}_{\mathbf{v}_\sigma} = 0 \quad \text{because} \quad \forall\sigma \lim_{\|\mathbf{v}_\sigma\|\to\infty} f_\sigma = 0 \tag{3.12}$$

By substituting the results (3.11) and (3.12) into the equation (3.10), we obtain the continuity equation:

$$\frac{\partial n_\sigma}{\partial t} + \nabla \cdot (n_\sigma \mathbf{u}_\sigma) = 0 \tag{3.13}$$

where $\mathbf{u}_\sigma$ represents the average velocity of particles of species σ at the position $\mathbf{x}_\sigma$. By multiplying equation (3.4) by the velocity $\mathbf{v}_\sigma$ and integrating over velocity space, we obtain the following:

$$\iiint_{\mathbb{R}^3} \left[\frac{\partial}{\partial t}(f_\sigma \mathbf{v}_\sigma) + \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot (f_\sigma \mathbf{v}_\sigma \mathbf{v}_\sigma) + \mathbf{v}_\sigma \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot (f_\sigma \mathbf{a}_\sigma)\right] \mathrm{d}^3\mathbf{v}_\sigma = \sum_\rho \iiint_{\mathbb{R}^3} \mathbf{v}_\sigma C_{\sigma\rho}(f_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma \tag{3.14}$$

The first term simply becomes the partial time derivative of $n_\sigma \mathbf{u}_\sigma$. The second term deserves a more careful analysis. We will make use of the following substitution:

$$\mathbf{v}_\sigma = \mathbf{u}_\sigma + \overline{\mathbf{v}}_\sigma \implies \mathrm{d}^3\mathbf{v}_\sigma = \mathrm{d}^3\overline{\mathbf{v}}_\sigma \quad \text{because} \quad \mathbf{u}_\sigma = \mathbf{u}_\sigma(\mathbf{x},t) \quad \text{then} \quad \iiint_{\mathbb{R}^3} f_\sigma \overline{\mathbf{v}}_\sigma \,\mathrm{d}^3\overline{\mathbf{v}}_\sigma = \mathbf{0} \tag{3.15}$$

$$\iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma \mathbf{v}_\sigma \,\mathrm{d}^3\mathbf{v}_\sigma = \iiint_{\mathbb{R}^3} f_\sigma \left(\mathbf{u}_\sigma\mathbf{u}_\sigma + \mathbf{u}_\sigma\overline{\mathbf{v}}_\sigma + \overline{\mathbf{v}}_\sigma\mathbf{u}_\sigma + \overline{\mathbf{v}}_\sigma\overline{\mathbf{v}}_\sigma\right) \mathrm{d}^3\overline{\mathbf{v}}_\sigma \tag{3.16}$$

The average velocity $\mathbf{u}_\sigma$ being independent of $\overline{\mathbf{v}}_\sigma$, the integration of the mixed terms leads to their cancellation. We will define the pressure tensor in the following way:

$$\mathbf{P}_\sigma = m_\sigma \iiint_{\mathbb{R}^3} f_\sigma \overline{\mathbf{v}}_\sigma \overline{\mathbf{v}}_\sigma \,\mathrm{d}^3\overline{\mathbf{v}}_\sigma \tag{3.17}$$

$$\frac{\partial}{\partial \mathbf{x}_\sigma} \cdot \iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma \mathbf{v}_\sigma \,\mathrm{d}^3\mathbf{v}_\sigma = \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot \left(n_\sigma \mathbf{u}_\sigma \mathbf{u}_\sigma + \frac{1}{m_\sigma}\mathbf{P}_\sigma\right) \tag{3.18}$$

The right-hand side of the equation (3.14) represents the friction force of species σ against species ρ:

$$\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma C_{\sigma\rho}(f_\sigma) \,\mathrm{d}^3\mathbf{v}_\sigma = -n_\sigma \nu_{\sigma\rho} \left(\mathbf{u}_\rho - \mathbf{u}_\sigma\right) \Longleftrightarrow \mathbf{R}_{\sigma\rho} = n_\sigma m_\sigma \nu_{\sigma\rho} \left(\mathbf{u}_\sigma - \mathbf{u}_\rho\right) \tag{3.19}$$

When it is only the electromagnetic fields that drive the non-relativistic acceleration of any particle:

$$\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma \mathbf{a}_\sigma \cdot \frac{\partial f_\sigma}{\partial \mathbf{v}_\sigma} \,\mathrm{d}^3\mathbf{v}_\sigma = \frac{q_\sigma}{m_\sigma} \iiint_{\mathbb{R}^3} \mathbf{v}_\sigma \left(\mathbf{E} + \mathbf{v}_\sigma \times \mathbf{B}\right) \cdot \frac{\partial f_\sigma}{\partial \mathbf{v}_\sigma} \,\mathrm{d}^3\mathbf{v}_\sigma \tag{3.20}$$

We will proceed to integrate by parts the last integral. The equation (3.15) allows to determine:

$$\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma \cdot \frac{\partial f_\sigma}{\partial \mathbf{v}_\sigma} \,\mathrm{d}^3\mathbf{v}_\sigma = \underbrace{f_\sigma \hat{\mathbf{n}} \cdot \mathbf{v}_\sigma\big|_{\partial\mathbb{R}^3}}_{0} - \iiint_{\mathbb{R}^3} f_\sigma \,\mathrm{d}^3\mathbf{v}_\sigma = -n_\sigma \tag{3.21}$$

$$\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma \mathbf{v}_\sigma \cdot \frac{\partial f_\sigma}{\partial \mathbf{v}_\sigma} \,\mathrm{d}^3\mathbf{v}_\sigma = \underbrace{f_\sigma \hat{\mathbf{n}} \cdot \mathbf{v}_\sigma \mathbf{v}_\sigma\big|_{\partial\mathbb{R}^3}}_{0} - \iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma \,\mathrm{d}^3\mathbf{v}_\sigma = -n_\sigma \mathbf{u}_\sigma \tag{3.22}$$

By gathering the previous equations, we recover the following form of the equation (3.14):

$$m_\sigma \left[\frac{\partial}{\partial t}\left(n_\sigma \mathbf{u}_\sigma\right) + \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot \left(n_\sigma \mathbf{u}_\sigma \mathbf{u}_\sigma\right)\right] = n_\sigma q_\sigma \left(\mathbf{E} + \mathbf{u}_\sigma \times \mathbf{B}\right) - \frac{\partial}{\partial \mathbf{x}_\sigma} \cdot \mathbf{P}_\sigma + \sum_\rho \mathbf{R}_{\sigma\rho} \tag{3.23}$$

We extend the left-hand side of the previous equation to arrive at the following:

$$\mathbf{u}_\sigma \frac{\partial n_\sigma}{\partial t} + n_\sigma \frac{\partial \mathbf{u}_\sigma}{\partial t} + \mathbf{u}_\sigma \left[\nabla \cdot \left(n_\sigma \mathbf{u}_\sigma\right)\right] + n_\sigma (\mathbf{u}_\sigma \cdot \nabla)\mathbf{u}_\sigma = n_\sigma \frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} \tag{3.24}$$

Since the equation (3.23) is no longer dependent on velocity space, we write it in a more familiar form. Using the continuity equation (3.13), we obtain the equation of motion:

$$n_\sigma m_\sigma \frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} = n_\sigma q_\sigma \left(\mathbf{E} + \mathbf{u}_\sigma \times \mathbf{B}\right) - \nabla \cdot \mathbf{P}_\sigma + \sum_\rho \mathbf{R}_{\sigma\rho} \tag{3.25}$$

We will then multiply the equation (3.4) by $m_\sigma \mathbf{v}_\sigma^2/2$ which will give us the energy evolution:

$$\begin{aligned} &\iiint_{\mathbb{R}^3} \left[\frac{\partial}{\partial t}\left(\frac{m_\sigma \mathbf{v}_\sigma^2}{2} f_\sigma\right) + \frac{\partial}{\partial \mathbf{x}} \cdot \left(\frac{m_\sigma \mathbf{v}_\sigma^2}{2} f_\sigma \mathbf{v}_\sigma\right) + \right. \\ &\left. + \frac{m_\sigma \mathbf{v}_\sigma^2}{2} \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot \left(f_\sigma \mathbf{a}_\sigma\right)\right] \mathrm{d}^3\mathbf{v}_\sigma = \sum_\rho \iiint_{\mathbb{R}^3} \frac{m_\sigma \mathbf{v}_\sigma^2}{2} C_{\sigma\rho}(f_\sigma) \,\mathrm{d}^3\mathbf{v}_\sigma \end{aligned} \tag{3.26}$$

We decompose the velocity $\mathbf{v}_\sigma$ into two terms: $\mathbf{v}_\sigma = \mathbf{u}_\sigma + \overline{\mathbf{v}}_\sigma$. The mixed term disappears, then:

$$\iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma^2 \,\mathrm{d}^3\mathbf{v}_\sigma = \iiint_{\mathbb{R}^3} f_\sigma \left(\mathbf{u}_\sigma + \overline{\mathbf{v}}_\sigma\right)^2 \mathrm{d}^3\overline{\mathbf{v}}_\sigma = n_\sigma \mathbf{u}_\sigma^2 + \frac{1}{m_\sigma} \mathrm{Tr}\,\mathbf{P}_\sigma \tag{3.27}$$

Using this decomposition again, we solve the second term:

$$\iiint_{\mathbb{R}^3} f_\sigma \mathbf{v}_\sigma^2 \mathbf{v}_\sigma \, \mathrm{d}\mathbf{v}_\sigma = \iiint_{\mathbb{R}^3} f_\sigma \left(\mathbf{u}_\sigma^2 + 2\mathbf{u}_\sigma \cdot \overline{\mathbf{v}}_\sigma + \overline{\mathbf{v}}_\sigma^2\right) \left(\mathbf{u}_\sigma + \overline{\mathbf{v}}_\sigma\right) \mathrm{d}^3\overline{\mathbf{v}}_\sigma =$$

$$= n_\sigma \mathbf{u}_\sigma^2 \mathbf{u}_\sigma + \frac{2}{m_\sigma} \mathbf{u}_\sigma \cdot \mathbf{P}_\sigma + \frac{1}{m_\sigma} \mathbf{u}_\sigma \mathrm{Tr}\, \mathbf{P}_\sigma + \frac{2}{m_\sigma} \mathbf{Q}_\sigma \tag{3.28}$$

where we have defined the heat flux:

$$\mathbf{Q}_\sigma = \iiint_{\mathbb{R}^3} \frac{1}{2} m_\sigma f_\sigma \overline{\mathbf{v}}_\sigma^2 \overline{\mathbf{v}}_\sigma \, \mathrm{d}^3\overline{\mathbf{v}}_\sigma \tag{3.29}$$

When the acceleration is induced only by electromagnetic fields, we solve the third term of the left-hand side of the equation (3.25):

$$\iiint_{\mathbb{R}^3} \frac{q_\sigma}{m_\sigma} \mathbf{v}_\sigma^2 \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot [f_\sigma \mathbf{E}] \, \mathrm{d}^3\mathbf{v}_\sigma = -\frac{q_\sigma}{m_\sigma} \mathbf{E} \cdot \iiint_{\mathbb{R}^3} 2 f_\sigma \mathbf{v}_\sigma \, \mathrm{d}^3\mathbf{v}_\sigma = -\frac{2q_\sigma}{m_\sigma} n_\sigma \mathbf{u}_\sigma \cdot \mathbf{E} \tag{3.30}$$

$$\iiint_{\mathbb{R}^3} \frac{q_\sigma}{m_\sigma} \mathbf{v}_\sigma^2 \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot \underbrace{(\mathbf{v}_\sigma \times \mathbf{B})}_{\perp \mathbf{v}_\sigma} \mathrm{d}^3\mathbf{v}_\sigma = 0 \tag{3.31}$$

The right-hand side of the equation (3.26) is only the rate of energy transfer by collisions between species σ and ρ. By gathering the previous equations, we recover equation (3.26) in the following form:

$$\frac{\partial}{\partial t} \left(\frac{n_\sigma m_\sigma \mathbf{u}_\sigma^2}{2} + \frac{1}{2} \mathrm{Tr}\, \mathbf{P}_\sigma \right) + \nabla \cdot \left(\frac{n_\sigma m_\sigma \mathbf{u}_\sigma^2}{2} \mathbf{u}_\sigma \right) +$$

$$+ \nabla \cdot \left(\mathbf{u}_\sigma \cdot \mathbf{P}_\sigma + \frac{1}{2} \mathbf{u}_\sigma \mathrm{Tr}\, \mathbf{P}_\sigma + \mathbf{Q}_\sigma \right) = n_\sigma q_\sigma \mathbf{u}_\sigma \cdot \mathbf{E} + \sum_\rho \left(\frac{\partial W}{\partial t} \right)_{\sigma\rho} \tag{3.32}$$

Following quite a few simplifications, we recover the continuity equation multiplied by $\mathbf{u}_\sigma^2$:

$$\frac{\partial}{\partial t} \left(n_\sigma \mathbf{u}_\sigma \cdot \mathbf{u}_\sigma \right) + \nabla \cdot \left(n_\sigma \mathbf{u}_\sigma^2 \mathbf{u}_\sigma \right) = n_\sigma \frac{\partial}{\partial t} \left(\mathbf{u}_\sigma^2 \right) + \mathbf{u}_\sigma^2 \left[\frac{\partial n_\sigma}{\partial t} + \nabla \cdot \left(n_\sigma \mathbf{u}_\sigma \right) \right] + n_\sigma (\mathbf{u}_\sigma \cdot \nabla) \left(\mathbf{u}_\sigma^2 \right) \tag{3.33}$$

The left-hand side of the equation (3.32) then becomes:

$$n_\sigma m_\sigma \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\mathbf{u}_\sigma^2}{2} \right) + \frac{\partial}{\partial t} \left(\frac{\mathrm{Tr}\, \mathbf{P}_\sigma}{2} \right) + \nabla \cdot (\mathbf{u}_\sigma \cdot \mathbf{P}_\sigma) + \nabla \cdot \left(\frac{\mathbf{u}_\sigma \mathrm{Tr}\, \mathbf{P}_\sigma}{2} \right) + \nabla \cdot \mathbf{Q}_\sigma \tag{3.34}$$

We recover the advective derivative of the trace of the pressure tensor by the following:

$$\nabla \cdot (\mathbf{u}_\sigma \mathrm{Tr}\, \mathbf{P}_\sigma) = \mathbf{u}_\sigma \cdot \nabla \left(\mathrm{Tr}\, \mathbf{P}_\sigma \right) + \mathrm{Tr}\, \mathbf{P}_\sigma (\nabla \cdot \mathbf{u}_\sigma) \tag{3.35}$$

Consequently, the left-hand side of the equation (3.32) becomes:

$$n_\sigma m_\sigma \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\mathbf{u}_\sigma^2}{2} \right) + \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\mathrm{Tr}\, \mathbf{P}_\sigma}{2} \right) + \frac{\mathrm{Tr}\, \mathbf{P}_\sigma}{2} (\nabla \cdot \mathbf{u}_\sigma) + \nabla \cdot (\mathbf{u}_\sigma \cdot \mathbf{P}_\sigma) + \nabla \cdot \mathbf{Q}_\sigma \tag{3.36}$$

By subtracting the product of the equation (3.25) with $\mathbf{u}_\sigma$ from the equation (3.32), we obtain:

$$\frac{1}{2} \left(\frac{\mathrm{d}}{\mathrm{d}t} \left(\mathrm{Tr}\, \mathbf{P}_\sigma \right) + (\nabla \cdot \mathbf{u}_\sigma) \mathrm{Tr}\, \mathbf{P}_\sigma \right) + \mathrm{Tr}\left(\mathbf{P}_\sigma \cdot \nabla \mathbf{u}_\sigma \right) = -\nabla \cdot \mathbf{Q}_\sigma + \sum_\rho \left[\mathbf{u}_\sigma \cdot \mathbf{R}_{\sigma\rho} + \left(\frac{\partial W}{\partial t} \right)_{\sigma\rho} \right] \tag{3.37}$$

The first term of the right-hand side of the previous equation (3.37) presents the rate of heat production within species σ, the second term presents the heat by friction between species σ and ρ, while the last term represents the rate of energy transfer between species σ and ρ.

3.2.1 Adiabatic and isothermal limits

When a specific phenomenon is the subject of our examination, it is appropriate to consider the propagation speed proper to this phenomenon. Let v_{ph} be the aforementioned speed:

1. The thermal speed of the species σ satisfies the strong inequality $\sqrt{2k_{\mathrm{B}}T_\sigma/m_\sigma} \gg v_{\mathrm{ph}}$. The plasma temperature is also very high. Then, the heat flux $\mathbf{Q}_\sigma$ dominates the other terms of the right-hand side of the equation (3.37), and the interparticle collisions are very rare. Consequently, the temperature of the species σ can be considered as uniform.

2. The thermal speed of the species σ satisfies the strong inequality $\sqrt{2k_{\mathrm{B}}T/m_\sigma} \ll v_{\mathrm{ph}}$, and the temperature of the plasma is very high. In this case, the heat flux is almost zero, and the interparticle collisions are also very rare. This applied to the equation (3.37) gives us:

$$\frac{1}{2}\left(\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{Tr}\,\mathbf{P}_\sigma\right) + (\nabla\cdot\mathbf{u}_\sigma)\mathrm{Tr}\,\mathbf{P}_\sigma\right) + \mathrm{Tr}\left(\mathbf{P}_\sigma\cdot\nabla\mathbf{u}_\sigma\right) = 0 \tag{3.38}$$

The continuity equation allows us to find the divergence of $\mathbf{u}_\sigma$:

$$\nabla\cdot\mathbf{u}_\sigma = -\frac{1}{n_\sigma}\frac{\mathrm{d}n_\sigma}{\mathrm{d}t} \tag{3.39}$$

We extend the term $\mathrm{Tr}\left(\mathbf{P}_\sigma\cdot\nabla\mathbf{u}_\sigma\right)$ in coordinates by the Einstein summation convention:

$$\mathrm{Tr}\left(\mathbf{P}_\sigma\cdot\nabla\mathbf{u}_\sigma\right) \equiv P^i{}_j\partial_i u^j \tag{3.40}$$

The pressure tensor is usually diagonal. Furthermore, its behaviour is often characterized by two components, one of which is parallel to the plasma motion, $P_\sigma^{\|}$, and the other corresponds to the directions perpendicular to this motion, $P_\sigma^{\perp}$. Under the assumption of such a tensor, we introduce the direction $\hat{\mathbf{s}}$ as the one that follows the velocity $\mathbf{u}_\sigma$:

$$\mathrm{Tr}\left(\mathbf{P}_\sigma\cdot\nabla\mathbf{u}_\sigma\right) = P_\sigma^{\perp}\nabla\cdot\mathbf{u}_\sigma + (P_\sigma^{\|} - P_\sigma^{\perp})\frac{\partial\mathbf{u}_\sigma}{\partial\mathbf{s}} \tag{3.41}$$

The equation (3.38) turns out in the following form in this specific case:

$$n_\sigma^2\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{P_\sigma^{\perp}}{n_\sigma^2}\right) + n_\sigma\frac{\mathrm{d}}{\mathrm{D}t}\left(\frac{P_\sigma^{\|}}{n_\sigma}\right) + (P_\sigma^{\|} - P_\sigma^{\perp})\frac{\partial\mathbf{u}_\sigma}{\partial\mathbf{s}} = 0 \tag{3.42}$$

3.3 Magnetohydrodynamics

In contrast to the two-fluid model, magnetohydrodynamics treats the entire plasma as a single fluid. Consequently, we introduce the following quantities: the total mass density, the center of mass velocity, the total charge density, and the total current density:

$$\rho = \sum_\sigma n_\sigma m_\sigma \quad \mathbf{U} = \frac{1}{\rho}\sum_\sigma n_\sigma m_\sigma\mathbf{u}_\sigma \quad \rho_q = \sum_\sigma n_\sigma q_\sigma \quad \mathbf{J} = \sum_\sigma n_\sigma q_\sigma\mathbf{u}_\sigma \tag{3.43}$$

By multiplying the continuity equation (3.13) by m_σ and summing over all species σ, the magnetohydrodynamic continuity equation turns out in the following form:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{U}) = 0 \tag{3.44}$$

To arrive at the magnetohydrodynamic representation of the equation of motion analogous to the equation (3.25), we introduce the magnetohydrodynamic pressure tensor:

$$\mathbf{P}_{\mathrm{MHD}} = \sum_\sigma\mathbf{P}_\sigma = \sum_\sigma\iiint_{\mathbb{R}^3} f_\sigma m_\sigma\overline{\mathbf{v}}_\sigma\overline{\mathbf{v}}_\sigma\,\mathrm{d}^3\overline{\mathbf{v}}_\sigma \tag{3.45}$$

By summing the equation (3.25) over all species σ present in the plasma, we obtain:

$$\rho \frac{\mathrm{d}\mathbf{U}}{\mathrm{d}t} = \rho_q \mathbf{E} + \mathbf{J} \times \mathbf{B} - \nabla \cdot \mathbf{P}_{\mathrm{MHD}} \tag{3.46}$$

where $\mathrm{d}_t = \partial_t + \mathbf{U} \cdot \nabla$. The friction terms are internal forces, thus disappear in the MHD model.

3.4 Conservative properties of a Vlasov-Maxwell system

A Vlasov-Maxwell system consists of a collisionless plasma whose distribution function is governed by the Vlasov equation, and of the electromagnetic field in which the plasma is submerged, described by Maxwell's equations. The diverse conservation laws exhibited by a Vlasov-Maxwell system include, trivially, the continuity, total momentum, and total energy conservation laws. The latter two, of course, refer to the momentum and, respectively, the energy carried by both matter and the electromagnetic fields. On the contrary, it is worth noting that the entropy of the Vlasov-Maxwell system is a conserved quantity as well. The entropy is formally defined as:

$$S[f_\sigma] = -\sum_\sigma \iiint_{\mathbb{R}^3} \iiint_{\mathbb{R}^3} f_\sigma(\mathbf{x}, \mathbf{v}, t) \ln f_\sigma(\mathbf{x}, \mathbf{v}, t) \,\mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{x} \tag{3.47}$$

where the necessary precautions regarding dimensions are taken into account when computing the logarithm of the distribution function, that is to say, it is simply normalised to be dimensionless. Conservation of entropy is proven by observing the following, where we omit the sum:

$$\frac{\mathrm{d}S[f_\sigma]}{\mathrm{d}t} = -\iiint_{\mathbb{R}^3} \iiint_{\mathbb{R}^3} \frac{\partial}{\partial t} \left[f_\sigma(\mathbf{x}, \mathbf{v}, t) \ln f_\sigma(\mathbf{x}, \mathbf{v}, t)\right] \,\mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{x} = \tag{3.48}$$

$$= -\iiint_{\mathbb{R}^3} \iiint_{\mathbb{R}^3} \frac{\partial f_\sigma(\mathbf{x}, \mathbf{v}, t)}{\partial t} \left[\ln f_\sigma(\mathbf{x}, \mathbf{v}, t) + 1\right] \mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{x} \tag{3.49}$$

We now employ the Vlasov equation and observe:

$$\frac{\mathrm{d}S_\sigma}{\mathrm{d}t} = \iiint_{\mathbb{R}^3} \iiint_{\mathbb{R}^3} \left[1 + \ln f_\sigma\right] \left(\frac{\partial}{\partial \mathbf{x}} \cdot (f_\sigma \mathbf{v}) + \frac{\partial}{\partial \mathbf{v}} \cdot (f_\sigma \mathbf{a})\right) \mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{v} = \tag{3.50}$$

$$= \iiint_{\mathbb{R}^3} \iiint_{\mathbb{R}^3} \left[\frac{\partial}{\partial \mathbf{x}} \cdot (f_\sigma \mathbf{v} \ln f_\sigma) + \frac{\partial}{\partial \mathbf{v}} \cdot (f_\sigma \mathbf{a} \ln f_\sigma)\right] \mathrm{d}^3\mathbf{v}\,\mathrm{d}^3\mathbf{x} = \tag{3.51}$$

$$= \oint_{\partial \mathbb{R}^6} (f_\sigma \mathbf{v} \ln f_\sigma + f_\sigma \mathbf{a} \ln f_\sigma) \cdot \mathrm{d}\mathbf{\Sigma} = 0 \tag{3.52}$$

for any non-pathological distribution function that rapidly decays to zero as $\mathbf{x}$ and $\mathbf{v}$ go to infinity.

Chapter 4

Oscillations and waves in plasmas

We begin our consideration of plasma oscillations and waves with an overview of the general procedure for studying oscillations, and proceed from the perspective of magnetohydrodynamics, first focusing on isotropic behaviour and small-amplitude waves. Later, we shall explore how the two-fluid and, respectively, the Vlasov models better describe anisotropic behaviour with respect to the magnetic field, and how they shed light on phenomena that are ignored by the magnetohydrodynamic description. We shall also explore the regimes where the descriptions obtained by different models are expected to agree.

4.1 General procedure in studying plasma oscillations

In this section, we shall lay the relevant groundwork and outline the blueprint to be followed throughout the chapter. All plasma phenomena, including plasma waves and oscillations, can be fully described by combining Maxwell's equations relating electromagnetic fields to charge and current distributions within the plasma, with the Lorentz force equation, represented by Vlasov, two-fluid, or magnetohydrodynamic equations of motion, depending on the perspective of study, that capture the response of charges and currents to imposed electromagnetic fields. In the ensuing chapter, dedicated to plasma oscillations and waves of small amplitude, the complexity of the aforementioned equations will be addressed by, in most cases, linearising them at the appropriate time. We employ the following notation: barred quantities shall denote perturbations from the equilibrium quantities with a superdexed square or indexed zero. We shall, in all cases, endeavour to determine the dispersion relation of the particular oscillation or wave in question and discuss the physical implications of specific boundary conditions. We shall proceed by applying the announced method to the case of a simple electrostatic wave in a collisionless isotropic plasma from the two-fluid perspective. The governing equations are thus the two-fluid continuity and momentum equations, along with the Poisson equation:

$$\frac{\partial n_\sigma}{\partial t} + \nabla \cdot (n_\sigma \mathbf{u}_\sigma) = 0 \qquad n_\sigma \frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} = n_\sigma q_\sigma \mathbf{E} - \nabla P_\sigma \qquad \nabla \cdot \mathbf{E} = \frac{1}{\varepsilon_0} \sum_\sigma n_\sigma q_\sigma \tag{4.1}$$

Isotropy allows us to express the pressure term as the gradient of the scalar pressure, $P_\sigma = \gamma n_\sigma k_\mathrm{B} T_\sigma$. We shall next linearise these equations about the equilibrium state of a quasi-neutral stationary plasma, thus $\mathbf{u}_\sigma^\square = \mathbf{0}$ and $\sum_\sigma n_\sigma^\square q_\sigma = 0$. The pressure term contains variations in both number density and temperature, and we now impose the reasonable condition that the temperature perturbations are negligible compared to those of other quantities. The governing equations become:

$$\frac{\partial \overline{n}_\sigma}{\partial t} + n_\sigma^\square \nabla \cdot \overline{\mathbf{u}}_\sigma = 0 \qquad n_\sigma^\square m_\sigma \frac{\partial \overline{\mathbf{u}}_\sigma}{\partial t} = n_\sigma^\square q_\sigma \overline{\mathbf{E}} - \gamma k_\mathrm{B} T_\sigma^\square \nabla \overline{n}_\sigma \qquad \nabla \cdot \overline{\mathbf{E}} = \frac{1}{\varepsilon_0} \sum_\sigma \overline{n}_\sigma q_\sigma \tag{4.2}$$

One's attention may have been drawn to the seemingly self-contradictory term of 'electrostatic waves'. Indeed, all waves exhibit mutually inducted electric and magnetic fields. In the case of the so-called electrostatic waves, we ignore the inductive behaviour of electromagnetic fields, because mechanical number-density perturbations dominate over induced fields. In that case, the electric field is given by the gradient $\overline{\mathbf{E}} = -\nabla \overline{\phi}$, so that, when taking the divergence of the momentum equation and employing the commutativity of partial differential operators to exploit the continuity equation, one obtains:

$$m_\sigma \frac{\partial^2 \overline{n}_\sigma}{\partial t^2} = n_\sigma^\square q_\sigma \nabla^2 \overline{\phi} + \gamma k_\mathrm{B} T_\sigma^\square \nabla^2 \overline{n}_\sigma \tag{4.3}$$

The obtained equation can be formally solved by Fourier transforming the equation, or simply by assuming a single Fourier mode $\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{r}-\mathrm{i}\omega t)$, so that expressing $\overline{n}_\sigma$ gives:

$$m_\sigma\omega^2\overline{n}_\sigma = n_\sigma^\square q_\sigma \mathbf{k}^2\overline{\phi} + \gamma k_\mathrm{B}T_\sigma^\square \mathbf{k}^2\overline{n}_\sigma \implies \overline{n}_\sigma = \frac{n_\sigma^\square q_\sigma}{m_\sigma}\frac{\mathbf{k}^2\overline{\phi}}{\omega^2-\frac{\gamma k_\mathrm{B}T_\sigma}{m_\sigma}\mathbf{k}^2} \tag{4.4}$$

We finally obtain the dispersion relation of an electrostatic wave via Poisson's equation:

$$\left[1-\sum_\sigma \frac{n_\sigma^\square q_\sigma^2}{\varepsilon_0 m_\sigma}\frac{1}{\omega^2-\mathbf{k}^2\gamma k_\mathrm{B}T_\sigma^\square/m_\sigma}\right]\overline{\phi}=0 \tag{4.5}$$

We introduce the plasma frequency of species σ, $\omega_{\mathrm{p}\sigma}^2 = n_\sigma^\square q_\sigma^2/\varepsilon_0 m_\sigma$, so that the dispersion relation reads:

$$\mathcal{D}(\omega,\mathbf{k}) = 1-\sum_\sigma \frac{\omega_{\mathrm{p}\sigma}^2}{\omega^2-\mathbf{k}^2\gamma k_\mathrm{B}T_\sigma^\square/m_\sigma}=0 \tag{4.6}$$

First, notice that the summed quantities in reality represent the electric susceptibilities of each species. We will use this interpretation later. In the case of a simple ion-electron plasma, one observes that the dispersion relation follows three strictly distinct regimes depending on the relation between the phase velocity $\omega^2/\mathbf{k}^2$ to the thermal velocities of ions and electrons, $k_\mathrm{B}T_\mathrm{e}^\square/m_\mathrm{e}$ and $k_\mathrm{B}T_\mathrm{i}^\square/m_\mathrm{i}$.

1. The fully adiabatic regime, where $\omega^2/\mathbf{k}^2 \gg k_\mathrm{B}T_\mathrm{i}^\square/m_\mathrm{i}, k_\mathrm{B}T_\mathrm{e}^\square/m_\mathrm{e}$ and $\gamma=3$ since the plasma is perturbed in a single direction parallel to $\mathbf{k}$, describes electrostatic phenomena whose phase velocity is much greater than thermal velocities of both species, so that ions and electrons negligibly exchange heat to modify their respective internal energies. We expand the denominators in the obtained dispersion relation to obtain:

$$\mathcal{D}(\omega,\mathbf{k}) \simeq 1-\frac{\omega_\mathrm{pe}^2}{\omega^2}\left(1+\frac{3k_\mathrm{B}T_\mathrm{e}^\square}{m_\mathrm{e}}\frac{\mathbf{k}^2}{\omega^2}\right)-\frac{\omega_\mathrm{pi}^2}{\omega^2}\left(1+\frac{3k_\mathrm{B}T_\mathrm{i}^\square}{m_\mathrm{i}}\frac{\mathbf{k}^2}{\omega^2}\right)=0 \tag{4.7}$$

Notice that due to negligible electron mass in comparison to that of ions, we allow ourselves to drop the ion contribution to obtain:

$$\mathcal{D}(\omega,\mathbf{k}) \simeq 1-\frac{\omega_\mathrm{pe}^2}{\omega^2}\left(1+\frac{3k_\mathrm{B}T_\mathrm{e}^\square}{m_\mathrm{e}}\frac{\mathbf{k}^2}{\omega^2}\right)=0 \tag{4.8}$$

One can solve the biquadratic equation for ω^2, neglecting the negative branch solution, and develop using the starting hypothesis:

$$\omega^2 \simeq \frac{\omega_\mathrm{pe}^2}{2}\left(1+\sqrt{1+\frac{12k_\mathrm{B}T_\mathrm{e}^\square}{m_\mathrm{e}}\frac{\mathbf{k}^2}{\omega_\mathrm{pe}^2}}\right)\simeq \omega_\mathrm{pe}^2+\frac{3k_\mathrm{B}T_\mathrm{e}^\square}{m_\mathrm{e}}\mathbf{k}^2 \tag{4.9}$$

This type of electrostatic wave is commonly referred to as the electron plasma wave, the Langmuir wave, or the Bohm-Gross wave.

2. The fully isothermal regime, where $\omega^2/\mathbf{k}^2 \ll k_\mathrm{B}T_\mathrm{i}^\square/m_\mathrm{i}, k_\mathrm{B}T_\mathrm{e}^\square/m_\mathrm{e}$ and $\gamma=1$, describes electrostatic phenomena propagating at a much lower speed than thermal velocities of both species, thus allowing the plasma to thermalise as the wave propagates. Expanding the denominator, we obtain:

$$\mathcal{D}(\omega,\mathbf{k}) \simeq 1+\frac{1}{\mathbf{k}^2\lambda_\mathrm{Di}^2}+\frac{1}{\mathbf{k}^2\lambda_\mathrm{De}^2}=1+\frac{1}{\mathbf{k}^2\lambda_\mathrm{d}^2}=0 \tag{4.10}$$

The obtained dispersion relation is independent of the phenomenon frequency, and solving for $\mathbf{k}$ gives imaginary solutions. This corresponds to Debye shielding of the phenomenon and is merely expected, since such phenomena allow the plasma to thermalise fully.

3. Electrons follow an isothermal regime and can therefore fully thermalise amongst each other, while ions follow an adiabatic regime, thus $k_\mathrm{B}T_\mathrm{i}^\square/m_\mathrm{i} \ll \omega^2/\mathbf{k}^2 \ll k_\mathrm{B}T_\mathrm{e}^\square/m_\mathrm{e}$. Expanding each denominator in the dispersion relation according to the respective regime, as in the above, we find:

$$\mathcal{D}(\omega, \mathbf{k}) \simeq 1 - \frac{\omega_{\mathrm{pi}}^2}{\omega^2}\left(1 + \frac{3k_{\mathrm{B}}T_{\mathrm{i}}^{\square}}{m_{\mathrm{i}}}\frac{\mathbf{k}^2}{\omega^2}\right) + \frac{1}{\mathbf{k}^2\lambda_{\mathrm{De}}^2} = 0 \tag{4.11}$$

One may recast the obtained dispersion relation into the following form:

$$\omega^2 = \frac{\omega_{\mathrm{pi}}^2}{1 + 1/\mathbf{k}^2\lambda_{\mathrm{De}}^2}\left(1 + \frac{3k_{\mathrm{B}}T_{\mathrm{i}}^{\square}}{m_{\mathrm{i}}}\frac{\mathbf{k}^2}{\omega^2}\right) = \frac{\mathbf{k^2}\omega_{\mathrm{pi}}^{\mathbf{2}}\lambda_{\mathrm{De}}^2}{1 + \mathbf{k}^2\lambda_{\mathrm{De}}^2}\left(1 + \frac{3k_{\mathrm{B}}T_{\mathrm{i}}^{\square}}{m_{\mathrm{i}}}\frac{\mathbf{k}^2}{\omega^2}\right) \tag{4.12}$$

It is worthwhile to interpret the numerator of the multiplying factor, notably physically:

$$\omega_{\mathrm{pi}}^2\lambda_{\mathrm{De}}^2 = \frac{n_{\mathrm{i}}^{\square}q_{\mathrm{i}}^2}{\varepsilon_0 m_{\mathrm{i}}}\frac{\varepsilon_0 k_{\mathrm{B}}T_{\mathrm{e}}^{\square}}{n_{\mathrm{e}}^{\square}e^2} = \frac{Zk_{\mathrm{B}}T_{\mathrm{e}}^{\square}}{m_{\mathrm{i}}} = c_{\mathrm{i}}^2 \tag{4.13}$$

The relevant parameter is commonly known as the ion-acoustic velocity; such electrostatic waves are called ion-acoustic waves. We shall explore its relevance in later considerations, in particular from the Vlasov perspective. The dispersion relation thus becomes:

$$\omega^2 \simeq \frac{c_{\mathrm{i}}^2\mathbf{k}^2}{1 + \mathbf{k}^2\lambda_{\mathrm{De}}^2} + \frac{3k_{\mathrm{B}}T_{\mathrm{i}}^{\square}}{m_{\mathrm{i}}}\mathbf{k}^2 \tag{4.14}$$

Note that, for self-consistency, it is necessary to have $c_{\mathrm{i}}^2 \gg k_{\mathrm{B}}T_{\mathrm{i}}^{\square}/m_{\mathrm{i}}$, otherwise, the dispersion relation would become $\omega^2 \simeq 3\mathbf{k^2}k_{\mathrm{B}}T_{\mathrm{i}}^{\square}/m_{\mathrm{i}}$, which violates the starting hypothesis. Note that this condition is equivalent to $T_{\mathrm{e}}^{\square} \gg T_{\mathrm{i}}^{\square}$, so that ion acoustic waves can only propagate in regimes where electrons are much hotter than ions.

Now that we have laid out the blueprint of finding the dispersion relation for a particular case and discussed the relevant regimes, we shall employ the same approach in our subsequent considerations.

4.2 Elementary plasma waves in ideal MHD description

It is first worthwhile to remind ourselves of the relevant regimes where the ideal magnetohydrodynamic description excels:

1. Macroscopic phenomena taking place on spatial scales much greater than typical Larmor radii of each plasma species, as well as the Debye length of the plasma under consideration. Thus, MHD cannot describe electrostatic waves and primarily focuses on electromagnetic induction waves.

2. Relatively slow phenomena whose typical duration is much greater than gyro-periods of each plasma species.

3. Fully isotropic and collisionless plasma phenomena, thus failing to describe phenomena characterised by a distinct preferred direction of propagation.

4. Plasma phenomena whose propagation velocity throughout space is greater than thermal velocities of all species present in the plasma, therefore all plasma species are in their respective adiabatic regimes.

Let us also remind ourselves of the relevant ideal MHD equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \qquad \frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) \qquad \rho\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = -\nabla P + \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{\mu_0} \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{P}{\rho^\gamma}\right) = 0 \tag{4.15}$$

4.2.1 Alfvén waves: fast and slow modes

We endeavour to describe small-amplitude waves and begin by linearising these equations about a stationary equilibrium state. Let us first focus on a regime with null macroscopic velocity, thus $\mathbf{u}_0 = \mathbf{0}$, so that the convective time derivative simplifies into a partial time derivative for the macroscopic velocity perturbation:

$$\frac{\partial \overline{\rho}}{\partial t} + \rho_0 \nabla \cdot \overline{\mathbf{u}} = 0 \qquad \frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times (\overline{\mathbf{u}} \times \mathbf{B}_0) \qquad \rho_0 \frac{\partial \overline{\mathbf{u}}}{\partial t} = -\nabla \overline{p} + \frac{(\nabla \times \overline{\mathbf{B}}) \times \mathbf{B}_0}{\mu_0} \qquad \overline{p} = \frac{\gamma p_0}{\rho_0} \overline{\rho} \tag{4.16}$$

We eliminate $\overline{p}$, and then apply the spatial and temporal Fourier transform by assuming that perturbed quantities follow a single Fourier mode, thus:

$$-\omega \overline{\rho} + \rho_0 \mathbf{k} \cdot \overline{\mathbf{u}} = 0 \qquad -\omega \overline{\mathbf{B}} = \mathbf{k} \times (\overline{\mathbf{u}} \times \mathbf{B}_0) \qquad -\omega \rho_0 \overline{\mathbf{u}} = -\overline{\rho} c_s^2 \mathbf{k} + \frac{(\mathbf{k} \times \overline{\mathbf{B}}) \times \mathbf{B}_0}{\mu_0} \tag{4.17}$$

where we have introduced the sound speed $c_s^2 = \gamma p_0 / \rho_0$. We proceed by eliminating $\overline{\rho}$ and $\overline{\mathbf{B}}$:

$$-\omega \rho_0 \overline{\mathbf{u}} = -\rho_0 c_s^2 \frac{\mathbf{k} \cdot \overline{\mathbf{u}}}{\omega} \mathbf{k} + \frac{[\mathbf{k} \times (\mathbf{k} \times (\mathbf{B}_0 \times \overline{\mathbf{u}}))] \times \mathbf{B}_0}{\mu_0 \omega} \tag{4.18}$$

We shall first develop the multiple vector product, and then impose $\overline{\mathbf{u}} = \overline{\mathbf{u}}^{\perp} + \overline{\mathbf{u}}^{\parallel}$, $\mathbf{k} = \mathbf{k}_{\perp} + \mathbf{k}_{\parallel}$, where perpendicular and parallel components are to be taken with respect to the equilibrium magnetic field $\mathbf{B}_0$. We shall also express our results in a simpler form by multiplying by ω / ρ_0, so that:

$$-\omega \rho_0 \overline{\mathbf{u}} = -\rho_0 c_s^2 \frac{\mathbf{k} \cdot \overline{\mathbf{u}}}{\omega} \mathbf{k} + \frac{(\mathbf{k} \cdot \overline{\mathbf{u}})(\mathbf{k} \cdot \mathbf{B}_0)\mathbf{B}_0 - (\mathbf{k} \cdot \overline{\mathbf{u}})\mathbf{B}_0^2 \mathbf{k} - (\mathbf{k} \cdot \mathbf{B}_0)^2 \overline{\mathbf{u}} + (\mathbf{k} \cdot \mathbf{B}_0)(\mathbf{B}_0 \cdot \overline{\mathbf{u}})\mathbf{k}}{\mu_0 \omega} \tag{4.19}$$

$$\begin{pmatrix} \omega^2 - c_{\mathrm{A}}^2 \mathbf{k}^2 - c_s^2 k_{\perp}^2 & -c_s^2 k_{\perp} k_{\parallel} \\ -c_s^2 k_{\perp} k_{\parallel} & \omega^2 - c_s^2 k_{\parallel}^2 \end{pmatrix} \begin{pmatrix} \overline{u}^{\perp} \\ \overline{u}^{\parallel} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \omega^4 - (c_{\mathrm{A}}^2 + c_s^2)\mathbf{k}^2 \omega^2 + c_{\mathrm{A}}^2 c_s^2 \mathbf{k}^2 k_{\parallel}^2 = 0 \tag{4.20}$$

where we have introduced the Alfvén speed as $c_{\mathrm{A}}^2 = \mathbf{B}_0^2 / \mu_0 \rho_0$. Let us introduce $\theta = \angle(\mathbf{B}_0, \mathbf{k})$. One may solve this biquadratic equation to obtain:

$$\frac{\omega^2}{\mathbf{k}^2} = \frac{c_{\mathrm{A}}^2 + c_s^2}{2} \pm \sqrt{\frac{(c_{\mathrm{A}}^2 + c_s^2)^2}{4} - c_{\mathrm{A}}^2 c_s^2 \cos^2 \theta} \tag{4.21}$$

It is now useful to introduce the pressure parameter β as follows:

$$\frac{c_s^2}{c_{\mathrm{A}}^2} = \frac{\gamma p_0}{\rho_0} \frac{\mu_0 \rho_0}{\mathbf{B}_0^2} = \frac{\gamma}{2} \frac{p_0}{\mathbf{B}_0^2 / 2\mu_0} = \frac{\gamma}{2} \beta \tag{4.22}$$

The solution to the dispersion relation thus becomes:

$$\frac{\omega^2}{c_{\mathrm{A}}^2 \mathbf{k}^2} = \left(\frac{1}{2} + \frac{\gamma}{4} \beta \right) \left[1 \pm \sqrt{1 - \frac{2\gamma \beta \cos^2 \theta}{\left(1 + \frac{\gamma}{2}\beta\right)^2}} \right] \tag{4.23}$$

Let us now investigate the two limiting regimes, that is, the regime of dominant magnetic pressure, where $\beta \ll 1$, and the regime of dominant hydrostatic pressure, where $\beta \gg 1$. In the regime of dominant magnetic pressure, and taking the positive branch of the square root term, we obtain:

$$\lim_{\beta \ll 1} \frac{\omega_+^2}{\mathbf{k}^2} \simeq c_{\mathrm{A}}^2 + c_s^2 \sin^2 \theta + \frac{c_s^4}{c_{\mathrm{A}}^2} \sin^2 \theta \cos^2 \theta + \mathcal{O}(\beta^3) \tag{4.24}$$

while for the negative branch of the square root term:

$$\lim_{\beta \ll 1} \frac{\omega_-^2}{\mathbf{k}^2} \simeq c_s^2 \cos^2 \theta - \frac{c_s^4}{c_{\mathrm{A}}^2} \sin^2 \theta \cos^2 \theta + \mathcal{O}(\beta^3) \tag{4.25}$$

Thus, in the dominant magnetic pressure regime, the positive-branch solution yields a plasma wave travelling at the Alfvén speed. Since in this regime $\beta \ll 1$, that is, $c_s^2 \ll c_{\mathrm{A}}^2$, the positive branch solution is often called the fast mode, while the negative branch solution is commonly known as the magnetosonic mode or the slow mode. In the dominant hydrostatic pressure regime, one can proceed by expanding the positive branch solution around $1/\beta = 0$ to obtain:

$$\lim_{\beta\gg 1} \frac{\omega^2}{\mathbf{k}^2} \simeq c_s^2 + c_{\mathrm{A}}^2 \sin^2\theta + \frac{c_{\mathrm{A}}^4}{c_s^2}\sin^2\theta\cos^2\theta + \mathcal{O}(\beta^{-2}) \tag{4.26}$$

while for the negative branch solution, one may arrive at:

$$\lim_{\beta\gg 1} \frac{\omega^2}{\mathbf{k}^2} \simeq c_{\mathrm{A}}^2 \cos^2\theta + \frac{c_{\mathrm{A}}^4}{c_s^2}\sin^2\theta\cos^2\theta + \mathcal{O}(\beta^{-2}) \tag{4.27}$$

Therefore, in the dominant hydrostatic pressure regime, the only relevant solution is the positive branch one corresponding to regular sound waves in plasmas.

4.2.2 Alfvén waves: shear and compressional modes

It is worthwhile to discuss the behaviour of plasma waves under certain initial conditions. Let us focus on the following linearised MHD equation:

$$-\omega\overline{\mathbf{B}} = \mathbf{k}\times(\overline{\mathbf{u}}\times\mathbf{B}_0) = (\mathbf{k}\cdot\mathbf{B}_0)\overline{\mathbf{u}} + (\mathbf{k}\cdot\overline{\mathbf{u}})\mathbf{B}_0 \tag{4.28}$$

Let the initial perturbation of the magnetic field, $\overline{\mathbf{B}}$, be perpendicular to $\mathbf{B}_0$. Inputting this into the previous equation, one finds $\mathbf{k}\cdot\overline{\mathbf{u}} = 0$, which gives $-\omega\overline{\rho} = 0$ for the continuity equation. Performing the inverse Fourier transform on the latter yields a constant perturbation to the density, which is therefore zero. The physical meaning of this result is that shear Alfvén waves are incompressible. Moreover, we substitute this expression into the equation. (4.19):

$$\omega^2\overline{\mathbf{u}} = \frac{(\mathbf{k}\cdot\mathbf{B}_0)^2}{\mu_0\rho_0}\overline{\mathbf{u}} \implies \omega^2 = c_{\mathrm{A}}^2\mathbf{k}^2\cos^2\theta \tag{4.29}$$

Notice that the dispersion relation of the shear Alfvén mode corresponds to the negative branch solution of the $\beta \gg 1$ regime, where the hydrostatic pressure dominates over magnetic pressure. Since, in this regime, $c_{\mathrm{A}}^2 \ll c_s^2$, the shear Alfvén mode is necessarily slow. If initial conditions are such that $\overline{\mathbf{B}}\cdot\mathbf{B}_0 \neq 0$, then Alfvén waves necessarily show compressional behaviour, and can be categorised as both fast and slow modes, depending on the pressure parameter. We shall later compare the MHD analysis with those of the two-fluid and Vlasov models and observe that the MHD model fails to capture many intricacies of plasma oscillations.

4.3 Elementary cold plasma waves in the two-fluid description

Let us reflect on the title of this section. To distinguish waves that depend on temperature from waves that do not, we shall employ the terminology of *cold* and *warm* plasma waves. Cold waves thus have a temperature-independent dispersion relation, such that the temperature of each plasma species can be set to zero without altering its kinetics. Meanwhile, warm plasma waves suggest that the dispersion relation depends on the temperature of certain species. Evidently, the terminology does not refer to the *temperature* of a wave, but simply the dependence or lack thereof on plasma species temperature. Nevertheless, regarding cold plasma waves as simply zero-temperature waves is misleading. Such waves are simply a consequence of a large number of plasma particles following identical Hamilton-Lagrangian dynamics, whereas warm plasma waves involve various groups of particles having different dynamics due to their specific initial velocity governed by their respective distribution function, and, therefore, require thermodynamic and statistical mechanical considerations.

Having thoroughly discussed the meaning behind the employed terminology, we shall start our considerations on cold plasma waves by eliminating the magnetic field in Maxwell's equations and performing a Fourier transform to obtain:

$$\nabla\times(\nabla\times\mathbf{E}) = -\mu_0\frac{\partial\mathbf{J}}{\partial t} - \frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} \implies \mathbf{k}\times(\mathbf{k}\times\mathbf{E}) = -\mathrm{i}\mu_0\omega\mathbf{J} + \frac{\omega^2}{c^2}\mathbf{E} \tag{4.30}$$

We shall moreover relate the current density to the electric field by means of the conductivity tensor, that is, $\mathbf{J} = \boldsymbol{\sigma}\cdot\mathbf{E}$, giving:

$$-\frac{c^2}{\omega^2}\mathbf{k}\times(\mathbf{k}\times\mathbf{E}) = \mathrm{i}\frac{c^2\mu_0}{\omega}\boldsymbol{\sigma}\cdot\mathbf{E}+\mathbf{E} = \left(\frac{\mathrm{i}\boldsymbol{\sigma}}{\varepsilon_0\omega}+\mathbf{1}\right)\cdot\mathbf{E} \tag{4.31}$$

The double vector product can be recast into tensor form via the following dyadic product:

$$\mathbf{k}\times(\mathbf{k}\times\mathbf{E}) = \mathbf{kk}\cdot\mathbf{E}-\mathbf{k}^2\mathbf{E} = \mathbf{k}^2\left(\frac{\mathbf{kk}}{\mathbf{k}^2}-\mathbf{1}\right)\cdot\mathbf{E} \tag{4.32}$$

Eq. (4.30) thus becomes:

$$\left[\frac{c^2\mathbf{k}^2}{\omega^2}\left(\frac{\mathbf{kk}}{\mathbf{k}^2}-\mathbf{1}\right)+\frac{\mathrm{i}\boldsymbol{\sigma}}{\varepsilon_0\omega}+\mathbf{1}\right]\cdot\mathbf{E} = \mathbf{0} \implies \det\left[\frac{c^2\mathbf{k}^2}{\omega^2}\left(\frac{\mathbf{kk}}{\mathbf{k}^2}-\mathbf{1}\right)+\frac{\mathrm{i}\boldsymbol{\sigma}}{\varepsilon_0\omega}+\mathbf{1}\right] = 0 \tag{4.33}$$

which provides the dispersion relation for any plasma wave once $\boldsymbol{\sigma}$ has been determined.

4.3.1 Obtaining dispersion relations via the dielectric tensor

In most cases, this form of the dispersion relation is used in the two-fluid perspective, since the conductivity tensor can be related to the mobilities of each species, which, for a stationary cold plasma subject to a uniform magnetic field, is given by the two-fluid momentum equations:

$$n_\sigma m_\sigma\frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} = n_\sigma q_\sigma(\mathbf{E}+\mathbf{u}_\sigma\times\mathbf{B}_0) \implies m_\sigma\frac{\partial\overline{\mathbf{u}}_\sigma}{\partial t} = q_\sigma(\overline{\mathbf{E}}+\overline{\mathbf{u}}_\sigma\times\mathbf{B}_0) \tag{4.34}$$

We perform a Fourier transform on the previous linearised equation to obtain:

$$-\mathrm{i}m_\sigma\omega\overline{\mathbf{u}}_\sigma = q_\sigma\overline{\mathbf{E}}+q_\sigma\overline{\mathbf{u}}_\sigma\times\mathbf{B}_0 \tag{4.35}$$

Let us choose $\mathbf{B}_0 = B_0\,\hat{\mathbf{z}}$, so that the previous equation becomes:

$$-\mathrm{i}\omega m_\sigma\begin{pmatrix}\overline{u}_\sigma^{\mathrm{x}}\\ \overline{u}_\sigma^{\mathrm{y}}\\ \overline{u}_\sigma^{\mathrm{z}}\end{pmatrix} = q_\sigma\overline{\mathbf{E}}+q_\sigma B_0\begin{pmatrix}\overline{u}_\sigma^{\mathrm{y}}\\ -\overline{u}_\sigma^{\mathrm{x}}\\ 0\end{pmatrix} \implies \begin{pmatrix}-\mathrm{i}\omega & -q_\sigma B_0/m_\sigma & 0\\ q_\sigma B_0/m_\sigma & -\mathrm{i}\omega & 0\\ 0 & 0 & -\mathrm{i}\omega\end{pmatrix}\cdot\overline{\mathbf{u}}_\sigma = \frac{q_\sigma}{m_\sigma}\overline{\mathbf{E}} \tag{4.36}$$

Introducing the mobility tensor of species σ as $\overline{\mathbf{u}}_\sigma = \boldsymbol{\mu}_\sigma\cdot\overline{\mathbf{E}}$, we obtain:

$$\boldsymbol{\mu}_\sigma = \frac{q_\sigma}{m_\sigma}\begin{pmatrix}-\mathrm{i}\omega & -q_\sigma B_0/m_\sigma & 0\\ q_\sigma B_0/m_\sigma & -\mathrm{i}\omega & 0\\ 0 & 0 & -\mathrm{i}\omega\end{pmatrix}^{-1} = \frac{q_\sigma}{m_\sigma}\begin{pmatrix}\frac{\mathrm{i}\omega}{\Omega_\sigma^2-\omega^2} & \frac{\Omega_\sigma}{\Omega_\sigma^2-\omega^2} & 0\\ -\frac{\Omega_\sigma}{\Omega_\sigma^2-\omega^2} & \frac{\mathrm{i}\omega}{\Omega_\sigma^2-\omega^2} & 0\\ 0 & 0 & \frac{\mathrm{i}}{\omega}\end{pmatrix} \tag{4.37}$$

The conductivity tensor thus reads:

$$\boldsymbol{\sigma} = \sum_\sigma n_\sigma^\square q_\sigma\boldsymbol{\mu}_\sigma = \sum_\sigma\frac{n_\sigma^\square q_\sigma^2}{m_\sigma}\begin{pmatrix}\frac{\mathrm{i}\omega}{\omega^2-\Omega_\sigma^2} & -\frac{\Omega_\sigma}{\Omega_\sigma^2-\omega^2} & 0\\ \frac{\Omega_\sigma}{\omega^2-\omega^2} & \frac{\mathrm{i}\omega}{\omega^2-\Omega_\sigma^2} & 0\\ 0 & 0 & \frac{\mathrm{i}}{\omega}\end{pmatrix} \tag{4.38}$$

It is now useful to introduce the electric permittivity tensor as follows:

$$\boldsymbol{\varepsilon} = \frac{\mathrm{i}\boldsymbol{\sigma}}{\varepsilon_0\omega}+\mathbf{1} = \begin{pmatrix}\varepsilon_1 & -\mathrm{i}\varepsilon_2 & 0\\ \mathrm{i}\varepsilon_2 & \varepsilon_1 & 0\\ 0 & 0 & \varepsilon_3\end{pmatrix} \tag{4.39}$$

$$\varepsilon_1 = S = 1-\sum_\sigma\frac{\omega_{\mathrm{p}\sigma}^2}{\omega^2-\Omega_\sigma^2} \qquad \varepsilon_2 = D = \sum_\sigma\frac{\Omega_\sigma}{\omega}\frac{\omega_{\mathrm{p}\sigma}^2}{\omega^2-\Omega_\sigma^2} \qquad \varepsilon_3 = P = 1-\sum_\sigma\frac{\omega_{\mathrm{p}\sigma}^2}{\omega^2} \tag{4.40}$$

where we have used the S, D, P notation as a mnemonic for sum, difference, and parallel, respectively, the reason for which shall become clear in what follows, recognise that the product $c^2\mathbf{k}^2/\omega^2 = n^2$, that is, the optical index of refraction. Let us choose a coordinate system in which $\mathbf{k} = k_\perp\,\hat{\mathbf{x}}+k_\parallel\,\hat{\mathbf{z}}$. This allows us to simplify the expressions and ignore certain components along $\hat{\mathbf{y}}$. Note that such simplifications are not possible in plasmas that are non-uniform in the $x-y$ plane. The dispersion relation can be cast into matrix form as follows:

$$\det\begin{pmatrix}\varepsilon_1 - n_\parallel^2 & -\mathrm{i}\varepsilon_2 & n_\parallel n_\perp \\ \mathrm{i}\varepsilon_2 & \varepsilon_1 - n^2 & 0 \\ n_\parallel n_\perp & 0 & \varepsilon_3 - n_\perp^2\end{pmatrix} \equiv \det\begin{pmatrix}\varepsilon_1 - n^2\cos^2\theta & -\mathrm{i}\varepsilon_2 & n\sin\theta\cos\theta \\ \mathrm{i}\varepsilon_2 & \varepsilon_1 - n^2 & 0 \\ n^2\sin\theta\cos\theta & 0 & \varepsilon_3 - n^2\sin^2\theta\end{pmatrix} = 0 \tag{4.41}$$

Let us divert our attention to the case $\theta = 0$, where the dispersion relation becomes:

$$\det\begin{pmatrix}\varepsilon_1 - n^2 & -\mathrm{i}\varepsilon_2 & 0 \\ \mathrm{i}\varepsilon_2 & \varepsilon_1 - n^2 & 0 \\ 0 & 0 & \varepsilon_3\end{pmatrix} = 0 \implies \left[(\varepsilon_1 - n^2)^2 - \varepsilon_2^2\right]\varepsilon_3 = 0 \iff \left[(S - n^2)^2 - D^2\right]P = 0 \tag{4.42}$$

Thus, either $P = 0$ or $n^2 = \varepsilon_1 \pm \varepsilon_2 \equiv S \pm D$. Let us expand these solutions:

$$R = S + D = 1 - \sum_\sigma \frac{\omega_{\mathrm{p}\sigma}^2}{\omega(\omega + \Omega_\sigma)} \qquad L = S - D = 1 - \sum_\sigma \frac{\omega_{\mathrm{p}\sigma}^2}{\omega(\omega - \Omega_\sigma)} \tag{4.43}$$

.

where we have introduced the R and L notation as mnemonics for right-hand polarised and left-hand polarised. The S, D, P mnemonic now becomes apparent since:

$$S = \frac{R + L}{2} \qquad D = \frac{R - L}{2} \tag{4.44}$$

We observe that R diverges at $\omega = -\Omega_\sigma$ and L diverges at $\omega = \Omega_\sigma$. These are, in fact, so-called resonance points, which we shall discuss later in further detail. Notice that here, we have defined the cyclotron frequencies as signed, so that R diverges at cyclotron frequencies of negatively charged plasma species, and L reciprocally diverges at cyclotron frequencies of positively charged plasma species. It is useful to dwell on the limiting conditions for ω, such as $\omega \to \infty$ and $\omega \to 0$. The former is easily determined, $\lim_{\omega\to\infty} R, L = 1$, while the latter requires some attention:

$$\lim_{\omega\to 0} R, L = \lim_{\omega\to 0}\left[1 - \frac{1}{\omega}\left(\frac{\omega_{\mathrm{pi}}^2}{\omega \pm \Omega_{\mathrm{i}}} + \frac{\omega_{\mathrm{pe}}^2}{\omega \pm \Omega_{\mathrm{e}}}\right)\right] = \lim_{\omega\to 0}\left[1 - \frac{1}{\omega}\frac{(\omega_{\mathrm{pi}}^2 + \omega_{\mathrm{pe}}^2)\omega \pm (\Omega_{\mathrm{e}}\omega_{\mathrm{pi}}^2 + \Omega_{\mathrm{i}}\omega_{\mathrm{pe}}^2)}{(\omega \pm \Omega_{\mathrm{i}})(\omega \pm \Omega_{\mathrm{e}})}\right] \tag{4.45}$$

$$\frac{\omega_{\mathrm{p}\sigma}^2}{\Omega_\sigma} = \frac{n_\sigma^\square q_\sigma^2}{\varepsilon_0 m_\sigma}\frac{m_\sigma}{q_\sigma B_0} = \frac{n_\sigma^\square q_\sigma}{\varepsilon_0 B_0} \implies \frac{\omega_{\mathrm{pi}}^2}{\Omega_{\mathrm{i}}} = -\frac{\omega_{\mathrm{pe}}^2}{\Omega_{\mathrm{e}}} \tag{4.46}$$

$$\lim_{\omega\to 0} R, L = 1 - \frac{\omega_{\mathrm{pe}}^2 + \omega_{\mathrm{pi}}^2}{\Omega_{\mathrm{i}}\Omega_{\mathrm{e}}} \simeq 1 - \frac{\omega_{\mathrm{pe}}^2}{\Omega_{\mathrm{e}}\Omega_{\mathrm{i}}} \simeq 1 + \frac{n_{\mathrm{e}}^\square e}{\varepsilon_0 B_0}\frac{m_{\mathrm{i}}}{ZeB_0} = 1 + \frac{c^2}{B_0^2/\mu_0 n_{\mathrm{i}}^\square m_{\mathrm{i}}} \simeq 1 + \frac{c^2}{c_{\mathrm{A}}^2} \tag{4.47}$$

Thus, at low frequencies, both R and L are related to Alfvén modes. The $n^2 = L$ is the slow mode since it gives larger $\mathbf{k}$, while $n^2 = R$ is the fast mode since it gives smaller $\mathbf{k}$. Let us now find the electric field eigenvector for the case $n^2 = R$:

$$\begin{pmatrix}-D & -\mathrm{i}D & 0 \\ \mathrm{i}D & -D & 0 \\ 0 & 0 & P\end{pmatrix}\cdot\overline{\mathbf{E}} = \lambda\overline{\mathbf{E}} \implies \frac{\overline{E}_{\mathrm{x}}}{\overline{E}_{\mathrm{y}}} = -\mathrm{i} \implies \text{Right-circulating wave, hence } R \tag{4.48}$$

Analogously, choosing $n^2 = L$ gives a left-circulating wave. Let us summarize the main principles of what we have developed so far. When the wave vector is strictly parallel to the magnetic field, two distinct modes exist: a right-circularly polarised wave with dispersion relation $n^2 = R$ such that its resonance point $n \to \infty$ is reached as $\omega \to \Omega_{\mathrm{e}}$, and a left-circularly polarised wave with dispersion relation $n^2 = L$ such that $n \to \infty$ as $\omega \to \Omega_{\mathrm{i}}$. It is worth noting that, because the ion cyclotron motion is left-handed, it seems reasonable that a left-circulating wave resonates with ions, and vice versa for electrons. At low frequencies, both left- and right-circulating modes degenerate into Alfvén modes with the dispersion relation $n^2 = 1 + c^2/c_{\mathrm{A}}^2$. Compared with the magnetohydrodynamic model, we observe an additive factor of 1. This is because the MHD model ignored displacement current. The displacement-current term shows that, for plasmas with sufficiently low number densities and/or a strong magnetic field, the Alfvén velocity may exceed the speed of light. For a plasma to exhibit significant Alfvénic behaviour, as described by magnetohydrodynamics, it must satisfy $c_{\mathrm{A}}^2 \ll c^2$, or equivalently, $\Omega_{\mathrm{i}} \ll \omega_{\mathrm{pi}}$.

4.3.2 Resonances and cut-offs

In our preceding considerations, we have encountered conditions on ω such that the index of refraction and certain terms of dispersion relations diverge, and have called them resonances. Such and similar conditions present great importance when exploring possible wave modes for a given incident angle $\theta = \angle(\mathbf{B}, \mathbf{k})$ within a given regime. Thus, we shall formally introduce the concept of resonances and cut-offs, as well as give physical meaning to the nomenclature of certain quantities, in particular, the index of refraction. Any general situation leading to $n^2 \to \infty$ is called a resonance and physically corresponds to the wavelength of the wave going to zero. Thus, any dissipative effect in such a situation will be strongly damped by internal plasma mechanisms. Reciprocally, any general situation described by $n^2 = 0$ is called a cut-off, and corresponds to complete wave reflection, since, in the complex plane, n changes from being purely imaginary to being purely real or vice versa depending on a given situation. In a general non-uniform plasma, there may exist multiple layers or regions of plasma where $n^2 \to \infty$ and/or $n^2 = 0$. At such layers, waves will be either completely reflected or fully absorbed. An example of resonance regimes was explored in the previous subsection; let us illustrate one cut-off regime by re-taking Eq. (4.41) and imposing $\theta = \pi/2$:

$$\det \begin{pmatrix} S & -\mathrm{i}D & 0 \\ \mathrm{i}D & S - n^2 & 0 \\ 0 & 0 & P - n^2 \end{pmatrix} = 0 \implies \left[S(S-n^2) - D^2\right](P - n^2) = 0 \tag{4.49}$$

We observe two distinct modes: one related to parallel motion along the stationary magnetic field, the other to motion perpendicular to it. The first mode gives the simple dispersion relation for any unmagnetised plasma, as expected, since the magnetic field has no effect on the parallel motion of charges and is thus called the ordinary mode with dispersion relation:

$$n^2 = \frac{S^2 - D^2}{S} = \frac{(S-D)(S+D)}{S} = \frac{RL}{S} \tag{4.50}$$

Cut-offs occur when $R = 0$ or $L = 0$ while resonances occur when $S = 0$. Since this mode depends on the applied magnetic field, it is called the extraordinary mode. It is worthwhile to investigate the resonance further. Since S depends on both the cyclotron and the plasma frequencies of all species, it is called a hybrid resonance. In a typical ion-electron plasma, $S = 0$ corresponds to:

$$S = 1 - \frac{\omega_{\mathrm{pi}}^2}{\omega^2 - \Omega_{\mathrm{i}}^2} - \frac{\omega_{\mathrm{pe}}^2}{\omega^2 - \Omega_{\mathrm{e}}^2} \tag{4.51}$$

We expect two distinct solutions for ω^2, one of the order of or greater than Ω_{e}, and another much smaller than Ω_{e}. These are respectively called the upper and lower hybrid resonant frequencies, ω_{UH} and ω_{LH}, whose expressions are:

$$\omega_{\mathrm{UH}}^2 \simeq \omega_{\mathrm{pe}}^2 + \Omega_{\mathrm{e}}^2 \quad \text{and} \quad \omega_{\mathrm{LH}}^2 \simeq \Omega_{\mathrm{i}}^2 + \frac{\omega_{\mathrm{pi}}^2}{1 + \frac{\omega_{\mathrm{pe}}^2}{\Omega_{\mathrm{e}}}^2} = \Omega_{\mathrm{i}}^2 + \Omega_{\mathrm{e}}^2 \frac{\omega_{\mathrm{pi}}^2}{\omega_{\mathrm{pe}}^2 + \Omega_{\mathrm{e}}^2} \tag{4.52}$$

We now return to our considerations of Alfvén modes in a cold plasma, in a more general setting. Our starting dispersion relation is the following:

$$\begin{pmatrix} S - n^2\cos^2\theta & -iD & n^2\sin\theta\cos\theta \\ iD & S - n^2 & 0 \\ n^2\sin\theta\cos\theta & 0 & P - n^2\sin^2\theta \end{pmatrix} \begin{pmatrix} E_{\mathrm{x}} \\ E_{\mathrm{y}} \\ E_{\mathrm{z}} \end{pmatrix} = 0 \tag{4.53}$$

where S, D, P, n, and θ retain their usual meanings, such that the dispersion relation becomes a biquadratic equation in n:

$$An^4 - Bn^2 + C = 0 \qquad \text{where} \tag{4.54}$$

$$A = S\sin^2\theta + P\cos^2\theta \qquad B = (S^2 - D^2)\sin^2\theta + PS(1+\cos^2\theta) \qquad C = PRL \tag{4.55}$$

We may evaluate the determinant of the matrix and set it equal to zero, whose solutions for n^2 are:

$$n^2 = \frac{B \pm \sqrt{B^2 - 4AC}}{2A} \tag{4.56}$$

Let us draw our attention to the discriminant:

$$B^2 - 4AC = (S^2 - D^2 - SP)^2 \sin^4\theta + 4P^2D^2\cos^2\theta \tag{4.57}$$

which is semipositive-definite for all θ. Therefore, n is either purely real, corresponding to a propagating wave, or purely imaginary, corresponding to an evanescent wave. Cut-offs correspond to $C = 0$, while resonances correspond to $A \to 0$, that is, $S\sin^2\theta + P\cos^2\theta \simeq 0$. The obtained biquadratic equation gives the general dispersion relation of a cold plasma for arbitrary θ.

The physical mechanism behind cutoffs is, however, a genuinely good question. Let us consider the case $\mathbf{k}\cdot\mathbf{B} = 0$, so that $\theta = \pi/2$, and let us, moreover, consider a simple Langmuir wave; hence, we neglect ion contributions to the dispersion relation. The cutoffs are determined by $PRL = 0$, that is:

$$P = 0 \implies \omega^2 = \omega_{\mathrm{pe}}^2 \qquad R = 0 \implies \omega(\omega + \Omega_\mathrm{e}) = \omega_{\mathrm{pe}}^2 \qquad L = 0 \implies \omega(\omega - \Omega_\mathrm{e}) = \omega_{\mathrm{pe}}^2 \tag{4.58}$$

Notice that $P = 0$ indicates a resonance, and thus shall not be considered. Let us concretely focus on the $R = 0$ case, since the $L = 0$ case will be shown later to be completely analogous. Multiplying the linearised Fourier transformed electron momentum equation by $\mathrm{i}\omega$ gives:

$$m_\mathrm{e}\omega^2\overline{\mathbf{u}}_\mathrm{e} = \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{E}} + \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} \times \mathbf{B}_0 \tag{4.59}$$

Inputting the solution for ω given by $R = 0$:

$$m_\mathrm{e}\omega_{\mathrm{pe}}^2\overline{\mathbf{u}}_\mathrm{e} - \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{E}} = m_\mathrm{e}\Omega_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} + \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} \times \mathbf{B}_0 \tag{4.60}$$

Expanding ω_{pe}^2 and Ω_e using their definitions gives:

$$m_\mathrm{e}\frac{n_\mathrm{e}^{\square}q_\mathrm{e}^2}{\varepsilon_0 m_\mathrm{e}}\overline{\mathbf{u}}_\mathrm{e} - \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{E}} = m_\mathrm{e}\frac{q_\mathrm{e}B_0}{m_\mathrm{e}}\omega\overline{\mathbf{u}}_\mathrm{e} + \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} \times \mathbf{B}_0 \tag{4.61}$$

Next, we take the divergence of the equation, or equivalently, take the dot product with $\mathrm{i}\mathbf{k}$:

$$\frac{q_\mathrm{e}^2}{\varepsilon_0}\nabla\cdot(n_\mathrm{e}^{\square}\overline{\mathbf{u}}_\mathrm{e}) - \mathrm{i}q_\mathrm{e}\omega\nabla\cdot\overline{\mathbf{E}} = \nabla\cdot[q_\mathrm{e}B_0\omega\overline{\mathbf{u}}_\mathrm{e} + \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} \times \mathbf{B}_0] \tag{4.62}$$

The linearised two-fluid continuity equation, along with the inverse Fourier transform, gives:

$$q_\mathrm{e}\left(-\frac{q_\mathrm{e}}{\varepsilon_0}\frac{\partial\overline{n}_\mathrm{e}}{\partial t} - \mathrm{i}\omega\nabla\cdot\overline{\mathbf{E}}\right) = q_\mathrm{e}\frac{\partial}{\partial t}\left(\nabla\cdot\overline{\mathbf{E}} - \frac{\overline{n}_\mathrm{e}q_\mathrm{e}}{\varepsilon_0}\right) = \nabla\cdot[q_\mathrm{e}B_0\omega\overline{\mathbf{u}}_\mathrm{e} + \mathrm{i}q_\mathrm{e}\omega\overline{\mathbf{u}}_\mathrm{e} \times \mathbf{B}_0] \tag{4.63}$$

The left-hand side of the equation is thus zero due to Gauss's law. The right-hand side of the equation thus ought to be zero as well:

$$\mathrm{i}\omega B_0\nabla\cdot\overline{\mathbf{u}}_\mathrm{e} + \omega\mathbf{B}_0\cdot(\nabla\times\overline{\mathbf{u}}_\mathrm{e}) = \omega\overline{\mathbf{u}}_\mathrm{e}\cdot\underbrace{(\nabla\times\mathbf{B}_0)}_{\mathbf{0}} = \mathbf{0} \tag{4.64}$$

The ω is dropped, since when inverse Fourier-transforming, it only contributes to the equation by an additive constant, which evaluates to zero when invoking equilibrium conditions. Since the stationary magnetic field $\mathbf{B}_0$ (whose curl is zero, being an externally applied field, so that the third term drops), and the macroscopic oscillatory electron velocity have only real components (it is precisely ω which determines the complex properties of the refractive index), either side of the equation is either purely imaginary or purely real, and when invoking equilibrium conditions, it must be $\overline{\mathbf{u}}_\mathrm{e} = \mathbf{0}$. Thus, at cutoffs of Langmuir waves, macroscopic electron motion comes to a halt, and the wave does not propagate since there is no net oscillatory charge motion to support electrostatic waves. The $L = 0$ case shall not receive further treatment other than recognising that the only difference is that it contributes a negative sign to the first term of the right-hand side of equation (4.60), which does not fundamentally change the result.

4.3.3 Clemmow-Mullaly-Allis diagram and taxonomy of modes

The CMA (Clemmow-Mullaly-Allis) diagram provides an elegant method for revealing and classifying the large number of modes embedded in the dispersion relation. If the ion species is known, the cyclotron and plasma frequencies are effectively linearly dependent; the only two truly free parameters are the number density and the magnetic field. The CMA diagram is thus constructed in a chart having $\ln{(\omega_{\mathrm{pe}}^2 + \omega_{\mathrm{pi}}^2)/\omega^2}$

as its horizontal axis and $\ln \Omega_e^2/\omega^2$ as its vertical axis, then plotting the wave normal surfaces as $1/n \sim \theta$ for a considerable number of points on the diagram to obtain a detailed description. It is, however, worth noting that the topology of the wave normal surfaces changes only at specific boundaries in the parameter space; otherwise, it merely deforms, but retains its topology. The parameter space is divided into a finite number of regions, called bounded volumes, separated by curves called bounding surfaces, across which modes change fundamentally, i.e., their topology changes. For example, if an Alfvén mode exists at a particular point inside a bounded volume, it exists everywhere within the bounded volume. The appropriate choice of bounding surfaces is the following:

1. The principal resonances, which are curves in parameter space where n^2 has a resonance at either $\theta = 0$ or $\theta = \pi/2$. These are precisely $R = \infty$ (electron cyclotron resonance), $L = \infty$ (ion cyclotron resonance), and $S = 0$ (upper and lower hybrid resonances).

2. The cutoffs $R = 0$, $L = 0$, and $P = 0$.

The behaviour of wave normal surfaces inside a bounded volume and when crossing a bounding surface can be deduced using the five following theorems as consequences of the considerations above:

1. Inside a bounded volume, n cannot vanish.

   Proof. $n = 0 \implies PRL = 0$, which are precisely the bounding surfaces.

2. If n^2 has a resonance and thus goes to infinity at any point in the bounded volume, then at every other point within the bounded volume there exists a resonance at a unique angle θ_{res} and its associated mirror angles, namely $-\theta_{\text{res}}$, $\pi - \theta_{\text{res}}$ and $-(\pi - \theta_{\text{res}})$, but at no other angles.

   Proof. $n^2 \to \infty \implies A \to 0 \implies \tan^2 \theta_{\text{res}} = -P/S$. Neither P nor S can change signs within the bounded volume and are single-valued functions of their location in parameter space. Thus, $-P/S$ changes sign only at bounding surfaces. This also corresponds to the vanishing of the radius of a wave normal surface.

3. For any point in parameter space and for a given interval in θ in which n is finite, n is either pure real or pure imaginary throughout that interval.

   Proof. n^2 is a continuous function of θ, so that changing sign means crossing a bounding surface.

4. n is symmetric about $\theta = 0$ and $\theta = \pi/2$.

   Proof. n^2 is a rational composition of $\sin^2 \theta$ and $\cos^2 \theta$.

5. Two modes may coincide only at $\theta = 0$, $\theta = \pi/2$ or in special cases where $PD = 0$ and $RL = PS$.

   Proof. For $0 < \theta < \pi/2$, the solution for n^2 involves the square root of the discriminant:

   $$\sqrt{B^2 - 4AC} = \sqrt{(RL - PS)^2 \sin^4 \theta + 4P^2 D^2 \cos^2 \theta}$$

   which can only vanish for $RL = PS$ and $PD = 0$ simultaneously.

These theorems provide sufficient information to characterise the topology of a wave normal surface throughout all of parameter space. These theorems show that only three types of wave normal surfaces exist: ellipsoid, dumbbell, and wheel, all rotationally symmetric about the z-axis. We consider the three following types of wave normal surfaces:

1. Bounded volume with no resonance and $n^2 > 0$ at some point within the volume. Hence, n^2 is positive and finite throughout the entire bounded volume at every θ. The wave normal surface is ellipsoidal with symmetry about $\theta = 0$ and $\theta = \pi/2$.

2. Bounded volume having a resonance at $0 < \theta_{\text{res}} < \pi/2$ and $n^2(\theta) > 0$ for $\theta < \theta_{\text{res}}$. At θ_{res}, $n \to \infty$, so the radius of the wave normal surfaces goes to zero.

3. Bounded volume having a resonance at $0 < \theta_{\text{res}} < \pi/2$ and $n^2(\theta) < 0$ for $\theta < \theta_{\text{res}}$. As in the previous case, now only the wave normal surface exists for angles greater than the resonant angle.

Consider now the relationship between two modes, corresponding to the two solutions for n^2. Since two modes cannot intersect other than at $\theta = 0$ and $\theta = \pi/2$ and at mirror angles, if one mode is an ellipsoid and the other contains a resonance, then the ellipsoid will always be outside of the other mode. Also, at most one mode can be resonant, so at most one mode is either dumbbell- or wheel-type. Observe why. Resonances occur when $A \to 0$, so that:

$$n^2 \simeq \frac{B \pm (B - 2AC/B)}{2A} \implies n_+^2 = \frac{B}{A} \quad \text{and} \quad n_-^2 = \frac{C}{A} \tag{4.65}$$

where $|n_+^2| \gg |n_-^2|$ since $B^2 \gg |4AC|$. The root n_+^2 has resonance while n_-^2 does not, so the latter corresponds to an ellipsoid. The ellipsoidal surfaces always have a greater value of ω/kc than a dumbbell or a wheel at all angles, so it will always be the fast mode. Thus, dumbbells and wheels are slow modes.

CONTINUE

4.4 Elementary warm plasma waves in the two-fluid description

In this section, we explore Alfvén waves from a two-fluid perspective and find that they exhibit three distinct forms depending on whether the Alfvén velocity exceeds or is less than the ion or electron thermal velocities. We first consider phenomena propagating perpendicular to the stationary magnetic field and use these results to study propagation parallel to the field.

4.4.1 Perpendicular propagation in warm magnetised plasmas

Applying the general procedure of studying plasma waves, we begin by linearising the two-fluid momentum equation about a stationary equilibrium state:

$$n_\sigma^\square m_\sigma \frac{\partial \overline{\mathbf{u}}_\sigma}{\partial t} = n_\sigma^\square q_\sigma (\overline{\mathbf{E}} + \overline{\mathbf{u}}_\sigma \times \mathbf{B}_0) - \nabla \cdot \overline{\mathbf{P}}_\sigma \tag{4.66}$$

where we have ignored the friction term, the pressure term shall be developed with respect to parallel and perpendicular directions to the stationary magnetic field, that is $\nabla \cdot \overline{\mathbf{P}}_\sigma = \nabla_\perp \overline{P}_\sigma^\perp + \nabla_\parallel \overline{P}_\sigma^\parallel$. Since quasi-neutrality is assumed, the perpendicular current given by the two-fluid theory is effectively identical to the one provided by the MHD description, consisting of a polarisation drift and a diamagnetic drift. Thus, in the first approximation, one can neglect the temporal change in $\overline{\mathbf{u}}_\sigma$ to obtain:

$$\overline{\mathbf{u}}_\sigma \times \mathbf{B}_0 \simeq -\overline{\mathbf{E}}^\perp + \frac{1}{n_\sigma^\square q_\sigma} \nabla_\perp \overline{P}_\sigma^\perp \tag{4.67}$$

Taking the cross product with $\mathbf{B}_0$ and adding drift corrections to account for the time-dependancy of $\overline{\mathbf{u}}_\sigma$, one obtains:

$$\overline{\mathbf{u}}_\sigma^\perp \simeq \frac{\overline{\mathbf{E}} \times \mathbf{B}_0}{\mathbf{B}_0^2} + \frac{\mathbf{B}_0 \times \nabla_\perp \overline{P}_\sigma^\perp}{n_\sigma^\square q_\sigma \mathbf{B}_0^2} + \frac{m_\sigma}{q_\sigma \mathbf{B}_0^2} \frac{\mathrm{d}\overline{\mathbf{E}}^\perp}{\mathrm{d}t} - \frac{m_\sigma}{n_\sigma^\square q_\sigma^2 \mathbf{B}_0^2} \frac{\mathrm{d}(\nabla_\perp \overline{P}_\sigma^\perp)}{\mathrm{d}t} \tag{4.68}$$

However, since ω is assumed to be much smaller than cyclotron frequencies, these are effectively second-order terms. However, we shall observe that the first-order contributions of the $\mathbf{E} \times \mathbf{B}$ drift cancel due to quasi-neutrality. It is therefore necessary to retain these terms. The perpendicular current density is:

$$\overline{\mathbf{J}}^\perp = \sum_\sigma n_\sigma^\square q_\sigma \overline{\mathbf{u}}_\sigma = \sum_\sigma n_\sigma q_\sigma \left(\frac{\overline{\mathbf{E}} \times \mathbf{B}}{\mathbf{B}^2} + \frac{\mathbf{B}_0 \times \nabla_\perp \overline{P}_\sigma^\perp}{n_\sigma q_\sigma \mathbf{B}_0^2} + \frac{m_\sigma}{q_\sigma \mathbf{B}_0^2} \frac{\mathrm{d}\overline{\mathbf{E}}^\perp}{\mathrm{d}t} - \frac{m_\sigma}{n_\sigma q_\sigma^2 \mathbf{B}_0^2} \nabla_\perp \frac{\mathrm{d}P_\sigma^\perp}{\mathrm{d}t} \right) \tag{4.69}$$

$$\overline{\mathbf{J}}^\perp = \sum_\sigma \left(\frac{\mathbf{B}_0 \times \nabla_\perp \overline{P}_\sigma^\perp}{\mathbf{B}_0^2} + \frac{n_\sigma m_\sigma}{\mathbf{B}_0^2} \frac{\mathrm{d}\overline{\mathbf{E}}^\perp}{\mathrm{d}t} - \frac{m_\sigma}{q_\sigma \mathbf{B}_0^2} \nabla_\perp \frac{\mathrm{d}P_\sigma^\perp}{\mathrm{d}t} \right) \simeq \frac{n_\mathrm{i} m_\mathrm{i} \dot{\overline{\mathbf{E}}}^\perp}{\mathbf{B}_0^2} - \frac{\mathbf{B}_0 \times \nabla_\perp \overline{P}^\perp}{\mathbf{B}_0^2} \tag{4.70}$$

where we have summed the pressure contributions from each species and used $m_\mathrm{e} \ll m_\mathrm{i}$ to ignore electron polarisation drift. Notice also that we have dropped the second-order pressure correction, since the first-order term survived the sum. The strong inequality equally suggests:

$$\rho_0 = \sum_\sigma n_\sigma^\square m_\sigma \simeq n_\mathrm{i}^\square m_\mathrm{i} \implies c_\mathrm{A}^2 = \frac{\mathbf{B}_0^2}{\mu_0\rho_0} \simeq \frac{\mathbf{B}_0^2}{\mu_0 n_\mathrm{i}^\square m_\mathrm{i}} \tag{4.71}$$

We now consider the linearised Maxwell-Faraday equation:

$$\nabla \times \overline{\mathbf{E}} = (\nabla_\perp + \nabla_\parallel) \times (\overline{\mathbf{E}}^\perp + \overline{\mathbf{E}}^\parallel) = \nabla_\perp \times \overline{\mathbf{E}}^\perp + \nabla_\perp \times \overline{\mathbf{E}}^\parallel + \nabla_\parallel \times \overline{\mathbf{E}}^\perp = -\frac{\partial \overline{\mathbf{B}}}{\partial t} \tag{4.72}$$

whose parallel and perpendicular components to the stationary magnetic field are:

$$\hat{\mathbf{B}}_0 \cdot (\nabla_\perp \times \overline{\mathbf{E}}^\perp) = \mathrm{i}\omega \overline{B}^\parallel \quad \text{and} \quad \hat{\mathbf{B}}_0 \times (\nabla_\perp \overline{E}^\parallel - \mathrm{i}k_\parallel \overline{E}^\perp) = -\mathrm{i}\omega \overline{B}^\perp \tag{4.73}$$

Maxwell-Ampère equation, on the other hand, gives:

$$\hat{\mathbf{B}}_0 \cdot (\nabla_\perp \times \overline{\mathbf{B}}^\perp) = \mu_0 \overline{J}^\parallel \quad \text{and} \quad \hat{\mathbf{B}}_0 \times (\nabla_\perp \overline{B}^\parallel - \mathrm{i}k_\parallel \overline{B}^\perp) = -\mu_0 \overline{\mathbf{J}}^\perp \tag{4.74}$$

We may now substitute eq. (4.70) into the previous equations to obtain:

$$\hat{\mathbf{B}}_0 \times (\nabla_\perp \overline{B}^\parallel - \mathrm{i}k_\parallel \overline{B}^\perp) = \frac{\mu_0 n_\mathrm{i} m_\mathrm{i}}{\mathbf{B}_0^2} \mathrm{i}\omega \overline{E}^\perp + \frac{\mu_0 \nabla \overline{P} \times \hat{\mathbf{B}}_0}{B_0} \tag{4.75}$$

which, upon rearranging, gives:

$$\nabla_\perp \left(\overline{B}^\parallel + \frac{\mu_0 \overline{P}^\perp}{B} \right) \times \hat{\mathbf{B}}_0 - \mathrm{i}k_\parallel \overline{\mathbf{B}}^\perp \times \hat{\mathbf{B}}_0 = -\frac{\mathrm{i}\omega}{c_A^2} \overline{\mathbf{E}}^\perp \tag{4.76}$$

We have thus obtained a coupled system of equations relating $\overline{\mathbf{B}}^\perp$ and $\overline{\mathbf{E}}^\perp$, namely:

$$\mathrm{i}\omega \overline{\mathbf{B}}^\perp \times \hat{\mathbf{B}}_0 - \mathrm{i}k_\parallel \overline{\mathbf{E}}^\perp = -\nabla_\perp \overline{E}^\parallel \quad \text{and} \quad -\mathrm{i}k_\parallel \overline{\mathbf{B}}^\perp \times \hat{\mathbf{B}}_0 + \frac{\mathrm{i}\omega}{c_A^2} \overline{\mathbf{E}}^\perp = -\nabla_\perp \left(\overline{B}^\parallel + \frac{\mu_0 \overline{P}^\perp}{B} \right) \times \hat{\mathbf{B}}_0 \tag{4.77}$$

We then solve the system algebraically to find:

$$\overline{\mathbf{E}}^\perp = \frac{\mathrm{i}k_\parallel \nabla_\perp \overline{E}^\parallel + \mathrm{i}\omega \nabla_\perp (\overline{B}^\parallel + \mu_0 \overline{P}^\perp / B) \times \hat{\mathbf{B}}_0}{\omega^2 / c_A^2 - k_\parallel^2} \tag{4.78}$$

$$\overline{\mathbf{B}}^\perp \times \hat{\mathbf{B}}_0 = \frac{\mathrm{i}(\omega / c_A^2) \nabla_\perp \overline{E}^\parallel + \mathrm{i}k_\parallel \nabla_\perp (\overline{B}^\parallel + \mu_0 \overline{P}^\perp / B_0) \times \hat{\mathbf{B}}_0}{\omega^2 / c_A^2 - k_\parallel^2} \tag{4.79}$$

Although the expressions seem frightening, they allow us to find the parallel components $\overline{\mathbf{B}}^\parallel$ and $\overline{\mathbf{E}}^\parallel$:

$$\nabla \cdot (\overline{\mathbf{E}}^\perp \times \hat{\mathbf{B}}_0) = \mathrm{i}\omega \overline{B}^\parallel \quad \text{and} \quad \nabla \cdot (\overline{\mathbf{B}}^\perp \times \hat{\mathbf{B}}_0) = \mu_0 \overline{J}^\parallel \tag{4.80}$$

It is now convenient to note that $\nabla \cdot (\nabla_\perp \psi \times \hat{\mathbf{x}}) = \nabla \cdot (\nabla \psi \times \hat{\mathbf{x}}) = \hat{\mathbf{x}} \cdot (\nabla \times \nabla \psi) = 0$, such that $\nabla \cdot (\nabla_\perp \overline{E}^\parallel \times \hat{\mathbf{B}}_0) = 0$ and $\nabla \cdot (\nabla_\perp (\overline{B}^\parallel + \mu_0 \overline{P}^\perp / B_0) \times \hat{\mathbf{B}}_0) = 0$ which results in:

$$\nabla \cdot \left(\frac{\nabla_\perp (\overline{B}^\parallel + \mu_0 \overline{P}^\perp / B_0)}{\omega^2 / c_A^2 - k_\parallel^2} \right) = -\overline{B}^\parallel \quad \text{and} \quad \nabla \cdot \left(\frac{\mathrm{i}\omega \nabla_\perp \overline{E}^\parallel}{\omega^2 / c_A^2 - k_\parallel^2} \right) = c_A^2 \mu_0 \overline{J}^\parallel \tag{4.81}$$

These two equations precisely distinguish the compressional and shear modes of a particular wave. This is verified by means of the linearised continuity equation:

$$\nabla \cdot \mathbf{u}_\sigma^\perp = \nabla \cdot \left(\frac{\overline{\mathbf{E}} \times \mathbf{B}_0}{\mathbf{B}_0^2} \right) = \frac{1}{\mathbf{B}_0^2} \nabla \cdot (\overline{\mathbf{E}} \times \mathbf{B}_0) = \frac{1}{\mathbf{B}_0^2} \mathbf{B}_0 \cdot (\nabla \times \overline{\mathbf{E}}) = -\frac{1}{\mathbf{B}_0^2} \mathbf{B}_0 \cdot \frac{\partial \overline{\mathbf{B}}}{\partial t} = \frac{\mathrm{i}\omega}{B_0} \overline{B}^\parallel \tag{4.82}$$

This demonstrates that modes with $\omega \overline{B}^\parallel = 0 \iff \overline{B}^\parallel = 0$ are incompressible. Now, we turn our attention to compressional modes, thus $\overline{B}^\parallel \neq 0$ and $\overline{E}^\parallel = 0$. Having $\overline{E}^\parallel = 0$ means $\overline{J}^\parallel = 0$ which effectively means that there is no parallel motion $\overline{\mathbf{u}}_\sigma = \overline{\mathbf{u}}_\sigma^\perp$, thus:

$$\frac{\partial \overline{n}_\sigma}{\partial t} + n_\sigma^\square \nabla \cdot \overline{\mathbf{u}}_\sigma^\perp = 0 \implies \frac{\partial \overline{n}_\sigma}{\partial t} + \frac{n_\sigma^\square}{B_0}\frac{\partial \overline{B}^\parallel}{\partial t} = 0 \implies \frac{\overline{n}_\sigma}{n_\sigma^\square} = \frac{\overline{B}^\parallel}{B_0} \tag{4.83}$$

In the adiabatic/isothermal regime, we can relate the magnetic field variation to the pressure variation:

$$\frac{\overline{P}^\perp}{P} = \frac{\sum_\sigma \overline{P}_\sigma^\perp}{P} = \frac{\sum_\sigma \gamma \overline{n}_\sigma P_\sigma^\perp / n_\sigma}{P} = \frac{\sum_\sigma \gamma \overline{B}^\parallel P_\sigma^\perp / B_0}{P} = \frac{\gamma \overline{B}^\parallel}{B_0 P} \sum_\sigma P_\sigma^\perp = \frac{\gamma \overline{B}^\perp}{B_0} \frac{P^\perp}{P} \simeq \frac{\gamma \overline{B}^\parallel}{B_0} \tag{4.84}$$

Thus, $\mu_0 \overline{P}^\perp / B_0 = \frac{\gamma \mu_0 P \overline{B}^\parallel}{B_0^2} = \frac{\gamma (n_\mathrm{i} k_\mathrm{B} T_\mathrm{i} + n_\mathrm{e} k_\mathrm{B} T_\mathrm{e})}{B_0^2/\mu_0} \overline{B}^\parallel = \frac{\gamma k_\mathrm{B}(T_\mathrm{i} + Z T_\mathrm{e})}{m_\mathrm{i} c_A^2} \overline{B}^\parallel \simeq c_s^2 / c_A^2 \overline{B}^\parallel$ and we obtain:

$$\nabla \cdot \left(\frac{c_s^2 + c_A^2}{\omega^2 - c_A^2 k_\parallel^2} \nabla_\perp \overline{B}^\parallel \right) + \overline{B}^\parallel = 0 \tag{4.85}$$

Upon exchanging $\nabla \leftrightarrow \nabla_\perp \leftrightarrow i\mathbf{k}_\perp$, we at last obtain the dispersion relation of the compressional mode:

$$1 - \frac{k_\perp^2 (c_A^2 + c_s^2)}{\omega^2 - k_\parallel^2 c_A^2} = 0 \Longleftrightarrow \omega^2 = (c_A^2 + c_s^2) k_\perp^2 + k_\parallel^2 c_A^2 = k^2 c_A^2 + k_\perp^2 c_s^2 = k^2 (c_A^2 + c_s^2 \sin^2 \theta) \tag{4.86}$$

4.4.2 Parallel propagation in warm magnetised plasmas

We now investigate the two-fluid shear modes of Alfvén waves. In this case, we use $\overline{B}^\parallel = 0$ so that $\nabla \cdot \overline{\mathbf{u}}_\sigma^\perp = 0$. The parallel component of the linearised equation of motion reads:

$$n_\sigma m_\sigma \frac{\partial \overline{u}_\sigma^\parallel}{\partial t} = n_\sigma q_\sigma \overline{E}^\parallel - \gamma k_\mathrm{B} T_\sigma \partial_\parallel \overline{n}_\sigma \tag{4.87}$$

The linearised continuity equation gives:

$$\frac{\partial \overline{n}_\sigma}{\partial t} + \nabla_\parallel \cdot (n_\sigma \overline{\mathbf{u}}_\sigma^\parallel) = 0 \tag{4.88}$$

We may take the time derivative of the equation of motion and invoke the previous:

$$n_\sigma m_\sigma \frac{\partial^2 \overline{u}^\parallel}{\partial t^2} = n_\sigma q_\sigma \frac{\partial \overline{E}^\parallel}{\partial t} - \gamma k_\mathrm{B} T_\sigma \partial_\parallel \frac{\partial \overline{n}_\sigma}{\partial t} = n_\sigma q_\sigma \frac{\partial \overline{E}^\parallel}{\partial t} - \gamma n_\sigma k_\mathrm{B} T_\sigma \nabla_\parallel^2 \overline{u}_\sigma^\parallel \tag{4.89}$$

$$\frac{\partial^2 \overline{u}^\parallel}{\partial t^2} + \frac{\gamma k_\mathrm{B} T_\sigma}{m_\sigma} \nabla_\parallel^2 \overline{u}^\parallel = \frac{q_\sigma}{m_\sigma} \frac{\partial \overline{E}^\parallel}{\partial t} \tag{4.90}$$

Performing Fourier transform and isolating $\overline{u}^\parallel$, we obtain:

$$\overline{u}_\sigma^\parallel = \frac{i\omega q_\sigma}{m_\sigma} \frac{\overline{E}^\parallel}{\omega^2 - (\gamma k_\mathrm{B} T_\sigma / m_\sigma) k_\parallel^2} \tag{4.91}$$

We may now express the parallel component of the current density:

$$\overline{J}^\parallel = \sum_\sigma n_\sigma q_\sigma \overline{u}_\sigma^\parallel = \mathrm{i}\varepsilon_0 \omega \overline{E}^\parallel \sum_\sigma \frac{n_\sigma q_\sigma^2 / \varepsilon_0 m_\sigma}{\omega^2 - (\gamma k_\mathrm{B} T_\sigma / m_\sigma) k_\parallel^2} \tag{4.92}$$

We can now insert this expression into one of the above, namely:

$$\nabla \cdot \left(\frac{\mathrm{i}\omega \nabla_\perp \overline{E}^\parallel}{\omega^2 / c_A^2 - k_\parallel^2} \right) = c_A^2 \mu_0 \overline{J}^\parallel \implies \nabla \cdot \left(\frac{\nabla_\perp \overline{E}^\parallel}{\omega^2 / c_A^2 - k_\parallel^2} \right) - \frac{c_A^2}{c^2} \overline{E}^\parallel \sum_\sigma \frac{n_\sigma q_\sigma^2 / \varepsilon_0 m_\sigma}{\omega^2 - (\gamma k_\mathrm{B} T_\sigma / m_\sigma) k_\parallel^2} = 0 \tag{4.93}$$

Upon again using Fourier to convert $\nabla \to \nabla_\perp \to i\mathbf{k}_\perp$, we obtain:

$$\frac{k_\perp^2}{\omega^2 - c_A^2 k_\parallel^2} + \sum_\sigma \frac{\omega_{\mathrm{p}\sigma}^2 / c^2}{\omega^2 - (\gamma k_\mathrm{B} T_\sigma / m_\sigma) k_\parallel^2} = 0 \tag{4.94}$$

1. Now, we investigate the limiting case $\omega^2/k_\parallel^2 \gg k_B T_e/m_e, k_B T_i/m_i$. Since $\omega_{pe} \gg \omega_{pi}$, the electron term dominates over the ion term, so that:

$$\frac{k_\perp^2}{\omega^2 - c_A^2 k_\parallel^2} + \frac{\omega_{pe}^2/c^2}{\omega^2 - (\gamma k_B T_e/m_e)k_\parallel^2} \implies \omega^2 = \frac{k_\parallel^2 c_A^2}{1 + k_\perp^2 c^2/\omega_{pe}^2} \tag{4.95}$$

   This limiting case is called the inertial Alfvén wave, and the quantity c/ω_{pe} is called the electron collisionless skin depth. If $k_\perp^2 c^2/\omega_{pe}^2$ is not too large, then $\omega/k_\parallel$ is of the order of the Alfvén velocity and the condition $\omega^2 \gg k_\parallel^2 k_B T_e/m_e$ corresponds to $c_A^2 \gg v_{Te}^2$ or $\beta_e = \frac{n_e k_B T_e}{B^2/\mu_0} \ll \frac{m_e}{m_i}$. Inertial Alfvén modes exist only in the ultra-low-pressure parameter regime, since their dispersion relation does not depend on temperature and is thus called a cold plasma wave.

2. In the situation where $k_B T_i/m_i \ll \omega^2/k_\parallel^2 \ll k_B T_e/m_e$, the equation above becomes:

$$\frac{k_\perp^2}{\omega^2 - c_A^2 k_\parallel^2} + \frac{\omega_{pe}^2/c^2}{(\gamma k_B T_e/m_e)k_\parallel^2} + \frac{\omega_{pi}^2/c^2}{\omega^2} = 0 \tag{4.96}$$

   Note that this equation is quadratic in ω^2, so that there are two distinct modes. We shall distinguish between the two by comparing them with the ion acoustic velocity. First, let $\omega^2/k_\parallel^2 \gg k_B T_e/m_i$. In this case, the last term can be dropped, and we obtain:

$$\omega^2 = c_A^2 k_\parallel^2 \left(1 + \frac{k_\perp^2}{c_A^2}\frac{k_B T_e}{m_e}\frac{c^2}{\omega_{pe}^2}\right) \tag{4.97}$$

   which is the dispersion relation of the so-called kinetic Alfvén wave. It is convenient to introduce the fictitious ion Larmor radius via the ion acoustic velocity:

$$\rho_i^2 = \frac{1}{c_A^2}\frac{k_B T_e}{m_e}\frac{c^2}{\omega_{pe}^2} = \frac{\mu_0 n_i m_i}{B^2}\frac{k_B T_e}{m_e}\frac{\varepsilon_0 c^2 m_e}{n_e q_e^2} = \frac{n_i m_i k_B T_e}{B^2 n_e q_e^2} = \frac{Z m_i^2}{B^2 q_i^2}\frac{k_B T_e}{m_i} \tag{4.98}$$

   If $k_\perp^2 \rho_i^2$ isn't too large, then $\omega/k_\parallel$ is of the order of the Alfvén velocity, and the condition $\omega^2 \ll k_\parallel^2 k_B T_e/m_e$ corresponds to having $\beta_e \gg m_e/m_i$. The assumed condition $\omega^2/k_\parallel^2 \gg k_B T_e/m_i$ corresponds to $\beta_e \ll 1$. The kinetic Alfvén wave is thus faithfully described by the previous dispersion relation in the regime $m_e/m_i \ll \beta_e \ll 1$.

3. We now assume $\omega^2/k_\parallel^2 \ll k_B T_e/m_e, k_B T_i/m_i$, that is, a plasma wave with phase velocity much smaller than both the ion and electron thermal velocities. The general dispersion relation reduces to:

$$\omega^2 = c_A^2 k_\parallel^2 \left(1 + k_\perp^2 \frac{\omega_{pi}^2}{\Omega_i^2}\lambda_D^2\right) = c_A^2 k_\parallel^2 \left(1 + \frac{k_\perp^2 k_B T_e}{(1 + T_e/T_i) m_i \Omega_i^2}\right) \tag{4.99}$$

   This situation corresponds to high-β_e shear modes.

To summarise: the shear mode has $\overline{B}^\parallel = 0$, $\overline{E}^\parallel \neq 0$ and $\overline{J}^\parallel \neq 0$. In the ultra-low pressure parameter regime, where $\beta_e \ll \frac{m_e}{m_i}$, the shear Alfvén wave exists as a cold plasma wave, and as one of two types of warm plasma waves in the regime $\beta_e \gg \frac{m_e}{m_i}$. The shear Alfvén mode isincompressiblee, that is $\frac{d\overline{n}_\sigma}{dt} = 0 \iff \nabla \cdot \overline{\mathbf{u}}_\sigma^\perp = 0$, which is essentially $\mathbf{k}_\perp \cdot \overline{\mathbf{u}}_\sigma^\perp = 0$, thus $\mathbf{k}_\perp$ is orthogonal to $\overline{\mathbf{u}}_\sigma^\perp$. The Alfvén wave in the ultra-low pressure parameter, that is, the inertial Alfvén wave, is independent of temperature and thus describes cold plasma waves. The shear-mode parallel dynamics is related to parallel-propagating ion-acoustic modes, since both Alfvén shear waves and parallel-propagating ion-acoustic modes involve parallel motion driven by a finite parallel oscillatory component of the electric field. This parallel dynamics is absent from the MHD description due to its fundamental isotropic approximation.

4.4.3 Non-linear cold unmagnetised waves in the two-fluid description

In this subsection, we shall discuss the non-linear corrections of the two-fluid description of cold plasma waves on a specific example, which shall illustrate the general procedure for studying non-linear wave dynamics in plasmas. The example considered is an ion-acoustic wave in a cold plasma comprising ions and electrons. Contrary to our previous considerations, where we have pragmatically neglected all terms of order higher than linear, we shall keep the non-linear advective term in the two-fluid momentum equations. It is worth noting that the non-linear behaviour considered here has a significant influence on the dynamics of plasmas in which particular species follow spatially non-uniform macroscopic velocities. Moreover, systems with particular initial conditions, governed by non-linear dynamics, are susceptible to developing singularities, suggesting that our theoretical models lack an adequate physical description of the dynamics. We will address these issues later.

We begin with two-fluid momentum and continuity equations for both ion and electron species:

$$n_{\mathrm{i}} m_{\mathrm{e}} \left[\frac{\partial \mathbf{u}_{\mathrm{i}}}{\partial t} + (\mathbf{u}_{\mathrm{i}} \cdot \nabla) \mathbf{u}_{\mathrm{i}} \right] = n_{\mathrm{i}} q_{\mathrm{i}} \mathbf{E} - k_{\mathrm{B}} \nabla (n_{\mathrm{i}} T_{\mathrm{i}}) - n_{\mathrm{i}} m_{\mathrm{i}} \nu_{\mathrm{i} \to \mathrm{e}} \mathbf{u}_{\mathrm{i}} \quad \text{and} \quad \frac{\partial n_{\mathrm{i}}}{\partial t} + \nabla \cdot (n_{\mathrm{i}} \mathbf{u}_{\mathrm{i}}) = 0 \tag{4.100}$$

$$n_{\mathrm{e}} m_{\mathrm{e}} \left[\frac{\partial \mathbf{u}_{\mathrm{e}}}{\partial t} + (\mathbf{u}_{\mathrm{e}} \cdot \nabla) \mathbf{u}_{\mathrm{e}} \right] = -n_{\mathrm{e}} e \mathbf{E} - k_{\mathrm{B}} \nabla (n_{\mathrm{e}} T_{\mathrm{e}}) - n_{\mathrm{e}} m_{\mathrm{e}} \nu_{\mathrm{e} \to \mathrm{i}} \mathbf{u}_{\mathrm{e}} \quad \text{and} \quad \frac{\partial n_{\mathrm{e}}}{\partial t} + \nabla \cdot (n_{\mathrm{e}} \mathbf{u}_{\mathrm{e}}) = 0 \tag{4.101}$$

Poisson equation also applies:

$$\nabla \cdot \mathbf{E} = \frac{n_{\mathrm{i}} q_{\mathrm{i}} - n_{\mathrm{e}} e}{\varepsilon_0} = \frac{e}{\varepsilon_0} \left(Z n_{\mathrm{i}} - n_{\mathrm{e}} \right) \tag{4.102}$$

We consider a cold isothermal regime for both electrons and ions with $T_{\mathrm{i}} \ll T_{\mathrm{e}}$. Since $m_{\mathrm{e}} \ll m_{\mathrm{i}}$, we can ignore the inertial term of the electron momentum equation. Let us also assume a collisionless regime, in which the drag force terms vanish. We now linearise the equations around a stationary regime where $\mathbf{u}_{\mathrm{e}}^{\square} = \mathbf{u}_{\mathrm{i}}^{\square} = \mathbf{0}$, $Z n_{\mathrm{i}}^{\square} = n_{\mathrm{e}}^{\square}$ and $\mathbf{E}^{\square} = \mathbf{0}$, so that the governing equations become:

$$n_{\mathrm{i}}^{\square} m_{\mathrm{i}} \left(\frac{\partial \overline{\mathbf{u}}_{\mathrm{i}}}{\partial t} + (\overline{\mathbf{u}}_{\mathrm{i}} \cdot \nabla) \overline{\mathbf{u}}_{\mathrm{i}} \right) = n_{\mathrm{i}}^{\square} q_{\mathrm{i}} \overline{\mathbf{E}} \quad \text{and} \quad \frac{\partial \overline{n}_{\mathrm{i}}}{\partial t} + \nabla \cdot (n_{\mathrm{i}}^{\square} \overline{\mathbf{u}}_{\mathrm{i}}) = 0 \tag{4.103}$$

$$\mathbf{0} = -n_{\mathrm{e}}^{\square} e \overline{\mathbf{E}} - k_{\mathrm{B}} T_{\mathrm{e}} \nabla \overline{n}_{\mathrm{e}} \quad \text{and} \quad \frac{\partial \overline{n}_{\mathrm{e}}}{\partial t} + \nabla \cdot (n_{\mathrm{e}}^{\square} \overline{\mathbf{u}}_{\mathrm{e}}) = 0 \tag{4.104}$$

while the Poisson equation simply stays as is. We now assume that the oscillatory perturbation is fully described by a single Fourier mode, so that $\partial_t \equiv -\mathrm{i}\omega$ and $\nabla \equiv \mathrm{i}\mathbf{k}$, giving:

$$-\mathrm{i} n_{\mathrm{i}}^{\square} m_{\mathrm{i}} \omega \overline{\mathbf{u}}_{\mathrm{i}} + \mathrm{i} n_{\mathrm{i}}^{\square} m_{\mathrm{i}} (\overline{\mathbf{u}}_{\mathrm{i}} \cdot \mathbf{k}) \overline{\mathbf{u}}_{\mathrm{i}} = n_{\mathrm{i}}^{\square} q_{\mathrm{i}} \overline{\mathbf{E}} \quad \text{and} \quad -\mathrm{i}\omega \overline{n}_{\mathrm{i}} + \mathrm{i} n_{\mathrm{i}}^{\square} \mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}} = 0 \tag{4.105}$$

$$\mathbf{0} = -n_{\mathrm{e}}^{\square} e \overline{\mathbf{E}} - \mathrm{i} k_{\mathrm{B}} T_{\mathrm{e}} \mathbf{k} \overline{n}_{\mathrm{e}} \quad \text{and} \quad -\mathrm{i}\omega \overline{n}_{\mathrm{e}} + \mathrm{i} n_{\mathrm{e}}^{\square} \mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{e}} = 0 \tag{4.106}$$

$$\mathrm{i}\mathbf{k} \cdot \overline{\mathbf{E}} = \frac{e}{\varepsilon_0} (Z \overline{n}_{\mathrm{i}} - \overline{n}_{\mathrm{e}}) \tag{4.107}$$

Solving the system of equations by expressing everything in terms of $\overline{\mathbf{u}}_{\mathrm{i}}$ gives:

$$\mathbf{k} \overline{n}_{\mathrm{e}} = \frac{\mathrm{i} n_{\mathrm{e}}^{\square} e}{k_{\mathrm{B}} T_{\mathrm{e}}} \overline{\mathbf{E}} \quad \text{and} \quad \overline{n}_{\mathrm{i}} = \frac{\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}}{\omega} n_{\mathrm{i}}^{\square} \tag{4.108}$$

$$\mathrm{i}(\mathbf{k} \cdot \overline{\mathbf{E}}) \mathbf{k} = \frac{n_{\mathrm{i}}^{\square} Z e}{\varepsilon_0} \frac{\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}}{\omega} \mathbf{k} - \frac{\mathrm{i} n_{\mathrm{e}}^{\square} e^2}{\varepsilon_0 k_{\mathrm{B}} T_{\mathrm{e}}} \overline{\mathbf{E}} \implies \overline{\mathbf{E}} = \frac{n_{\mathrm{i}}^{\square} Z e \lambda_{D\mathrm{e}}^2}{\mathrm{i} \varepsilon_0 \omega} \frac{\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}}{\mathbf{k}^2 \lambda_{D\mathrm{e}}^2 + 1} \mathbf{k} \tag{4.109}$$

$$-\mathrm{i}\omega \overline{\mathbf{u}}_{\mathrm{i}} + \mathrm{i}(\overline{\mathbf{u}}_{\mathrm{i}} \cdot \mathbf{k}) \overline{\mathbf{u}}_{\mathrm{i}} = \frac{Ze}{m_{\mathrm{i}}} \overline{\mathbf{E}} = -\mathrm{i} \frac{n_{\mathrm{i}}^{\square} (Ze)^2 \lambda_{D\mathrm{e}}^2}{\varepsilon_0 m_{\mathrm{i}} \omega} \frac{\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}}{\mathbf{k}^2 \lambda_{D\mathrm{e}}^2 + 1} \mathbf{k} \tag{4.110}$$

In the limit $k \lambda_{D\mathrm{e}} \ll 1$, we may expand the denominator of the right-hand side:

$$-\mathrm{i}\omega \overline{\mathbf{u}}_{\mathrm{i}} + \mathrm{i}(\overline{\mathbf{u}}_{\mathrm{i}} \cdot \mathbf{k}) \overline{\mathbf{u}}_{\mathrm{i}} + \mathrm{i} \frac{\omega_{\mathrm{pi}}^2 \lambda_{D\mathrm{e}}^2}{\omega} (\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}) \mathbf{k} - \mathrm{i} \frac{\omega_{\mathrm{pi}}^2 \lambda_{D\mathrm{e}}^4}{\omega} \mathbf{k}^2 (\mathbf{k} \cdot \overline{\mathbf{u}}_{\mathrm{i}}) \mathbf{k} = \mathbf{0} \tag{4.111}$$

Notice that $\omega_{\mathrm{pi}}\lambda_{De} = c_\mathrm{i}$ and in the given limit, $\omega \simeq c_\mathrm{i}k(1-k^2\lambda_{De}^2/2)$, so that:

$$-\mathrm{i}\omega\overline{\mathbf{u}}_\mathrm{i} + \mathrm{i}(\overline{\mathbf{u}}_\mathrm{i}\cdot\mathbf{k})\overline{\mathbf{u}}_\mathrm{i} + \mathrm{i}\mathbf{c}_\mathrm{i}(\mathbf{k}\cdot\overline{\mathbf{u}}_\mathrm{i}) + (i)^3\frac{\lambda_{De}^2 c_\mathrm{i}}{2}\mathbf{k}^2(\mathbf{k}\cdot\overline{\mathbf{u}}_\mathrm{i}) = \mathbf{0} \tag{4.112}$$

Inverse Fourier-transforming gives the non-linear ion acoustic equation:

$$\frac{\partial\overline{\mathbf{u}}_\mathrm{i}}{\partial t} + \left[(\mathbf{u}_\mathrm{i}+\mathbf{c}_\mathrm{i})\cdot\nabla\right]\overline{\mathbf{u}}_\mathrm{i} + \frac{\lambda_{De}^2\mathbf{c}_\mathrm{i}}{2}\nabla^2(\nabla\cdot\overline{\mathbf{u}}_\mathrm{i}) = \mathbf{0} \tag{4.113}$$

4.4.3.1 Soliton solution of a non-linear ion acoustic wave

We retake the last equation from the previous question and impose unidimensionality:

$$\frac{\partial\overline{u}_\mathrm{i}}{\partial t} + (\overline{u}_\mathrm{i}+c_\mathrm{i})\frac{\partial\overline{u}_\mathrm{i}}{\partial x} + \frac{\lambda_{De}^2 c_\mathrm{i}}{2}\frac{\partial^3\overline{u}_\mathrm{i}}{\partial x^3} = 0 \tag{4.114}$$

We may discuss the roles of each term in the differential equation. The modified advective term is prone to introducing singularities near characteristic breaking points dependant on the initial velocity profile, while the diffusive term tends to relax the singularities and imposes global smoothness of the solution. It is interesting to consider the so-called soliton solutions, that is, convectively invariant solutions. By imposing the substitution $z = x - vt$, we find:

$$\frac{\partial\overline{u}_\mathrm{i}}{\partial t} + (\overline{u}_\mathrm{i}+c_\mathrm{i}-v)\frac{\partial\overline{u}_\mathrm{i}}{\partial z} + \frac{c_\mathrm{i}\lambda_{De}^2}{2}\frac{\partial^3\overline{u}_\mathrm{i}}{\partial z^3} = 0 \tag{4.115}$$

We search for stationary localised solutions, so that $\partial\overline{u}_\mathrm{i}/\partial t = 0$, so that:

$$(\overline{u}_\mathrm{i}+c_\mathrm{i}-v)\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z} + \frac{c_\mathrm{i}\lambda_{De}^2}{2}\frac{\mathrm{d}^3\overline{u}_\mathrm{i}}{\mathrm{d}z^3} = 0 \tag{4.116}$$

Integrating once with respect to z gives:

$$(c_\mathrm{i}-v)\overline{u}_\mathrm{i} + \frac{\overline{u}_\mathrm{i}^2}{2} + \frac{c_\mathrm{i}\lambda_{De}^2}{2}\frac{\mathrm{d}^2\overline{u}_\mathrm{i}}{\mathrm{d}z^2} = C \tag{4.117}$$

where $C = 0$ because we are seeking a localised solution. Multiplying the equation by $\overline{u}_\mathrm{i}'$, we obtain:

$$(c_\mathrm{i}-v)\overline{u}_\mathrm{i}\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z} + \frac{1}{2}\overline{u}_\mathrm{i}^2\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z} + \frac{c_\mathrm{i}\lambda_{De}^2}{2}\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z}\frac{\mathrm{d}^2\overline{u}_\mathrm{i}}{\mathrm{d}z^2} = 0 \tag{4.118}$$

Integrating gives:

$$(c_\mathrm{i}-v)\frac{\overline{u}_\mathrm{i}^2}{2} + \frac{\overline{u}_\mathrm{i}^3}{6} + \frac{c_\mathrm{i}\lambda_{De}^2}{4}\left(\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z}\right)^2 = C \tag{4.119}$$

where, again, $C = 0$ since we are searching for a localised solution. Isolating the derivative:

$$\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\mathrm{d}z} = \sqrt{\frac{2(v-c_\mathrm{i})}{c_\mathrm{i}\lambda_{De}^2}\overline{u}_\mathrm{i}^2 - \frac{2\overline{u}_\mathrm{i}^3}{3c_\mathrm{i}\lambda_{De}^2}} \tag{4.120}$$

Separating the equation and integrating:

$$\int\left(\frac{\mathrm{d}\overline{u}_\mathrm{i}}{\overline{u}_\mathrm{i}\sqrt{\frac{2(v-c_\mathrm{i})}{c_\mathrm{i}\lambda_{De}^2} - \frac{2\overline{u}_\mathrm{i}}{3c_\mathrm{i}\lambda_{De}^2}}} = \mathrm{d}z\right) \quad \text{with} \quad \overline{u}_\mathrm{i} = 3(v-c_\mathrm{i})\xi \implies \mathrm{d}\overline{u}_\mathrm{i} = 3(v-c_\mathrm{i})\,\mathrm{d}\xi \tag{4.121}$$

$$\int\frac{\mathrm{d}\xi}{\xi\sqrt{\frac{2(v-c_\mathrm{i})}{c_\mathrm{i}\lambda_{De}^2} - \frac{2(v-c_\mathrm{i})}{c_\mathrm{i}\lambda_{De}^2}\xi}} = \sqrt{\frac{c_\mathrm{i}\lambda_{De}^2}{2(v-c_\mathrm{i})}}\int\frac{\mathrm{d}\xi}{\xi\sqrt{1-\xi}} = z + C \tag{4.122}$$

We now evaluate the integral using the substitution $1-\xi = u^2$:

$$\int\frac{\mathrm{d}\xi}{\xi\sqrt{1-\xi}} = -2\int\frac{\mathrm{d}u}{1-u^2} = -2\,\mathrm{artanh}\,(u) \tag{4.123}$$

$$-2\,\mathrm{artanh}(u) = z\sqrt{\frac{2(v-c_\mathrm{i})}{c_\mathrm{i}\lambda_{De}^2}} + C \implies u = \tanh\left[C - z\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] \tag{4.124}$$

$$\xi = 1 - u^2 = 1 - \tanh^2\left[C - z\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] = \mathrm{sech}^2\left[C - z\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] \tag{4.125}$$

$$\overline{u}_\mathrm{i}(z) = 3(v-c_\mathrm{i})\,\mathrm{sech}^2\left[C - z\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] \tag{4.126}$$

It's easy to see that C is simply a translation, so we might as well drop it. The hyperbolic secant function is symmetric about $z = 0$, so that:

$$\overline{u}_\mathrm{i}(z) = 3(v-c_\mathrm{i})\,\mathrm{sech}^2\left[z\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] \implies \overline{u}_\mathrm{i}(x,t) = 3(v-c_\mathrm{i})\,\mathrm{sech}^2\left[(x-vt)\sqrt{\frac{v-c_\mathrm{i}}{2c_\mathrm{i}\lambda_{De}^2}}\right] \tag{4.127}$$

which is the soliton solution to the ion acoustic equation. It is clear that v must satisfy $v > c_\mathrm{i}$ in order to have a localised and convergent solution for all $z = x - vt$.

4.5 Vlasov-Poisson model of warm electrostatic waves

The Vlasov-Maxwell model discussed in section 3.4 of chapter 3 in the case of an unmagnetised plasma reduces to the Vlasov-Poisson model studied below. This model simply involves applying the Vlasov and Poisson equations to specific initial conditions and represents the most general model of electrostatic oscillations discussed here:

$$\frac{\partial f_\sigma}{\partial t} + \mathbf{v}\cdot\frac{\partial f_\sigma}{\partial \mathbf{x}} + \frac{q_\sigma}{m_\sigma}\mathbf{E}\cdot\frac{\partial f_\sigma}{\partial \mathbf{v}} = 0 \quad \text{and} \quad \nabla\cdot\mathbf{E} = \frac{\rho}{\varepsilon_0} = \frac{1}{\varepsilon_0}\sum_\sigma n_\sigma q_\sigma \tag{4.128}$$

We may linearise these equations to obtain:

$$\frac{\partial \overline{f}_\sigma}{\partial t} + \mathbf{v}\cdot\frac{\partial \overline{f}_\sigma}{\partial \mathbf{x}} + \frac{q_\sigma}{m_\sigma}\overline{\mathbf{E}}\cdot\frac{\partial f_\sigma^\square}{\partial \mathbf{v}} = 0 \quad \text{and} \quad \nabla\cdot\overline{\mathbf{E}} = \frac{1}{\varepsilon_0}\sum_\sigma \iiint_{\mathbb{R}^3} q_\sigma \overline{f}_\sigma\,\mathrm{d}^3\mathbf{v} + \frac{\rho_\mathrm{ext}}{\varepsilon_0} \tag{4.129}$$

where ρ_ext is some extraneous charge density causing the electrostatic wave. Fourier-transforming:

$$-\mathrm{i}\omega\overline{f}_\sigma + \mathrm{i}\mathbf{k}\cdot\mathbf{v}\overline{f}_\sigma + \frac{q_\sigma}{m_\sigma}\overline{\mathbf{E}}\cdot\frac{\partial f_\sigma^\square}{\partial \mathbf{v}} = 0 \quad \text{and} \quad \mathrm{i}\mathbf{k}\cdot\overline{\mathbf{E}} = \frac{1}{\varepsilon_0}\sum_\sigma \iiint_{\mathbb{R}^3} q_\sigma \overline{f}_\sigma\,\mathrm{d}^3\mathbf{v} + \frac{\rho_\mathrm{ext}}{\varepsilon_0} \tag{4.130}$$

Notice that $\overline{\mathbf{E}} = -i\mathbf{k}\overline{\phi}$ so that we may express $\overline{f}_\sigma$ from the first equation as follows:

$$\overline{f}_\sigma = -\frac{q_\sigma}{m_\sigma}\phi\frac{\mathbf{k}\cdot\frac{\partial f_\sigma^\square}{\partial \mathbf{v}}}{\omega - \mathbf{k}\cdot\mathbf{v}} \implies \mathbf{k}^2\phi = -\frac{1}{\varepsilon_0}\sum_\sigma\frac{q_\sigma^2}{m_\sigma}\phi\iiint_{\mathbb{R}^3}\frac{\mathbf{k}\cdot\frac{\partial f_\sigma^\square}{\partial \mathbf{v}}}{\omega - \mathbf{k}\cdot\mathbf{v}}\,\mathrm{d}^3\mathbf{v} + \frac{\rho_\mathrm{ext}}{\varepsilon_0} \tag{4.131}$$

A resonant mode is clearly produced when $\omega = \mathbf{k}\cdot\mathbf{v}$. It is convenient to rewrite the previous equation as follows to determine the dispersion relation effectively:

$$\underbrace{1 + \sum_\sigma\frac{q_\sigma^2}{\varepsilon_0 m_\sigma}\frac{1}{\mathbf{k}^2}\iiint_{\mathbb{R}^3}\frac{\mathbf{k}}{\omega - \mathbf{k}\cdot\mathbf{v}}\cdot\frac{\partial f_\sigma^\square}{\partial \mathbf{v}}\,\mathrm{d}^3\mathbf{v}}_{\mathcal{D}(\omega,\mathbf{k})} = \frac{\rho_\mathrm{ext}}{\varepsilon_0\mathbf{k}^2\phi} \tag{4.132}$$

Hence, for no extraneous charge density, $\mathcal{D}(\omega,\mathbf{k}) = 0$ precisely gives the dispersion relation equivalent to the one obtained by setting the dielectric tensor determinant to zero. If the extraneous charge density is not zero, such systems are reduced to solving:

$$\phi(\omega,\mathbf{k}) = \frac{\rho_\mathrm{ext}}{\varepsilon_0\mathbf{k}^2\mathcal{D}(\omega,\mathbf{k})} \tag{4.133}$$

Denoting $\mathbf{v}^\parallel$ as the parallel velocity component to the wave vector, we obtain for the zero extraneous charge density case:

$$F_\sigma^\square = \iint_{\mathbb{R}^2} f_\sigma^\square \, \mathrm{d}^2\mathbf{v}^\perp \implies 1 + \sum_\sigma \frac{q_\sigma^2}{\varepsilon_0 m_\sigma} \frac{1}{k} \int_{\mathbb{R}} \frac{1}{\omega - kv^\parallel} \frac{\mathrm{d}F_\sigma^\square}{\mathrm{d}v^\parallel} \, \mathrm{d}v^\parallel = 0 \tag{4.134}$$

As the dispersion relation, the integral in the previous equation is commonly known as Landau's integral, which Landau observed cannot be approximated by a single Fourier mode, since it requires initial conditions regarding the distribution function under consideration.

4.5.1 Two-stream instability

Before further developing the Vlasov-Poisson model, we shall first reflect, by way of a simple illustrative example, on the physical relevance of the term representing the initial conditions of the distribution function. Consider a cold plasma consisting of two species, electrons and ions, having different macroscopic streaming velocities $\mathbf{u}_\sigma$. The linearised equation of motion, the continuity equation, and Poisson's equation are all given as follows:

$$\frac{\partial \overline{\mathbf{u}}_\sigma}{\partial t} + (\mathbf{u}_\sigma^\square \cdot \nabla)\overline{\mathbf{u}}_\sigma = -\frac{q_\sigma}{m_\sigma}\nabla\overline{\phi} \quad \text{and} \quad \frac{\partial \overline{n}_\sigma}{\partial t} + \mathbf{u}_\sigma^\square \cdot \nabla \overline{n}_\sigma = -n_\sigma^\square \nabla \cdot \overline{\mathbf{u}}_\sigma \quad \text{and} \quad \nabla^2\overline{\phi} = -\frac{1}{\varepsilon_0}\sum_\sigma q_\sigma \overline{n}_\sigma \tag{4.135}$$

Without loss of generality in respect to our illustrative endeavours, we assume that the wave is fully described by a single Fourier mode:

$$\overline{n}_\sigma = n_\sigma^\square \frac{\mathbf{k}^2}{(\omega - \mathbf{k}\cdot\mathbf{u}_\sigma^\square)} \frac{q_\sigma}{m_\sigma} \overline{\phi} \tag{4.136}$$

from which we obtain the dispersion relation:

$$1 - \sum_\sigma \frac{\omega_{\mathrm{p}\sigma}^2}{(\omega - \mathbf{k}\cdot\mathbf{u}_\sigma)^2} = 0 \tag{4.137}$$

This is a quartic equation in ω. It is hence possible, for some $\mathbf{k}$, that two or four solutions for ω are complex, each pair of complex solutions being complex conjugates, so that the wave may exhibit exponential growth in time. We will now consider with much more detail the ion-electron streaming instability, specifically in the case where electrons drift at velocity $\mathbf{u}_\mathrm{e}^\square$ through a background of stationary ions. The dispersion relation in the cold plasma limit is thus:

$$1 - \frac{\omega_{\mathrm{pi}}^2}{\omega^2} - \frac{\omega_{\mathrm{pe}}^2}{\left(\omega - \mathbf{k}\cdot\mathbf{u}_\mathrm{e}^\square\right)^2} = 0 \tag{4.138}$$

We cast this equation into a dimensionless one by imposing $\varepsilon = Zm_\mathrm{e}/m_\mathrm{i}$, $z = \omega/\omega_\mathrm{pe}$, and $\lambda = \mathbf{k}\cdot\mathbf{u}_\mathrm{e}^\square/\omega_\mathrm{pe}$:

$$1 - \frac{\varepsilon}{z^2} - \frac{1}{(z-\lambda)^2} = 0 \tag{4.139}$$

The two terms diverge at $z = 0$ and $z = \lambda$, respectively. This equation, consisting of continuous functions, therefore has a minimum between the aforementioned. If the minimum value of the two non-constant terms evaluates to below unity, then there will be two complex solutions of the dispersion relation. Since both terms decay to zero in the limits $z \to \infty$ and $z \to -\infty$, there are always two solutions for $z < 0$ and $z > \lambda$. Notice that, since all terms defining the dispersion relation are real, in the case of two complex solutions between the two singularities, those are necessarily a pair of complex conjugates, one containing a positive imaginary component, while the other is negative. The existence of a solution with a positive imaginary component implies a growing exponential term within the full solution, so that instability occurs. We shall find this condition in the considered case by defining the function:

$$f(z) = \frac{\varepsilon}{z^2} + \frac{1}{(z-\lambda)^2} \implies \frac{\partial f}{\partial z} = -\frac{2\varepsilon}{z^3} - \frac{2}{(z-\lambda)^3} \tag{4.140}$$

The derivative is equal to zero at the local minimum, so that:

$$(z-\lambda)\sqrt[3]{\varepsilon} = -z \implies z_{\min} = \frac{\lambda\sqrt[3]{\varepsilon}}{\sqrt[3]{\varepsilon}+1} \tag{4.141}$$

$$f(z_{\min}) = \frac{\sqrt[3]{\varepsilon}(\sqrt[3]{\varepsilon}+1)^2}{\lambda^2} + \frac{(\sqrt[3]{\varepsilon}+1)^2}{\lambda^2} = \frac{(\sqrt[3]{\varepsilon}+1)^3}{\lambda^2} \overset{!}{>} 1 \tag{4.142}$$

Thus, we obtain:

$$\lambda^2 < \left(1+\sqrt[3]{\varepsilon}\right)^3 \Longleftrightarrow \left(\frac{\mathbf{k}\cdot\mathbf{u}_{\mathrm{e}}^{\square}}{\omega_{\mathrm{pe}}}\right)^2 < \left(1+\sqrt[3]{\frac{Zm_{\mathrm{e}}}{m_{\mathrm{i}}}}\right)^3 \tag{4.143}$$

Streaming instabilities are a reason why many simple proposed methods for attaining thermonuclear fusion do not work. These include, e.g., directing a highly energetic deuterium beam toward a tritium beam, with the expectation that thermonuclear reactions will occur. The two-stream system exhibits an instability whose growth rate can become sufficiently large that the two mono-energetic beams dissipate before any significant nuclear reactions occur.

4.5.2 Treatment of electrostatic waves using Landau approach

Consider a stationary, spatially uniform, quasi-neutral, and unmagnetised plasma consisting of two species, electrons and ions. The possibly shifted Maxwellian distribution gives the equilibrium velocity distributions of both species. The equilibrium electrostatic field within the plasma is assumed to be zero, so that we may as well set the constant initial electrostatic potential to zero. We now believe that, at the initial time $t = 0$, the distribution function of each species exhibits a small perturbation from the equilibrium distribution, so that:

$$f_\sigma(t,\mathbf{x},\mathbf{v}) = f_\sigma^{\square}(\mathbf{v}) + \overline{f}_\sigma(t,\mathbf{x},\mathbf{v}) \tag{4.144}$$

The linearised Vlasov equation is thus:

$$\frac{\partial \overline{f}_\sigma}{\partial t} + \mathbf{v}\cdot\frac{\partial \overline{f}_\sigma}{\partial \mathbf{x}} - \frac{q_\sigma}{m_\sigma}\frac{\partial\overline{\phi}}{\partial\mathbf{x}}\cdot\frac{\partial f_\sigma^{\square}}{\partial\mathbf{v}} = 0 \tag{4.145}$$

All perturbed quantities are assumed to have spatial dependence as $\exp\left[\mathrm{i}\mathbf{k}\cdot\mathbf{x}\right]$, that is, a single Fourier mode, so that, when Fourier-transforming, we obtain:

$$\frac{\partial \overline{f}_\sigma}{\partial t} + \mathrm{i}\mathbf{k}\cdot\mathbf{v}\overline{f}_\sigma - \frac{q_\sigma\overline{\phi}}{m_\sigma}\mathrm{i}\mathbf{k}\cdot\frac{\partial f_\sigma^{\square}}{\partial\mathbf{v}} = 0 \tag{4.146}$$

Laplace-transforming in time gives:

$$(s+\mathrm{i}\mathbf{k}\cdot\mathbf{v})\overline{f}_\sigma(s,\mathbf{v}) - \overline{f}_\sigma(0,\mathbf{v}) - \frac{q_\sigma\overline{\phi}(s)}{m_\sigma}\mathrm{i}\mathbf{k}\cdot\frac{\partial f_\sigma^{\square}}{\partial\mathbf{v}} = 0 \tag{4.147}$$

Solving for the perturbation in the distribution function, we obtain:

$$\overline{f}_\sigma(s,\mathbf{v}) = \frac{1}{s+\mathrm{i}\mathbf{k}\cdot\mathbf{v}}\left(\overline{f}_\sigma(0,\mathbf{v}) + \frac{q_\sigma\overline{\phi}(s)}{m_\sigma}\mathrm{i}\mathbf{k}\cdot\frac{\partial f_\sigma^{\square}}{\partial\mathbf{v}}\right) \tag{4.148}$$

Poisson's equation gives:

$$\nabla^2\overline{\phi} = -\frac{1}{\varepsilon_0}\sum_\sigma \overline{n}_\sigma q_\sigma = -\frac{1}{\varepsilon_0}\sum_\sigma q_\sigma\iiint_{\mathbb{R}^3}\overline{f}_\sigma(t,\mathbf{x},\mathbf{v})\,\mathrm{d}^3\mathbf{v} \tag{4.149}$$

Fourier-transforming, then Laplace-transforming the previous equation gives:

$$\mathbf{k}^2\overline{\phi}(s) = \frac{1}{\varepsilon_0}\sum_\sigma q_\sigma\iiint_{\mathbb{R}^3}\overline{f}_\sigma(s,\mathbf{v})\,\mathrm{d}^3\mathbf{v} = \frac{1}{\varepsilon_0}\sum_\sigma q_\sigma\iiint_{\mathbb{R}^3}\frac{1}{s+\mathrm{i}\mathbf{k}\cdot\mathbf{v}}\left(\overline{f}_\sigma(0,\mathbf{v}) + \frac{q_\sigma\overline{\phi}(s)}{m_\sigma}\mathrm{i}\mathbf{k}\cdot\frac{\partial f_\sigma^{\square}}{\partial\mathbf{v}}\right)\mathrm{d}^3\mathbf{v} \tag{4.150}$$

Isolating $\overline{\phi}$ gives:

$$\overline{\phi}(s) = \frac{N(s)}{D(s)} \quad \text{where} \quad N(s) = \frac{1}{\varepsilon_0\mathbf{k}^2}\sum_\sigma q_\sigma\iiint_{\mathbb{R}^3}\frac{\overline{f}_\sigma(0,\mathbf{v})}{s+\mathrm{i}\mathbf{k}\cdot\mathbf{v}}\,\mathrm{d}^3\mathbf{v} \quad \text{and} \tag{4.151}$$

$$D(s) = 1 - \frac{1}{\mathbf{k}^2} \sum_\sigma \frac{q_\sigma^2}{\varepsilon_0 m_\sigma} \iiint_{\mathbb{R}^3} \mathrm{i}\mathbf{k} \cdot \frac{\partial f_\sigma^\square}{\partial \mathbf{v}} \frac{\mathrm{d}^3\mathbf{v}}{s + \mathrm{i}\mathbf{k} \cdot \mathbf{v}} \tag{4.152}$$

To obtain the solution, we apply the inverse Laplace transform:

$$\overline{\phi}(t) = \frac{1}{2\pi\mathrm{i}} \int_{\beta-\mathrm{i}\infty}^{\beta+\mathrm{i}\infty} \frac{N(s)}{D(s)} \exp(st)\, \mathrm{d}s \tag{4.153}$$

where β is chosen to be larger than the fastest growing exponential term in $\overline{\phi}(s)$, notice that, due to the complexity of $N(s)$ and $D(s)$, direct integration leading to an analytical solution is unfeasible. The real parameter β in question mathematically needs to fulfil $\beta > \mathrm{Re}\ \gamma$ where γ is the fastest growing exponential term. In the complex plane, this corresponds to the line $\beta - \mathrm{i}\infty \to \beta + \mathrm{i}\infty$ being to the right of γ, so that the physically damped waves aren't considered when Laplace-transforming $\overline{\phi}(s)$ back into the time domain. In effect, when performing the Laplace transform, we use the so-called Jordan curve, which is here a semicircle whose base is parallel to the real axis. When choosing the axis of integration to be left of some exponentially growing terms, effectively one needs to cut out $\beta + [\mathrm{Im}\ \gamma_i - \delta, \mathrm{Im}\ \gamma_i + \delta]$ and create a keyhole contour around each of the poles γ_i. Effectively, all poles of the Laplace-transformed potential perturbation are precisely either poles of the numerator $N(s)$ or zeroes of the denominator $D(s)$.

The problem of singularities associated with the wave-particle resonance was specifically the fact that Laplace-transforming the equations involved the restriction that, in the complex plane, we integrate over an axis that is to the right of all poles of the transformed perturbed potential, thus ignoring all damped waves. Let us now consider the analytic continuations of N and D so that:

$$N(s) = \frac{1}{\varepsilon_0 \mathbf{k}^2} \sum_\sigma q_\sigma \int_{\mathbb{R}} \frac{\overline{F}_\sigma(0, v^\parallel)}{s + \mathrm{i}kv^\parallel} \mathrm{d}v^\parallel \quad \text{and} \quad D(s) = 1 - \frac{1}{\mathbf{k}^2} \sum_\sigma \frac{q_\sigma^2}{\varepsilon_0 m_\sigma} \int_{\mathbb{R}} \frac{\mathrm{i}k}{s + \mathrm{i}kv^\parallel} \frac{\partial F_\sigma^\square}{\partial v^\parallel} \mathrm{d}v^\parallel \tag{4.154}$$

where F_σ is simply the distribution function integrated over the two dimensions perpendicular to $\mathbf{k}$; however, since the perturbed distribution function has to be integrable, it doesn't contribute a pole to $N(s)$. Effectively, all poles of the perturbed potential come from zeroes of the denominator function. Hence, for some zero γ_i of the denominator function, we can perform the integration with either a straight axis with $\beta > \mathrm{Re}\ \gamma$, a straight axis with a semicircle about the zero, so $\beta = \mathrm{Re}\ \gamma$, and a straight axis with a keyhole construction $\beta < \mathrm{Re}\ \gamma$. This is commonly known as Landau's rule:

$$\int_{\mathbb{R}} \frac{g(v^\parallel)}{s + \mathrm{i}kv^\parallel} \mathrm{d}v^\parallel = \begin{cases} \int_{-\infty}^{\infty} \frac{g(v^\parallel)}{s + \mathrm{i}kv^\parallel} \mathrm{d}v^\parallel, & \mathrm{Re}\ s > 0, \\ \mathrm{P.V.} \int_{-\infty}^{\infty} \frac{g(v^\parallel)}{s + \mathrm{i}kv^\parallel} \mathrm{d}v^\parallel + \pi \mathrm{i} g\left(\frac{\mathrm{i}s}{k}\right), & \mathrm{Re}\ s = 0, \\ \int_{-\infty}^{\infty} \frac{g(v^\parallel)}{s + \mathrm{i}kv^\parallel} \mathrm{d}v^\parallel + 2\pi \mathrm{i} g\left(\frac{is}{k}\right), & \mathrm{Re}\ s < 0. \end{cases} \tag{4.155}$$

where $g(v^\parallel)$ is either $\overline{F}_\sigma(0, v^\parallel)$ or $\mathrm{i}k\partial_{v^\parallel} F_\sigma^\square$, and where the principal value is formally defined by the following limit about the pole of the function in question, so the sum of limits to the left and right. The final perturbed potential thus depends on the positions of the poles in the complex plane. If the pole γ_i is on the right of the imaginary axis, so $\mathrm{Re}\ \gamma_i > 0$, the wave is unstable and exponentially grows. On the contrary, poles for which $\mathrm{Re}\ \gamma_i < 0$ are stable and exponentially decay.

We shall now proceed to find the dispersion relation:

$$D(s) = 1 - \frac{1}{\mathbf{k}^2} \sum_\sigma \frac{q_\sigma^2}{\varepsilon_0 m_\sigma} \int_{\mathbb{R}} \frac{\mathrm{i}k}{s + \mathrm{i}kv^\parallel} \frac{\partial F_\sigma^\square}{\partial v^\parallel} \mathrm{d}v^\parallel \tag{4.156}$$

For a Maxwellian distribution, we have:

$$F_\sigma^\square(v^\parallel) = n_\sigma \sqrt{\frac{m_\sigma}{2\pi k_\mathrm{B} T_\sigma}} \exp\left[-\frac{m_\sigma (v^\parallel)^2}{2k_\mathrm{B} T_\sigma}\right] \implies \frac{\partial F_\sigma^\square}{\partial v^\parallel} = -\frac{n_\sigma v^\parallel}{\sqrt{2\pi}} \left(\frac{m_\sigma}{k_\mathrm{B} T_\sigma}\right)^{3/2} \exp\left[-\frac{m_\sigma (v^\parallel)^2}{2k_\mathrm{B} T_\sigma}\right] \tag{4.157}$$

We may introduce $\xi = v^\parallel \sqrt{\frac{m_\sigma}{2k_\mathrm{B} T_\sigma}}$ to obtain rather legible equations. Notice that each term in the sum corresponds to the electric susceptibility of species σ:

$$D(s) = 1 - \frac{n_\sigma^\square q_\sigma^2}{\varepsilon_0 \mathbf{k}^2 \sqrt{\pi} k_\mathrm{B} T_\sigma} \int_{-\infty}^{\infty} \frac{\xi \exp\left[-\xi^2\right]}{\sqrt{2k_\mathrm{B} T_\sigma / m_\sigma} s/\mathrm{i}k + \xi} \mathrm{d}\xi \tag{4.158}$$

$$\chi_\sigma = \frac{1}{\mathbf{k}^2\lambda_{D\sigma}^2\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\xi - \sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}}\frac{\mathrm{i}s}{k} + \sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}}\frac{\mathrm{i}s}{k}}{\xi - \sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}}\frac{\mathrm{i}s}{k}}\exp\left[-\xi^2\right]\mathrm{d}\xi = \tag{4.159}$$

$$= \frac{1}{\mathbf{k}^2\lambda_{D\sigma}^2}\left[1 + \frac{\alpha_\sigma}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp\left[-\xi^2\right]}{\xi - \alpha_\sigma}\mathrm{d}\xi\right] = \frac{1}{\mathbf{k}^2\lambda_D^2}\left[1 + \frac{\alpha_\sigma}{\sqrt{\pi}}Z(\alpha_\sigma)\right] \quad \text{where} \quad \alpha_\sigma = \frac{\mathrm{i}s}{k}\sqrt{\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}} \tag{4.160}$$

The function $Z(\alpha_\sigma)$ is commonly called the plasma dispersion function. We expect a clear correspondence with the two-fluid model in the adiabatic limit, where $|\alpha_\sigma| \gg 1$, as well as in the isothermal limit, where $|\alpha_\sigma| \ll 1$. In both limits, the two-fluid model experienced no damping, so that we may use the limit where Re $s \to 0$, which corresponds to Landau's rule where Re $s = 0$:

$$Z(\alpha_\sigma) = \frac{1}{\sqrt{\pi}}\mathrm{P.V.}\int_{-\infty}^{\infty}\frac{\exp\left[-\xi^2\right]}{\xi - \alpha_\sigma}\mathrm{d}\xi + \mathrm{i}\sqrt{\pi}\exp\left[-\alpha_\sigma^2\right] \tag{4.161}$$

We may execute the Taylor series of $1/(\xi - \alpha_\sigma)$ in the limit $|\alpha_\sigma| \gg 1$ to obtain:

$$\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp\left[-\xi^2\right]}{\xi - \alpha_\sigma}\mathrm{d}\xi = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\left[\sum_{n=0}^{\infty}\left(\frac{\xi}{\alpha_\sigma}\right)^n\right]\exp\left[-\xi^2\right]\mathrm{d}\xi \simeq -\frac{1}{\alpha_\sigma}\left(1 + \frac{1}{2\alpha_\sigma^2} + \frac{3}{4\alpha_\sigma^4} + \cdots\right) \tag{4.162}$$

In the other limit, $|\alpha_\sigma| \ll 1$, we introduce a new variable $\eta = \xi - \alpha_\sigma$:

$$\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp\left[-\xi^2\right]}{\xi - \alpha_\sigma}\mathrm{d}\xi = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp\left[-\eta^2 - 2\alpha_\sigma\eta - \alpha_\sigma^2\right]}{\eta}\mathrm{d}\eta = \tag{4.163}$$

$$= \frac{\exp\left[-\alpha_\sigma^2\right]}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp\left[-\eta^2\right]}{\eta}\left[\sum_{n=0}^{\infty}\frac{(-2\alpha_\sigma\eta)^n}{n!}\right]\mathrm{d}\eta \simeq -2\alpha_\sigma\left(1 - \frac{2\alpha_\sigma^2}{3} + \cdots\right) \tag{4.164}$$

We may now proceed by evaluating the plasma susceptibilities. For $|\alpha_\sigma| \gg 1$, we introduce the *frequency* $\omega = \mathrm{i}s$ so that $\alpha_\sigma = \sqrt{\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}}\omega/k$:

$$\chi_\sigma = \frac{1}{\mathbf{k}^2\lambda_D^2}\left[1 + \alpha_\sigma\left(-\frac{1}{\alpha_\sigma}\left[1 + \frac{1}{2\alpha_\sigma^2} + \frac{3}{4\alpha_\sigma^4} + \cdots\right] + \mathrm{i}\sqrt{\pi}\exp\left[-\alpha_\sigma^2\right]\right)\right] = \tag{4.165}$$

$$= \frac{1}{\mathbf{k}^2\lambda_D^2}\left[-\left(\frac{1}{2\alpha_\sigma^2} + \frac{3}{4\alpha_\sigma^4} + \cdots\right) + \mathrm{i}\alpha_\sigma\sqrt{\pi}\exp\left[-\alpha_\sigma^2\right]\right] = \tag{4.166}$$

$$= -\frac{\omega_{\mathrm{p}\sigma}^2}{\omega^2}\left(1 + \frac{3\mathbf{k}^2}{\omega^2}\frac{k_\mathrm{B}T_\sigma}{m_\sigma} + \cdots\right) + \frac{\mathrm{i}\omega}{\|\mathbf{k}\|^3\lambda_{\mathrm{d}\sigma}^2}\sqrt{\frac{\pi m_\sigma}{2k_\mathrm{B}T_\sigma}}\exp\left[-\frac{m_\sigma\omega^2}{2k_\mathrm{B}T_\sigma\mathbf{k}^2}\right] \tag{4.167}$$

Notice that due to $\omega_{\mathrm{pe}}^2 \gg \omega_{\mathrm{pi}}^2$, we may drop the ion contributions. If there exists a solution such that $|\alpha_\sigma| \gg 1$, we may only keep the first non-constant term in the infinite sum:

$$\mathcal{D}(\omega, \mathbf{k}) = 1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2}\left(1 + \frac{3\mathbf{k}^2}{\omega^2}\frac{k_\mathrm{B}T_\mathrm{e}}{m_\mathrm{e}} + \cdots\right) + \frac{\mathrm{i}\omega}{\|\mathbf{k}\|^3\lambda_{\mathrm{De}}^2}\sqrt{\frac{\pi m_\mathrm{e}}{2k_\mathrm{B}T_\mathrm{e}}}\exp\left[-\frac{m_\mathrm{e}\omega^2}{2k_\mathrm{B}T_\mathrm{e}\mathbf{k}^2}\right] = 0 \tag{4.168}$$

Although we can solve the equation by plugging $\omega = \mathrm{Re}\ \omega + \mathrm{i}\,\mathrm{Im}\ \omega$, it is better to divide the dispersion relation into its real and imaginary parts as follows:

$$\mathcal{D}(\omega, \mathbf{k}) = \mathrm{Re}\ \mathcal{D}(\omega_\mathrm{Re} + i\omega_\mathrm{Im}) + \mathrm{i}\,\mathrm{Im}\ \mathcal{D}(\omega_\mathrm{Re} + \mathrm{i}\omega_\mathrm{Im}) = 0 \tag{4.169}$$

$$\mathcal{D}_\mathrm{Re}(\omega, \mathbf{k}) = 1 - \frac{\omega_{\mathrm{pe}}^2}{\omega^2}\left(1 + \frac{3\mathbf{k}^2}{\omega^2}\frac{k_\mathrm{B}T_\mathrm{e}}{m_\mathrm{e}} + \cdots\right) \quad \text{and} \quad \mathcal{D}_\mathrm{Im}(\omega, \mathbf{k}) = \frac{\omega}{\|\mathbf{k}\|^3\lambda_{\mathrm{De}}^2}\sqrt{\frac{\pi m_\mathrm{e}}{2k_\mathrm{B}T_\mathrm{e}}}\exp\left[-\frac{m_\mathrm{e}\omega^2}{2k_\mathrm{B}T_\mathrm{e}\mathbf{k}^2}\right] = 0 \tag{4.170}$$

Assuming a weakly damped oscillation where $\omega_\mathrm{Im} \ll \omega_\mathrm{Re}$, we may Taylor-expand the equation:

$$\mathcal{D}_\mathrm{Re}(\omega_\mathrm{Re}, \mathbf{k}) + \mathrm{i}\omega\left(\frac{\partial\mathcal{D}_\mathrm{Re}}{\partial\omega}\right)\Big|_{\omega_\mathrm{Re}} + \mathrm{i}\left[\mathcal{D}_\mathrm{Im}(\omega_\mathrm{Re}, \mathbf{k}) + \mathrm{i}\omega_\mathrm{Im}\left(\frac{\partial\mathcal{D}_\mathrm{Re}}{\partial\omega}\right)\Big|_{\omega_\mathrm{Re}}\right] = 0 \tag{4.171}$$

Since $\omega_{\mathrm{Re}} \gg \omega_{\mathrm{Im}}$, the real part is $\mathcal{D}_{\mathrm{Re}}(\omega_{\mathrm{Re}}, \mathbf{k}) \simeq 0$ The two imaginary terms in the expanded equation give:

$$\omega_{\mathrm{Im}} = -\mathcal{D}_{\mathrm{Im}}(\omega_{\mathrm{Re}}) \Big/ \left.\left(\frac{\partial \mathcal{D}_{\mathrm{Re}}}{\partial \omega}\right)\right|_{\omega_{\mathrm{Re}}} \tag{4.172}$$

The equation $\mathcal{D}_{\mathrm{Re}}(\omega_{\mathrm{Re}}, \mathbf{k}) = 0$ gives:

$$\omega_{\mathrm{Re}}^2 = \omega_{\mathrm{pe}}^2 \left(1 + \frac{3\mathbf{k}^2}{\omega_{\mathrm{Re}}^2} \frac{k_{\mathrm{B}} T_{\mathrm{e}}}{m_{\mathrm{e}}}\right) \simeq \omega_{\mathrm{pe}}^2 \left(1 + 3\mathbf{k}^2 \lambda_{D\mathrm{e}}^2\right) \tag{4.173}$$

so that the imaginary component of the oscillation frequency becomes:

$$\omega_{\mathrm{Im}} = -\sqrt{\frac{\pi}{8}} \frac{\omega_{\mathrm{pe}}^2}{\mathbf{k}^3 \lambda_{D\mathrm{e}}^3} \exp\left[-\frac{1 + 3\mathbf{k}^2 \lambda_{D\mathrm{e}}^2}{2\mathbf{k}^2 \lambda_{D\mathrm{e}}^2}\right] \tag{4.174}$$

The negative value ensures that the oscillation is damped. Even though we had assumed $\omega_{\mathrm{Im}} \ll \omega_{\mathrm{Re}}$, the damping caused by ω_{Im} is significant even after a single period.

4.5.3 Stability of distribution functions with a single global maximum

As it has been demonstrated in subsection 4.5.1, a system whose distribution function exhibits two local maxima is susceptible to presenting an instability. It is thus only reasonable to expect systems whose distribution function presents a single global maximum to always exhibit stable oscillations. This is indeed the case. The first stability criterion provides a necessary condition for instability via the so-called Norton-Gardner Theorem. We start with the dispersion relation:

$$\mathcal{D}(is, \mathbf{k}) = 1 - \frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}}{\mathrm{d}v^{\parallel}} \left[F_{\mathrm{e}}^{\square} + \sum_{\sigma \neq \mathrm{e}} \frac{Z_\sigma^2 m_{\mathrm{e}}}{m_\sigma} F_\sigma^{\square}\right] \frac{\mathrm{d}v^{\parallel}}{v^{\parallel} - \mathrm{i}s/k} \tag{4.175}$$

We may introduce the total distribution function as the weighted sum of distribution functions given in the square brackets, motivated by quasi-neutrality, so that:

$$\mathcal{D}(is, \mathbf{k}) = 1 - \frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}F^{\square}}{\mathrm{d}v^{\parallel}} \frac{\mathrm{d}v^{\parallel}}{v^{\parallel} - \mathrm{i}s/k} \tag{4.176}$$

where $\mathcal{L}$ denotes integration with respect to the according Landau's rule, and we assume that the total distribution function has a single global maximum. We proceed by introducing $\mathrm{i}s = \mathrm{i}(-\mathrm{i}\omega) = \omega_{\mathrm{Re}} - \mathrm{i}\gamma$ where ω_{Re} is the oscillatory frequency and γ the damping coefficient. We proceed by contradicting that the dispersion relation is satisfied in this case, thus effectively evaluating on the real axis:

$$\mathcal{D}(\omega_{\mathrm{Re}}, \gamma, \mathbf{k}) = 1 - \frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}F^{\square}}{\mathrm{d}v^{\parallel}} \frac{v^{\parallel} - \frac{\omega_{\mathrm{Re}}}{k} + \mathrm{i}\frac{\gamma}{k}}{\left(v^{\parallel} - \frac{\omega_{\mathrm{Re}}}{k}\right)^2 + \left(\frac{\gamma}{k}\right)^2} \mathrm{d}v^{\parallel} \tag{4.177}$$

We may expand the integral into its real and imaginary components and impose that they satisfy the dispersion relation:

$$1 - \frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}F^{\square}}{\mathrm{d}v^{\parallel}} \frac{v^{\parallel} - \omega_{\mathrm{Re}}/k}{\left(v^{\parallel} - \omega_{\mathrm{Re}}/k\right)^2 + (\gamma/k)^2} \mathrm{d}v^{\parallel} = 0 \tag{4.178}$$

$$\frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}F^{\square}}{\mathrm{d}v^{\parallel}} \frac{\gamma/k}{\left(v^{\parallel} - \omega_{\mathrm{Re}}/k\right)^2 + (\gamma/k)^2} \mathrm{d}v^{\parallel} = 0 \tag{4.179}$$

The total distribution function having a single maximum at some value $v_{\mathrm{max}}^{\parallel}$, we impose $\overline{v}^{\parallel} = v^{\parallel} - v_{\mathrm{max}}^{\parallel}$:

$$1 - \frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square} \mathbf{k}^2} \int_{\mathcal{L}} \frac{\mathrm{d}F^{\square}}{\mathrm{d}\overline{v}^{\parallel}} \left[\frac{v_{\mathrm{max}}^{\parallel} - \omega_{\mathrm{Re}}/k}{\left(v_{\mathrm{max}}^{\parallel} - \omega_{\mathrm{Re}}/k + \overline{v}^{\parallel}\right)^2 + (\gamma/k)^2} + \frac{\overline{v}^{\parallel}}{\left(v_{\mathrm{max}}^{\parallel} - \omega_{\mathrm{Re}}/k + \overline{v}^{\parallel}\right)^2 + (\gamma/k)^2}\right] \mathrm{d}v^{\parallel} = 0 \tag{4.180}$$

$$\frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square}\mathbf{k}^2}\int_{\mathcal{L}}\frac{\mathrm{d}F^{\square}}{\mathrm{d}\overline{v}^{\parallel}}\frac{\gamma/k}{\left(v^{\parallel}-\omega_{\mathrm{Re}}/k\right)^2+\left(\gamma/k\right)^2}\,\mathrm{d}v^{\parallel}=0 \tag{4.181}$$

The first term in the brackets of the first equation, being equivalent to the second equation, evaluates to zero. We hence obtain:

$$1-\frac{\omega_{\mathrm{pe}}^2}{n_{\mathrm{e}}^{\square}\mathbf{k}^2}\int_{\mathcal{L}}\frac{\mathrm{d}F^{\square}}{\mathrm{d}\overline{v}^{\parallel}}\frac{\overline{v}^{\parallel}}{\left(v^{\parallel}-\omega_{\mathrm{Re}}/k\right)^2+\left(\gamma/k\right)^2}\,\mathrm{d}v^{\parallel}=0 \tag{4.182}$$

Notice that the total distribution function in question, having a single global maximum, satisfies:

$$\forall\overline{v}^{\parallel}\in\mathbb{R},\quad \overline{v}^{\parallel}\frac{\mathrm{d}F^{\square}}{\mathrm{d}\overline{v}^{\parallel}}\leq 0 \tag{4.183}$$

Thus, the integral necessarily evaluates to a strictly negative term for any non-pathological distribution function. The dispersion relation is therefore not satisfied, which precisely concludes the contradiction. Therefore, any total distribution function attaining a single maximum exhibits stable oscillations.

4.5.4 Landau damping of an ion-acoustic wave

In this subsection, we shall illustrate the procedure for studying Landau damping of an ion-acoustic wave. We first begin by setting up the integrals in the dispersion relation given by equation (4.156), and noting that the derivative of the unperturbed distribution functions is:

$$\frac{\mathrm{d}F_{\alpha}^{\square}}{\mathrm{d}v^{\parallel}}=n_{\alpha}^{\square}\sqrt{\frac{m_{\alpha}}{2\pi k_{\mathrm{B}}T_{\alpha}}}\frac{\mathrm{d}}{\mathrm{d}v^{\parallel}}\left(\exp\left[-\frac{m_{\alpha}(v^{\parallel})^2}{2k_{\mathrm{B}}T_{\alpha}}\right]\right)=-\frac{n_{\alpha}^{\square}m_{\alpha}v^{\parallel}}{k_{\mathrm{B}}T_{\alpha}}\sqrt{\frac{m_{\alpha}}{2\pi k_{\mathrm{B}}T_{\alpha}}}\exp\left[-\frac{m_{\alpha}(v^{\parallel})^2}{2k_{\mathrm{B}}T_{\alpha}}\right] \tag{4.184}$$

For ions, we introduce the substitution $\zeta_{\mathrm{i}}=v^{\parallel}/v_{\mathrm{thi}}$ and $\beta_{\mathrm{i}}=\omega/kv_{\mathrm{thi}}\gg 1$. We decompose $\omega=\omega_{\mathrm{Re}}+\mathrm{i}\omega_{\mathrm{Im}}$. We know that in the given limits, the two-fluid model predicts no damping, so that we may assume $\omega_{\mathrm{Im}}\ll\omega_{\mathrm{Re}}$, that is $\beta_{\mathrm{i}}=\beta_{\mathrm{i}}^{\mathrm{Re}}+\beta_{\mathrm{i}}^{\mathrm{Im}}$ with $\beta_{\mathrm{i}}^{\mathrm{Re}}\gg\beta_{\mathrm{i}}^{\mathrm{Im}}$:

$$\omega_{\mathrm{Im}}(\omega_{\mathrm{Re}},k)\simeq-\mathcal{D}_{\mathrm{Im}}(\omega_{\mathrm{Re}},k)\Bigg/\frac{\partial\mathcal{D}_{\mathrm{Re}}}{\partial\omega_{\mathrm{Re}}} \tag{4.185}$$

The real part of the ion integral in the dispersion relation evaluates to:

$$-\frac{n_{\mathrm{i}}^{\square}m_{\mathrm{i}}}{k_{\mathrm{B}}T_{\mathrm{i}}k\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\zeta_{\mathrm{i}}\exp\left(-\zeta_{\mathrm{i}}^2\right)}{\beta_{\mathrm{i}}-\zeta_{\mathrm{i}}}\,\mathrm{d}\zeta_{\mathrm{i}}\simeq-\frac{n_{\mathrm{i}}^{\square}m_{\mathrm{i}}}{k_{\mathrm{B}}T_{\mathrm{i}}k\sqrt{\pi}}\int_{-\infty}^{\infty}\left(\frac{\zeta_{\mathrm{i}}}{\beta_{\mathrm{i}}}+\frac{\zeta_{\mathrm{i}}^2}{\beta_{\mathrm{i}}^2}+\cdots\right)\exp\left(-\zeta_{\mathrm{i}}^2\right)\mathrm{d}\zeta_{\mathrm{i}}\simeq \tag{4.186}$$

$$\simeq-\frac{n_{\mathrm{i}}^{\square}m_{\mathrm{i}}}{k_{\mathrm{B}}T_{\mathrm{i}}\sqrt{\pi}}\,\mathrm{P.V.}\int_{-\infty}^{\infty}\frac{\zeta_{\mathrm{i}}(\zeta_{\mathrm{i}}-\beta_{\mathrm{i}}^{\mathrm{Re}})}{(\zeta_{\mathrm{i}}-\beta_{\mathrm{i}}^{\mathrm{Re}})^2+(\beta_{\mathrm{i}}^{\mathrm{Im}})^2}\,\mathrm{d}\zeta_{\mathrm{i}}\simeq\frac{n_{\mathrm{i}}^{\square}k^2}{\omega_{\mathrm{Re}}^2} \tag{4.187}$$

For electrons, we introduce $\zeta_{\mathrm{e}}=u/v_{\mathrm{the}}$ and $\beta_{\mathrm{e}}=\omega/kv_{\mathrm{the}}\ll 1$, so that:

$$-\frac{n_{\mathrm{e}}^{\square}m_{\mathrm{e}}}{k_{\mathrm{B}}T_{\mathrm{e}}\sqrt{\pi}}\mathrm{P.V.}\int_{-\infty}^{\infty}\frac{\zeta_{\mathrm{e}}(\zeta_{\mathrm{e}}-\beta_{\mathrm{e}}^{\mathrm{Re}})}{(\zeta_{\mathrm{e}}-\beta_{\mathrm{e}}^{\mathrm{Re}})^2+(\beta_{\mathrm{e}}^{\mathrm{Im}})^2}\,\mathrm{d}\zeta_{\mathrm{e}}\simeq-\frac{n_{\mathrm{e}}^{\square}m_{\mathrm{e}}}{k_{\mathrm{B}}T_{\mathrm{e}}} \tag{4.188}$$

The real part of the dispersion relation is thus:

$$\mathcal{D}_{\mathrm{Re}}(\omega_{\mathrm{Re}},k)=1-\frac{\omega_{\mathrm{pi}}^2}{\omega_{\mathrm{Re}}^2}+\frac{1}{k^2\lambda_{De}^2}\implies\frac{\partial\mathcal{D}_{\mathrm{Re}}}{\partial\omega_{\mathrm{Re}}}=\frac{2\omega_{\mathrm{pi}}^2}{\omega_{\mathrm{Re}}^3} \tag{4.189}$$

The imaginary part of the dispersion relation is, however:

$$\mathcal{D}_{\mathrm{Im}}(\omega_{\mathrm{Re}},k)=\frac{n_{\mathrm{i}}^{\square}q_{\mathrm{i}}^2\omega_{\mathrm{Re}}}{\varepsilon_0k_{\mathrm{B}}T_{\mathrm{i}}k^3}\sqrt{\frac{\pi m_{\mathrm{i}}}{2k_{\mathrm{B}}T_{\mathrm{i}}}}\exp\left[-\frac{m_{\mathrm{i}}\omega_{\mathrm{Re}}^2}{2k_{\mathrm{B}}T_{\mathrm{i}}k^2}\right]+\frac{n_{\mathrm{e}}^{\square}e^2\omega_{\mathrm{Re}}}{\varepsilon_0k_{\mathrm{B}}T_{\mathrm{e}}k^3}\sqrt{\frac{\pi m_{\mathrm{e}}}{2k_{\mathrm{B}}T_{\mathrm{e}}}}\exp\left[-\frac{m_{\mathrm{e}}\omega_{\mathrm{Re}}^2}{2k_{\mathrm{B}}T_{\mathrm{e}}k^2}\right] \tag{4.190}$$

We can now evaluate the damping coefficient:

$$\omega_{\mathrm{Im}}(\omega_{\mathrm{Re}},k)\simeq-\frac{\omega_{\mathrm{Re}}^4}{\omega_{\mathrm{pi}}^2k^3\lambda_{Di}^2}\sqrt{\frac{\pi m_{\mathrm{i}}}{8k_{\mathrm{B}}T_{\mathrm{i}}}}\exp\left[-\frac{m_{\mathrm{i}}\omega_{\mathrm{Re}}^2}{2k_{\mathrm{B}}T_{\mathrm{i}}k^2}\right]+\frac{\omega_{\mathrm{Re}}^4}{\omega_{\mathrm{pi}}^2k^3\lambda_{De}^2}\sqrt{\frac{\pi m_{\mathrm{e}}}{8k_{\mathrm{B}}T_{\mathrm{e}}}}\exp\left[-\frac{m_{\mathrm{e}}\omega_{\mathrm{Re}}^2}{2k_{\mathrm{B}}T_{\mathrm{e}}k^2}\right] \tag{4.191}$$

We know that $\omega_{\text{Re}}^2 = k^2 c_{\text{i}}^2$ where $c_{\text{i}}^2 = k_{\text{B}} T_{\text{e}}/m_{\text{i}}$, $n_{\text{i}}^{\square} q_{\text{i}} \simeq n_{\text{e}}^{\square} e$, $q_{\text{i}} = Ze$, and $\lambda_{D\text{e}}^2 \omega_{\text{pi}}^2 = c_{\text{i}}^2$:

$$\omega_{\text{Im}} \simeq -\sqrt{\frac{\pi}{8}} \|\mathbf{k}\| c_{\text{i}} \left(\frac{T_{\text{e}}}{T_{\text{i}}}\right)^{3/2} \exp\left[-\frac{T_{\text{e}}}{2T_{\text{i}}}\right] - \sqrt{\frac{\pi}{8}} \frac{\|\mathbf{k}\| c_{\text{i}}}{Z} \sqrt{\frac{m_{\text{e}}}{m_{\text{i}}}} \exp\left[-\frac{m_{\text{e}}}{2m_{\text{i}}}\right] \tag{4.192}$$

where we can drop the exponential factor $\exp\left[-m_{\text{e}}/2m_{\text{i}}\right] \simeq 1$.

4.6 Kinetic description of waves in magnetised plasmas

Behind all considerations above, it has been tacitly assumed that the wave phase experienced by a particle is the phase the particle would have experienced if it had not deviated from its original position. This approximation is valid when $|\mathbf{k} \cdot (\mathbf{x}(t) - \mathbf{x}_0)| \ll \pi/2$. Two situations in which this condition is satisfied are large-amplitude waves that cause particle displacement comparable to the wavelength, and small-amplitude waves propagating over particles with sufficiently high initial velocities, thereby moving substantially during a wave period. We will now discuss the latter, that is, waves in warm magnetised plasmas. We consider an electrostatic potential wave $\overline{\phi}(\mathbf{x}, t) = \overline{\phi} \exp\left[\mathrm{i}\mathbf{k} \cdot \mathbf{x} - \mathrm{i}\omega t\right]$:

$$\mathbf{k}^2 \overline{\phi} = \frac{1}{\varepsilon_0} \sum_\sigma \overline{n}_\sigma q_\sigma \quad \text{where} \quad \overline{n}_\sigma = \iiint_{\mathbb{R}^3} \overline{f}_\sigma(\mathbf{v}) \, \mathrm{d}^3\mathbf{v} \tag{4.193}$$

In the presence of a uniform magnetic field, the linearised Vlasov equation reads:

$$\frac{\partial \overline{f}_\sigma}{\partial t} + \mathbf{v} \cdot \frac{\partial \overline{f}_\sigma}{\partial \mathbf{x}} + \frac{q_\sigma}{m_\sigma} (\mathbf{v} \times \mathbf{B}) \cdot \frac{\partial \overline{f}_\sigma}{\partial \mathbf{v}} = \frac{q_\sigma}{m_\sigma} \nabla \overline{\phi} \cdot \frac{\partial f_\sigma^{\square}}{\partial \mathbf{v}} \tag{4.194}$$

Along an unperturbed trajectory, by Liouville's theorem, we find:

$$\overline{f}_\sigma(\mathbf{x}, \mathbf{v}, t) = \frac{q_\sigma}{m_\sigma} \int_{-\infty}^{t} \left[\nabla \overline{\phi} \cdot \frac{\partial f_\sigma^{\square}}{\partial \mathbf{v}}\right]_{\mathbf{x}(t'), \mathbf{v}(t')} \mathrm{d}t' \tag{4.195}$$

Let us now consider the typical Maxwellian distribution, so that:

$$\overline{f}_\sigma(\mathbf{x}, \mathbf{v}, t) = -\frac{2n_\sigma^{\square} q_\sigma \overline{\phi}}{m_\sigma} \left(\frac{m_\sigma}{2\pi k_{\text{B}} T_\sigma}\right)^{3/2} \exp\left[-\frac{m_\sigma \mathbf{v}^2}{2k_{\text{B}} T_\sigma}\right] \int_{-\infty}^{t} \left[\frac{\mathrm{i}\mathbf{k} \cdot m_\sigma \mathbf{v}}{2k_{\text{B}} T_\sigma} \exp\left[\mathrm{i}\mathbf{k} \cdot \mathbf{x}(t') - \mathrm{i}\omega t'\right]\right] \mathrm{d}t' \tag{4.196}$$

The exponential velocity term has been factored out since, along an unperturbed orbit, the kinetic energy is a constant of motion. Note that $(\mathrm{i}\mathbf{k} \cdot \mathbf{x})' = (\mathrm{i}\mathbf{k} \cdot \mathbf{v})$:

$$\overline{f}_\sigma(\mathbf{x}, \mathbf{v}, t) = -\frac{q_\sigma \overline{\phi}}{k_{\text{B}} T_\sigma} f_\sigma^{\square} \int_{-\infty}^{t} \left(\exp\left[-\mathrm{i}\omega t'\right] \frac{\mathrm{d}}{\mathrm{d}t'} \left[\exp\left[\mathrm{i}\mathbf{k} \cdot \mathbf{x}(t')\right]\right]\right) \mathrm{d}t' = \tag{4.197}$$

$$= -\frac{q_\sigma \overline{\phi}}{k_{\text{B}} T_\sigma} f_\sigma^{\square} \left(\exp\left[\mathrm{i}\mathbf{k} \cdot \mathbf{x}(t') - \mathrm{i}\omega t'\right] \Big|_{-\infty}^{t} + \mathrm{i}\omega I_{\text{phase}}(\mathbf{x}, t)\right) \tag{4.198}$$

where we have defined the so-called phase history integral as follows:

$$I_{\text{phase}}(\mathbf{x}, t) = \int_{-\infty}^{t} \exp\left[\mathrm{i}\mathbf{k} \cdot \mathbf{x}(t') - \mathrm{i}\omega t'\right] \mathrm{d}t' \tag{4.199}$$

The evaluation of this integral requires knowledge of the unperturbed trajectory $\mathbf{x}$. In this case, we may first decompose the particle's velocity as follows:

$$\mathbf{v}(t') = v^{\parallel} \,\hat{\mathbf{B}} + \mathbf{v}^{\perp} \cos\left[\Omega_\sigma(t' - t)\right] - \hat{\mathbf{B}} \times \mathbf{v}^{\perp} \sin\left[\Omega_\sigma(t' - t)\right] \tag{4.200}$$

which satisfies a particular particle's dynamics and the boundary condition $\mathbf{v}(t) = \mathbf{v}$. Integrating gives:

$$\mathbf{x}(t') = \mathbf{x} + v^{\parallel}(t' - t) \,\hat{\mathbf{B}} + \frac{1}{\Omega_\sigma} \left[\mathbf{v}^{\perp} \sin\left[\Omega_\sigma(t' - t)\right] + \hat{\mathbf{B}} \times \mathbf{v}^{\perp} \left(\cos\left[\Omega_\sigma(t' - t)\right] - 1\right)\right] \tag{4.201}$$

which satisfies $\mathbf{x}(t) = \mathbf{x}$. Let us introduce the angle variable $\varphi = \angle(\mathbf{k}_\perp, \mathbf{v}^\perp)$, so that:

$$\mathbf{k}\cdot\mathbf{x}(t') = \mathbf{k}\cdot\mathbf{x} + k_\parallel v^\parallel (t'-t) + \frac{k_\perp v^\perp}{\Omega_\sigma}\left(\sin\left[\Omega_\sigma(t'-t)+\varphi\right] - \sin\varphi\right) \tag{4.202}$$

We perform the translation $t = t' - \tau$, so that:

$$I_{\text{phase}}(\mathbf{x},t) = \exp\left[\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t\right]\int_{-\infty}^{0}\exp\left[\mathrm{i}(k_\parallel v^\parallel - \omega)\tau + \frac{\mathrm{i}k_\perp v^\perp}{\Omega_\sigma}\left[\sin\left[\Omega_\sigma\tau + \varphi\right] - \sin\varphi\right]\right]\mathrm{d}\tau \tag{4.203}$$

Now, we employ the Jacobi-Anger expansion:

$$\exp\left[\mathrm{i}z\sin\theta\right] = \sum_{n=-\infty}^{\infty} J_n(z)\exp\left(\mathrm{i}n\theta\right) \tag{4.204}$$

The phase history integral thus becomes:

$$I_{\text{phase}}(\mathbf{x},t) = e^{\left[\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t - \frac{\mathrm{i}k_\perp v^\perp \sin\varphi}{\Omega_\sigma}\right]}\sum_{n=-\infty}^{\infty} J_n\left(\frac{k_\perp v^\perp}{\Omega_\sigma}\right)\frac{\exp\left[\mathrm{i}(k_\parallel v^\parallel - \omega)\tau + \mathrm{i}n(\Omega_\sigma\tau + \varphi))\right]\Big|_{-\infty}^{0}}{-\mathrm{i}(\omega - k_\parallel v^\parallel - n\Omega_\sigma)} \tag{4.205}$$

The lower limit corresponds to initial values in the distant past, and will hereby be dropped:

$$\overline{f}_\sigma(\mathbf{x},\mathbf{v},t) = -\frac{q_\sigma\overline{\phi}}{k_\mathrm{B}T_\sigma}f_\sigma^\square\exp\left[\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t\right]\left(1 - \exp\left[-\frac{k_\perp v^\perp}{\Omega_\sigma}\right]\sum_{n=-\infty}^{\infty}\frac{J_n\left(k_\perp v^\perp/\Omega_\sigma\right)\omega\exp\left[\mathrm{i}n\varphi\right]}{\omega - k_\parallel v^\parallel - n\Omega_\sigma}\right) \tag{4.206}$$

It is now useful to exploit the following integral identity:

$$\int_0^\infty zJ_n^2(\beta z)\exp\left[-\alpha^2 z^2\right]\mathrm{d}z = \frac{\exp\left[-\beta^2/2\alpha^2\right]}{2\alpha^2}I_n\left(\frac{\beta^2}{2\alpha^2}\right) \tag{4.207}$$

$$\overline{n}_\sigma = -\frac{n_\sigma^\square q_\sigma\overline{\phi}}{k_\mathrm{B}T_\sigma}\left[1 + \alpha_\sigma^{(0)}\exp\left(-k_\perp^2 r_\sigma^2\right)\sum_{n=-\infty}^{\infty} I_n\left(k_\perp^2 r_\sigma^2\right)\int_{-\infty}^{\infty}\frac{\exp\left[-\xi^2\right]}{\xi - \alpha_\sigma^{(n)}}\mathrm{d}\xi\right] \tag{4.208}$$

where $r_\sigma = \sqrt{k_\mathrm{B}T_\sigma/m_\sigma\Omega_\sigma^2}$ and $\alpha_\sigma^{(n)} = (\omega - n\Omega_\sigma)\sqrt{\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}}/k_\parallel$. The notation used for $\alpha_\sigma^{(n)}$ should not be confused with higher derivative notation. Substituting this into Poisson's equation yields the dispersion relation for the electrostatic wave. It is notable to remark that the found dispersion relation can be manipulated to show that Landau damping appears both at the wave frequency ω and at cyclotron harmonics. It is appropriate to verify whether our considerations regarding magnetic fields reduce to pure electrostatic waves, as described by the Vlasov-Poisson model. In the limit $\|\mathbf{B}\| \to 0$, we find $\forall n \in \mathbb{Z}, \alpha_\sigma^{(n)} \to 0$. The integral above can thus be taken out of the sum. Furthermore, $\sum_{n\in\mathbb{Z}} I_n(\lambda) = \exp\left[\lambda\right]$. Our considerations thus reduce to the description of the Vlasov-Poisson model in the unmagnetised case. We find that the dispersion relation is of the following form:

$$\mathcal{D}(\omega,\mathbf{k}) = 1 + \sum_\sigma \frac{1}{\mathbf{k}^2\lambda_{\mathrm{d}\sigma}^2}\left(1 + \alpha_\sigma^{(0)}\exp\left[-k_\perp^2 r_\sigma^2\right]\sum_{n=-\infty}^{\infty} I_n(k_\perp^2 r_\sigma^2)Z\left(\alpha_\sigma^{(n)}\right)\right) \tag{4.209}$$

Because the modified Bessel functions of the first kind are even with respect to the parameter n, we may recast as follows:

$$\mathcal{D}(\omega,\mathbf{k}) = 1 + \sum_\sigma \frac{\exp\left[-k_\perp^2 r_\sigma^2\right]}{\mathbf{k}^2\lambda_{\mathrm{d}\sigma}^2}\left[I_0(k_\perp^2 r_\sigma^2)\left[1 + \alpha_\sigma^{(0)}Z\left(\alpha_\sigma^{(0)}\right)\right] + \right. \tag{4.210}$$

$$\left. + \sum_{n=1}^{\infty} I_n(k_\perp^2 r_\sigma^2)\left(2 + \alpha_\sigma^{(n)}\left[Z\left(\alpha_\sigma^{(n)}\right) + Z\left(\alpha_\sigma^{(-n)}\right)\right]\right)\right] \tag{4.211}$$

This dispersion relation ought to reduce to our considerations of cold waves in magnetised plasma in the two-fluid description. This corresponds to the case $\alpha_\sigma^{(0)} = \sqrt{\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}}\omega/k_\parallel \gg 1$. Notice that, due to the harmonic nature of $\alpha_\sigma^{(n)}$ with respect to cyclotron frequencies, we shall furthermore assume that ω

indeed does not coincide with any cyclotron harmonic, so that we may perform expansions of the plasma dispersion functions as follows:

$$1+\alpha_\sigma^{(0)} Z\left(\alpha_\sigma^{(0)}\right)=1+\alpha_\sigma^{(0)}\left[-\frac{1}{\alpha_\sigma^{(0)}}\left(1+\frac{1}{2\left[\alpha_\sigma^{(0)}\right]^2}+\cdots\right)+\mathrm{i}\sqrt{\pi}\exp\left[-\left(\alpha_\sigma^{(0)}\right)^2\right]\right] \tag{4.212}$$

$$1+\alpha_\sigma^{(0)} Z\left(\alpha_\sigma^{(0)}\right)\simeq-\frac{1}{2\left(\alpha_\sigma^{(0)}\right)^2}+\mathrm{i}\alpha_\sigma^{(n)}\sqrt{\pi}\exp\left[-\left(\alpha_\sigma^{(0)}\right)^2\right] \tag{4.213}$$

$$2+\alpha_\sigma^{(0)}\left[Z\left(\alpha_\sigma^{(n)}\right)+Z\left(\alpha_\sigma^{(-n)}\right)\right]=-\frac{2n^2\Omega_\sigma^2}{\omega^2-n^2\Omega_\sigma^2}+\mathrm{i}\alpha_\sigma^{(0)}\sqrt{\pi}\left(e^{-\left(\alpha_\sigma^{(n)}\right)^2}+e^{-\left(\alpha_\sigma^{(-n)}\right)^2}\right) \tag{4.214}$$

Upon inputting these expansions into the dispersion relation, it is now clear that electrostatic waves exhibit resonances at cyclotron frequencies. Moreover, the imaginary terms indicate that Landau damping takes place at both the wave frequency ω, but also at cyclotron harmonics. Notice that, due to the transcendental nature of the dispersion relation, its solution $(\omega, \mathbf{k})$ has infinitely many values for ω for any given $k_\perp$.

4.6.1 Bernstein waves

We consider the situation where the wave phase is uniform in the direction of the magnetic field, so that $\mathbf{k}\cdot\mathbf{B}=0$. Such a situation is thus equivalent to the limit $k_\parallel \to 0$, in which case the Landau damping terms and the inverse square terms, $1/(\alpha_\sigma^{(n)})^2$ vanish. The full dispersion relation is thus reduced to:

$$\mathcal{D}(\omega,\mathbf{k})=1-\sum_\sigma \frac{\exp\left[-k_\perp^2 r_\sigma^2\right]}{k_\perp^2 r_\sigma^2}\sum_{n=1}^{\infty}\frac{2n^2\omega_{\mathrm{p}\sigma}^2}{\omega^2-n^2\Omega_\sigma^2}I_n(k_\perp^2 r_\sigma^2) \tag{4.215}$$

Waves described by this dispersion relation are commonly referred to as Bernstein waves or hot plasma waves since their existence depends on the plasma having a finite temperature. For given $k_\perp$, there are infinitely many solutions for ω, that is, each solution is associated with a particular cyclotron harmonic. This is precisely because, as ω is increased, it will inevitably cross a cyclotron harmonic, where the non-trivial piecewise-continuous term of the dispersion relation oscillates between negative and positive infinity. We shall now consider the case of electron Bernstein waves and drop the ion contribution to the dispersion relation:

$$1=2\frac{\exp\left[-k_\perp^2 r_\sigma^2\right]}{k_\perp^2 r_\mathrm{e}^2}\sum_{n=1}^{\infty}\frac{n^2\omega_{\mathrm{pe}}^2}{\omega^2-n^2\Omega_\mathrm{e}^2}I_n(k_\perp^2 r_\mathrm{e}^2) \tag{4.216}$$

which exhibits distinct behaviour in the two following limits:

1. The magnetic field is sufficiently strong to satisfy $\omega_{\mathrm{pe}}^2 \ll \Omega_\mathrm{e}^2$. Each factor in the resonant term is negligible compared to unity except when $\omega^2 \sim \mathcal{O}(n^2\Omega_\mathrm{e}^2)$. Thus, only a single term in the sum is near resonance, and the dispersion relation is satisfied under the following condition:

$$\omega^2=n^2\Omega_\mathrm{e}^2+2\frac{\exp\left[k_\perp^2 r_\mathrm{e}^2\right]}{k_\perp^2 r_\mathrm{e}^2}n^2\omega_{\mathrm{pe}}^2 I_n(k_\perp^2 r_\mathrm{e}^2) \qquad n\in\mathbb{N}_+ \tag{4.217}$$

The small and large limits of $k_\perp^2 r_\mathrm{e}^2$ are determined using the asymptotic of the modified Bessel functions of the first kind, namely:

$$\lim_{\xi\ll 1} I_n(\xi)=\frac{1}{n!}\left(\frac{\xi}{2}\right)^n \quad \text{and} \quad \lim_{\xi\gg 1} I_n(\xi)=\frac{\exp[\xi]}{\sqrt{2\pi\xi}} \tag{4.218}$$

These results indeed form a generalisation of the upper hybrid resonances for warm magnetised plasmas.

2. The magnetic field is sufficiently weak to satisfy $\omega_{\mathrm{pe}}^2 \gg \Omega_{\mathrm{e}}^2$. For large $k_\perp^2 r_{\mathrm{e}}^2$, the product of the exponential and the modified Bessel function is unity, thus leaving $k_\perp^3 r_{\mathrm{e}}^3$ in the denominator, so that the dispersion relation again reduces to cyclotron harmonics. The other limit for $k_\perp^2 r_{\mathrm{e}}^2$ is, however, more convoluted, as the $n = 1$ term is independent of $k_\perp^2 r_{\mathrm{e}}^2$ in the lowest order. One may show that, in this limit, the n-th mode starts at $\omega \simeq (n+1)\Omega_{\mathrm{e}}$ for small $k_\perp^2 r_{\mathrm{e}}^2$.

It is important to note the rather restrictive condition $k_\parallel = 0$. In cylindrical or planar geometries, such a situation is only possible if the considered oscillations were produced via infinitely long antennas parallel to the magnetic field. This is, in reality, simply not the case. In any regions where $\mathbf{k} \cdot \mathbf{B} \neq 0$, evaluating to even infinitesimal values, Bernstein waves as described above would experience vastly different mode behaviour, as well as Landau damping.

Chapter 5

Magnetic field structure

In this chapter, an in-depth analysis of the geometric properties of magnetic fields will be presented, with emphasis on field-lines. The latter are commonly used in practice to define specific coordinate systems. The concepts developed in the chapter will provide a foundation for constructing the aforementioned coordinate systems, which, in turn, underpin the theory of magnetic confinement. The necessity of magnetically confining plasmas arises from the fact that their temperatures frequently reach extremely high values. These temperatures can be high enough to destroy or even melt container walls, measuring instruments, and any other equipment. In plasma physics, the main types of plasma confinement are magnetic and inertial. The latter originates in atomic physics, whereas magnetic confinement relies on the theory of charged-particle motion in magnetic fields. It is, thus, the objective of this chapter to introduce the fundamental geometric properties of general magnetic field structures.

5.1 Magnetic field-lines

A magnetic field-line is, by definition, a curve whose tangent, at each point, is parallel to the magnetic field $\mathbf{B}$. Let $\mathrm{d}\boldsymbol{\ell}$ be the following equivalent expressions that describe an infinitesimal element of the curve, then a field-line:

$$\mathbf{B} \propto \mathrm{d}\boldsymbol{\ell} \qquad \text{or} \qquad \mathbf{B} \times \mathrm{d}\boldsymbol{\ell} = \mathbf{0} \tag{5.1}$$

Using the natural parametrization, i.e., arc length, we obtain the following equation:

$$\frac{\mathbf{B}}{\|\mathbf{B}\|} = \frac{\mathrm{d}\boldsymbol{\ell}}{\|\mathrm{d}\boldsymbol{\ell}\|} \tag{5.2}$$

It is possible to determine the magnetic field-lines by solving equation (5.2) in components. Let us then consider a magnetic field-line parametrized by its arc length ℓ. The velocity vector of the curve is then the unit length vector, whose direction is $\hat{\mathbf{B}}$. By taking the scalar product of the velocity vector with itself, it follows:

$$\hat{\mathbf{B}}^2 = 1 \implies \hat{\mathbf{B}} \cdot \frac{\mathrm{d}\hat{\mathbf{B}}}{\mathrm{d}\ell} = 0 \implies \hat{\mathbf{B}} \perp \frac{\mathrm{d}\hat{\mathbf{B}}}{\mathrm{d}\ell} \tag{5.3}$$

The curvature vector of the considered magnetic field-line is, by definition, given by:

$$\boldsymbol{\kappa} = \left.\frac{\mathrm{d}\hat{\mathbf{B}}}{\mathrm{d}\ell}\right|_{\text{along } \hat{\mathbf{B}}} \tag{5.4}$$

It should be noted that the partial derivative with respect to ℓ is equivalent to the directional derivative along the magnetic field-line, i.e., along $\hat{\mathbf{B}}$:

$$\frac{\partial}{\partial \ell} \equiv \hat{\mathbf{B}} \cdot \nabla = \left.\frac{\mathrm{d}}{\mathrm{d}\ell}\right|_{\text{along } \hat{\mathbf{B}}} \tag{5.5}$$

It follows that the following expression gives the curvature vector:

$$\boldsymbol{\kappa} = (\hat{\mathbf{B}} \cdot \nabla)\hat{\mathbf{B}} = -\hat{\mathbf{B}} \times (\nabla \times \hat{\mathbf{B}}) \implies \boldsymbol{\kappa} \perp \mathbf{B} \tag{5.6}$$

So far, we have considered the curvature of a magnetic field-line in the absence of any other field. When a field-line belongs entirely to a surface, it is appropriate to distinguish the components of the curvature.

We will define the normal curvature as the perpendicular component of the curvature $\boldsymbol{\kappa}$ to the considered surface, and the geodesic curvature as the tangential component of $\boldsymbol{\kappa}$ to the surface. In plasma systems, it is only convenient to employ magnetic fields in a manner to construct a useful coordinate system. We will thus construct a contravariant coordinate system (u, v, w) in such a way that w designates the coordinate of any point along a field-line $\mathbf{B}$. To make the following approach less abstract, this third coordinate can be represented by the arc length ℓ of the field-line in question. The surfaces to which magnetic field-lines naturally belong are precisely the magnetic surfaces, of which we will later provide a rigorous definition. These magnetic surfaces can be envisaged as surfaces of $u = \text{const}$. Then, it is often desirable to know the components of the curvature perpendicular and tangent to a surface of $u = \text{const}$. Let $\mathbf{n}$ be the unit vector in the direction in which the curvature $\boldsymbol{\kappa}$ lies. The vectors $\mathbf{n}$, ∇u, and ∇v are then all perpendicular to the magnetic field $\mathbf{B}$, and therefore belong to the same surface. We want to decompose the normal vector $\mathbf{n}$ into two components, one of which is along ∇u (perpendicular to the surfaces of $u = \text{const.}$) and the other is tangent to the aforementioned surfaces. The choice of ∇v is not sufficient, because the unit vectors of the coordinate system do not necessarily generate an orthonormal basis. We will find the component of ∇v parallel to ∇u, then subtract it from ∇v. Let $\hat{\mathbf{u}} = \nabla u / \|\nabla u\|$:

$$(\nabla v)_{\parallel} = (\hat{\mathbf{u}} \cdot \nabla v)\, \hat{\mathbf{u}} \qquad \text{and} \qquad (\nabla v)_{\perp} = (\hat{\mathbf{u}} \times \nabla v) \times \hat{\mathbf{u}} \tag{5.7}$$

The curvature $\boldsymbol{\kappa}$ can be easily found under the following expression:

$$\boldsymbol{\kappa} = \kappa_u \nabla u + \kappa_v \nabla v \qquad \text{since} \qquad \boldsymbol{\kappa} \perp \mathbf{B} \implies \boldsymbol{\kappa} \perp \nabla w \tag{5.8}$$

By making use of the equation (5.7), we recover the desired components:

$$\boldsymbol{\kappa} = \left[\kappa_u \|\nabla u\| + \kappa_v (\hat{\mathbf{u}} \cdot \nabla v)\right] \hat{\mathbf{u}} + \kappa_v (\hat{\mathbf{u}} \times \nabla v) \times \hat{\mathbf{u}} \tag{5.9}$$

The first term corresponds to the normal curvature, along $\hat{\mathbf{N}}$, and the second to the geodesic curvature:

$$\kappa_{\mathrm{N}} = \kappa_u \|\nabla u\| + \kappa_v \|\nabla v\| \cos \angle\left(\nabla u, \nabla v\right) \qquad \text{and} \qquad \kappa_{\mathrm{G}} = \kappa_v \|\nabla v\| \sin \angle\left(\nabla u, \nabla v\right) \tag{5.10}$$

5.1.1 Magnetic pressure and tension

It proves instructive to examine the way to attribute certain fictitious forces to magnetic field-lines in a medium traversed by a current. This will serve as a foundation when considering the concepts of magnetic pressure and tension. In a neutral, or even quasi-neutral, medium, the Lorentz force density and the Maxwell-Ampère equation allow us to write:

$$\mathbf{f}_{\mathrm{L}} = \frac{\mathrm{d}\mathbf{F}_{\mathrm{L}}}{\mathrm{d}V} = \mathbf{J} \times \mathbf{B} = \frac{1}{\mu_0} \left(\nabla \times \mathbf{B}\right) \times \mathbf{B} = \frac{1}{\mu_0} \left(\mathbf{B} \cdot \nabla\right) \mathbf{B} - \frac{1}{2\mu_0} \nabla \mathbf{B}^2 \tag{5.11}$$

This is equivalent to the following expression by employing the results of the previous section:

$$\mathbf{f}_{\mathrm{L}} = \frac{\mathbf{B}^2}{\mu_0} \boldsymbol{\kappa} + \frac{\partial}{\partial \ell} \left(\frac{\mathbf{B}^2}{2\mu_0}\right) \hat{\mathbf{B}} - \nabla \left(\frac{\mathbf{B}^2}{2\mu_0}\right) \tag{5.12}$$

We will then introduce the Frenet-Serret frame attached to the considered line in such a way that:

$$\mathbf{t} = \hat{\mathbf{B}}, \quad \mathbf{n} = \hat{\boldsymbol{\kappa}} \quad \text{and} \quad \mathbf{b} = \mathbf{t} \times \mathbf{n} \tag{5.13}$$

where $\mathbf{b}$ designates the unit binormal vector to the magnetic field-line in question:

$$\mathbf{f} = \frac{\mathbf{B}^2}{\mu_0} \kappa\, \mathbf{n} - \frac{\partial}{\partial \mathbf{n}} \left(\frac{\mathbf{B}^2}{2\mu_0}\right) - \frac{\partial}{\partial \mathbf{b}} \left(\frac{\mathbf{B}^2}{2\mu_0}\right) \tag{5.14}$$

The contribution of magnetic pressure corresponds to the last two terms above. The magnetic pressure tends to compress the magnetic field-lines in the normal plane, i.e., generated by $\mathbf{n}$ and $\mathbf{b}$:

$$\nabla p_{\mathrm{mag}} = \frac{\partial}{\partial \mathbf{n}} \left(\frac{\mathbf{B}^2}{2\mu_0}\right) + \frac{\partial}{\partial \mathbf{b}} \left(\frac{\mathbf{B}^2}{2\mu_0}\right) \tag{5.15}$$

The first term of the expression (5.14) represents the magnetic tension. This tension is indeed a fictitious force that pulls on a magnetic field-line in the tangential direction to this line. This is more evident when considering the length of the field-line. We will represent the number of magnetic field-lines per unit area by the magnitude of the magnetic field. The surface belonging to a single field-line is qualitatively of the order of $1/|\mathbf{B}|$. The tension per unit length on a single magnetic field-line is thus given by:

$$\frac{\mathrm{d}\mathbf{T}_{\mathrm{mag}}}{\mathrm{d}\ell} = \frac{\mathbf{B}^2}{\mu_0}\frac{\boldsymbol{\kappa}}{\|\mathbf{B}\|} = \frac{\|\mathbf{B}\|}{\mu_0}\boldsymbol{\kappa} \tag{5.16}$$

5.2 Magnetic surfaces

Any surface or, more precisely, two-dimensional subdomain, contoured by a particular set of magnetic field-lines that ergodically cover the subdomain, is called a *magnetic surface*. We distinguish between magnetic surfaces in toroidal and open systems. The equation of a magnetic field-line can be expressed in Hamiltonian form. Hamiltonian theory guarantees the existence of magnetic surfaces provided there is a direction of symmetry. Moreover, it is worth discussing why, instead of closed systems, we employed the terminology of toroidal systems. Notice that the most interesting form of a magnetic surface is that of a two-dimensional manifold, in which case there exists a bijective, diffeomorphic parametrisation. If such a surface exists and has no edges, that is, it does not intersect its domain of definition, and if the magnetic field vanishes nowhere on it, the manifold must, by a non-trivial result of topology, be a toroid. In other words, it must be topologically equivalent to a torus. This type of topology is employed specifically in tokamaks and stellarators.

Even without much mathematical rigour, only using our considerations of motion of charged particles in magnetic fields, let us consider the situation of confining a plasma in a circular magnetic field, as illustrated in Figure (5.1):

$$q\mathbf{v}_{\mathrm{d}} = \left(mv_\parallel^2 + \frac{1}{2}mv_\perp^2\right)\frac{\mathbf{R}\times\mathbf{B}}{\mathbf{R}^2\mathbf{B}^2} = \left(mv_\parallel^2 + \frac{1}{2}mv_\perp^2\right)\frac{\mathbf{B}\times\boldsymbol{\kappa}}{\mathbf{B}^2} \tag{5.17}$$

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 5.1: Circular **B** field.

The drift velocity depends on the sign of the charge. Under the hypothesis of a purely circular, or even cylindrical, magnetic field, it appears clearly that a non-uniform charge distribution will develop promptly, with the ion current being progressively pushed upward. In contrast, electrons undergo downward displacement. This non-uniform charge distribution will generate an electric field. This field, in turn, induces an $\mathbf{E}\times\mathbf{B}$ drift in the radial direction, and thus a net radial movement for all charges, causing the plasma to escape the imposed confinement. It is therefore necessary to design a magnetic field shape that does not allow for the appearance of a non-uniform distribution. One plausible approach is to modify the magnetic field-lines by winding them around a torus. Taking into account expression (5.17), it is easy to see that the drift velocity tends to confine all particles to the surface of the torus, as illustrated by Figure **??**.

5.2.1 Confinement in open systems via mirrors and cusps

The results of subsections (2.9.1) and (2.9.2) indicate a potential method for confining a plasma between two zones characterized by high magnetic field densities, induced by electric currents. These are commonly produced by two Helmholtz coils, as illustrated in Figure (**??**). The two particle-deflection zones are often referred to as *magnetic mirrors*. Under the assumption of a magnetic field that varies adiabatically in space—for example, the one examined in subsection (2.5.2)—the maintenance of plasma confinement exerted by the two mirrors is satisfactory for short time scales.

When considering a single particle trapped between the two zones, confinement will be maintained continuously by the two mirrors due to symmetry. However, in the case of a plasma with a high abundance of particles, they will progressively collide. The targeted particles undergo a velocity change, which can lead to their entry into the loss cone. Indeed, interparticle collisions tend to equalize the distribution of particles in phase space $(\mathbf{p}, \mathbf{x})$. Consequently, an increasing number of particles enter the loss cone, leading to the plasma's progressive escape from confinement.

Just as with magnetic mirrors, characterized by zones of high magnetic field density, it is possible to design magnetic *cusps*. An example was examined in sub-subsection (2.10.5). The condition for the passage of a particle approaching the cusp is given by inequality (2.126). Thus, particles whose velocity does not satisfy this inequality undergo deflection. This suggests a method of plasma confinement that creates two cusp zones, as illustrated in Figure **??**. A particle will not penetrate the cusp zones provided that the velocity component parallel to the magnetic field in the centre is not large enough to satisfy inequality (2.126).

5.2.2 Constructing magnetic surfaces in open systems

Mirror-type magnetic devices do not have magnetic surfaces contoured by a single field-line, so it is necessary to define the shape of these surfaces. The natural choice of the cross-section is precisely that of the circular cross-section at the centre of the device.

5.2.3 Constructing magnetic surfaces in toroidal systems

Toroidal systems are characterized by cylindrical or toroidal symmetry, depending on the type of confinement device. We will then name the symmetry coordinate *ignorable*. Thanks to this symmetry, Hamiltonian theory guarantees us the existence of surfaces. It should be noted that, in practice, toroidal confinement devices do not necessarily have to be symmetric. In this case, magnetic surfaces can be approximated by neglecting the perturbations that cause the asymmetry. In principle, magnetic surfaces in asymmetric devices can be determined experimentally or numerically. It should be noted that, in a tokamak, a toroidal/parallel plasma current is required to determine the existence of magnetic surfaces, since, in this case, the magnetic field is (partially) created by this current. However, in a stellarator, the magnetic field is produced by coils located outside the device, so the plasma plays no role in the structure of the magnetic fields.

Among all the magnetic surfaces contoured by magnetic field-lines, it is possible to define the surfaces to which belong the field-lines that close on themselves after one or more transits around the device. In this case, it is necessary to construct or define these magnetic surfaces because a single field-line does not describe them. These magnetic surfaces are called rational surfaces because the ratio of the number of transits in the poloidal direction and in the toroidal direction is a sensible number. A set of these field-lines then defines a rational magnetic surface. Since the set of real numbers is dense, any irrational number can be approximated more and more closely by a rational number, and vice versa. The same applies to magnetic surfaces. To construct a coordinate system from magnetic surfaces, one must choose these surfaces appropriately. The typical approach, that is, the zeroth approximation of the actual magnetic field structure, relies on the isotropic magnetohydrodynamic equilibrium equation $\mathbf{J} \times \mathbf{B} = \nabla P_{\mathrm{MHD}}$. The constancy of the magnetohydrodynamic pressure characterizes these magnetic surfaces, because $\mathbf{B} \cdot \nabla P_{\mathrm{MHD}} = 0$, and the plasma currents are defined there, because $\mathbf{J} \cdot \nabla P_{\mathrm{MHD}} = 0$. In a neutral plasma, charge conservation reduces to $\nabla \cdot \mathbf{J} = 0$. The same goes: $\nabla \cdot \mathbf{B} = 0$. Moreover, the isotropic magnetohydrodynamic equilibrium gives us:

$$\nabla \oint_\Gamma \frac{\mathrm{d}\ell}{\|\mathbf{B}\|} = \oint_\Gamma \nabla \left(\frac{1}{\|\mathbf{B}\|}\right) \mathrm{d}\ell = -\oint_\Gamma \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\|\mathbf{B}\|^3}\,\mathrm{d}\ell \tag{5.18}$$

where Γ is a magnetic field-line belonging to a rational surface $\Sigma_{\mathbb{Q}}$. It follows:

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = -\mathbf{B} \times (\nabla \times \mathbf{B}) = \mu_0 \mathbf{J} \times \mathbf{B} = \mu_0 \nabla P_{\mathrm{MHD}} \perp \Sigma_{\mathbb{Q}} \tag{5.19}$$

Therefore, the integral of the equation (5.18) is uniform on a rational magnetic surface, independently of the magnetic field-line on which it is evaluated. When considering the field-lines associated with irrational surfaces, this integral does not exist because these lines do not close on themselves. The violation of this uniformity can occur even when considering rational surfaces. This is due to the device's asymmetry, which induces island or magnetic-chain formation that we, for now, ignore but will address later. In an asymmetric device, according to ideal magnetohydrodynamics, rational surfaces are unstable and tend to tear, leading to the formation of fibre-like structures characterized by multiple magnetic axes.

It is debatable to claim that simple magnetohydrodynamic equilibrium is not too gross an approximation. This model ignores the effects of resistivity, viscosity, inertia, and the finite Larmor radius.

In principle, one must solve a time-evolution problem. Even if the system under consideration tends to evolve towards a coherent stationary state characterized by certain magnetic surfaces, these surfaces need not be identical to those predicted by ideal magnetohydrodynamics. From a pragmatic point of view, we will assume the existence of an adequate set of magnetic surfaces whose cross-sections form a set of smooth, closed curves. This indeed yields the zero-order approximation to the system's real magnetic structure. We will refer to the degenerate magnetic surfaces with zero volume as magnetic axes. It is indeed a magnetic field-line that closes after exactly one toroidal transit. Magnetic axes do not necessarily have to be circular; they can be helical, as in the case of a stellarator.

5.3 Elementary curvilinear coordinate systems

A coordinate system is a framework for specifying the location of a point in space using a unique n-tuple. In three-dimensional space, a coordinate system consists of three families of adequate coordinate surfaces, so that a point is defined by the intersection of three surfaces, one for each family. The coordinate curves are intersections of two coordinate surfaces. Moreover, we will use only direct coordinate systems where the product of unit vectors $\hat{\mathbf{e}}_i \cdot (\hat{\mathbf{e}}_j \times \hat{\mathbf{e}}_k)$ is positive when (i, j, k) is a cyclic permutation of $(1, 2, 3)$.

5.3.1 General non-orthogonal coordinate systems

For non-orthogonal coordinates, there are two sets of natural basis vectors to represent general vector quantities. The first set consists of $(\nabla\varrho, \nabla\theta, \nabla\zeta)$ where:

$$\nabla\varrho = \frac{\partial\varrho}{\partial x}\,\hat{\mathbf{e}}_{\mathrm{x}} + \frac{\partial\varrho}{\partial y}\,\hat{\mathbf{e}}_{\mathrm{y}} + \frac{\partial\varrho}{\partial z}\,\hat{\mathbf{e}}_{\mathrm{z}} \tag{5.20}$$

The definitions of $\nabla\theta$ and $\nabla\zeta$ are analogous. The coordinate system (x, y, z) with the associated triple of basis vectors $(\hat{\mathbf{e}}_{\mathrm{x}}, \hat{\mathbf{e}}_{\mathrm{y}}, \hat{\mathbf{e}}_{\mathrm{z}})$ corresponds to the standard Cartesian coordinate system. The second set consists of $(\partial\mathbf{r}/\partial\varrho, \partial\mathbf{r}/\partial\theta, \partial\mathbf{r}/\partial\zeta)$ where:

$$\frac{\partial\mathbf{r}}{\partial\varrho} = \frac{\partial x}{\partial\varrho}\,\hat{\mathbf{e}}_{\mathrm{x}} + \frac{\partial y}{\partial\varrho}\,\hat{\mathbf{e}}_{\mathrm{y}} + \frac{\partial z}{\partial\varrho}\,\hat{\mathbf{e}}_{\mathrm{z}} \tag{5.21}$$

The definitions of $\partial\mathbf{r}/\partial\theta$ and $\partial\mathbf{r}/\partial\zeta$ are analogous. If the coordinates (ϱ, θ, ζ) were orthogonal, then $\nabla\varrho$ and $\partial\mathbf{r}/\partial\varrho$ would be parallel. These directions are generally not parallel. We can decompose any vector into components with respect to these two sets of basis vectors so that:

$$\mathbf{B} = B_\varrho\nabla\varrho + B_\theta\nabla\theta + B_\zeta\nabla\zeta \quad \text{and} \quad \mathbf{B} = B^\varrho\frac{\partial\mathbf{r}}{\partial\varrho} + B^\theta\frac{\partial\mathbf{r}}{\partial\theta} + B^\zeta\frac{\partial\mathbf{r}}{\partial\zeta} \tag{5.22}$$

We can establish the link between the two sets of basis vectors as follows. First note that the vector $\partial\mathbf{r}/\partial\varrho$ points, by definition, in a direction along which the coordinates θ and ζ do not increase, then:

$$\frac{\partial\mathbf{r}}{\partial\varrho}\cdot\nabla\theta = 0 \quad \text{and} \quad \frac{\partial\mathbf{r}}{\partial\varrho}\cdot\nabla\zeta = 0 \implies \frac{\partial\mathbf{r}}{\partial\varrho} = J\nabla\theta\times\nabla\zeta \tag{5.23}$$

where J is a coefficient. To determine it, we consider a step $\mathrm{d}\mathbf{x}$ at fixed θ and ζ, then $\mathrm{d}\varrho = \nabla\varrho\cdot\mathrm{d}\mathbf{x}$. This gives us $\nabla\varrho\cdot(\partial\mathbf{x}/\partial\varrho) = 1$. By taking the scalar product of the equation (5.23) with $\nabla\varrho$:

$$J = \frac{1}{\nabla\varrho\cdot(\nabla\theta\times\nabla\zeta)} = \frac{\partial(x, y, z)}{\partial(\varrho, \theta, \zeta)} \tag{5.24}$$

Thus, J is precisely the Jacobian generated by the transformation of coordinates. It is evident that we will obtain the same for the other coordinates, then:

$$\mathbf{B} = \frac{1}{J}\left(B^\varrho\nabla\theta\times\nabla\zeta + B^\theta\nabla\zeta\times\nabla\varrho + B^\zeta\nabla\varrho\times\nabla\theta\right) \tag{5.25}$$

So far, the vector $\mathbf{B}$ was any vector. Let us rely on the case of the magnetic field. If there are *good* magnetic surfaces, then $\mathbf{B}\cdot\nabla\varrho = 0$. By taking the scalar product of the equation (5.25) with $\nabla\varrho$, we obtain that $B^\varrho = 0$. There are then five non-zero components (B^θ, B^ζ, B_ϱ, B_θ, B_ζ) of $\mathbf{B}$ in the two decompositions.

5.3.1.1 Cylindrical-toroidal or pseudo-toroidal system

In toroidal plasma confinement, coordinate systems are defined by toroidal topology. The *good* toroidal coordinates form a three-dimensional version of bipolar coordinates and have been used only occasionally in plasma equilibrium calculations. The cylindrical-toroidal coordinate system is actually a cylindrical coordinate system, characterized by the triple (r, φ, z), which has been toroidalized by replacing the cylindrical coordinates with (ρ, θ, ζ). The coordinate surfaces $\rho =$ const. are a perfect torus of major radius R. The circle of radius R in the horizontal plane is called the minor axis, and the axis $\hat{\mathbf{z}}$ is called the central axis. The radius ρ measures the distance from a point to the minor axis. The coordinate surfaces $\theta =$ const. are partial cones that touch the minor axis. The third family of coordinate surfaces, $\zeta =$ const., consists of planes that touch the central axis. It should be noted that it was necessary to reverse the direction of the azimuthal angle, i.e. $\zeta = \pi/2 - \varphi$ so that the triple (ρ, θ, ζ) is a direct coordinate system where $\hat{\boldsymbol{\rho}} \cdot (\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\zeta}}) > 0$. The angle θ is also called the poloidal angle, and ζ the toroidal angle. The transformation from the cylindrical-toroidal coordinate system to the Cartesian one is trivially given by the following:

$$x = (R + \rho\cos(\theta))\sin(\zeta) \tag{5.26}$$

$$y = (R + \rho\cos(\theta))\cos(\zeta) \tag{5.27}$$

$$z = \rho\sin(\theta) \tag{5.28}$$

When $R = 0$, the pseudo-toroidal coordinate system degenerates into a spherical coordinate system.

5.3.1.2 Generalized cylindrical-toroidal system

In theoretical plasma physics, it is convenient to choose magnetic surfaces as the first family of coordinate surfaces, regardless of their shape. We label these surfaces by $\varrho =$ const.. We will soon identify ϱ with certain more precise measurable quantities, such as, for example, the magnetic flux enclosed by the surface, its volume, or the pressure. The coordinate surfaces $\theta =$ const. and $\zeta =$ const. are also generalized. The former are linked to the magnetic axis and generally form surfaces similar to partial cones. The coordinate surfaces $\theta =$ const. can be envisioned by continuously folding, stretching, or squeezing the partial cones of the elementary system from the previous sub-subsection. Since the topology of the $\theta =$ const. surfaces is identical to that of the elementary system, we use the same notation. The coordinate θ then represents a generalized poloidal angle. The same applies to the $\zeta =$ const. surfaces where ζ represents a generalized toroidal angle. It should be noted that the terms *poloidal* and *toroidal* no longer present the corresponding *elementary* angles, but adequate surfaces whose shape depends on the structure of the magnetic fields under consideration. The reason why we would like to work with these coordinates is obvious. For analytical calculations, it is convenient to choose natural coordinates so that the geometry of the problem is simple. Thus, the complex geometric characteristics of the problem that exist in a neutral coordinate system, for example, in the Cartesian system, are handled by coordinate transformations. In theoretical plasma physics, it is convenient to choose the angular coordinates θ and ζ such that the magnetic field-lines appear straight, that is to say, that $\mathrm{d}\theta/\mathrm{d}\zeta$ is a constant along a field-line, or, more precisely, a flux function. There exists a choice of coordinates θ and ζ for which the field-lines $\mathbf{B}$, and the currents $\mathbf{J}$ are, at least one of the two, straight lines. It is customary to designate these coordinates as *magnetic coordinates* or *flux coordinates*. They will be denoted by θ_f and ζ_f and will be the subject of the following chapter. To recapitulate, if the magnetic field $\mathbf{B}$ is known everywhere, the same is true for the current $\mathbf{J}$; it is then appropriate to find θ_f and ζ_f such that both are straight lines in the coordinate system $(\varrho, \theta_\mathrm{f}, \zeta_\mathrm{f})$. It is possible to construct an intermediate coordinate system $(\varrho, \theta_\mathrm{e}, \zeta_\mathrm{e})$, called *elementary*, like the pseudo-toroidal system, and then deform the surfaces $\theta_\mathrm{e} =$ const. and $\zeta_\mathrm{e} =$ const. of the system until the natural system presents the desired properties.

5.3.1.3 Illustrative example of the generalized cylindrical-toroidal system

We consider a magnetic field in a circular, axisymmetric tokamak, in which the surfaces of constant flux form perfect, non-concentric tori. The minor axis coincides with the system's magnetic axis. The magnetic field-lines are assumed to traverse the magnetic surfaces with a constant pitch. We will also presume that the rotational transform (defined rigorously below) $\iota\!\!-= 3$, which is to say that a magnetic field-line makes three poloidal transits for every toroidal transit. If we note the points where $\mathbf{B}$ crosses a surface at $\theta_{\mathrm{e}} = \text{const.}$ and transfer them into coordinates $(\theta_{\mathrm{e}}, \zeta_{\mathrm{e}})$, we observe a non-linear behaviour of the magnetic field-lines. To make the magnetic field-lines appear as straight lines in the coordinate system $(\varrho, \theta_{\mathrm{f}}, \zeta_{\mathrm{f}})$, the arc length of a considered field-line must follow the angle θ_{f} linearly. This approach is equivalent to isolating a magnetic surface resembling a torus and constructing the angles θ_{f} from the centre of the intersection of the magnetic surface with a poloidal plane, which is indeed a circle. It should be noted that the magnetic axis does not coincide with this centre. The magnetic field-lines now appear as straight lines in terms of coordinates θ_{f} and ζ_{f} on the surface $\varrho = \text{const.}$ It should be remarked that the transformation between θ_{e} and θ_{f} is not linear.

5.3.1.4 Interlude on the Clebsch coordinate systems

An alternative flux coordinate system, though more complicated, in toroidal devices is the Clebsch coordinate system, (ϱ, β, ℓ). Again, ϱ represents the magnetic surfaces, while β represents the magnetic field-lines within a magnetic surface, and ℓ is the arc length of a magnetic field-line. A surface $\beta = \text{const.}$ is indeed a ribbon extending from the magnetic axis and crossing the magnetic surfaces $\varrho = \text{const.}$ following the magnetic field-lines.

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 5.2: Magnetic field in a circular tokamak. The torus represents a surface $\varrho = \text{const.}$ The ribbon is indeed a surface $\beta = \text{const.}$ The intersection of the two is a curve of the coordinate ℓ which follows a magnetic field-line.

The potential troubles of the surfaces of $\ell = \text{const.}$ are not evident at first glance. First, one could choose an arbitrary origin where $\ell = 0$ for all field-lines. This could be a common poloidal plane, then the surfaces of $\ell = \text{const.}$ must be constructed. The arc length ℓ as well as the surfaces $\ell = \text{const.}$ are affected by the pitch of the magnetic field-lines, which is generally not uniform on a magnetic surface. A necessary consequence of such surfaces: $\ell = \text{const.}$ is the non-correspondence between $\mathbf{B}$ and $\nabla\ell$, that is to say, the two need not be parallel. However, it is essential to emphasize that the operator $\mathbf{B}\cdot\nabla$ reduces to $B(\partial/\partial\ell)$ in this coordinate system. We will encounter another problem when considering irrational magnetic surfaces. Such a surface can be envisioned as being constructed from a single field-line, because the latter covers the surface in question ergodically. Thus, on this surface, there will exist only a single point to which we could attribute the value $\ell = 0$. Also, since a single field-line covers the magnetic surface and field-lines reside on surfaces $\beta = \text{const.}$, it might seem that $\beta = 0$ everywhere on the magnetic surface. Although irrational surfaces can, in principle, be parametrised solely by the arc length ℓ, this is not useful in practice. Rational surfaces also pose a problem. If a field-line closes upon itself after a certain number of toroidal transits, then there exist intervals of β that intertwine. A possible exit from this problematic situation is to introduce *cuts* in the magnetic surface and restrict the coordinate ℓ to a single toroidal transit. All these troubles are still not resolved, because the coordinate β is no longer specified as single-valued according to this convention, because the curve of the coordinate ℓ does not close upon itself after a toroidal transit, unlike the angles θ and ζ in the pseudo-toroidal coordinate system. Although the Clebsch coordinate system can be very useful for theoretical developments, it poses several practical challenges.

Unlike toroidal devices, the Clebsch system presents itself as helpful as it is well-defined once applied to open devices. Single-valued specification problems are avoided because open systems have two ends and are therefore aperiodic.

5.4 Labelling magnetic surfaces

Before developing flux coordinate systems in detail, it is necessary to identify appropriate measures that represent the magnetic surfaces. We will again distinguish between the two cases: toroidal systems and open systems. Regarding toroidal systems, we first consider the magnetic fluxes enclosed by and outside of the magnetic surfaces. We will increasingly distinguish between toroidal flux, poloidal ribbon flux, and poloidal disk flux. We will also rely on measures such as the enclosed volume and the pressure exerted on the magnetic surfaces. We will finally give a precise definition of the magnetic axis.

5.4.1 Labelling magnetic surfaces in toroidal systems

A suitable way to label magnetic surfaces is to use magnetic flux. Since magnetic field-lines are tangent to magnetic surfaces, the flux inside or outside a magnetic surface is conserved. We then denominate magnetic surfaces as surfaces of constant flux, or briefly, flux surfaces. We assume that a coordinate system (ϱ, θ, ζ) has been established, where $\theta = \theta(x, y, z)$ and $\zeta = \zeta(x, y, z)$. There exist several magnetic fluxes that can serve to label a magnetic surface with a unique value. First, the toroidal flux:

$$\Phi_{\mathrm{tor}} = \iint_{S_{\mathrm{tor}}} \mathbf{B} \cdot \mathrm{d}\mathbf{S} \tag{5.29}$$

where S_{tor} is a surface produced by cutting a magnetic surface with a poloidal surface of ζ = const, then, the poloidal fluxes inside and outside the magnetic surfaces can serve as labels. The poloidal flux inside a magnetic surface is measured through a ribbon of θ = const. between the magnetic axis and the surface considered:

$$\Phi^{\mathrm{r}}_{\mathrm{pol}} = \iint_{S^{\mathrm{r}}_{\mathrm{pol}}} \mathbf{B} \cdot \mathrm{d}\mathbf{S} \tag{5.30}$$

The flux outside a magnetic surface is given by integrating $\mathbf{B}$ on a disk-like surface tangent everywhere to the surface and bounded by a curve of coordinate ζ = const. on this surface:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}} = \iint_{S^{\mathrm{d}}_{\mathrm{pol}}} \mathbf{B} \cdot \mathrm{d}S \tag{5.31}$$

The latter includes the tokamak transformer flux and does not require reference to the magnetic axis. To compare the two poloidal fluxes, it is convenient to choose the disk forming part of the approximately circular surface bounded by the magnetic axis. It is evident that the two fluxes are complementary with respect to the net flux in the system, so we have the following equality:

$$\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}} + \mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}} = 0 \implies \nabla\Phi^{\mathrm{r}}_{\mathrm{pol}} + \nabla\Phi^{\mathrm{d}}_{\mathrm{pol}} = \mathbf{0} \tag{5.32}$$

We will see later that, when a transformer flux acts on the surfaces $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ = const., the latter contract at a velocity $\mathbf{v}$, such that $\mathbf{v} \cdot \nabla\Phi^{\mathrm{d}}_{\mathrm{pol}} + \partial\Phi^{\mathrm{d}}_{\mathrm{pol}}/\partial t = 0$. This velocity is indeed equal to the $\mathbf{E} \times \mathbf{B}$ drift in the case of a fully ionized ideal plasma. From the point of view of poloidal ribbons, we draw the same conclusion, so the choice of poloidal flux is arbitrary.

To develop the formalism of flux coordinates, we need more specific expressions for toroidal and poloidal fluxes. We consider the following integral:

$$\iiint_V \mathbf{B} \cdot \nabla\zeta \,\mathrm{d}^3\mathbf{r} = \iint_V \nabla \cdot (\mathbf{B}\zeta) \,\mathrm{d}^3\mathbf{r} \tag{5.33}$$

where V denotes the volume enclosed by a magnetic surface. The second form follows easily because $\nabla \cdot \mathbf{B} = 0$. We further consider a toroidal surface, cut by a poloidal surface at ζ = const. Let $\zeta = 0$, then the volume is enclosed by three surfaces: S_{torus}, $S_{\zeta=0}$ and $S_{\zeta=2\pi}$:

$$\iiint_V \nabla \cdot (\mathbf{B}\zeta) \,\mathrm{d}^3\mathbf{r} = \oiint_{\partial V} \mathbf{B}\zeta \cdot \mathrm{d}\mathbf{S} = \iint_{S_{\mathrm{torus}}} \underbrace{\mathbf{B}\zeta \cdot \mathrm{d}\mathbf{S}}_{\mathbf{B}\perp\mathrm{d}\mathbf{S}} + \iint_{S_{\zeta=0}} \underbrace{\mathbf{B}\zeta}_{0} \cdot \mathrm{d}\mathbf{S} + \iint_{S_{\zeta=2\pi}} \underbrace{\mathbf{B}\zeta}_{2\pi\mathbf{B}} \cdot \mathrm{d}\mathbf{S} \tag{5.34}$$

First note that the flux through $S_{\zeta=2\pi}$ corresponds to the toroidal flux. An expression follows then of a form invariant to the choice of functions θ and ζ. Analogously, we also find the expression for the poloidal flux through the ribbons:

$$\Phi_{\mathrm{tor}} = \frac{1}{2\pi} \iiint_V \mathbf{B} \cdot \nabla\zeta \,\mathrm{d}^3\mathbf{r} \quad \text{and} \quad \Phi^{\mathrm{r}}_{\mathrm{pol}} = \frac{1}{2\pi} \iiint_V \mathbf{B} \cdot \nabla\theta \,\mathrm{d}^3\mathbf{r} \tag{5.35}$$

There are various labels for magnetic surfaces other than fluxes. We designate by *flux function* any function or variable uniform on a magnetic surface. These functions are independent of coordinates θ and ζ and only depend on the label of the magnetic surfaces. A prominent example of a flux function is precisely the volume enclosed by a magnetic surface:

$$V = \iiint_V \mathrm{d}^3\mathbf{r} = \iiint_V J \,\mathrm{d}\varrho \,\mathrm{d}\theta \,\mathrm{d}\zeta = \iiint_V \frac{\mathrm{d}\varrho \,\mathrm{d}\theta \,\mathrm{d}\zeta}{\nabla\varrho \cdot (\nabla\theta \times \nabla\zeta)} \tag{5.36}$$

Another flux quantity is the scalar pressure in ideal magnetohydrodynamic equilibrium, since $\nabla p = \mathbf{J} \times \mathbf{B} \implies \mathbf{B} \cdot \nabla p = 0$, thus $p = p(\varrho)$. This can always pose a problem when $\partial p / \partial \varrho = 0$.

Certain flux quantities are more adequate than others. For example, the pressure p and the poloidal flux through the *disks* $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ decrease with distance from the centre of the magnetic axis. To have a direct coordinate system, one would then have to reverse the direction of another coordinate, for example, ζ. This modification introduces additional issues with integration paths; therefore, it is convenient to add a negative sign to these two flux quantities. Thus, the following flux quantities can represent the first flux coordinate: $\varrho \equiv \Phi_{\mathrm{tor}}$, $\varrho \equiv \Phi^{\mathrm{r}}_{\mathrm{pol}}$, $\varrho \equiv -\Phi^{\mathrm{d}}_{\mathrm{pol}}$, $\varrho \equiv V$, and $\varrho \equiv -p$.

5.4.2 Rotational transform in toroidal systems

It is pertinent to introduce a parameter describing the rotation of the magnetic field in a toroidal system. We have implicitly introduced this parameter when considering the toroidal confinement method. We denominate the net poloidal angle traversed $\Delta\theta$ after a toroidal transit of $\Delta\zeta = 2\pi$ as the rotational transform angle ι. This angle is defined for a magnetic surface by the average of the rotational transforms of each independent magnetic field-line:

$$\iota = \lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} \iota_k \tag{5.37}$$

The angle of the rotational transform is, by definition, a flux quantity. It is clear that, on an irrational surface, $n = 1$ because there exists a single field-line that covers the surface ergodically. It is customary to introduce another quantity called *rotational transform*, equal to the number of poloidal transits per toroidal transit:

$$\iota\!\!\!- = \frac{\iota}{2\pi} \tag{5.38}$$

It is evident that $\iota\!\!\!- \in \mathbb{Q}$ on rational surfaces and that $\iota\!\!\!- \in \mathbb{R} \setminus \mathbb{Q}$ on irrational surfaces. We can also introduce the safety factor commonly encountered in literature regarding tokamaks:

$$q = \frac{1}{\iota\!\!\!-} \tag{5.39}$$

In coordinate systems where magnetic field-lines are straight lines, the slope of these lines on a magnetic surface is given by:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\zeta} = \iota\!\!\!- \quad \text{or else} \quad \frac{\mathrm{d}\zeta}{\mathrm{d}\theta} = q \tag{5.40}$$

The rotational transform can be related to the poloidal and toroidal components of the magnetic field and to their respective fluxes. The expression is given in equation 5.57, since a complete demonstration requires understanding flux coordinates. In general plasma literature, stellarators additionally use $\varrho \equiv \iota\!\!\!-$, $\varrho \equiv -q$ along with $\varrho \equiv -p$ and $\varrho \equiv -\Phi^{\mathrm{d}}_{\mathrm{pol}}$ as the first flux coordinate, while tokamaks usually use $\varrho \equiv q$ and $\varrho \equiv -\iota\!\!\!-$. In any case, it is only appropriate to choose a radially increasing flux function as the first flux coordinate to ensure a direct coordinate system.

5.4.3 Magnetic axis

A magnetic surface parametrised by a flux quantity is defined by the equation $\mathbf{B} \cdot \nabla \varrho = 0$. Degenerate magnetic surfaces correspond precisely to the magnetic axes defined by $\nabla \varrho = \mathbf{0}$. The preceding equation means that there is no singularity of the gradient of the considered flux quantity when the volume of the surface tends towards zero. The physical meaning is obvious. When $V \to 0$, $\Phi \propto r^2$, where r represents the distance to the centre of the magnetic axis. Therefore, $\nabla \Phi \propto r$ when $V \to 0$ so that $\nabla \Phi \to \mathbf{0}$. For any other flux quantity, provided that $\mathrm{d}\varrho / \mathrm{d}\Phi$ is non-singular at the magnetic axis, one must use:

$$\nabla \varrho(\Phi) = \frac{\mathrm{d}\varrho}{\mathrm{d}\Phi} \nabla \Phi \tag{5.41}$$

5.4.4 Labelling magnetic surfaces in open systems

For magnetic mirrors, no additional remark on the volumes enclosed by magnetic surfaces is necessary. In open systems, there exist two magnetic field fluxes, namely the axial flux Φ_{ax} and the azimuthal flux Φ_{az}. Pressure is not as helpful in this case because plasma phenomena are often anisotropic with respect to directions along and across a magnetic field-line.

5.5 Flux coordinate systems

The following section is entirely dedicated to the theoretical development of flux coordinate systems. These coordinate systems are established in such a manner that magnetic field-lines are *straight* lines when expressed in terms of flux coordinates. Magnetic field-lines are evidently not straight lines in most cases; it is only that the field-line equation simplifies to that of a straight line in these coordinates.

5.5.1 Boozer flux coordinate system

We return to the contravariant expression of a magnetic field (5.22). By a previous result, we eliminated the component along ϱ, thus:

$$\mathbf{B} = B^\theta \frac{\partial \mathbf{r}}{\partial \theta} + B^\zeta \frac{\partial \mathbf{r}}{\partial \zeta} = B^\theta J \nabla\zeta \times \nabla\varrho + B^\zeta J \nabla\varrho \times \nabla\theta \tag{5.42}$$

The divergence of the magnetic field gives us a link between the two components:

$$\nabla \cdot \mathbf{B} = \frac{\partial}{\partial \theta}(JB^\theta) + \frac{\partial}{\partial \zeta}(JB^\zeta) = 0 \tag{5.43}$$

This being an implicit equation linking the two components, there exists a function $\nu(\varrho, \theta, \zeta) \in \mathcal{C}^2$:

$$B^\theta = -\frac{1}{J}\frac{\partial \nu}{\partial \zeta} \quad \text{and} \quad B^\zeta = \frac{1}{J}\frac{\partial \nu}{\partial \theta} \implies -\frac{\partial^2 \nu}{\partial \zeta \partial \theta} + \frac{\partial^2 \nu}{\partial \theta \partial \zeta} = 0 \tag{5.44}$$

We can now introduce the following expression of the magnetic field:

$$\mathbf{B} = \nabla\varrho \times \left(\frac{\partial \nu}{\partial \theta}\nabla\theta + \frac{\partial \nu}{\partial \zeta}\nabla\zeta\right) \tag{5.45}$$

Since $\nabla\varrho \times \nabla\varrho = \mathbf{0}$, we find a new expression often called *Clebsch form*:

$$\mathbf{B} = \nabla\varrho \times \nabla\nu \implies \mathbf{B} \cdot \nabla\nu = 0 \tag{5.46}$$

Certain magnetic field-lines therefore belong to the surfaces of $\nu = \text{const.}$ Since surfaces $\varrho = \text{const.}$ also contain field-lines, we thus obtain an implicit equation describing a field-line for a chosen ϱ:

$$\nu(\varrho = \text{const}, \theta, \zeta) = \text{const.} \tag{5.47}$$

Since the magnetic field is a single-valued function for all (ϱ, θ, ζ), the same applies to $\nabla\varrho \times \nabla\nu$. This indeed imposes certain constraints on the form of the function $\nu(\varrho, \theta, \zeta)$. It is then necessary that $\nabla\varrho\times\nabla\nu$ be periodic in θ and in ζ. Otherwise, $\mathbf{B}$ would not be defined as single-valued. It is evident that the general form of ν is given as follows:

$$\nu(\varrho, \theta, \zeta) = P(\varrho)\theta + Q(\varrho)\zeta + \tilde{\nu}(\varrho, \theta, \zeta) \tag{5.48}$$

where $\tilde{\nu}(\varrho, \theta, \zeta)$ is a function purely periodic in θ and ζ. Returning to equation (5.35):

$$\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} = \frac{1}{2\pi}\frac{\mathrm{d}}{\mathrm{d}\varrho}\iiint_V J\mathbf{B}\cdot\nabla\zeta \,\mathrm{d}\varrho\,\mathrm{d}\theta\,\mathrm{d}\zeta = \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} J\mathbf{B}\cdot\nabla\zeta\,\mathrm{d}\theta\,\mathrm{d}\zeta =$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} JB^\zeta\,\mathrm{d}\theta\,\mathrm{d}\zeta = \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi}\frac{\partial \nu}{\partial \theta}\,\mathrm{d}\theta\,\mathrm{d}\zeta = \left\{\frac{\partial \nu}{\partial \theta} = P(\varrho) + \frac{\partial \tilde{\nu}}{\partial \theta}\right\} =$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \left(P(\varrho) + \frac{\partial \tilde{\nu}}{\partial \theta} \right) \mathrm{d}\theta \, \mathrm{d}\zeta = 2\pi P(\varrho) + \underbrace{\tilde{\nu}\Big|_0^{2\pi}}_{0} \implies \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} = 2\pi P(\varrho) \tag{5.49}$$

An entirely analogous approach gives us the derivative of the poloidal flux with respect to ϱ:

$$\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} = -2\pi Q(\varrho) \tag{5.50}$$

We return to the expression of the auxiliary function $\nu(\varrho, \theta, \zeta)$:

$$\nu(\varrho, \theta, \zeta) = \frac{1}{2\pi} \left(\theta \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} - \zeta \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \right) + \tilde{\nu}(\varrho, \theta, \zeta) \tag{5.51}$$

In the case where $\tilde{\nu}$ is zero or a flux function, the coordinates θ and ζ introduced at the beginning are already flux coordinates. Recall that the magnetic field-lines lie on the surfaces of $\nu = \text{const}$. Thus, the previous equation provides us with an equation of a field-line as a straight line because the derivatives of the fluxes are flux functions:

$$\theta \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} - \zeta \frac{\mathrm{d}\Phi^{r}_{\mathrm{pol}}}{\mathrm{d}\varrho} = \text{const.} \tag{5.52}$$

We will henceforth pursue the following notation: $\theta = \theta_{\mathrm{f}}$ and $\zeta = \zeta_{\mathrm{f}}$. If the function $\tilde{\nu}$ is not zero or a flux function, it suffices to introduce a change of variables:

$$\theta_{\mathrm{f}} = \theta + 2\pi\tilde{\nu} \Bigg/ \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \quad \text{and} \quad \zeta_{\mathrm{f}} = \zeta \qquad \text{or else} \qquad \theta_{\mathrm{f}} = \theta \quad \text{and} \quad \zeta_{\mathrm{f}} = \zeta - 2\pi\tilde{\nu} \Bigg/ \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \tag{5.53}$$

This allows us to write the auxiliary function ν as follows:

$$\nu(\varrho, \theta_{\mathrm{f}}, \zeta_{\mathrm{f}}) = \frac{\theta_{\mathrm{f}}}{2\pi} \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} - \frac{\zeta_{\mathrm{f}}}{2\pi} \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \tag{5.54}$$

The components of the magnetic field are thus given in an exquisite form:

$$\mathbf{B} = \frac{1}{2\pi J_{\mathrm{f}}} \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \nabla\theta_{\mathrm{f}} + \frac{1}{2\pi J_{\mathrm{f}}} \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \nabla\zeta_{\mathrm{f}} \tag{5.55}$$

We can now return to the expression of the rotational transform:

$$\iota\!\!\!- = \frac{\mathrm{d}\theta_{\mathrm{f}}}{\mathrm{d}\zeta_{\mathrm{f}}} = \frac{\mathbf{B} \cdot \nabla\theta_{\mathrm{f}}}{\mathbf{B} \cdot \nabla\zeta_{\mathrm{f}}} = \frac{B^{\theta_{\mathrm{f}}}}{B^{\zeta_{\mathrm{f}}}} = \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \Bigg/ \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} = \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\Phi_{\mathrm{tor}}} \tag{5.56}$$

When considering magnetic fields present in tokamaks, it is common to retain the symmetry angle $\zeta = \zeta_{\mathrm{f}}$ and, consequently, to change the poloidal coordinate. We modify expression (6.54) using the rotational transform to obtain a simple contravariant expression of the magnetic field:

$$\mathbf{B} = \frac{1}{2\pi J_{\mathrm{f}}} \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \left(\iota\!\!\!- \nabla\theta_{\mathrm{f}} + \nabla\zeta_{\mathrm{f}} \right) \tag{5.57}$$

Boozer coordinates are defined by fixing the components $B_{\theta_{\mathrm{f}}}$ and $B_{\zeta_{\mathrm{f}}}$ of the magnetic field on magnetic surfaces. They are precisely given by equation (5.57). We could introduce a new variable $\alpha_{\mathrm{f}} = \iota\!\!\!- \theta_{\mathrm{f}} + \zeta_{\mathrm{f}}$ to simplify the previous expression:

$$\mathbf{B} = \frac{1}{2\pi J_{\mathrm{f}}} \frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \nabla\alpha_{\mathrm{f}} \tag{5.58}$$

We seek the expression of the current $\mathbf{J}$ in these coordinates. The condition of ideal magnetohydrodynamic equilibrium gives us:

$$\nabla p = \mathbf{J} \times \mathbf{B} \implies \mathbf{J} \cdot \nabla p = 0 \implies \mathbf{J} \perp \nabla p \tag{5.59}$$

Since pressure is a flux function, this suggests that the expression of the current turns out to be of the following form:

$$\mathbf{J} = J^{\theta} J \nabla\varrho \times \nabla\theta + J^{\zeta} J \nabla\varrho \times \nabla\zeta \tag{5.60}$$

We evoke the hypothesis of quasi-neutrality and apply it to the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0 \overset{\rho=0}{\Longrightarrow} \nabla \cdot \mathbf{J} = \frac{\partial}{\partial \theta}(JJ^\theta) + \frac{\partial}{\partial \zeta}(JJ^\zeta) = 0 \tag{5.61}$$

This gives us the same conditions as for the magnetic field, namely equations (5.42) and (5.43). There exists then a function $\mu(\varrho, \theta, \zeta)$ such that:

$$J^\theta = -\frac{1}{J}\frac{\partial \mu}{\partial \zeta} \quad \text{and} \quad J^\zeta = \frac{1}{J}\frac{\partial \mu}{\partial \zeta} \tag{5.62}$$

The expression of the current thus takes the familiar form like expression (5.45):

$$\mathbf{J} = \nabla \varrho \times \left(\frac{\partial \mu}{\partial \theta} \nabla \theta + \frac{\partial \mu}{\partial \zeta} \nabla \zeta \right) \tag{5.63}$$

By a previous reasoning, specifically the one that allowed us to obtain equation (5.48), we write:

$$\mu(\varrho, \theta, \zeta) = R(\varrho)\theta + S(\varrho)\zeta + \tilde{\mu}(\varrho, \theta, \zeta) \tag{5.64}$$

where $\tilde{\mu}(\varrho, \theta, \zeta)$ is a periodic function in θ and ζ. The Maxwell-Ampère equation allows us to link functions R, S and $\tilde{\mu}$ to those of the magnetic field, namely $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. However, we omit this part, as it consists only of linking the corresponding components. We then return to the expression of the magnetic field (5.22) and search for its covariant components. We return to expression (5.42):

$$\mathbf{B} = \frac{\nabla \varrho}{2\pi} \times \nabla \left(\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \theta_{\mathrm{f}} - \frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \zeta_{\mathrm{f}} \right) = \frac{\nabla \varrho}{2\pi} \times \nabla \left(\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \theta_{\mathrm{f}} + \frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \zeta_{\mathrm{f}} \right) \tag{5.65}$$

In tokamaks, it is customary to use the safety factor as well as the poloidal flux $\Phi^{\mathrm{r}}_{\mathrm{pol}}$ as the flux label:

$$\mathbf{B} = \frac{\nabla \Phi^{\mathrm{r}}_{\mathrm{pol}}}{2\pi} \times \nabla \left(q\theta_{\mathrm{f}} - \zeta_{\mathrm{f}} \right) \tag{5.66}$$

In contrast, stellarator literature uses toroidal flux as the flux label as well as the rotational transform:

$$\mathbf{B} = \frac{\nabla \Phi_{\mathrm{tor}}}{2\pi} \times \nabla \left(\theta_{\mathrm{f}} - \iota\zeta_{\mathrm{f}} \right) \tag{5.67}$$

The general Clebsch form of the magnetic field can be written as a sum of two components:

$$\mathbf{B} = \frac{1}{2\pi}\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \nabla \zeta_{\mathrm{f}} \times \nabla \varrho + \frac{1}{2\pi}\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho} \nabla \varrho \times \nabla \theta_{\mathrm{f}} = \mathbf{B}_{\mathrm{pol}} + \mathbf{B}_{\mathrm{tor}} \tag{5.68}$$

where $\mathbf{B}_{\mathrm{pol}}$ and $\mathbf{B}_{\mathrm{tor}}$ denote the poloidal and toroidal vectorial components of the magnetic field. It is important to emphasize that $\mathbf{B}_{\mathrm{pol}}$ is parallel to $\hat{\mathbf{e}}_{\theta_{\mathrm{f}}}$, but $\mathbf{B}_{\mathrm{pol}}$ is not parallel to $\nabla \theta_{\mathrm{f}}$. The same applies to $\mathbf{B}_{\mathrm{tor}}$. To find the covariant components, it is appropriate to return to equation (5.65):

$$J_{\mathrm{f}} = \frac{1}{\nabla \theta_{\mathrm{f}} \cdot (\nabla \zeta_{\mathrm{f}} \times \nabla \varrho)} = \frac{1}{2\pi}\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho}\frac{1}{\mathbf{B} \cdot \nabla \theta_{\mathrm{f}}} = \frac{1}{2\pi B^{\theta_{\mathrm{f}}}}\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho} \tag{5.69}$$

We can easily find these components using the metric tensor of the flux coordinate system:

$$B_i = (g_{ij})_{\mathrm{f}} B^j = g_{i\theta_{\mathrm{f}}} B^{\theta_{\mathrm{f}}} + g_{i\zeta_{\mathrm{f}}} B^{\zeta_{\mathrm{f}}} \tag{5.70}$$

In tokamaks, it is customary to choose $\Psi = \Phi^{\mathrm{r}}_{\mathrm{pol}}/2\pi$ as the first flux coordinate, so that the Jacobian is then simply $J_{\mathrm{f}} = 1/B^{\theta_{\mathrm{f}}} = 1/(\mathbf{B} \cdot \nabla \theta_{\mathrm{f}})$.

5.5.2 Flux-surface average

The flux-surface average of some function $f(\mathbf{r})$ is defined by the volume average over an infinitesimally small shell of volume ΔV between two magnetic surfaces of volumes V and $V + \Delta V$, thus:

$$\langle f(\mathbf{r}) \rangle = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \iiint_{\Delta V} f(\mathbf{r})\, \mathrm{d}^3\mathbf{r} \tag{5.71}$$

The elementary volume element $\mathrm{d}^3\mathbf{r}$ is, by definition, equal to $\mathrm{d}^3\mathbf{r} = |\mathrm{d}\mathbf{r}(\varrho) \cdot \mathrm{d}\mathbf{S}(\varrho)|$, where $\mathrm{d}\mathbf{r}(\varrho)$ is the infinitesimal change in the position along the flux coordinate ϱ. In contrast, $\mathrm{d}\mathbf{S}(\varrho)$ is the oriented infinitesimal surface vector; thus, with adequate orientation, we may drop the absolute value brackets. Because the differential surface element is tangential to the considered magnetic surface, we find:

$$\mathrm{d}\mathbf{S}(\varrho) = \frac{\nabla\varrho}{\|\nabla\varrho\|}\,\mathrm{d}S(\varrho) \implies \mathrm{d}^3\mathbf{r} = \mathrm{d}\mathbf{r}(\varrho)\cdot\frac{\nabla\varrho}{\|\nabla\varrho\|}\,\mathrm{d}S(\varrho) \equiv \frac{\mathrm{d}\varrho\,\mathrm{d}S(\varrho)}{\|\nabla\varrho\|} \tag{5.72}$$

This, in turn, results in the following equivalence:

$$\langle f(\mathbf{r})\rangle = \frac{\mathrm{d}\varrho}{\mathrm{d}V}\iint_{S(\varrho)}\frac{f(\mathbf{r})}{\|\nabla\varrho\|}\,\mathrm{d}S(\varrho) \tag{5.73}$$

Of course, simplifications arise when choosing the volume enclosed by magnetic surfaces as the flux coordinate. In arbitrary generalised toroidal coordinates (ϱ,θ,ζ), where the Jacobian is given by $J = [\nabla\varrho\cdot(\nabla\theta\times\nabla\zeta)]^{-1}$, the infinitesimal surface element is given by $\mathrm{d}S = J\|\nabla\varrho\|\,\mathrm{d}\theta\,\mathrm{d}\zeta$. The flux-surface average, which is now evaluated over a closed magnetic surface, simplifies to:

$$\langle f(\mathbf{r})\rangle = \frac{\mathrm{d}\varrho}{\mathrm{d}V}\oiint_{S(\varrho)} J f(\mathbf{r})\,\mathrm{d}\theta\,\mathrm{d}\zeta \tag{5.74}$$

Notice that setting $f(\mathbf{r}) = 1$ gives:

$$\frac{\mathrm{d}V}{\mathrm{d}\varrho} = \int_0^{2\pi}\int_0^{2\pi} J\,\mathrm{d}\theta\,\mathrm{d}\zeta \tag{5.75}$$

which reduces to unity for the choice $\varrho = V$.

5.5.2.1 Flux-surface average in axisymmetric systems

In axisymmetric systems, the flux-surface average further simplifies because there is no ζ-dependence, so that the integral over ζ contributes a multiplicative factor of 2π to both the numerator and the denominator in equation (5.71); thus, integration over θ suffices.

Furthermore, in axisymmetric systems, the flux-surface average is sometimes defined as the closed loop integral along a magnetic field-line as follows:

$$\langle f(\mathbf{r})\rangle = \frac{\oint f(\mathbf{r})\frac{\mathrm{d}\ell}{\|\mathbf{B}\|}}{\oint\frac{\mathrm{d}\ell}{\|\mathbf{B}\|}} \tag{5.76}$$

To show that the two definitions are equivalent, we shall resort to a Clebsch coordinate system where magnetic field-lines are straight lines formed by the intersection of magnetic surfaces given by $\varrho = \text{const.}$ and of surfaces $q\theta - \zeta = \text{const.}$ where q is the safety factor. The magnetic field-line, that is, $\mathbf{B}$, is thus locally determined as follows:

$$\mathbf{B} \propto \nabla\varrho\times\nabla(q\theta-\zeta) \tag{5.77}$$

where the proportionality constant shall be determined below. Taking the dot product with $\nabla\theta$:

$$\mathbf{B}\cdot\nabla\theta \propto [\nabla\varrho\times\nabla(q\theta) + \nabla\zeta\times\nabla\varrho]\cdot\nabla\theta = \left[\nabla\varrho\times\frac{\mathrm{d}q}{\mathrm{d}\varrho}\theta\nabla\varrho + q\nabla\varrho\times\nabla\theta + \nabla\varrho\cdot(\nabla\theta\times\nabla\zeta)\right]\cdot\nabla\theta \tag{5.78}$$

The first term vanishes since $\nabla\varrho\times\nabla\varrho = 0$, the second term vanishes since $\nabla\varrho\times\nabla\theta\perp\nabla\theta$, while the residual term is simply the Jacobian. The equation of a magnetic field-line by components gives:

$$\frac{\mathrm{d}\ell}{\|\mathbf{B}\|} = \frac{\mathrm{d}\theta}{\mathbf{B}\cdot\nabla\theta} \tag{5.79}$$

We thus see that, in axisymmetric devices, the integrands of the two definitions of the flux-surface average are equivalent up to a proportionality factor, which we shall determine as follows. Recall equation (5.66). Since $\Phi^{\mathrm{r}}_{\mathrm{pol}}$ is a flux function, it depends only on the flux coordinate; therefore, applying the chain rule for derivatives gives:

$$\mathbf{B}(\varrho,\theta,\zeta) = \frac{1}{2\pi}\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho}\nabla\varrho\times\nabla(q\theta-\zeta) \tag{5.80}$$

The proportionality factor is thus a flux function and does not contribute to the integral over θ, as it cancels out in the numerator and the denominator. The two definitions are thus equivalent in axisymmetric systems. Notice that the $\oint \frac{\mathrm{d}l}{\|\mathbf{B}\|}$ flux-surface average being equivalent to the flux-surface average over a single period of θ, it is not necessary to evaluate the integral over a closed loop, but over the arc spanning a period of θ. In the case of distinguishing rational from irrational magnetic surfaces, one evidently cannot evaluate the complete closed-loop integral on irrational surfaces. Our previous discussions of the restrictions imposed by rational and irrational surfaces, respectively, remain valid.

5.5.3 Magnetic differential equation

We refer to equations of the following type as magnetic differential equations:

$$\mathbf{B} \cdot \nabla f = S \tag{5.81}$$

where S is a source term, and f is an unknown single-valued function as the sought-after solution of the differential equation, both assumed to be scalar fields, $f = f(\mathbf{r})$, $S = S(\mathbf{r})$. Our subsequent discussions of the magnetic differential equations will primarily concern toroidal systems, as open systems impose fewer restrictions on their properties.

1. The single-valuedness of f determines a first condition on S:

$$\mathbf{B} \cdot \nabla f = \|\mathbf{B}\| \frac{\partial f}{\partial \ell} = S \implies f(\ell) = f(0) + \int_0^\ell \frac{S}{\|\mathbf{B}\|}\, \mathrm{d}\ell \tag{5.82}$$

   The solution f is assumed to be single-valued, so the source term must obey the condition:

$$\oint S \frac{\mathrm{d}\ell}{\|\mathbf{B}\|} = 0 \tag{5.83}$$

   The closed-loop integral is, naturally, taken on rational magnetic surfaces. The condition above must be satisfied for every closed field-line, automatically belonging to some rational surface.

2. A second condition is that the flux-surface average of the equation is zero:

$$\langle S \rangle = \frac{\oint S \frac{\mathrm{d}\ell}{\|\mathbf{B}\|}}{\oint \frac{\mathrm{d}\ell}{\|\mathbf{B}\|}} = \frac{\mathrm{d}\varrho}{\mathrm{d}V} \oiint_{\Sigma(\varrho)} JS \, \mathrm{d}\theta \, \mathrm{d}\zeta = 0 \tag{5.84}$$

These two conditions are, in principle, *satisfiably* dependent, but also independent. The former implies the latter only if rational magnetic surfaces form a dense set in the system under consideration. The second condition is necessary to account for regularity on irrational magnetic surfaces if the rational surfaces did in fact not form a dense set. The regularity of the magnetic differential equation requires both conditions, since the second condition does not imply the first, while the first is not well-defined on irrational magnetic surfaces.

The general solution of the magnetic differential equation is thus given as a sum of the particular solution, f_p, and a homogeneous solution, f_h, which, in particular, satisfies:

$$\mathbf{B} \cdot \nabla f_\mathrm{h} = 0 \implies \frac{\partial f_\mathrm{h}}{\partial \ell} = 0 \tag{5.85}$$

Thus, the homogeneous solution is a flux function, but only in toroidal systems. To find the particular solution, we resort to rewriting the equation in terms of Boozer coordinates, $(\varrho, \theta_\mathrm{f}, \zeta_\mathrm{f})$, which is particularly motivated by the fact that magnetic field-lines are straight in these coordinate systems. The equation is thus equivalent to:

$$B^i \frac{\partial f_\mathrm{p}}{\partial u^i} = S \implies \underbrace{B^\varrho}_{0} \frac{\partial f_\mathrm{p}}{\partial \varrho} + B^{\theta_\mathrm{f}} \frac{\partial f_\mathrm{p}}{\partial \theta_\mathrm{f}} + B^{\zeta_\mathrm{f}} \frac{\partial f_\mathrm{p}}{\partial \zeta_\mathrm{f}} = S \tag{5.86}$$

Notice $B^\varrho = 0$. The other contravariant components of $\mathbf{B}$ contain the Jacobian in the denominator. Thus, the differential equation can be Fourier-transformed over θ_f and ζ_f. Let $a_{mn}(\varrho)$ be the expansion coefficients in the Fourier series of $J_\mathrm{f} S$. The flux-average integral condition requires that $a_{00}(\varrho) = 0$ on all flux surfaces. The first condition, however, fixes $\sum_{m,n} a_{mn}(\varrho \in \mathbb{Q})/(m\iota\!\!\!- - n) = 0$, so that $a_{mn}(\varrho \in \mathbb{Q})$

must vanish on their respective rational surfaces where $m\iota\!\!\!- - n = 0$. A swift conclusion is that f can be reconstructed from its harmonics f_{mn} pondered by coefficients $\tilde{a}_{mn}(\varrho) = a_{mn}(\varrho)/(m\iota\!\!\!- - n)$ along with a constant factor independent of m or n. Lastly, it shall be worth noting that in tokamaks, f and S are generally independent of ζ, so that their Fourier expansions are executed only with respect to θ.

5.5.4 Ideal-MHD-equilibrium conditions for toroidal systems

This subsection is dedicated to further examine the periodic function $\tilde{\mu}(\varrho, \theta, \zeta)$ in equation (5.64). We begin by inputting the contravariant expressions of the current density and magnetic field, namely expressions given in equations (6.54) and (5.140). Note that the latter can be readily derived from our previous considerations; it is only convenient to use a result we will obtain later.

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho}\nabla\varrho = J^{\theta_\mathrm{f}} B^{\zeta_\mathrm{f}} \mathbf{e}_{\theta_\mathrm{f}} \times \mathbf{e}_{\zeta_\mathrm{f}} + J^{\zeta_\mathrm{f}} B^{\theta_\mathrm{f}} \mathbf{e}_{\zeta_\mathrm{f}} \times \mathbf{e}_{\theta_\mathrm{f}} \iff \frac{\mathrm{d}p}{\mathrm{d}\varrho} = J_\mathrm{f}\left(J^{\theta_\mathrm{f}} B^{\zeta_\mathrm{f}} - J^{\zeta_\mathrm{f}} B^{\theta_\mathrm{f}}\right) \tag{5.87}$$

The other components of the ideal magnetohydrodynamic equilibrium equation are trivially zero. The contravariant components are given as follows:

$$B^i = \frac{1}{2\pi J_\mathrm{f}}\begin{pmatrix} 0 \\ \dfrac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} \\ \dfrac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} \end{pmatrix} \quad \text{and} \quad J^\mathrm{i} = \frac{1}{2\pi J_\mathrm{f}}\begin{pmatrix} 0 \\ \dfrac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} - 2\pi\dfrac{\partial\tilde{\mu}}{\partial\zeta_\mathrm{f}} \\ \dfrac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho} + 2\pi\dfrac{\partial\tilde{\mu}}{\partial\theta_\mathrm{f}} \end{pmatrix} \tag{5.88}$$

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho} = \frac{1}{4\pi^2 J_\mathrm{f}}\left(-\frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} + \frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} - 2\pi\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\theta_\mathrm{f}} - 2\pi\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\zeta_\mathrm{f}}\right) \tag{5.89}$$

Notice that the last two terms in the brackets correspond to the dot product $4\pi^2 J_\mathrm{f}\mathbf{B}\cdot\nabla\tilde{\mu}$, giving:

$$\frac{1}{4\pi^2 J_\mathrm{f}}\left(-\frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} + \frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}\right) - \mathbf{B}\cdot\nabla\tilde{\mu} = \frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{5.90}$$

Notice that this is, in fact, a magnetic differential equation for $\tilde{\mu}$. Rearranging:

$$\mathbf{B}\cdot\nabla\tilde{\mu} = \frac{1}{4\pi^2 J_\mathrm{f}}\left(-\frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} + \frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}\right) - \frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{5.91}$$

Thus, the source term must satisfy the two conditions discussed in subsection 5.5.3:

$$\mathrm{d}\varrho\int_0^{2\pi}\int_0^{2\pi}\left[\frac{1}{4\pi^2 J_\mathrm{f}}\left(-\frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} + \frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}\right) - \frac{\mathrm{d}p}{\mathrm{d}\varrho}\right] J_\mathrm{f}\,\mathrm{d}\theta_\mathrm{f}\,\mathrm{d}\zeta_\mathrm{f} = 0 \tag{5.92}$$

All quantities in the integrand are flux functions:

$$\frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} - \frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho} = \frac{\mathrm{d}p}{\mathrm{d}\varrho}\int_0^{2\pi}\int_0^{2\pi} J_\mathrm{f}\,\mathrm{d}\theta_\mathrm{f}\,\mathrm{d}\zeta_\mathrm{f} = \frac{\mathrm{d}p}{\mathrm{d}\varrho}\frac{\mathrm{d}V}{\mathrm{d}\varrho} \tag{5.93}$$

The other solvability condition is that the closed-loop integral along a magnetic field-line vanishes. Note that we shall employ the equation of a magnetic field-line, namely:

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho}\oint\frac{\mathrm{d}\ell}{\|\mathbf{B}\|} = \left\{\frac{\mathrm{d}\ell}{\|\mathbf{B}\|} = \frac{\mathrm{d}\zeta_\mathrm{f}}{B^{\zeta_\mathrm{f}}}, B^{\zeta_\mathrm{f}} = \frac{1}{2\pi J_\mathrm{f}}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}\right\} = \oint\left(\frac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} - \frac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\frac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\right)\frac{\mathrm{d}\zeta_\mathrm{f}}{B^{\zeta_\mathrm{f}}} = \tag{5.94}$$

$$= \frac{\dfrac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\dfrac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} - \dfrac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\dfrac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}}{2\pi\dfrac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}}\int_0^{2N\pi}\mathrm{d}\zeta_\mathrm{f} = \frac{\dfrac{\mathrm{d}I^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}\dfrac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho} - \dfrac{\mathrm{d}I_\mathrm{tor}}{\mathrm{d}\varrho}\dfrac{\mathrm{d}\Phi^\mathrm{r}_\mathrm{pol}}{\mathrm{d}\varrho}}{\dfrac{\mathrm{d}\Phi_\mathrm{tor}}{\mathrm{d}\varrho}}N \tag{5.95}$$

where N denotes the number of toroidal transits necessary for the magnetic field-line to close upon itself. Using the result given by equation (5.93), we find:

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho}\oint\frac{\mathrm{d}\ell}{\|\mathbf{B}\|}=N\frac{\mathrm{d}V}{\mathrm{d}\varrho}\frac{\mathrm{d}p}{\mathrm{d}\varrho}\Big/\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho}\iff\frac{\mathrm{d}p}{\mathrm{d}\varrho}\left(\frac{\mathrm{d}V}{\mathrm{d}\Phi_{\mathrm{tor}}}-\frac{1}{N}\oint\frac{\mathrm{d}\ell}{\|\mathbf{B}\|}\right)=0 \tag{5.96}$$

Note that the loop integral in the equation above is commonly referred to as the *proper length* of a magnetic field-line. For magnetic surfaces where $\frac{\mathrm{d}p}{\mathrm{d}\varrho}\neq 0$, we find that the loop integral in the previous expression must be the same for all magnetic field-lines belonging to the same magnetic surface. Thus, on a common magnetic surface, all magnetic field-lines must have the same proper length. This condition is often referred to as the *current-closure condition* in the plasma physics literature. If it is satisfied, the system is said to possess true magnetic surfaces. In all axisymmetric systems, e.g., tokamaks, this condition is, indeed, satisfied. In non-axisymmetric systems, like stellarators, it is not. In a more advanced context, when the current closure condition is violated, the Fourier coefficients of $\tilde{\mu}$ will exhibit resonant denominators of the form $m\iota\!\!\!-n$ leading to divergent currents and magnetic island formation.

The magnetic differential equation for $\tilde{\mu}$ is thus found in a simpler form:

$$\mathbf{B}\cdot\nabla\tilde{\mu}=\frac{\mathrm{d}p}{\mathrm{d}\varrho}\left(\frac{1}{4\pi^2J_{\mathrm{f}}}\frac{\mathrm{d}V}{\mathrm{d}\varrho}-1\right)\iff\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\theta_{\mathrm{f}}}+\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\zeta_{\mathrm{f}}}=\frac{1}{2\pi}\frac{\mathrm{d}p}{\mathrm{d}\varrho}\left(\frac{\mathrm{d}V}{\mathrm{d}\varrho}-4\pi^2J_{\mathrm{f}}\right) \tag{5.97}$$

We return our attention to the flux-surface average of the inverse Jacobian, $\mathcal{J}_{\mathrm{f}}=J_{\mathrm{f}}^{-1}$ that is:

$$\langle\mathcal{J}_{\mathrm{f}}\rangle=\frac{\int_0^{2\pi}\int_0^{2\pi}\mathrm{d}\theta_{\mathrm{f}}\,\mathrm{d}\zeta_{\mathrm{f}}}{\int_0^{2\pi}\int_0^{2\pi}J_{\mathrm{f}}\,\mathrm{d}\theta_{\mathrm{f}}\,\mathrm{d}\zeta_{\mathrm{f}}}=4\pi^2\Big/\frac{\mathrm{d}V}{\mathrm{d}\varrho}\implies\frac{\mathrm{d}V}{\mathrm{d}\varrho}=\frac{4\pi^2}{\langle\mathcal{J}_{\mathrm{f}}\rangle} \tag{5.98}$$

Consequently, the magnetic differential equation becomes:

$$\mathbf{B}\cdot\nabla\tilde{\mu}=\frac{\mathrm{d}p}{\mathrm{d}\varrho}\left(\frac{1}{\langle\mathcal{J}_{\mathrm{f}}\rangle}-J_{\mathrm{f}}\right)\iff\frac{\mathrm{d}\Phi^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\theta_{\mathrm{f}}}+\frac{\mathrm{d}\Phi_{\mathrm{tor}}}{\mathrm{d}\varrho}\frac{\partial\tilde{\mu}}{\partial\zeta_{\mathrm{f}}}=2\pi\frac{\mathrm{d}p}{\mathrm{d}\varrho}\left(\frac{1}{\langle\mathcal{J}_{\mathrm{f}}\rangle}-J_{\mathrm{f}}\right) \tag{5.99}$$

To solve for the unknown function $\tilde{\mu}(\varrho,\theta_{\mathrm{f}},\zeta_{\mathrm{f}})$, one must solve its magnetic differential equation given by (5.99), which is usually done using Fourier expansions of $\tilde{\mu}(\varrho,\theta_{\mathrm{f}},\zeta_{\mathrm{f}})$ and J_{f}.

5.5.5 Symmetric flux coordinates in a tokamak

In an ideal axisymmetric tokamak, it seems natural to choose the symmetric angle as the generalized toroidal coordinate. The surfaces $\zeta=\text{const.}$ are vertical planes and $\nabla\zeta$ is oriented in the direction of symmetry. We will name this angle $\zeta=\zeta_{\mathrm{o}}$ to highlight its symmetric properties. Toroidal symmetry allows us to conclude that flux surfaces are orthogonal to surfaces $\zeta_{\mathrm{o}}=\text{const}$, that is $\nabla\Phi\cdot\nabla\zeta_{\mathrm{o}}=0$. We want to use the symmetry coordinate ζ_{o} to make the most of the symmetry. It is necessary to construct surfaces $\theta_{\mathrm{o}}=\text{const.}$ so that $\nabla\theta_{\mathrm{o}}\cdot\nabla\zeta_{\mathrm{o}}=0$. These two conditions are equivalent to the curves of coordinate ζ_{o} being circles in the toroidal plane.

In general, we cannot say $\nabla\varrho\cdot\nabla\theta_{\mathrm{o}}=0$, because toroidal symmetry does not allow imposing such a condition. We return to expression (5.66) of the magnetic field:

$$\mathbf{B}=\frac{q}{2\pi}\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}\times\nabla\theta_{\mathrm{o}}+\frac{1}{2\pi}\nabla\zeta_{\mathrm{o}}\times\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}} \tag{5.100}$$

Since $\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}\cdot\nabla\zeta_0=0=\nabla\theta_{\mathrm{o}}\cdot\nabla\zeta_{\mathrm{o}}$, then $\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}\times\nabla\theta_{\mathrm{o}}\parallel\nabla\zeta_{\mathrm{o}}$, hence $\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}\times\nabla\theta_{\mathrm{o}}=\kappa\nabla\zeta_{\mathrm{o}}$:

$$\mathbf{B}=\frac{q\kappa}{2\pi}\nabla\zeta_{\mathrm{o}}+\frac{1}{2\pi}\nabla\zeta_{\mathrm{o}}\times\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}} \tag{5.101}$$

Equation (5.24) gives us the value of κ:

$$\kappa=\frac{\nabla\zeta_{\mathrm{o}}\cdot(\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}\times\nabla\theta_{\mathrm{o}})}{\|\nabla\zeta_{\mathrm{o}}\|^2}=\frac{1}{J_{\mathrm{o}}\|\nabla\zeta_{\mathrm{o}}\|^2} \tag{5.102}$$

We can introduce the radius of curvature R of a curve of the coordinate ζ_{o} as $\|\nabla\zeta_{\mathrm{o}}\|=1/R$:

$$\kappa=\frac{R^2}{J_{\mathrm{o}}}\implies\mathbf{B}_{\mathrm{tor}}=\frac{qR^2}{2\pi J_{\mathrm{o}}}\nabla\zeta_{\mathrm{o}}=\frac{I}{2\pi}\nabla\zeta_{\mathrm{o}} \tag{5.103}$$

where we have defined a function I. An important property of this function is that it is a flux function, i.e. $I=I(\Phi^{\mathrm{r}}_{\mathrm{pol}})$. We begin the demonstration with the Maxwell-Ampère equation:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \implies \mu_0 J^i = \varepsilon^{ijk}\partial_j B_k \implies \mu_0 J^{\Phi^{\mathrm{r}}_{\mathrm{pol}}} = \frac{1}{J_{\mathrm{o}}}\left(\frac{\partial B_{\zeta_{\mathrm{o}}}}{\partial \theta_{\mathrm{o}}} - \frac{\partial B_{\theta_{\mathrm{o}}}}{\partial \zeta_{\mathrm{o}}}\right) = \frac{1}{J_{\mathrm{o}}}\frac{\partial B_{\zeta_{\mathrm{o}}}}{\partial \theta_{\mathrm{o}}} \tag{5.104}$$

where the disappearing term vanishes due to toroidal symmetry. Recall that, from the point of view of ideal magnetohydrodynamics, pressure is a function of flux. Then, the ideal magnetohydrodynamic equilibrium equation imposes the following condition:

$$\mathbf{J} \times \mathbf{B} = \nabla p \implies \mathbf{J} \cdot \nabla p = 0 \implies \mathbf{J} \cdot \nabla \Phi^{\mathrm{r}}_{\mathrm{pol}} = J^{\Phi^{\mathrm{r}}_{\mathrm{pol}}} = 0 \implies \frac{\partial B_{\zeta_{\mathrm{o}}}}{\partial \theta_0} = 0 \tag{5.105}$$

Consequently, the toroidal component of the magnetic field does not depend on the poloidal angle θ_{o}. It is thus a flux function $B_{\zeta_{\mathrm{o}}} = B_{\zeta_{\mathrm{o}}}(\Phi^{\mathrm{r}}_{\mathrm{pol}})$. This concludes the proof.

$$B_{\zeta_{\mathrm{o}}} = \mathbf{B} \cdot \hat{\mathbf{e}}_{\zeta_{\mathrm{o}}} = \frac{q\kappa}{2\pi}\underbrace{\nabla \zeta_{\mathrm{o}} \cdot \hat{\mathbf{e}}_{\zeta_{\mathrm{o}}}}_{1} + \frac{1}{2\pi}\underbrace{\overbrace{(\nabla \zeta_{\mathrm{o}} \times \nabla \Phi^{\mathrm{r}}_{\mathrm{pol}})}^{\perp \hat{\mathbf{e}}_{\zeta_{\mathrm{o}}}} \cdot \hat{\mathbf{e}}_{\zeta_{\mathrm{o}}}}_{0} = \frac{I}{2\pi} \implies I = I(\Phi^{\mathrm{r}}_{\mathrm{pol}}) \tag{5.106}$$

It turns out to be opportune to evaluate the Jacobian of transformation between Cartesian coordinates and those of Boozer. The magnetic field vector can be written in covariant form:

$$\mathbf{B} = B_{\Phi^{\mathrm{r}}_{\mathrm{pol}}}\nabla \Phi^{\mathrm{r}}_{\mathrm{pol}} + B_{\theta_{\mathrm{o}}}\nabla \theta_{\mathrm{o}} + B_{\zeta_{\mathrm{o}}}\nabla \zeta_{\mathrm{o}} = \beta(\Phi^{\mathrm{r}}_{\mathrm{pol}}, \theta_{\mathrm{o}}, \zeta_{\mathrm{o}})\nabla \Phi^{\mathrm{r}}_{\mathrm{pol}} + I(\Phi^{\mathrm{r}}_{\mathrm{pol}})\nabla \theta_{\mathrm{o}} + G(\Phi^{\mathrm{r}}_{\mathrm{pol}})\nabla \zeta_{\mathrm{o}} \tag{5.107}$$

Taking the dot product of the previous equation with equation (5.100), it follows:

$$\mathbf{B}^2 = \frac{q}{2\pi}(\nabla \Phi^{\mathrm{r}}_{\mathrm{pol}} \times \nabla \theta_{\mathrm{o}}) \cdot B_{\zeta_{\mathrm{o}}}\nabla \zeta_{\mathrm{o}} + \frac{1}{2\pi}(\nabla \zeta_{\mathrm{o}} \times \nabla \Phi^{\mathrm{r}}_{\mathrm{pol}}) \cdot B_{\theta_{\mathrm{o}}}\nabla \theta_{\mathrm{o}} = \frac{1}{2\pi J}\left(G(\Phi^{\mathrm{r}}_{\mathrm{tor}}) + \iota I(\Phi^{\mathrm{r}}_{\mathrm{tor}})\right) \tag{5.108}$$

It is customary to define the first flux coordinate as $\Psi = \Phi^{\mathrm{r}}_{\mathrm{tor}}/2\pi$, then:

$$J = \frac{1}{\nabla \Psi \cdot (\nabla \theta_{\mathrm{o}} \times \nabla \zeta_{\mathrm{o}})} = \frac{G(\Psi) + \iota I(\Psi)}{\mathbf{B}^2} \tag{5.109}$$

Consequently, on a magnetic surface $\Psi = \text{const.}$, the Jacobian is a function of the magnetic field.

5.5.6 General Clebsch-type flux coordinate systems

We shall now formally develop specific useful properties of flux coordinate systems of Clebsch-type, in particular, retaking subsection 5.3.1.4, and specifically, equation (5.46). As it has not been discussed above, the auxiliary function $\nu(\varrho, \theta_{\mathrm{f}}, \zeta_{\mathrm{f}})$ is, in fact, a stream function. Notice also that the Clebsch form exhibits a specific property analogous to gauge invariance. Namely, let f be an arbitrary differentiable function of a single argument, then:

$$\nabla \varrho \times \nabla(\nu + f(\varrho)) = \nabla \varrho \times \nabla \nu + \nabla \varrho \times \frac{\mathrm{d}f}{\mathrm{d}\varrho}\nabla \varrho = \nabla \varrho \times \nabla \nu \tag{5.110}$$

and similarly for $\varrho \to \varrho + f(\nu)$. This property is only the consequence of the equivalence between the magnetic vector potential $\mathbf{A}$ and $\varrho \nabla \nu$:

$$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times (\varrho \nabla \nu) = \nabla \varrho \times \nabla \nu + \varrho \nabla \times \nabla \nu = \nabla \varrho \times \nabla \nu \tag{5.111}$$

where evidently the curl of the gradient is zero. The construction of Clebsch-type flux coordinate systems and their overall usefulness are fully discussed in subsection 5.3.1.4, and here, we also choose ℓ to be the third flux coordinate. Notice that the Jacobian in this coordinate system is precisely:

$$J = \frac{1}{\nabla \varrho \cdot (\nabla \nu \times \nabla \ell)} = \frac{1}{\nabla \ell \cdot (\nabla \varrho \times \nabla \nu)} = \frac{1}{\nabla \ell \cdot \mathbf{B}} = \frac{1}{\|\mathbf{B}\|} \tag{5.112}$$

since, by definition of a magnetic field-line, $\mathbf{B} \cdot \nabla \ell = \|\mathbf{B}\|$. It is now convenient to express the magnetic flux through some surface S as follows:

$$\iint_S \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \iint_S J\mathbf{B} \cdot \nabla \ell \, \mathrm{d}\varrho \, \mathrm{d}\nu = \iint_S J\|\mathbf{B}\| \, \mathrm{d}\varrho \, \mathrm{d}\nu = \iint_S \mathrm{d}\varrho \, \mathrm{d}\nu \tag{5.113}$$

Thus, Clebsch-type flux coordinate systems are only elegant when considering magnetic fluxes in particular throughout toroidal systems.

5.5.7 Boozer-Grad flux coordinate system

The idea of Boozer-Grad flux coordinate systems is to find a covariant representation for the magnetic field $\mathbf{B}$ such that it reduces to the gradient of a scalar function in current-free regions. Rather than taking the arc-length of a magnetic field-line as the third flux coordinate, the Boozer-Grad coordinate system introduces a scalar function χ such that it replaces $\hat{\mathbf{B}}(\mathbf{B}\cdot\nabla)\ell$ in Clebsch coordinate systems:

$$\mathbf{B} = B_\varrho \nabla\varrho + B_\nu \nabla\nu + \|\mathbf{B}\|\nabla\ell = B'_\varrho \nabla\varrho + B'_\nu \nabla\nu + \nabla\chi \tag{5.114}$$

where we have used B'_ϱ and B'_ν to differentiate the two pairs of seemingly identical covariant components of $\mathbf{B}$, but are, in general, not the same. Thus $\|\mathbf{B}\|\cdot\nabla\ell$ and $\nabla\chi$ are in fact not parallel in general. The Jacobian in these coordinates reads:

$$J = \frac{1}{\nabla\varrho\cdot(\nabla\nu\times\nabla\chi)} = \frac{1}{\nabla\chi\cdot(\nabla\varrho\times\nabla\nu)} = \frac{1}{\mathbf{B}^2} \tag{5.115}$$

The sought-after result of Boozer-Grad coordinates is to obtain an expression of the following type:

$$\mathbf{B} = \lambda\nabla\varrho + \nabla\chi \tag{5.116}$$

In other words, we wish that the term along $\nabla\nu$ vanishes. We find motivation in the condition of the ideal MHD equilibrium:

$$\nabla p = \mathbf{J}\times\mathbf{B} \implies \mathbf{J}\cdot\nabla p = 0 \implies \mathbf{J}\cdot\frac{\mathrm{d}p}{\mathrm{d}\varrho}\nabla\varrho = 0 \tag{5.117}$$

since $p = p(\varrho)$ is a flux function. Thus, $\mathbf{J}\cdot\nabla\varrho = 0$, unless possibly $\dfrac{\mathrm{d}p}{\mathrm{d}\varrho} = 0$, which we shall ignore and deem as anomalous flux function behaviour. Maxwell-Ampère equation gives:

$$\mu_0\mathbf{J} = \nabla\times\mathbf{B} = \nabla\times(\lambda\nabla\varrho + \nabla\chi) = \nabla\lambda\times\nabla\varrho \tag{5.118}$$

Thus, $\mathbf{J}$ is a Clebsch form in these coordinates. We thus ought to find λ as follows:

$$\nabla\lambda = \frac{\partial\lambda}{\partial\varrho}\nabla\varrho + \frac{\partial\lambda}{\partial\nu}\nabla\nu + \frac{\partial\lambda}{\partial\chi}\nabla\chi \implies \mu_0\mathbf{J} = \frac{\partial\lambda}{\partial\nu}\nabla\nu\times\nabla\varrho + \frac{\partial\lambda}{\partial\chi}\nabla\chi\times\nabla\varrho \tag{5.119}$$

We recognise the Clebsch form for the magnetic field, thus:

$$\mu_0\mathbf{J} = -\frac{\partial\lambda}{\partial\nu}\mathbf{B} + \frac{\partial\lambda}{\partial\chi}\nabla\chi\times\nabla\varrho \tag{5.120}$$

Taking the dot product with $\mathbf{B}$ gives:

$$\mu_0\mathbf{J}\cdot\mathbf{B} = -\frac{\partial\lambda}{\partial\nu}\mathbf{B}^2 - \frac{\partial\lambda}{\partial\chi}\mathbf{B}\cdot[\nabla\varrho\times(\mathbf{B}-\lambda\nabla\varrho)] \tag{5.121}$$

Notice that the last term evaluates to zero. Thus, we find that the parallel current density contributes to λ as follows:

$$\frac{\partial\lambda}{\partial\nu} = -\frac{\mu_0\mathbf{J}\cdot\mathbf{B}}{\mathbf{B}^2} \tag{5.122}$$

Substituting expression (5.120) back into the ideal MHD equilibrium condition, we find:

$$\nabla p = \frac{\mathrm{d}p}{\mathrm{d}\varrho}\nabla\varrho = \frac{1}{\mu_0}\left(\frac{\partial\lambda}{\partial\chi}\nabla\chi\times\nabla\varrho - \frac{\partial\lambda}{\partial\nu}\mathbf{B}\right)\times\mathbf{B} \tag{5.123}$$

which, expanding, gives:

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho}\nabla\varrho = \frac{1}{\mu_0}\frac{\partial\lambda}{\partial\chi}\mathbf{B}\times(\nabla\varrho\times\nabla\chi) = \frac{1}{\mu_0}\frac{\partial\lambda}{\partial\chi}\left[(\mathbf{B}\cdot\nabla\chi)\nabla\varrho - (\mathbf{B}\cdot\nabla\varrho)\nabla\chi\right] \tag{5.124}$$

We identify the contravariant component, which is, by definition, $\mathbf{B}\cdot\nabla\varrho = B^\varrho = 0$. Thus:

$$\frac{\mathrm{d}p}{\mathrm{d}\varrho}\nabla\varrho = \frac{1}{\mu_0}\frac{\partial\lambda}{\partial\chi}\left[\underbrace{(\nabla\varrho\times\nabla\nu)\cdot\nabla\chi}_{\mathbf{B}^2}\right]\nabla\varrho = \frac{\mathbf{B}^2}{\mu_0}\frac{\partial\lambda}{\partial\chi}\nabla\varrho \tag{5.125}$$

Having chosen good magnetic surfaces, $\nabla\varrho = \mathbf{0}$ only on the magnetic axis:

$$\frac{\partial\lambda}{\partial\chi} = \frac{\mu_0}{\mathbf{B}^2}\frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{5.126}$$

The factor λ is thus entirely determined in ν and χ via the two partial differential equations above, namely (5.122) and (5.126). Its dependency on ϱ is, however, arbitrary. We shall thus impose a type of gauge invariance by defining:

$$\lambda(\varrho, \nu = 0, \chi = 0) = 0 \tag{5.127}$$

Notice that this implies $\mathbf{J} = \mathbf{0}$ when $\lambda = 0$, which is indeed the case, since it makes $\mathbf{J}\cdot\mathbf{B}$ and ∇p vanish. Thus, λ is simply a flux-function for free-space, that is, in regions of zero current density. By the arbitrariness of λ, by imposing that $\lambda = 0$ when $\mathbf{J} = \mathbf{0}$, we obtain:

$$\mathbf{B} = \nabla\chi \qquad \mathbf{J} = \mathbf{0} \tag{5.128}$$

which makes χ precisely the magnetic scalar potential in magnetostatics. The Boozer-Grad coordinates remain susceptible to arbitrariness due to the arbitrariness of ν, since they are determined up to an additive flux function. This forces the following condition of gauge invariance:

$$\chi \to \chi + f(\varrho) \implies \lambda \to \lambda - \frac{\mathrm{d}f}{\mathrm{d}\varrho} \tag{5.129}$$

5.5.8 Hamada flux coordinate system

In ideal magnetohydrodynamic equilibrium conditions, the current density satisfies functionally identical equations to those of the magnetic field:

$$\mathbf{J}\cdot\nabla\varrho = 0 \qquad \nabla\cdot\mathbf{J} = 0 \iff \mathbf{B}\cdot\nabla\varrho = 0 \qquad \nabla\cdot\mathbf{B} = 0 \tag{5.130}$$

Thus, it seems only possible to find a coordinate system in which the current density is *straight*. Therefore, we shall first endeavour to find a coordinate system in which the current density field-lines are straight, regardless of the shape of the magnetic field-lines. Then, we shall introduce the Hamada flux coordinate system in which both the magnetic field-lines and the current density field-lines are *straight* lines. A first necessary assumption is that the current density field is known. Note that the reader is asked to tolerate our choice of notation, as the current density and the Jacobian of a particular coordinate transformation are denoted almost the same. For this reason, J shall denote the Jacobian. At the same time, $\|\mathbf{J}\|$ and J with any subscript or superscript shall refer to the norm and contravariant and covariant components of the current density, respectively. The zero divergence equation above gives:

$$\frac{\partial}{\partial\theta}\left(JJ^\theta\right) + \frac{\partial}{\partial\zeta}\left(JJ^\zeta\right) = 0 \quad \text{since} \quad J^\varrho = \mathbf{J}\cdot\nabla\varrho = 0 \tag{5.131}$$

We are thus motivated to introduce a stream function $\eta(\varrho, \theta, \zeta)$ so that:

$$J^\theta = -\frac{1}{J}\frac{\partial\eta}{\partial\zeta} \qquad J^\zeta = \frac{1}{J}\frac{\partial\eta}{\partial\theta} \tag{5.132}$$

Analogously to the treatment of Boozer flux coordinates in subsection 5.5.1, namely equation 5.44:

$$\mathbf{J} = \nabla\varrho\times\nabla\eta \tag{5.133}$$

such that the current density field-lines on magnetic surfaces of $\varrho = $ const. are defined by $\nu(\varrho =$ const.$, \theta, \zeta) = $ const. Analogously to expression 5.51, the auxiliary stream function is of the form:

$$\eta(\varrho, \theta, \zeta) = \frac{1}{2\pi}\left(\frac{\mathrm{d}I_{\mathrm{tor}}}{\mathrm{d}\varrho}\theta - \frac{\mathrm{d}I^{\mathrm{r}}_{\mathrm{pol}}}{\mathrm{d}\varrho}\zeta\right) + \tilde{\eta}(\varrho, \theta, \zeta) \tag{5.134}$$

where $\tilde{\eta}(\varrho, \theta\zeta)$ is a purely periodic function of θ and ζ. The flux function $I^{\mathrm{r}}_{\mathrm{pol}}(\varrho)$ represents the poloidal ribbon current, that is, the poloidal ribbon current-density flux, flowing within the magnetic surface labelled by ϱ, given by:

$$I^{\mathrm{r}}_{\mathrm{pol}}(\varrho) = \frac{1}{2\pi}\iiint_{V(\varrho)}\mathbf{J}\cdot\nabla\theta\,\mathrm{d}^3\mathbf{r} = \iint_{S^{\mathrm{r}}_{\mathrm{pol}}(\varrho)}\mathbf{J}\cdot\mathrm{d}\mathbf{S} \tag{5.135}$$

The flux function $I_{\rm tor}(\varrho)$ analogously represents the toroidal current-density flux flowing within a magnetic surface labelled by ϱ, given by:

$$I_{\rm tor}(\varrho) = \frac{1}{2\pi} \iiint_{V(\varrho)} \mathbf{J} \cdot \nabla\zeta \, \mathrm{d}^3\mathbf{r} = \iint_{S_{\rm tor}(\varrho)} \mathbf{J} \cdot \mathrm{d}\mathbf{S} \tag{5.136}$$

To make the current density field-lines straight, we ought to express the periodic function $\tilde{\eta}(\varrho, \theta, \zeta)$ as a flux function or eliminate it. The transformations similar to those in equation (5.53) suffice:

$$\theta_{\rm J} = \theta + 2\pi\tilde{\nu} \Big/ \frac{\mathrm{d}I_{\rm tor}}{\mathrm{d}\varrho} \quad \text{and} \quad \zeta_{\rm J} = \zeta \qquad \text{or else} \qquad \theta_{\rm J} = \theta \quad \text{and} \quad \zeta_{\rm J} = \zeta - 2\pi\tilde{\nu} \Big/ \frac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}\varrho} \tag{5.137}$$

where the subscript J precisely highlights that these are coordinates in which the current density field-lines are straight, as it is given by the following equation:

$$\frac{1}{2\pi}\frac{\mathrm{d}I_{\rm tor}}{\mathrm{d}\varrho}\theta_{\rm J} - \frac{1}{2\pi}\frac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}\varrho}\zeta_{\rm J} = \text{const.} \tag{5.138}$$

The associated Clebsch form thus reads:

$$\mathbf{J} = \nabla\varrho \times \nabla\left(\frac{1}{2\pi}\frac{\mathrm{d}I_{\rm tor}}{\mathrm{d}\varrho}\theta_{\rm J} - \frac{1}{2\pi}\frac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}\varrho}\zeta_{\rm J}\right) \tag{5.139}$$

The non-trivial contravariant components of the current density are thus:

$$J^{\theta_{\rm J}} = \frac{1}{2\pi J_{\rm J}}\frac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}\varrho} \qquad J^{\zeta_{\rm J}} = \frac{1}{2\pi J_{\rm J}}\frac{\mathrm{d}I_{\rm tor}}{\mathrm{d}\varrho} \tag{5.140}$$

where $J_{\rm J}$ denotes the Jacobian of the transformation under consideration. Note that, in these coordinates, $J_{\rm J}$ might, however, not have a generally simple expression, since we had no regard to the magnetic field. It is only desirable to find a coordinate system in which both the current density field-lines and the magnetic field-lines are straight lines, and the Jacobian of the transformation is given by an uncomplicated expression. This is precisely achieved by the Hamada flux coordinate system, whose self-explanatory alternative name is *natural* coordinate system.

Let us precisely focus on the main objective that is to be resolved to produce the desired nature of Hamada flux coordinates. It is precisely the requirement that $\tilde{\eta}(\varrho, \theta, \zeta)$ be a flux function, that is, $\tilde{\eta}(\varrho)$, or identically zero, both equivalent to $\mathbf{B} \cdot \nabla\tilde{\eta} = 0$. Retaking our results from subsection 5.5.4, namely equation (5.99), in cases where $\mathrm{d}p/\mathrm{d}\varrho \neq 0$, the flux-surface average of the Jacobian must be equal to the Jacobian of the transformation. Furthermore, equation (5.98) motivates us to label magnetic surfaces via their enclosed volume, that is, $\varrho \equiv V$, since, in that case, the ever-present $\mathrm{d}V/\mathrm{d}\varrho$ is unity. Note that it is all the more possible to work with an arbitrary first flux coordinate; however, one has to drag this previous term throughout the following procedure. Let us denote the Hamada coordinates by $\varrho_{\rm H} \equiv V$, $\theta_{\rm H}$, and $\zeta_{\rm H}$, and prudently remind ourselves:

$$\langle \mathcal{J}_{\rm H} \rangle = 4\pi^2 = \frac{1}{J_{\rm H}} \tag{5.141}$$

One is thus confident that the Hamada coordinate system exists for a particular toroidal system when the two solvability conditions of the magnetic differential equation for $\tilde{\eta}(\varrho_{\rm H}, \theta_{\rm H}, \zeta_{\rm H})$, given by equation (5.99), are met. Moreover, the solvability conditions reduce to the conditions for proper ideal magnetohydrodynamic equilibrium. In terms of Hamada coordinates, the contravariant components of the magnetic field and the current density are given as follows:

$$B^i = 2\pi \begin{pmatrix} 0 \\ \dfrac{\mathrm{d}\Phi^{\rm r}_{\rm pol}}{\mathrm{d}V} \\ \dfrac{\mathrm{d}\Phi_{\rm tor}}{\mathrm{d}V} \end{pmatrix} \qquad J^i = 2\pi \begin{pmatrix} 0 \\ \dfrac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}V} \\ \dfrac{\mathrm{d}I_{\rm tor}}{\mathrm{d}V} \end{pmatrix} \qquad \mathbf{J} = \frac{\nabla V}{2\pi} \times \nabla\left(\frac{\mathrm{d}I_{\rm tor}}{\mathrm{d}V}\theta_{\rm H} - \frac{\mathrm{d}I^{\rm r}_{\rm pol}}{\mathrm{d}\zeta_{\rm H}}\right) \tag{5.142}$$

All other quantities can easily be determined using the fact that $J_{\rm H} = 1/4\pi^2$.

5.6 Magnetic field structure in helical systems

Helical confinement systems deserve particular treatment among confinement systems characterised by symmetry. Namely, symmetrical helical systems present the most general type of symmetry - any helically symmetric quantity depends at most on the radial coordinate, r, and on the mixed quantity $u = \ell\varphi + kz$, where ℓ is the screw lead of the symmetry-generating helix, while k is a number. Note that here, r, φ, and z represent the elementary cylindrical coordinates. Translation symmetry can thus be generated by letting $k \to 0$, making φ an ignorable coordinate, while axial symmetry is obtained via letting $\ell \to 0$, since, in that case, z is the ignorable coordinate. Let us investigate the relation between r and u:

$$\nabla u = \ell\nabla\varphi + k\nabla z \implies \nabla r \times \nabla u = \ell\nabla r \times \nabla\varphi + k\nabla r \times \nabla z = r\ell\nabla z - kr^2\nabla\varphi \tag{5.143}$$

$$\mathbf{h}(r,\varphi,z) = \frac{\ell\nabla z - kr^2\nabla\varphi}{\ell^2 + k^2r^2} = \frac{r}{\ell^2 + k^2r^2}\nabla r \times \nabla u \implies \mathbf{h}^2 = \frac{1}{\ell^2 + k^2r^2} \tag{5.144}$$

The vector field $\mathbf{h}$ is notably tangent to the helix defined by the intersection of surfaces $r =$ const. and $u =$ const. Thus, for any helically symmetric quantity, $\varrho(r,u)$ characterised by the same helical parameters, it follows from equation (5.144) that:

$$\mathbf{h} \cdot \nabla\varrho(r,u) = 0 \tag{5.145}$$

for a magnetic surface label. Moreover, the vector quantity $\mathbf{h}$ presents the following properties:

$$\nabla \cdot \mathbf{h} = \frac{\ell\nabla^2 z - kr^2\nabla^2\varphi}{\ell^2 + k^2r^2} - \left(\ell\nabla z - kr^2\nabla\varphi\right) \cdot \frac{2k^2r\nabla r}{(\ell^2 + k^2r^2)^2} = 0 \tag{5.146}$$

where the non-trivial scalar product part evaluates to zero due to the orthogonality of elementary cylindrical coordinates. Furthermore:

$$\nabla \times \mathbf{h} = \frac{-2kr\nabla r \times \nabla\varphi}{\ell^2 + k^2r^2} - \frac{2k^2r^2\nabla r}{(\ell^2 + k^2r^2)^2} \times (\nabla r \times \nabla u) =$$

$$= -\frac{2kr\nabla r \times \nabla\varphi}{\ell^2 + k^2r^2} + \frac{2k^2r^2}{(\ell^2 + k^2r^2)^2}\nabla u = -\frac{2k(\ell^2 + k^2r^2)\nabla z - 2k^2r^2\ell\nabla\varphi - 2k^3r^2\nabla z}{(\ell^2 + k^2r^2)^2} =$$

$$= -\frac{2k\ell^2\nabla z - 2k^2r^2\ell\nabla\varphi}{(\ell^2 + k^2r^2)} = -\frac{2k\ell}{\ell^2 + k^2r^2}\mathbf{h} \tag{5.147}$$

We have thus found a vector quantity satisfying the functionally analogous equations to the magnetic field, namely zero divergence and, in the force-free regime, its own curl parallel to itself. The magnetic field can therefore be written in the following form:

$$\mathbf{B} = \mathbf{h} \times \nabla\varrho(r,u) + \mathbf{h}f(r,u) \tag{5.148}$$

where $\varrho(r,u)$ is a magnetic surface label, and $f(r,u)$ is a force-free contribution. In stellarator literature, $\mathbf{h}f(r,u)$ is referred to as the *helical* magnetic field, whereas, in tokamak literature, it is the first term in the right-hand side of equation (5.148) that is referred to as such. Furthermore, note:

$$\mathbf{h} \cdot \nabla f = \nabla \cdot (f\mathbf{h}) - f(\nabla \cdot \mathbf{h}) = \nabla \cdot (\mathbf{B} - \mathbf{h} \times \nabla\varrho) = \nabla \cdot (\nabla\varrho \times \mathbf{h}) = -(\nabla \times \mathbf{h}) \cdot \nabla\varrho \propto \mathbf{h} \cdot \nabla\varrho = 0 \tag{5.149}$$

Choosing $\varrho \equiv \Phi^{\mathrm{r}}_{\mathrm{hel}}/2\pi$, the magnetic flux through a ribbon-like surface of $u =$ const. obtained by surgically cutting off at a distance r within a cylindrical volume around the magnetic axis, gives:

$$\nabla \times \mathbf{A} = \frac{1}{2\pi}\left[\Phi^{\mathrm{r}}_{\mathrm{hel}}(\nabla \times \mathbf{h}) - \nabla \times (\Phi^{\mathrm{r}}_{\mathrm{hel}}\mathbf{h})\right] + \mathbf{h}f = -\frac{1}{2\pi}\nabla \times (\Phi^{\mathrm{r}}_{\mathrm{hel}}\mathbf{h}) + \frac{\Phi^{\mathrm{r}}_{\mathrm{hel}}}{2\pi}\nabla \times \mathbf{h} + f\mathbf{h} \tag{5.150}$$

Notice that applying the divergence on the preceding gives:

$$\nabla \cdot (\nabla \times \mathbf{A}) = -\frac{1}{2\pi}\underbrace{\nabla \cdot (\nabla \times (\Phi^{\mathrm{r}}_{\mathrm{hel}}\mathbf{h}))}_{0} + \nabla \cdot \left[\frac{\Phi^{\mathrm{r}}_{\mathrm{hel}}}{2\pi}\nabla \times \mathbf{h} + f\mathbf{h}\right] = 0 \tag{5.151}$$

Hence, the latter part can be written as the curl of some vector potential $\mathbf{S}$, that is:

$$\mathbf{A} = -\frac{\Phi^{\mathrm{r}}_{\mathrm{hel}}}{2\pi}\mathbf{h} + \mathbf{S} + \nabla g \quad \text{where } g \text{ is a gauge term} \tag{5.152}$$

Namely, one can always choose such $\mathbf{S}$ that it cancels with the gauge term, so that we find the relation between the magnetic vector potential and the magnetic helical ribbon flux:

$$\frac{\Phi^{\mathrm{r}}_{\mathrm{hel}}}{2\pi} = -\frac{\mathbf{A}\cdot\mathbf{h}}{\mathbf{h}^2} \tag{5.153}$$

Furthermore, the current density from the given form (5.148) via the Maxwell-Ampère equation reads:

$$\mu_0\mathbf{J} = \nabla\times(\mathbf{h}\times\nabla\varrho + \mathbf{h}f) = \nabla\times(\mathbf{h}\times\nabla\varrho) + f(\nabla\times\mathbf{h}) + (\nabla f)\times\mathbf{h} \tag{5.154}$$

We shall first compute the component of the term $\nabla\times(\mathbf{h}\times\nabla\varrho)$ along $\mathbf{h}$ by taking the dot product:

$$\mathbf{h}\cdot[\nabla\times(\mathbf{h}\times\nabla\varrho)] = \nabla\cdot[(\mathbf{h}\times\nabla\varrho)\times\mathbf{h}] + (\mathbf{h}\times\nabla\varrho)\cdot(\nabla\times\mathbf{h}) \tag{5.155}$$

The second term evaluates analytically to zero since $\nabla\times\mathbf{h} \parallel \mathbf{h}$. Continuing:

$$(\mathbf{h}\times\nabla\varrho)\times\mathbf{h} = \mathbf{h}^2\nabla\varrho - (\mathbf{h}\cdot\nabla\varrho)\mathbf{h} = \mathbf{h}^2\nabla\varrho \tag{5.156}$$

where the mixed term has been eliminated since $\mathbf{h}$ is along the symmetry direction, and $\varrho = \varrho(r,u)$ only. Using equation (5.144), we find:

$$\mathbf{h}\cdot[\nabla\times(\mathbf{h}\times\nabla\varrho)] = \nabla\cdot\left(\frac{1}{\ell^2+k^2r^2}\nabla\varrho\right) \tag{5.157}$$

Expanding the del operators and using helical symmetry, we find:

$$\nabla\varrho = \frac{\partial\varrho}{\partial r}\mathbf{e}^r + \frac{\partial\varrho}{\partial\varphi}\mathbf{e}^\varphi + \frac{\partial\varrho}{\partial z}\mathbf{e}^z = \frac{\partial\varrho}{\partial r}\hat{\mathbf{r}} + \frac{\ell}{r}\frac{\partial\varrho}{\partial u}\hat{\boldsymbol{\varphi}} + k\frac{\partial\varrho}{\partial u}\hat{\mathbf{z}} \tag{5.158}$$

$$\begin{aligned}\nabla\cdot\left(\frac{1}{\ell^2+k^2r^2}\nabla\varrho\right) &= \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{r}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial r}\right) + \frac{1}{r}\frac{\partial}{\partial\varphi}\left(\frac{\ell}{r(\ell^2+k^2r^2)}\frac{\partial\varrho}{\partial\varphi}\right) + \frac{\partial}{\partial z}\left(\frac{k}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial z}\right) = \\ &= \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{r}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial r}\right) + \left[\frac{\ell^2}{r^2(\ell^2+k^2r^2)} + \frac{k^2}{\ell^2+k^2r^2}\right]\frac{\partial^2\varrho}{\partial u^2}\end{aligned} \tag{5.159}$$

Therefore, the component of $\nabla\times(\mathbf{h}\times\nabla\varrho)$ along $\mathbf{h}$ reads:

$$\mathbf{h}\cdot[\nabla\times(\mathbf{h}\times\nabla\varrho)] = \mathbf{h}\cdot\frac{\ell^2+k^2r^2}{r}\mathbf{h}\left[\frac{\partial}{\partial r}\left(\frac{r}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial r}\right) + \frac{1}{r}\frac{\partial^2\varrho}{\partial u^2}\right] \tag{5.160}$$

The component perpendicular to $\mathbf{h}$ turns out to be identically zero. To verify this fact, we perform the dot products of the quantity $\nabla\times(\mathbf{h}\times\nabla\varrho)$ with both ∇r and ∇u:

$$\begin{aligned}\nabla r\cdot[\nabla\times(\mathbf{h}\times\nabla\varrho)] &= \nabla\cdot[(\mathbf{h}\times\nabla\varrho)\times\nabla r] - (\mathbf{h}\times\nabla\varrho)\cdot[\nabla\times(\nabla r)] = \nabla\cdot[(\mathbf{h}\times\nabla\varrho)\times\nabla r] = \\ &= \nabla\cdot[(\mathbf{h}\cdot\nabla r)\nabla\varrho - (\nabla\varrho\cdot\nabla r)\mathbf{h}] = \nabla\cdot[(\nabla\varrho\cdot\nabla r)\mathbf{h}] = (\mathbf{h}\cdot\nabla)\left(\frac{\partial\varrho}{\partial r}\right) = 0\end{aligned} \tag{5.161}$$

where $\mathbf{h}\cdot\nabla r = 0$ by definition and $\nabla\cdot\mathbf{h} = 0$ as determined above. Since $\varrho = \varrho(r,u)$, its derivative is also helically symmetric, so that its gradient is perpendicular to the symmetry direction defined via $\mathbf{h}$. The treatment for the dot product with ∇u is analogous, and evaluates to zero by the same argument. Returning to the expression for the current density, we finally find:

$$\mu_0\mathbf{J} = (\nabla f)\times\mathbf{h} + \left(\frac{\ell^2+k^2r^2}{r}\left[\frac{\partial}{\partial r}\left(\frac{r}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial r}\right) + \frac{1}{r}\frac{\partial^2\varrho}{\partial u^2}\right] - \frac{2k\ell}{\ell^2+k^2r^2}f\right)\mathbf{h} \tag{5.162}$$

where we have used the expression for the curl $\nabla\times\mathbf{h}$ determined earlier. Moreover, we shall now determine the nature of the function $f(r,u)$ using a somewhat provident approach. Namely, consider the vector product $\nabla f\times\nabla\varrho$. Let us decompose it into components along $\mathbf{h}$ and perpendicular to it. In particular, the perpendicular component relates to:

$$\mathbf{h}\times(\nabla\varrho\times\nabla f) = (\mathbf{h}\cdot\nabla f)\nabla\varrho - (\mathbf{h}\cdot\nabla\varrho)\nabla f = \mathbf{0} \tag{5.163}$$

since $\mathbf{h} \cdot \nabla f = 0$, and by helical symmetry $\mathbf{h} \cdot \nabla \varrho = 0$, so that $\nabla \varrho \times \nabla f$ is along $\mathbf{h}$. Consider now the triple scalar product of ∇f and $\nabla \varrho$ with $\mathbf{h}$ as follows:

$$(\mathbf{h} \times \nabla f) \cdot (\mathbf{h} \times \nabla \varrho) = (\mathbf{h} \cdot \mathbf{h})(\nabla f \cdot \nabla \varrho) - (\mathbf{h} \cdot \nabla f)(\mathbf{h} \cdot \nabla \varrho) = \mathbf{h}^2 (\nabla f \cdot \nabla \varrho) \tag{5.164}$$

Therefore, since both ∇f and $\nabla \varrho$ are perpendicular to $\mathbf{h}$ while their vector product is strictly along $\mathbf{h}$, and their vector products with $\mathbf{h}$ uniquely determine their dot product, ∇f and $\nabla \varrho$ are hence collinear, that is, $\nabla f \times \nabla \varrho = \mathbf{0}$. Since $\mathbf{B} \cdot \nabla \varrho = 0$ by the definition of magnetic surfaces, it follows that $\mathbf{B} \cdot \nabla f = 0$, so that f is indeed a flux function. Moreover, these considerations allow us to obtain an exact expression for the rotational transform in the case of pure axisymmetry, that is, when we are allowed to set $k = 1$ and $\ell = 0$. The toroidal flux can be written as:

$$\Phi_{\text{tor}} = \iint_{\varrho} \mathbf{B}_{\text{tor}} \cdot \mathrm{d}\mathbf{S} = \iint_{\varrho} f \nabla \varphi \cdot \mathrm{d}\mathbf{S} = \iint_{\varrho} \frac{f}{r} \, \mathrm{d}r \, \mathrm{d}z \tag{5.165}$$

where the toroidal component of the magnetic field was expressed as $\mathbf{B}_{\text{tor}} = f \nabla \varphi$ by choosing $k = 1$ and $\ell = 0$ in equation (**??**). Choosing the poloidal ribbon flux as the magnetic surface label gives:

$$q(\Phi^{\mathrm{r}}_{\text{pol}}) = \frac{\mathrm{d}\Phi_{\text{tor}}}{\mathrm{d}\Phi^{\mathrm{r}}_{\text{pol}}} = \oint \frac{\|\mathbf{B}_{\text{tor}}\|}{\|\nabla \Phi^{\mathrm{r}}_{\text{pol}}\|} \mathrm{d}\ell = \oint \frac{\|\mathbf{B}_{\text{tor}}\|}{\|\mathbf{B}_{\text{pol}}\|} \frac{\mathrm{d}\ell}{2\pi r} = \frac{1}{\iota\!\!\!-(\Phi^{\mathrm{r}}_{\text{pol}})} \tag{5.166}$$

which completes our considerations on the rotational transform in subsection 5.4.2.

5.7 Magnetic field-line Hamiltonian

After all the introductory considerations on flux coordinate systems, it is convenient to discuss the concept of canonical coordinates for magnetic fields. The term *canonical* precisely relates to the Hamiltonian character of magnetic fields. Such coordinates are worth exploring since canonical coordinates for magnetic fields exist even when magnetic surfaces do not. By contrast, flux coordinate systems rely on the existence of magnetic surfaces. It ought to be emphasised that canonical coordinates are not synonymous with flux coordinates.

The guaranteed existence of canonical coordinates for magnetic fields is a wonderful result, since it allows us to use multiple methods emerging from Hamiltonian theory, namely canonical transformations, Hamilton-Jacobi theory, canonical perturbation theory, and others.

5.7.1 Flux coordinate systems revisited

Following our discussions on flux coordinate systems, we recall, in particular, equation (5.65):

$$\mathbf{B} = \frac{\nabla \varrho}{2\pi} \times \nabla \left(\frac{\mathrm{d}\Phi_{\text{tor}}}{\mathrm{d}\varrho} \theta_{\mathrm{f}} + \frac{\mathrm{d}\Phi^{\mathrm{d}}_{\text{pol}}}{\mathrm{d}\varrho} \zeta_{\mathrm{f}} \right) \tag{5.167}$$

where ϱ is a chosen magnetic surface label, while Φ_{tor} and $\Phi^{\mathrm{d}}_{\text{pol}}$ are the toroidal and poloidal disk magnetic fluxes, respectively, which are indeed flux functions. Introducing the substitution $\psi_{\text{tor}} = \Phi_{\text{tor}}/2\pi$ and $\psi^{\mathrm{d}}_{\text{pol}} = \Phi^{\mathrm{d}}_{\text{pol}}/2\pi$, we may express the magnetic field as follows:

$$\mathbf{B} = \nabla \psi_{\text{tor}} \times \nabla \theta_{\mathrm{f}} + \nabla \psi^{\mathrm{d}}_{\text{pol}} \times \nabla \zeta_{\mathrm{f}} \tag{5.168}$$

The rotational transform allows us to neatly express:

$$\iota\!\!\!- = -\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\text{pol}}}{\mathrm{d}\varrho} \Big/ \frac{\mathrm{d}\Phi_{\text{tor}}}{\mathrm{d}\varrho} = -\frac{\mathrm{d}\psi^{\mathrm{d}}_{\text{pol}}}{\mathrm{d}\psi_{\text{tor}}} \implies \mathbf{B} = \nabla \psi_{\text{tor}} \times \nabla (\theta_{\mathrm{f}} - \iota\!\!\!- \zeta_{\mathrm{f}}) \tag{5.169}$$

Reminder that the rotational transform is a flux function. We shall now employ the definitions of covariant and contravariant basis vectors, namely $\mathbf{e}^i = \nabla u^i$, $\mathbf{e}_i = \partial \mathbf{r}/\partial u^i$ along with $\mathbf{e}_i = J \mathbf{e}^j \times \mathbf{e}^k$ with (i, j, k) being an even permutation of $(1, 2, 3)$:

$$\mathbf{B} = \frac{1}{J_{\mathrm{f}}} \left(\frac{\partial \mathbf{r}}{\partial \zeta_{\mathrm{f}}} + \iota\!\!\!-(\psi_{\text{tor}}) \frac{\partial \mathbf{r}}{\partial \theta_{\mathrm{f}}} \right) = \frac{\iota\!\!\!-}{J_{\mathrm{f}}} \mathbf{e}_{\theta_{\mathrm{f}}} + \frac{1}{J_{\mathrm{f}}} \mathbf{e}_{\zeta_{\mathrm{f}}} \implies J = \frac{1}{\mathbf{B} \cdot \nabla \zeta_{\mathrm{f}}} \tag{5.170}$$

Notice the difference from our previous result for the Jacobian derived in subsection 5.5.1. This is precisely due to the fact that we have chosen the toroidal flux instead of the poloidal ribbon flux as the first flux coordinate. The magnetic field is thus uniquely determined if ψ_tor, $\iota\!\!\!-(\psi_\mathrm{tor})$, θ_f and ζ_f are known functions of position $\mathbf{r}$ in the considered toroidal system. Note that this particular case requires the toroidal magnetic field to be non-zero. In the opposite case, we employ our results for the poloidal magnetic field. Since the latter two flux coordinates, θ_f and ζ_f, are both periodic, the Cartesian components, (x, y, z), of the position vector $\mathbf{r}$ can be written as a double Fourier series. Treating the three components simultaneously by way of the position vector, we denote:

$$\mathbf{r}(\psi_\mathrm{tor}, \theta_\mathrm{f}, \zeta_\mathrm{f}) = \sum_{m,n} \mathbf{r}_{mn}(\psi_\mathrm{tor}) \exp\left[\mathrm{i}m\theta_\mathrm{f} - \mathrm{i}n\zeta_\mathrm{f}\right] \tag{5.171}$$

The Boozer flux coordinate system introduced a function χ, as such to make magnetic field-lines *straight* and to reduce to the magnetic scalar potential for current-less plasma, which is precisely given by:

$$\chi = \frac{\mu_0}{2\pi}\left(I_\mathrm{tor}\theta_\mathrm{f} + I^\mathrm{d}_\mathrm{pol}\zeta_\mathrm{f}\right) \tag{5.172}$$

Since the equation of a magnetic field-line is given by $\theta_\mathrm{f} - \iota\!\!\!-\zeta_\mathrm{f} = \text{const.} = \theta_\mathrm{f}^\square$, we may rewrite the exponents in the Fourier series above as follows:

$$\mathbf{r}(\psi_\mathrm{tor}, \theta_\mathrm{f}, \zeta_\mathrm{f}) = \sum_{m,n} \mathbf{r}_{mn}(\psi_\mathrm{tor}) \exp\left[\mathrm{i}\frac{\left(mI^\mathrm{d}_\mathrm{pol} + nI_\mathrm{tor}\right)\theta_\mathrm{f}^\square + (m\iota\!\!\!- - n)\, 2\pi\chi/\mu_0}{I_\mathrm{tor}\iota\!\!\!- + I^\mathrm{d}_\mathrm{pol}}\right] \tag{5.173}$$

To evaluate the Cartesian coordinate coefficients inside $\mathbf{r}_{mn}(\psi_\mathrm{tor})$, one performs integration over a field-line, starting from an arbitrary point, for which, conveniently, χ and $\theta_\mathrm{f}^\square$ can be chosen as zero, the former of which, $\theta_\mathrm{f}^\square$, is constant along its respective field-line, so it remains zero along the path of integration. It is thus possible to determine $\mathbf{r}(\psi_\mathrm{tor}, \theta_\mathrm{f}^\square = 0, \chi)$ via the coefficients in the expansion to give:

$$\mathbf{r}(\psi_\mathrm{tor}, \theta_\mathrm{f}, \zeta_\mathrm{f}) = \sum_{m,n} \mathbf{r}_{mn}(\psi_\mathrm{tor}) \exp\left[\mathrm{i}\frac{2\pi(m\iota\!\!\!- - n)\chi}{\mu_0(I_\mathrm{tor}\iota\!\!\!- + I^\mathrm{d}_\mathrm{pol})}\right] \tag{5.174}$$

Note that such expansions can be acquired using equivalent approaches without the construction of the function χ.

5.7.2 Canonical magnetic coordinates

We shall start our proof of the existence of canonical coordinates for any general magnetic field using the magnetic vector potential, since $\mathbf{B} = \nabla \times \mathbf{A}$ uniquely determines the magnetic field. Expressing the magnetic vector potential in the covariant form:

$$\mathbf{A} = A_\varrho \nabla\varrho + A_\theta \nabla\theta + A_\zeta \nabla\zeta \tag{5.175}$$

where (ϱ, θ, ζ) now form a generic coordinate system and ϱ does not necessarily label magnetic surfaces since they do not necessarily exist. Furthermore, the magnetic vector potential is uniquely determined up to a gradient of a scalar function, here, $\nabla G(\varrho, \theta, \zeta)$. Hence, we could choose G such that:

$$\frac{\partial G}{\partial \varrho} = A_\varrho \qquad G(\varrho = 0) = 0 \qquad \frac{\partial G}{\partial \theta} = A_\theta - \Psi \qquad \frac{\partial G}{\partial \zeta} = A_\zeta - \Phi \tag{5.176}$$

where Ψ and Φ are two single-valued functions that shall be presented later. This choice of G allows us to write:

$$\mathbf{A} = \Psi\nabla\theta + \Phi\nabla\zeta + \nabla G \implies \mathbf{B} = \nabla\Psi \times \nabla\theta + \nabla\Phi \times \nabla\zeta \tag{5.177}$$

This representation of the magnetic field is commonly referred to as its canonical form, which is reminiscent of the usual definition via the flux-surface label ϱ for existing magnetic surfaces. From the similarity with equation (6.54), the toroidal flux through a constant Ψ surface is precisely $2\pi\Psi$, while the poloidal disk flux through a constant Φ surface measures precisely $2\pi\Phi$. The formal proof of this analogy is derived using the canonical form and evaluating the magnetic flux through a surface of constant Ψ and Φ, respectively, along the directions of $\nabla\theta$ and $\nabla\zeta$. We shall illustrate the proof for the toroidal function:

$$\iint_{S(\Psi)} \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \iint_{S(\Psi)} (\mathbf{B} \cdot \nabla\zeta) J \,\mathrm{d}\varrho\,\mathrm{d}\theta = \iint_{S(\Psi)} \nabla\zeta \cdot (\nabla\Psi \times \nabla\theta) J \,\mathrm{d}\varrho\,\mathrm{d}\theta \tag{5.178}$$

$$\iint_{S(\Psi)} \nabla\Psi \cdot (\nabla\theta \times \nabla\zeta) J \,\mathrm{d}\varrho \,\mathrm{d}\theta = \iint_{S(\Psi)} \frac{\nabla\Psi \cdot (\nabla\theta \times \nabla\zeta)}{\nabla\varrho \cdot (\nabla\theta \times \nabla\zeta)} \mathrm{d}\varrho \,\mathrm{d}\theta = \tag{5.179}$$

$$\iint_{S(\Psi)} \left[\frac{\partial\Psi}{\partial\varrho}\nabla\varrho + \frac{\partial\Psi}{\partial\theta}\nabla\theta + \frac{\partial\Psi}{\partial\zeta}\nabla\zeta\right] \cdot \frac{\nabla\theta \times \nabla\zeta}{\nabla\varrho \cdot (\nabla\theta \times \nabla\zeta)} = \iint_{S(\Psi)} \frac{\partial\Psi}{\partial\varrho} \mathrm{d}\varrho \,\mathrm{d}\theta = 2\pi\Psi \tag{5.180}$$

where, in the last equality, we used the fact that the domain of integration is precisely $\Psi(\varrho, \theta, \zeta) =$ const. The demonstration for Φ representing the poloidal disk flux follows an analogous procedure. It is important to note that surfaces $\Psi(\varrho, \theta, \zeta) =$ const. and surfaces $\Phi(\varrho, \theta, \zeta) =$ const. do not coincide in general. If they were to coincide, they form, in fact, magnetic surfaces and can thus be related via a common flux surface label. In systems characterised by magnetic surfaces, these are represented by ψ_{tor} and $\psi^{\mathrm{d}}_{\mathrm{pol}}$, respectively. This is entirely equivalent to $\mathbf{B} \cdot \nabla\Psi \neq 0 \neq \mathbf{B} \cdot \nabla\Phi$ and $\mathbf{B} \cdot \nabla\Psi \neq \mathbf{B} \cdot \nabla\Phi$.

It is precisely these four variables, Ψ, Φ, θ, ζ, that can be considered as canonical variables in the Hamiltonian context. To show this, it suffices to show that they obey Hamilton's equations, which implies the existence of an associated Hamiltonian. We shall first eliminate the dependency of the poloidal stream function, Φ on the first flux-coordinate, ϱ, by expressing it in terms of the toroidal stream function, thus $\Phi(\varrho, \theta, \zeta) \equiv \Phi(\Psi, \theta, \zeta)$. In terms of the coordinates (Ψ, θ, ζ), the equation of a magnetic field-line reads:

$$\frac{\mathrm{d}\Psi}{\mathbf{B} \cdot \nabla\Psi} = \frac{\mathrm{d}\theta}{\mathbf{B} \cdot \nabla\theta} = \frac{\mathrm{d}\zeta}{\mathbf{B} \cdot \nabla\zeta} \tag{5.181}$$

Taking the dot product of equation (5.177) with $\nabla\Psi$, we find:

$$\mathbf{B} \cdot \nabla\Psi = \nabla\Psi \cdot (\nabla\Phi \times \nabla\zeta) = \nabla\Psi \cdot \left[\left(\frac{\partial\Phi}{\partial\Psi}\nabla\Psi + \frac{\partial\Phi}{\partial\theta}\nabla\theta + \frac{\partial\Phi}{\partial\zeta}\nabla\zeta\right) \times \nabla\zeta\right] = \frac{\partial\Phi}{\partial\theta}\nabla\Psi \cdot (\nabla\theta \times \nabla\zeta) \tag{5.182}$$

Plugging this result into the magnetic field-line equation, we find:

$$\frac{\mathrm{d}\Psi}{\mathrm{d}\zeta} = \frac{\mathbf{B} \cdot \nabla\Psi}{\mathbf{B} \cdot \nabla\zeta} = \frac{\partial\Phi}{\partial\theta} \frac{\nabla\Psi \cdot (\nabla\theta \times \nabla\zeta)}{\nabla\zeta \cdot (\nabla\Psi \times \nabla\theta)} = \frac{\partial\Phi}{\partial\theta} \tag{5.183}$$

Analogously, we find:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\zeta} = \frac{\mathbf{B} \cdot \nabla\theta}{\mathbf{B} \cdot \nabla\zeta} = \frac{\nabla\theta \cdot (\nabla\Phi \times \nabla\zeta)}{\nabla\zeta \cdot (\nabla\Psi \times \nabla\theta)} = \frac{\partial\Phi}{\partial\Psi} \frac{\nabla\theta \cdot (\nabla\Psi \times \nabla\zeta)}{\nabla\zeta \cdot (\nabla\Psi \times \nabla\theta)} = -\frac{\partial\Phi}{\partial\Psi} \tag{5.184}$$

We may thus identify the toroidal angle ζ with time, the poloidal stream function Φ with the negative Hamiltonian, $-\mathcal{H}$, the poloidal angle θ with a canonical coordinate, and the toroidal stream function Ψ with a canonical momentum. The function $-\Phi$ is commonly referred to as the magnetic field-line Hamiltonian. Furthermore, the toroidal stream function Ψ and the poloidal angle θ are conjugate variables.

In the case of a time-dependent magnetic field, the time-dependence can be transformed *away* via a canonical transformation, so to consider a Hamiltonian system with two degrees of freedom, namely the toroidal stream function Ψ and the poloidal angle θ as a first pair of conjugate variables, and the poloidal stream function Φ and the toroidal angle ζ as a second pair of conjugate variables. The new Hamiltonian is, therefore, time-independent and thus conservative. This allows us to use a wide range of mathematical and computational tools developed for conservative systems. In a more advanced context, the complexity of magnetic field-lines as a Hamiltonian system with two degrees of freedom implies the existence of stochastic field-line behaviour.

It is only appropriate to see how the Hamiltonian representation of magnetic field-lines reproduces all of our considerations about flux-coordinates in the case of existing magnetic surfaces. In such systems, the Hamiltonian analogue $-\Phi(\Psi, \theta, \zeta) = -\Phi(\Psi)$ only, so that the Hamiltonian is in its action-angle form. The toroidal stream function is the action of the system, hence a conserved quantity, and the poloidal angle depends linearly on the toroidal angle, whose relation is precisely determined by the rotational transform and an additive constant that uniquely determines a single magnetic field-line belonging to a particular magnetic surface. In this perspective, the rotational transform represents the system's angular frequency. Thus, a system with existing, *perfect* magnetic surfaces is integrable. In practice, realistic systems are usually characterised by *reasonably good* magnetic surfaces, which somewhat corresponds to near-integrable Hamiltonian systems. The common approach to such systems is precisely canonical perturbation theory.

5.8 Quasi-symmetry of magnetic fields

Quasi-symmetry is a remarkable concept in plasma physics and can be summarized in a single sentence: if the magnitude of the magnetic field $B = \|\mathbf{B}\|$ presents a continuous symmetry in specific coordinate systems, such that B is independent of one of the coordinates, any guiding centre follows a trajectory similar to that observed in the presence of a truly symmetric magnetic field. It is important to emphasize that the magnetic field vector $\mathbf{B}$ need not be symmetric. The idea presented here is promising, as it combines the advantageous confinement characteristics of tokamaks with the stability and durability of stellarators. Particle trajectories in the presence of drifts produced by $\nabla\mathbf{B}$ and by the curvature of fields: quasi-symmetry constitutes a method allowing to guarantee particle confinement, including in the presence of these drifts. Quasi-symmetry is closely associated with the Boozer coordinate system, which consists of a coordinate linked to magnetic surfaces, a poloidal angle coordinate θ, and a toroidal angle coordinate ζ in a toroidal magnetohydrodynamic equilibrium. It is necessary to emphasize that quasi-symmetry only arises in the case of the Boozer coordinate system, because any other choice of toroidal coordinates does not present the same consequences. We distinguish the following three cases:

1. Axial quasi-symmetry: $B = B(\varrho, \theta)$
2. Helical quasi-symmetry: $B = B(\varrho, m\theta - n\zeta)$ where $m, n \in \mathbb{N}^*$
3. Poloidal quasi-symmetry: $B = B(\varrho, \zeta)$

Magnetic fields characterized by a type of quasi-symmetry can be roughly described as follows: particle trajectories, as well as neoclassical transport in quasi-symmetric magnetic fields, are analogous to those in truly axisymmetric magnetic fields. The term *neoclassical* refers to the motion of guiding centres and to collisions in the absence of turbulence, and will be the subject of our study later. There exist several precise statements of the preceding fact:

1. The equations of motion of a guiding centre, expressed in Boozer coordinates, depend only on B and ϱ.
2. The average Lagrangian over a gyration period, expressed in Boozer coordinates, depends only on B and ϱ.
3. In quasi-symmetric magnetic fields, there exists a conserved momentum quantity, analogous to canonical angular momentum, which guarantees the confinement of charged particles.
4. When B is expressed in Boozer coordinates, radial neoclassical fluxes and parallel fluxes depend only on B and functions of ϱ.

We now demonstrate the simultaneous existence of the three types of quasi-symmetry. Recall that the result sought boils down to the following sentence: the existence of a symmetry in $\|\mathbf{B}\|$ implies the presence of the same symmetry in β (the covariant component of the magnetic field). We return to the ideal magnetohydrodynamic equilibrium equation $\mathbf{J} \times \mathbf{B} = \nabla p$ where $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$, then by applying the curl on equation (5.107), we find:

$$\nabla \times \mathbf{B} = \nabla\beta \times \nabla\Psi + \frac{\mathrm{d}I}{\mathrm{d}\Psi}\nabla\Psi \times \nabla\theta_\mathrm{o} + \frac{\mathrm{d}G}{\mathrm{d}\Psi}\nabla\Psi \times \nabla\zeta_\mathrm{o} \tag{5.185}$$

It should be noted that $\nabla\beta = \partial_\Psi\beta\nabla\Psi + \partial_{\theta_\mathrm{o}}\beta\nabla\theta_\mathrm{o} + \partial_{\zeta_\mathrm{o}}\beta\nabla\zeta_\mathrm{o}$, which gives us:

$$\left(\frac{\partial\beta}{\partial\zeta_\mathrm{o}} + \iota\frac{\partial\beta}{\partial\theta_\mathrm{o}} - \frac{\mathrm{d}G}{\mathrm{d}\Psi} - \iota\frac{\mathrm{d}I}{\mathrm{d}\Psi}\right)(\nabla\Psi \cdot (\nabla\theta_\mathrm{o} \times \nabla\zeta_\mathrm{o}))\nabla\Psi = \mu_0\frac{\mathrm{d}p}{\mathrm{d}\Psi}\nabla\Psi \tag{5.186}$$

Using a previous result, notably (5.109), we find:

$$\frac{\partial\beta}{\partial\zeta_\mathrm{o}} + \iota\frac{\partial\beta}{\partial\theta_\mathrm{o}} - \frac{\mathrm{d}G}{\mathrm{d}\Psi} - \iota\frac{\mathrm{d}I}{\mathrm{d}\Psi} = \mu_0\frac{G + \iota I}{\mathbf{B}^2}\frac{\mathrm{d}p}{\mathrm{d}\Psi} \tag{5.187}$$

Writing the Fourier transforms of the magnetic field and of the function β, one finds:

$$\frac{1}{\mathbf{B}^2} = \sum_{m,n}\mathcal{B}_{m,n}\exp(im\theta_\mathrm{o} - in\zeta_\mathrm{o}) \quad \text{and} \quad \beta = \sum_{m,n}\beta_{m,n}\exp(im\theta_\mathrm{o} - in\zeta_\mathrm{o}) \tag{5.188}$$

$$\beta_{m,n} = \frac{\mu_0 \mathcal{B}_{m,n}(G + \iota\!\!\!-I)}{\mathrm{i}(\iota\!\!\!-m - n)} \frac{\mathrm{d}p}{\mathrm{d}\Psi} \tag{5.189}$$

Assuming that the magnetic field depends on angles θ_o and ζ_o via a function $\chi = M\theta_\mathrm{o} - N\zeta_\mathrm{o}$, where M and N are integers, the same applies to $1/\mathbf{B}^2$. Then, $\mathcal{B}_{m,n}$ vanishes unless $m = kM$ and $n = kN$ for some $k \in \mathbb{Z}$. The same applies to $\beta_{m,n}$, so β depends on angles θ_o and ζ_o via the same function χ, as is the case for the magnetic field $\|\mathbf{B}\|$.

5.8.1 Axisymmetric confinement

In the case of axial symmetry, particle motions are strongly restricted by a conservation law generated by the symmetry. Here, axial symmetry is indeed a continuous rotational symmetry. The conservation law and its result are commonly referred to as *Tamm's theorem*.

5.8.1.1 Canonical angular momentum

The Lagrangian of a non-relativistic charged particle in a magnetic field is given as follows:

$$\mathcal{L}(\mathbf{r}, \dot{\mathbf{r}}, t) = q\mathbf{A}(\mathbf{r}, t) \cdot \dot{\mathbf{r}} + \frac{1}{2} m \dot{\mathbf{r}}^2 \quad \text{where} \quad \mathbf{B} = \nabla \times \mathbf{A} \tag{5.190}$$

The preceding equation expressed in a cylindrical coordinate system (ρ, φ, z) is found as follows:

$$\mathcal{L}(\rho, \varphi, z, \dot{\rho}, \dot{\varphi}, \dot{z}, t) = q\left(\dot{\rho} A_\rho + \rho\dot{\varphi} A_\varphi + \dot{z} A_\mathrm{z}\right) + \frac{1}{2} m \left(\dot{\rho}^2 + \rho^2 \dot{\varphi}^2 + \dot{z}^2\right) \tag{5.191}$$

We assume the existence of a gauge such that:

$$\frac{\partial A_\rho}{\partial \varphi} = \frac{\partial A_\varphi}{\partial \varphi} = \frac{\partial A_\mathrm{z}}{\partial \varphi} = 0 \quad \text{i.e. axisymmetry} \tag{5.192}$$

Then, $\partial\mathcal{L}/\partial\varphi = 0$. Noether's theorem guarantees the existence of a conserved quantity that we henceforth call canonical angular momentum:

$$\mathcal{P}_\varphi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = q\rho A_\rho + m\rho^2 \dot{\varphi} = q\rho A_\rho + m\rho v_\varphi = \text{const.} \tag{5.193}$$

Since the Lorentz force does no work on particles in the absence of electric fields, the azimuthal component of particle velocity is bounded by energy conservation. Furthermore, the constituent term of the vector potential can be made sufficiently larger than the one with v_φ by applying a strong magnetic field:

$$\frac{m\rho v_\varphi}{q\rho A_\varphi} \sim \frac{m v_\varphi}{qBL} \sim \frac{r_\mathrm{L}}{L} \quad \text{where } L \text{ represents the characteristic length of the system} \tag{5.194}$$

In a strong magnetic field, $r_\mathrm{L} \ll L$, so the ratio is of the same order as $r_\mathrm{L}/L \ll 1$. In this case, particles approximately conserve the product ρA_φ, which means they are confined to surfaces where $\rho A_\varphi = \text{const.}$ Even if one considers the kinetic term, the escape rate of particles from these surfaces is bounded from above by energy conservation. It should be emphasized that the surfaces for which $\rho A_\varphi = \text{const.}$ are precisely the magnetic surfaces. We begin by expanding the curl of an axisymmetric magnetic field:

$$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\partial A_\varphi}{\partial z}\,\hat{\boldsymbol{\rho}} + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_\mathrm{z}}{\partial \rho}\right)\hat{\boldsymbol{\varphi}} + \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\varphi)\,\hat{\mathbf{z}} \tag{5.195}$$

The result is immediate. The surfaces of $\rho A_\varphi = \text{const.}$ are indeed the magnetic surfaces:

$$\mathbf{B} \cdot \nabla(\rho A_\varphi) = B_\rho \frac{\partial}{\partial \rho}(\rho A_\varphi) + B_\mathrm{z} \frac{\partial}{\partial z}(\rho A_\varphi) = B_\rho \rho B_\mathrm{z} - B_\mathrm{z} \rho B_\rho = 0 \tag{5.196}$$

5.8.1.2 Difficulties with axisymmetric confinements

The magnetic field in a toroidal system must be maintained by a current distribution inside the magnetic surfaces, as shown in subsection (2.10.3.2). This distribution is concretized by equation (2.107). The current inside the system entails several problems. First, the production and maintenance of a stable current consume an enormous amount of energy. Similarly, the current tends to be unstable and can lead to other instabilities, which can result in the rupture of the magnetic surfaces defined by $\rho A_\varphi = \text{const.}$

Chapter 6

Magnetohydrodynamic equilibria

This chapter is entirely dedicated to the theoretical development of magnetohydrodynamic equilibria. Considering our previous results on the magnetohydrodynamic model, we conclude that it fails to capture the diverse intricacies predicted by other models. It is only natural to contest the validity of using the magnetohydrodynamic model to describe macroscopic plasma equilibria. In fact, the magnetohydrodynamic model, providing an entirely macroscopic perspective on plasma behaviour, allows us to efficiently assess situations in which the greater detail and accuracy of the two-fluid or Vlasov models are not our primary concern. Since both the two-fluid and the Vlasov model focus on microscopic behaviour more than global behaviour, they prove to be unsuitable for systems characterised by complex geometry, especially since those are often of utmost importance in the magnetohydrodynamic description.

6.1 Magnetic field structure revisited

The simplest non-trivial magnetic field that we shall first consider is a static free-space magnetic field produced by external currents that are physically outside of our system. The Maxwell-Ampère equation thus describes a curl-free magnetic field, which can consequently be expressed as a gradient of a scalar field, more precisely, of a magnetostatic potential.

$$\nabla \times \mathbf{B} = \mathbf{0} \implies \mathbf{B} = \nabla\chi \qquad \nabla \cdot \mathbf{B} = 0 \implies \nabla^2\chi = 0 \tag{6.1}$$

The magnetostatic potential thus satisfies the Laplace equation, and the system is entirely analogous to an electrostatic system. Thus, the magnetostatic potential inside an evacuated volume is uniquely determined by either the Dirichlet or the Neumann boundary conditions for χ. Moreover, if the system's magnetic configuration is continuously symmetric along a certain coordinate, then the linearised equations governing the magnetostatic potential can be Fourier-transformed along that coordinate.

Non-vacuum fields are more complicated, first because certain current distributions exist within the system, and second because they exhibit inelegant dependencies on boundary conditions. Furthermore, vacuum fields are peculiarly distinguished from non-vacuum fields by their property of being the lowest-energy fields that satisfy any given boundary conditions. Let us prove this statement. Consider an evacuated volume Ω whose boundary is $\partial\Omega$. Let $\mathbf{B}_{\text{min}}$ be the vacuum field describing the lowest-energy configuration that satisfies the necessary boundary conditions imposed on $\partial\Omega$. Let $\mathbf{B} \neq \mathbf{B}_{\text{min}}$ be another arbitrary field satisfying the same boundary conditions. Let $\delta\mathbf{B} = \mathbf{B} - \mathbf{B}_{\text{min}}$; hence, it vanishes on the boundary. The magnetic energy associated with the arbitrary field is given by:

$$W = \frac{1}{2\mu_0} \iiint_\Omega \mathbf{B}^2 \,\mathrm{d}^3\mathbf{r} = \frac{1}{2\mu_0} \iiint_\Omega \left[\mathbf{B}_{\text{min}}^2 + \mathbf{B}_{\text{min}} \cdot \delta\mathbf{B} + (\delta\mathbf{B})^2\right] \,\mathrm{d}^3\mathbf{r} = \tag{6.2}$$

$$= W_{\text{min}} + \frac{1}{2\mu_0} \iiint_\Omega \mathbf{B}_{\text{min}} \cdot (\delta\mathbf{B}) \,\mathrm{d}^3\mathbf{r} + \frac{1}{2\mu_0} \iiint_\Omega (\delta\mathbf{B})^2 \,\mathrm{d}^3\mathbf{r} \tag{6.3}$$

Let us focus on the middle term in the previous expression, since it is precisely this term that can contribute negatively to the total associated magnetic energy. Using $\delta\mathbf{B} = \nabla \times (\delta\mathbf{A}) + \nabla(\delta\Phi)$, the middle term can be recast as follows:

$$\iiint_\Omega \mathbf{B}_{\text{min}} \cdot (\delta\mathbf{B}) \,\mathrm{d}^3\mathbf{r} = \iiint_\Omega \left[\nabla \cdot (\delta\mathbf{A} \times \mathbf{B}_{\text{min}}) + \nabla \cdot (\mathbf{B}_{\text{min}}\delta\Phi)\right] \,\mathrm{d}^3\mathbf{r} + \iiint_\Omega \delta\mathbf{A} \cdot (\nabla \times \mathbf{B}_{\text{min}}) \,\mathrm{d}^3\mathbf{r} \tag{6.4}$$

The first term vanishes because $\delta\mathbf{A}$ must vanish on the boundary since both $\mathbf{B}$ and $\mathbf{B}_{\min}$ are assumed to satisfy the same boundary conditions. Moreover, the potential contribution vanishes since $\mathbf{B}$ satisfies $\nabla\cdot\mathbf{B}=0$. The last term also vanishes since $\mathbf{B}_{\min}$ is a curl-free field. The middle term in equation (6.3) evaluates to zero, so that it is only the third term that can, evidently, contribute only positively to the total magnetic energy. Thus, $\mathbf{B}_{\min}$ indeed describes the lowest-energy configuration. An important corollary is that all configurations satisfying the imposed boundary conditions and having finite current within the system do not represent the lowest-energy configuration and can, in principle, use the free energy to drive instabilities with respect to the imposed boundary conditions.

6.1.1 Force-free magnetic fields

Although we have shown that non-vacuum magnetic fields necessarily represent configurations of higher energy, it is not necessary that such configurations present instabilities or, more precisely, that they decay towards the associated vacuum configuration. This is possible for configurations referred to as force-free, in which the current distribution is not identically zero, but obeys $\mathbf{J}\times\mathbf{B}=\mathbf{0}$, that is, it is parallel to the magnetic field. If the initial conditions imposed on a plasma are non-force-free, and, during its subsequent evolution, the plasma transitions into a force-free configuration, it can get trapped in that particular configuration since, in such a state, no force drives its evolution further.

Our considerations on the concepts of magnetic pressure and tension from subsection 5.1.1 allow us to gain further insight regarding the evolution of non-force-free configurations. Let us consider the example of a simple circular loop current. The magnetic tension, as given in equation (5.16), acts along the magnetic field's curvature vector, thereby maintaining the loop's constricted shape and pinching it into a thinner loop. Meanwhile, the magnetic pressure force, expressed in equation (5.15), acts in the direction of smaller $\mathbf{B}^2$. This is illustrated in our example of a current loop by the fact that currents flow in opposite directions at diametrically opposite points of the loop and thus mutually repel. Since, due to the circular geometry, $\mathbf{B}^2$ is much larger inside the circular loop than outside, the loop experiences a force trying to expand the loop. Thus, if the considered circular loop carrying the current were elastic and freely deformable, magnetic forces would elongate into a thinner, larger loop.

6.1.2 Magnetohydrodynamic virial theorem

We re-examine the considerations on magnetic tension and pressure in the previous subsection, or, more precisely, the force density given by equation (5.11), to define the stress tensor as follows:

$$\mathbf{f}=-\nabla\cdot\mathbf{P}_{\mathrm{MHD}}+\frac{1}{\mu_0}(\mathbf{B}\cdot\nabla)\mathbf{B}-\nabla\left(\frac{\mathbf{B}^2}{2\mu_0}\right)=-\nabla\cdot\left(\mathbf{P}_{\mathrm{MHD}}+\frac{\mathbf{B}^2}{2\mu_0}\mathbf{1}-\frac{\mathbf{B}\mathbf{B}}{\mu_0}\right)\tag{6.5}$$

This allows us to express magnetohydrostatic equilibria as follows elegantly:

$$\nabla\cdot\mathbf{T}_{\mathrm{MHD}}=\nabla\cdot\left(\mathbf{P}_{\mathrm{MHD}}+\frac{\mathbf{B}^2}{2\mu_0}\mathbf{1}-\frac{\mathbf{B}\mathbf{B}}{\mu_0}\right)=\mathbf{0}\tag{6.6}$$

The magnetohydrodynamic stress tensor $\mathbf{T}_{\mathrm{MHD}}$ enables us to derive the magnetohydrodynamic virial theorem, which has fundamentally fuelled the theoretical and experimental development of magnetic confinement. Let us suppose that there exists a finite three-dimensional system Ω characterised by finite pressure and self-confined via currents at its interior. There are no currents in the vacuum region, that is, in the system's exterior. Such a system is thus in equilibrium, as governed by the condition above. Let us now consider the following expression:

$$\nabla\cdot(\mathbf{T}_{\mathrm{MHD}}\cdot\mathbf{r})=\mathbf{r}\cdot(\nabla\cdot\mathbf{T}_{\mathrm{MHD}})+\operatorname{Tr}\mathbf{T}_{\mathrm{MHD}}\tag{6.7}$$

where, by hypothesis, the first term vanishes. Let us expand the trace:

$$\operatorname{Tr}\mathbf{T}_{\mathrm{MHD}}\big|_{\Omega}=\left[\operatorname{Tr}\mathbf{P}_{\mathrm{MHD}}+\frac{\mathbf{B}^2}{2\mu_0}\right]\Bigg|_{\Omega}>0\tag{6.8}$$

The virial $\nabla\cdot(\mathbf{T}_{\mathrm{MHD}}\cdot\mathbf{r})$ is positive-definite inside the system, and certainly semi-positive-definite outside the system. Its integral over all space ought to evaluate to a positive quantity:

$$\iiint_{\mathbb{R}^3}\nabla\cdot(\mathbf{T}_{\mathrm{MHD}}\cdot\mathbf{r})\,\mathrm{d}^3\mathbf{r}=\oiint_{\partial\mathbb{R}^3}(\mathbf{T}_{\mathrm{MHD}}\cdot\mathbf{r})\cdot\mathrm{d}\mathbf{S}\tag{6.9}$$

By assumption, the plasma is of finite extent, so the pressure term vanishes at infinity. The magnetic field also vanishes at infinity since localised currents within the plasma produce it. The entire integral evaluates to zero identically, which contradicts the positive-definiteness of the introduced virial. That system cannot be. All three-dimensional static plasma confinement is necessarily maintained by currents external to the plasma, but also held by some mechanical structure. If the former would not be supported by mechanical means, these currents could be considered as part of the system, and the virial theorem above would be violated. Therefore, any plasma system of finite pressure and finite extent must have a tangible exterior surface on which the magnetohydrodynamic pressure can act.

6.1.3 Flux preservation and inductance

Another useful way to visualise magnetic-field behaviour is through magnetic self-inductance. Self-inductance of some system carrying a total current I is defined as the ratio between the magnetic flux through the system and this current that produces it. We shall further explore the consequences of such a definition via the following considerations. Let Σ denote some cross-section of a system Ω:

$$\Phi = \iint_{\Sigma} \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \iint_{\Sigma} (\nabla \times \mathbf{A}) \cdot \mathrm{d}\mathbf{S} = \oint_{\partial\Sigma} \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} \tag{6.10}$$

The magnetic energy stored within the system is expressed as follows:

$$W = \frac{1}{2\mu_0} \iiint_{\Omega} \mathbf{B}^2 \, \mathrm{d}^3\mathbf{r} = \frac{1}{2\mu_0} \iiint_{\Omega} \mathbf{B} \cdot (\nabla \times \mathbf{A}) \, \mathrm{d}^3\mathbf{r} = \frac{1}{2\mu_0} \iiint_{\Omega} \nabla \cdot (\mathbf{A} \times \mathbf{B}) + \mathbf{A} \cdot (\nabla \times \mathbf{B}) \, \mathrm{d}^3\mathbf{r} \tag{6.11}$$

$$W = \frac{1}{2\mu_0} \oint\!\!\!\oint_{\partial\Omega} (\mathbf{A} \times \mathbf{B}) \cdot \mathrm{d}\mathbf{S} + \frac{1}{2} \iiint_{\Omega} \mathbf{A} \cdot \mathbf{J} \, \mathrm{d}^3\mathbf{r} \tag{6.12}$$

The first term vanishes if the boundary were at infinity, so that we shall focus on the latter.

$$W = \frac{1}{2} \iiint_{\Omega} \mathbf{A} \cdot \mathbf{J} \, \mathrm{d}^3\mathbf{r} = \frac{1}{2} \iiint_{\Omega} \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} \, \mathbf{J} \cdot \mathrm{d}\mathbf{S} \tag{6.13}$$

where $\mathrm{d}\boldsymbol{\ell}$ and $\mathrm{d}\mathbf{S}$ are the differential length and surface elements parallel and perpendicular to the current density $\mathbf{J}$, respectively. For a circular loop carrying a total current I, the previous expression simplifies:

$$W = \frac{1}{2} I \oint_{\partial\Sigma} \mathbf{A} \cdot \mathrm{d}\boldsymbol{\ell} = \frac{1}{2} \Phi I = \frac{1}{2} L I^2 = \frac{1}{2} \frac{\Phi^2}{L} \tag{6.14}$$

For a perfectly conducting system, the magnetic flux produced is conserved; if the magnetic flux varied, a non-zero voltage distribution would occur in the loop, contradicting the perfect-conductivity assumption. In the last expression, we find that the system evolves to increase its self-inductance.

6.2 Basic properties of ideal magnetohydrodynamics

We shall summarise the magnetohydrodynamic equations and use them to derive certain elementary properties of ideal magnetohydrodynamic systems. Such systems are, among other laws, governed by the imposed zero resistivity, so that Ohm's law and Maxwell-Faraday equation in conjunction give:

$$\eta \mathbf{J} = \mathbf{E} + \mathbf{U} \times \mathbf{B} \implies \frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{U} \times \mathbf{B}) \tag{6.15}$$

We shall employ this result in our subsequent considerations. The reader is advised to subconsciously keep track of the number of results that use this equation, so that when we explore the resistive magnetohydrodynamic model, the reader gains insight into just how many of these concepts fail to retain their validity.

6.2.1 Magnetic flux tubes

In the magnetohydrodynamic model, magnetic flux tubes are of fundamental importance. The term is somewhat synonymous with the concept of magnetic surfaces, but the two are not interchangeable. More precisely, in a simple toroidal system with a single magnetic axis, flux tubes and magnetic surfaces designate the same concept. We shall nevertheless use the following definition.

In toroidal systems with multiple magnetic axes, we define a *flux tube* as the magnetic surface of maximal volume associated with a single magnetic axis. Emphasis on the word *maximal*; it may be possible that the maximum volume is not finite, as we shall see in some illustrative examples. A well-defined definition of a flux tube in a multiply-axed system is thus the supremum over volume of nested magnetic surfaces associated with a common magnetic axis. The supremum is taken in a manner to preserve the topology of the nested magnetic surfaces. In confined systems of finite extent, the supremum is indeed physically reached. We call *domain* of a flux tube the three-dimensional domain where its associated magnetic field is non-zero. The magnetic field structure within a system characterised by multiple magnetic axes is topologically described by the interrelation among associated magnetic flux tubes, which is physically manifested by their self-twisting and mutual linkage.

The ideal magnetohydrodynamic model provides us with an essential result, the frozen flux theorem. Consider the magnetic flux through the cross-section Σ of a flux tube:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\iint_{\Sigma} \mathbf{B}\cdot \mathrm{d}\mathbf{S} = \iint_{\Sigma}\left[\frac{\partial \mathbf{B}}{\partial t} + (\nabla\cdot\mathbf{B})\mathbf{U}\right]\cdot \mathrm{d}\mathbf{S} + \oint_{\partial\Sigma}(\mathbf{B}\times\mathbf{U})\cdot \mathrm{d}\boldsymbol{\ell} \tag{6.16}$$

where the second term inside the surface integral identically vanishes for $\nabla\cdot\mathbf{B} = 0$. The first term gives:

$$\iint_{\Sigma}\frac{\partial \mathbf{B}}{\partial t}\cdot \mathrm{d}\mathbf{S} = \iint_{\Sigma}(\nabla\times(\mathbf{U}\times\mathbf{B}))\cdot \mathrm{d}\mathbf{S} = \oint_{\partial\Sigma}(\mathbf{U}\times\mathbf{B})\cdot \mathrm{d}\boldsymbol{\ell} \tag{6.17}$$

$$\frac{\mathrm{d}\Phi}{\mathrm{d}t} = \iint_{\Sigma}\frac{\partial \mathbf{B}}{\partial t}\cdot \mathrm{d}\mathbf{S} + \oint(\mathbf{B}\times\mathbf{U})\cdot \mathrm{d}\boldsymbol{\ell} = \oint(\mathbf{U}\times\mathbf{B})\cdot \mathrm{d}\boldsymbol{\ell} + \oint(\mathbf{B}\times\mathbf{U})\cdot \mathrm{d}\boldsymbol{\ell} = 0 \tag{6.18}$$

This property of ideal magnetohydrodynamics is commonly referred to as the *frozen flux theorem.*

6.2.2 Magnetic helicity: interlude to the topology of magnetic field structure

The helicity of a particular magnetic flux tube, whose domain is Ω, is a measure of the extent to which magnetic field-lines wrap and coil around one another. It is formally defined via the following integral:

$$H = \iiint_{\Omega}\mathbf{A}\cdot\mathbf{B}\,\mathrm{d}^3\mathbf{r} \tag{6.19}$$

The validity of such a definition is indeed questionable since $\mathbf{A}$ can be subjected to a gauge transformation via $\mathbf{A}\to\mathbf{A}+\nabla f$ for some scalar function f. Thus, magnetic helicity would be of little practical use if it were itself gauge-dependent, since gauge has no intrinsic physical significance. Let us now consider the case of a volume enclosed by a magnetic surface, such that $\mathbf{B}\cdot\mathbf{n} = 0$, and let H' be the helicity within the same surface with the aforementioned associated gauge:

$$H' - H = \iiint_{\Omega}\mathbf{B}\cdot\nabla f\,\mathrm{d}^3\mathbf{r} = \iiint_{\Omega}\nabla\cdot(f\mathbf{B})\,\mathrm{d}^3\mathbf{r} = \iint_{\Omega} f\mathbf{B}\cdot \mathrm{d}\mathbf{S} = 0 \tag{6.20}$$

For volumes enclosed by magnetic surfaces, their associated magnetic helicities are gauge-independent.

Let us demonstrate that the magnetic helicity of any magnetic surface is an invariant:

$$\frac{\mathrm{d}}{\mathrm{d}t}\iiint_{\Omega}\mathbf{A}\cdot\mathbf{B}\,\mathrm{d}^3\mathbf{r} = \iiint_{\Omega}\frac{\partial}{\partial t}(\mathbf{A}\cdot\mathbf{B})\,\mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega}(\mathbf{A}\cdot\mathbf{B})\mathbf{U}\cdot \mathrm{d}\mathbf{S} = \tag{6.21}$$

$$= \iiint_{\Omega}\left(\frac{\partial \mathbf{A}}{\partial t}\cdot\mathbf{B} + \mathbf{A}\cdot\frac{\partial \mathbf{B}}{\partial t}\right)\mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega}(\mathbf{A}\cdot\mathbf{B})\mathbf{U}\cdot \mathrm{d}\mathbf{S} \tag{6.22}$$

Using equation (6.15), we find that, for some gauge f:

$$\frac{\partial \mathbf{A}}{\partial t} = \mathbf{U}\times\mathbf{B} + \nabla f \implies \iiint_{\Omega}\frac{\partial \mathbf{A}}{\partial t}\cdot\mathbf{B}\,\mathrm{d}^3\mathbf{r} = \iiint_{\Omega}(\mathbf{U}\times\mathbf{B}+\nabla f)\cdot\mathbf{B}\,\mathrm{d}^3\mathbf{r} = 0 \tag{6.23}$$

since the first term is perpendicular to the magnetic field, and the second vanishes because we have chosen a volume enclosed by a magnetic surface.

$$\iiint_\Omega \mathbf{A}\cdot\frac{\partial \mathbf{B}}{\partial t}\,\mathrm{d}^3\mathbf{r} = \iiint_\Omega \mathbf{A}\cdot\left[\nabla\times\frac{\partial \mathbf{A}}{\partial t}\right]\mathrm{d}^3\mathbf{r} = \iiint_\Omega \left[-\nabla\cdot\left(\mathbf{A}\times\frac{\partial \mathbf{A}}{\partial t}\right)+(\nabla\times\mathbf{A})\cdot\frac{\partial \mathbf{A}}{\partial t}\right]\mathrm{d}^3\mathbf{r} \tag{6.24}$$

where we recognise that the second integral is equal to that in equation (6.23), and therefore vanishes. The first integral is evaluated by means of the divergence theorem, giving:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \oiint_{\partial\Omega}\left[-\left(\mathbf{A}\times\frac{\partial \mathbf{A}}{\partial t}\right)+(\mathbf{A}\cdot\mathbf{B})\mathbf{U}\right]\cdot\mathrm{d}\mathbf{S} = \oiint_{\partial\Omega}\left[-\left(\mathbf{A}\times(\mathbf{U}\times\mathbf{B}+\nabla f)\right)+(\mathbf{A}\cdot\mathbf{B})\mathbf{U}\right]\cdot\mathrm{d}\mathbf{S} = \tag{6.25}$$

$$= \oiint_{\partial\Omega}\left[(\mathbf{A}\cdot\mathbf{U})\mathbf{B}-(\mathbf{A}\cdot\mathbf{B})\mathbf{U}+(\mathbf{A}\cdot\mathbf{B})\mathbf{U}\right]\cdot\mathrm{d}\mathbf{S} - \oiint_{\partial\Omega}(\mathbf{A}\times\nabla f)\cdot\mathrm{d}\mathbf{S} \tag{6.26}$$

where the first integral entirely evaluates to zero since, again, $\partial\Omega$ is a magnetic surface, thus $\mathbf{B}\perp\mathrm{d}\mathbf{S}$, and the latter two terms cancel out. The single remaining integral also evaluates to zero:

$$\oiint_{\partial\Omega}(\mathbf{A}\times\nabla f)\cdot\mathrm{d}\mathbf{S} = \oiint_{\partial\Omega}\left[f\nabla\times\mathbf{A}-\nabla\times(f\mathbf{A})\right]\cdot\mathrm{d}\mathbf{S} = \oiint_{\partial\Omega}\left[f\mathbf{B}-\nabla\times(f\mathbf{A})\right]\cdot\mathrm{d}\mathbf{S} = 0 \tag{6.27}$$

where we recognise that the first term vanishes since we have, yet again, chosen a magnetic surface, and the second term vanishes by Stokes's theorem since a closed surface has no boundary. We have thus fully demonstrated that the magnetic helicity, along with the magnetic flux, within a volume enclosed by a magnetic surface is an invariant in ideal magnetohydrodynamics.

We shall now briefly discuss how magnetic helicities of magnetic flux tubes describe a system's topology of magnetic surfaces, firstly with an example of a single flux tube within a toroidal system:

$$H_{\text{self}} = \iiint_\Omega \mathbf{A}\cdot\mathbf{B}\,\mathrm{d}^3\mathbf{r} = \iiint_\Omega \mathbf{B}\cdot\mathrm{d}\mathbf{S}\,\mathbf{A}\cdot\mathrm{d}\boldsymbol{\ell} \tag{6.28}$$

It is now opportune to discuss what the differential term $\mathbf{B}\cdot\mathrm{d}\mathbf{S}$ truly designates. Following our considerations on magnetic field structures, and envisioning the flux tube as a toroid, we can consider $\mathrm{d}\mathbf{S}$ to be the oriented differential element of the toroidal cross-section of the flux tube, thus $\mathbf{B}\cdot\mathrm{d}\mathbf{S} = \mathrm{d}\Phi_{\text{tor}}$. Consequently, $\mathrm{d}\boldsymbol{\ell}$ thus represents the differential length element of some curve $\mathcal{C}$ in the toroidal direction. Upon applying Stokes's theorem, we obtain the magnetic flux through the poloidal disk surface. Notice that the choice of curve $\mathcal{C}$ matters not, so long as the topology of the magnetic flux tube remains considered. One can, in ideal magnetohydrodynamics at least, always connect the curve $\mathcal{C}$ to the magnetic axis via a ribbon-like surface, and create a surface whose boundary is $\mathcal{C}$ via the disk enclosed by the axis. The flux enclosed by the curve $\mathcal{C}$ is thus $\Phi^{\text{d}}_{\text{pol}}+\Phi^{\text{r}}_{\text{pol}}$, taken with the sign. This sum is indeed constant, as shown in equation (5.32), so that we may write:

$$H_{\text{self}} = \Phi_{\text{tor}}\Phi^{\text{r}}_{\text{pol}} = \iota\!\!\!-\Phi^2_{\text{tor}} \tag{6.29}$$

where the rotational transform may be negative, and Φ_{tor} and $\Phi^{\text{r}}_{\text{pol}}$ denote the toroidal and the poloidal disk fluxes of the whole flux tube, respectively. Notice that the self-helicity of a magnetic surface is indeed a flux function, and its definition can also be extended to general magnetic surfaces.

We shall now introduce the *mutual* magnetic helicity associated with two magnetic flux tubes. Let $\mathbf{A}_1$ and $\mathbf{A}_2$ be the magnetic vector potentials of the first and the second flux tube, respectively, whose domains are Ω_1 and Ω_2. Denote $\Omega = \Omega_1\cup\Omega_2$. We shall define their mutual helicity as follows:

$$H_{1\to2} = \iiint_\Omega \mathbf{A}_1\cdot\mathbf{B}_2\,\mathrm{d}^3\mathbf{r} = \iiint_\Omega \mathbf{A}_1\cdot(\nabla\times\mathbf{A}_2)\,\mathrm{d}^3\mathbf{r} = \tag{6.30}$$

$$= \iiint_\Omega\left[-\nabla\cdot(\mathbf{A}_1\times\mathbf{A}_2)\right]\mathrm{d}^3\mathbf{r} + \iiint_\Omega(\nabla\times\mathbf{A}_1)\cdot\mathbf{A}_2\,\mathrm{d}^3\mathbf{r} = -\oiint_{\partial\Omega}(\mathbf{A}_1\times\mathbf{A}_2)\cdot\mathrm{d}\mathbf{S} + H_{2\to1} \tag{6.31}$$

The excessive term necessarily vanishes by the definition of the domain of a flux tube, so $H_{1\to2} = H_{2\to1}$. Furthermore, by an analogous argument, the definition above is gauge-independent, so that mutual helicity is well-defined.

For two flux tubes interlinked once, denoted as the i-th and j-th tubes, we perform an analogous procedure; however, now, we shall distinguish the two magnetic axes corresponding to their associated flux tubes. Notice that, since both the magnetic flux and the helicity are invariant inside each flux tube, we may freely deform the flux tubes to make their cross-sections within the linkage region perpendicular:

$$H_{i,j} = \pm \Phi^{(i)}_{\text{tor}} \Phi^{(j)}_{\text{tor}} \tag{6.32}$$

where the exact sign depends on the chosen orientation of both cross-sections.

Our considerations so far concerned only toroidal magnetic field structures. It is worth noting that, in helical magnetic field structures, e.g., in stellarators, the magnetic helicity of a flux tube contains two contributions, the *twist*, embodying the same concept as the rotational transform, as well as the *writhe*, which represents the winding and kinking of the magnetic axis itself. In toroidal systems, the magnetic axis has identically zero writhe. We shall return to these considerations once we have further developed our understanding of helical magnetic field structures.

Yet another analogous procedure, as the above, shows that, in general, having a total of N flux tubes, each with its respective rotational transform, and knowing the number of linkages for each pair of flux tubes, the total magnetic helicity of the system is given, each term up to a sign, as follows:

$$H_{\text{tot}} = \sum_{i=1}^{N} \iota^{(i)} \left[\Phi^{(i)}_{\text{tor}}\right]^2 + \sum_{i=1}^{N}\sum_{j=1}^{N} L_{ij} \Phi^{(i)}_{\text{tor}} \Phi^{(j)}_{\text{tor}} \tag{6.33}$$

We notice that the total helicity is a quadratic form in terms of the toroidal fluxes within each flux tube, so that the entire system's topology can be represented via an associated linkage matrix. In systems with a great number of linkages, the self-helicity may be neglected in certain circumstances.

In systems whose magnetic field structure is characterised by at least two magnetic axes, the magnetic field cannot globally be described via many, if any, of the flux coordinate systems explored in chapter 5, since these magnetic surfaces do not form a nested family of toroidal surfaces, and, throughout chapter 5, we have tacitly assumed the existence of a unique magnetic axis. Accent on the word *globally*; our considerations on flux coordinate systems were certainly not for naught. We can always construct such flux coordinate systems within the volume of a certain flux tube.

Let us gain much greater insight into the geometry of magnetic field structures with interlinked magnetic flux tubes by considering two non-intersecting smooth curves, $\gamma_1, \gamma_2 \in \mathcal{C}^1(\mathcal{S}^1, \mathbb{R}^3)$. We introduce the Gauss map from the torus to the unit sphere as follows:

$$\Gamma = \frac{\gamma_1(s) - \gamma_2(t)}{\|\gamma_1(s) - \gamma_2(t)\|} \tag{6.34}$$

By picking a point P on the unit sphere $\mathcal{S}^1$ and projecting a link orthogonally to the tangent plane of the point, we may create the so-called *link diagram.* Picking a point (s, t) on the unit sphere that goes to the point P corresponds to a crossing in the link diagram, where γ_1 winds around γ_2. By the definition of the Gauss map, the neighbourhood of such a point either preserves or reverses the orientation, depending on the sign of crossing, which is determined via the right-hand rule. To compute the linking number, it suffices to count the *signed* number of times that the Gauss map covers the point P. This formulation of the linking number allows us to represent the linking number between the curves γ_1 and γ_2 as follows:

$$\text{link}(\gamma_1, \gamma_2) = \frac{1}{4\pi} \oint_{\gamma_1} \oint_{\gamma_2} \frac{\mathbf{r_1} - \mathbf{r_2}}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3} \cdot (\mathrm{d}\boldsymbol{\ell}_1 \times \mathrm{d}\boldsymbol{\ell}_2) \tag{6.35}$$

which is commonly referred to as the *Gauss linking integral*, specifically computes the total signed area of the image of the Gauss map, and is then divided by the area of the unit sphere. Let us examine how this definition precisely relates to the mutual magnetic helicity of two flux tubes. Let γ_1 and γ_{j} now represent each a single magnetic field line belonging to two distinct magnetic flux tubes, Ω_1 and Ω_2, such that $\Omega = \Omega_1 \cup \Omega_2$, whose cross-sections shall be denoted as Σ_1 and Σ_2, respectively. The equation of a magnetic field line applies to each curve. Let us now show that the following integral is equivalent to the above definition of mutual magnetic helicity.

$$H_{1\leftrightarrow 2} = \iiint_{\Omega} \iiint_{\Omega} \underbrace{(\mathbf{B}_1 \cdot \mathrm{d}\boldsymbol{\Sigma}_1)}_{\mathrm{d}\Phi_1^{\text{tor}}} \underbrace{(\mathbf{B}_2 \cdot \mathrm{d}\boldsymbol{\Sigma}_2)}_{\mathrm{d}\Phi_2^{\text{tor}}} \frac{\mathbf{r}_1 - \mathbf{r}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3} \cdot (\mathrm{d}\boldsymbol{\ell}_1 \times \mathrm{d}\boldsymbol{\ell}_2) \tag{6.36}$$

where we have chosen a parametrisation of both Ω_1 and Ω_2 such that $\mathrm{d}\boldsymbol{\Sigma}$ denotes the oriented surface differential at all points orthogonal to the magnetic field-lines whose length differential is $\mathrm{d}\boldsymbol{\ell}$. Isolating independent terms gives:

$$H_{1\leftrightarrow 2} = \iiint_\Omega (\mathbf{B}_1 \cdot \mathrm{d}\mathbf{\Sigma}_1)\, \mathrm{d}\boldsymbol{\ell}_1 \cdot \iiint_\Omega (\mathbf{B}_2 \cdot \mathrm{d}\mathbf{\Sigma}_2) \frac{\mathrm{d}\boldsymbol{\ell}_2 \times (\mathbf{r}_1 - \mathbf{r}_2)}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3} \tag{6.37}$$

Having chosen $\mathbf{B} \cdot \mathrm{d}\mathbf{\Sigma} = \|\mathbf{B}\|\, \mathrm{d}\Sigma$, and the magnetic differential equation gives $\mathrm{d}\boldsymbol{\ell} \times \mathbf{B} = \mathbf{0}$, we find:

$$H_{1\leftrightarrow 2} = \iiint_\Omega \mathbf{B}_1(\mathbf{r}_1)\, \mathrm{d}^3\mathbf{r}_1 \cdot \iiint_\Omega \frac{\mathbf{B_2}(\mathbf{r}_2) \times (\mathbf{r}_1 - \mathbf{r}_2)}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3}\, \mathrm{d}^3\mathbf{r}_2 \tag{6.38}$$

We recognise that the inner integral gives precisely the magnetic vector potential via a convolution:

$$\iiint_\Omega \mathbf{B}_2 \times \nabla_2 \left(\frac{1}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right) \mathrm{d}^3\mathbf{r}_2 = \iiint_\Omega \left[\frac{\nabla_2 \times \mathbf{B}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} - \nabla_2 \times \left(\frac{\mathbf{B}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right) \right] \mathrm{d}^3\mathbf{r}_2 = \tag{6.39}$$

$$= \iiint_\Omega \frac{\mu_0 \mathbf{J_2}(\mathbf{r}_2)}{\|\mathbf{r}_1 - \mathbf{r}_2\|}\, \mathrm{d}^3\mathbf{r}_2 + \oiint_{\partial\Omega} \left(\frac{\mathbf{B}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right) \times \mathrm{d}\mathbf{S} = \mathbf{A}_2(\mathbf{r}_1) + \oiint_{\partial\Omega} \left(\frac{\mathbf{B}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|} \right) \times \mathrm{d}\mathbf{S} \tag{6.40}$$

where the remaining integral vanishes by the definition of a domain of a magnetic flux tube. These two definitions for mutual magnetic helicity are thus equivalent:

$$H_{1\leftrightarrow 2} = \iiint_\Omega \mathbf{A}_2 \cdot \mathbf{B}_1\, \mathrm{d}^3\mathbf{r} = \iiint_\Omega \mathbf{A}_1 \cdot \mathbf{B}_2\, \mathrm{d}^3\mathbf{r} \tag{6.41}$$

where we place emphasis on the last equality, since we could've chosen to integrate first with respect to the other flux tube. Notice that in equation (6.40), we have used the Coulomb gauge. However, this choice is arbitrary, since the boundary of the domain of integration is the boundary of a flux tube, and the gauge term vanishes identically. It was only convenient to employ the Coulomb gauge.

The conclusion of this parallel to a purely mathematical concept is that helicity fundamentally measures the linking number over all pairs of two magnetic field-lines, each belonging to its respective magnetic flux-tube. Moreover, we are now absolutely certain that the linking matrix given in equation (6.33) contains exclusively integer entries, since, in Euclidean space, the linking number is always an integer.

6.2.2.1 Transport of magnetic helicity

In this short subsection, we shall explore the transport properties of magnetic helicities under general considerations of resistive magnetohydrodynamics. The time derivative of magnetic helicity reads:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \iiint_\Omega \frac{\partial}{\partial t}(\mathbf{A} \cdot \mathbf{B})\, \mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega} (\mathbf{A} \cdot \mathbf{B}) \mathbf{U} \cdot \mathrm{d}\mathbf{S} = \iiint_\Omega \left(\frac{\partial \mathbf{A}}{\partial t} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{\partial \mathbf{B}}{\partial t} \right) \mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega} (\mathbf{A} \cdot \mathbf{B}) \mathbf{U} \cdot \mathrm{d}\mathbf{S} \tag{6.42}$$

$$\frac{\partial \mathbf{A}}{\partial t} = -\mathbf{E} - \nabla\phi \qquad \frac{\partial \mathbf{B}}{\partial t} = -(\nabla \times \mathbf{E}) \tag{6.43}$$

$$\frac{\mathrm{d}H}{\mathrm{d}t} = -\iiint_\Omega \left[(\nabla \times \mathbf{A}) \cdot \mathbf{E} + \mathbf{A} \cdot (\nabla \times \mathbf{E}) \right] \mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega} (\mathbf{A} \cdot \mathbf{B}) \mathbf{U} \cdot \mathrm{d}\mathbf{S} \tag{6.44}$$

where the potential term vanishes since Ω is a flux tube. The first term simplifies into:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \iiint_\Omega \nabla \cdot (\mathbf{A} \times \mathbf{E})\, \mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega} (\mathbf{A} \cdot \mathbf{B}) \mathbf{U} \cdot \mathrm{d}\mathbf{S} = \oiint_{\partial\Omega} \left[\mathbf{A} \times \mathbf{E} + (\mathbf{A} \cdot \mathbf{B}) \mathbf{U} \right] \cdot \mathrm{d}\mathbf{S} \tag{6.45}$$

The general resistive magnetohydrodynamic model describes the current via a resistivity tensor:

$$\boldsymbol{\eta} \cdot \mathbf{J} = \mathbf{E} + \mathbf{U} \times \mathbf{B} \tag{6.46}$$

Here, we omit mathematical labour, since it merely consists of expanding the double vector product, which cancels out with the other boundary contribution, while the term $(\mathbf{A} \cdot \mathbf{U})\mathbf{B}$ does not contribute since we have chosen a flux tube:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = -2 \iiint_\Omega \mathbf{B} \cdot \boldsymbol{\eta} \cdot \mathbf{J}\, \mathrm{d}^3\mathbf{r} + \oiint_{\partial\Omega} \left[\mathbf{A} \times (\boldsymbol{\eta} \cdot \mathbf{J}) \right] \cdot \mathrm{d}\mathbf{S} \tag{6.47}$$

Although it is difficult to evaluate the previous integral in practice, it is worth comparing the magnetic helicity dissipation to the Ohmic resistive dissipation rate, given by:

$$\left(\frac{\mathrm{d}W}{\mathrm{d}t}\right)_{\mathrm{Ohm}} = \iiint_\Omega \mathbf{J}\cdot\boldsymbol{\eta}\cdot\mathbf{J}\,\mathrm{d}^3\mathbf{r} \tag{6.48}$$

The Cauchy-Schwarz inequality for the volume integral gives:

$$\left(\iiint_\Omega \mathbf{B}\cdot\boldsymbol{\eta}\cdot\mathbf{J}\,\mathrm{d}^3\mathbf{r}\right)^2 \leq \left(\iiint_\Omega \mathbf{B}^2\,\mathrm{d}^3\mathbf{r}\right)\left(\iiint_\Omega \mathbf{J}\cdot(\boldsymbol{\eta}^\top\boldsymbol{\eta})\cdot\mathbf{J}\,\mathrm{d}^3\mathbf{r}\right) \tag{6.49}$$

where we recognise the total magnetic energy within the system, $W_\mathbf{B}$, while the latter term cannot be further simplified without approximations. In the case of scalar resistivity, $\boldsymbol{\eta}\cdot\mathbf{J}\equiv\eta\mathbf{J}$, we find:

$$\left(\frac{\mathrm{d}H}{\mathrm{d}t}\right)^2 \lesssim 8\mu_0\langle\eta\rangle W_\mathbf{B}\left(\frac{\mathrm{d}W}{\mathrm{d}t}\right)_{\mathrm{Ohm}} \tag{6.50}$$

where we allow the resistivity η to be variable within the magnetic flux tube.

6.3 Ideal magnetohydrostatic equilibria

6.3.1 Grad-Shafranov equation

The Grad-Shafranov equation is obtained via the ideal magnetohydrostatic equilibrium condition applied to an axisymmetric toroidal system; we reiterate multiple results derived in chapter 5. The magnetic field in such a system may be expressed as:

$$\mathbf{B} = \frac{1}{2\pi}\left(\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta + \mu_0 I^{\mathrm{d}}_{\mathrm{pol}}\nabla\zeta\right) \tag{6.51}$$

The first term is only familiar and simply denotes the poloidal magnetic field, while the second term is obtained via the Maxwell-Ampère equation for the axisymmetric system. We reuse the Maxwell-Ampère equation to evaluate the poloidal current density:

$$\mathbf{J}_{\mathrm{pol}} = \frac{1}{\mu_0}\nabla\times\mathbf{B}_{\mathrm{tor}} = \frac{1}{2\pi\mu_0}\nabla\times\left(\mu_0 I^{\mathrm{d}}_{\mathrm{pol}}\nabla\zeta\right) = \frac{\nabla I^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta}{2\pi} \tag{6.52}$$

We shall use a less straightforward approach to finding the toroidal current density. Notice:

$$\nabla\zeta\cdot(\nabla\times\mathbf{B}_{\mathrm{pol}}) = \nabla\cdot(\mathbf{B}_{\mathrm{pol}}\times\nabla\zeta) = \frac{1}{2\pi}\nabla\cdot\left[\left(\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta\right)\times\nabla\zeta\right] = -\frac{1}{2\pi}\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right) \tag{6.53}$$

where we have used a cylindrical coordinate system, so $r\nabla\zeta = \hat{\boldsymbol{\zeta}}$. Since $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ is a flux function, its gradient is perpendicular to magnetic surfaces, and since the toroidal system is axisymmetric, these are necessarily tangent to the toroidal gradient $\nabla\zeta$, hence $\nabla\zeta\cdot\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}} = 0$:

$$\mathbf{J}_{\mathrm{tor}} = \frac{1}{\mu_0}\left[(\nabla\times\mathbf{B}_{\mathrm{tor}})\cdot\hat{\boldsymbol{\zeta}}\right]\hat{\boldsymbol{\zeta}} = -\frac{r^2}{2\pi\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)\nabla\zeta \tag{6.54}$$

We may now input these expressions into:

$$\mathbf{J}\times\mathbf{B} = \frac{1}{4\pi^2}\left[\nabla I^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta - \frac{r^2}{2\pi\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)\nabla\zeta\right]\times\left(\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta + \mu_0 I^{\mathrm{d}}_{\mathrm{pol}}\nabla\zeta\right) = \tag{6.55}$$

Notice that both $I^{\mathrm{d}}_{\mathrm{pol}}$ and $\Phi^{\mathrm{r}}_{\mathrm{pol}}$ are flux functions, so the vector product of the first term in the first bracket and the first term in the other bracket is identically zero. The other terms give:

$$\nabla\zeta\times\left(\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\times\nabla\zeta\right) = \frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}} \qquad \nabla\zeta\times(\nabla\zeta\times\nabla I^{\mathrm{d}}_{\mathrm{pol}}) = -\frac{1}{r^2}\nabla I^{\mathrm{d}}_{\mathrm{pol}} \tag{6.56}$$

which, in return, gives the ideal magnetohydrostatic equilibrium condition as follows:

$$\nabla p + \frac{I^{\mathrm{d}}_{\mathrm{pol}}}{4\pi^2 r^2}\nabla I^{\mathrm{d}}_{\mathrm{pol}} + \frac{1}{\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}} = \mathbf{0} \tag{6.57}$$

Since both the pressure and the poloidal disk current are flux functions, we may apply the derivative chain rule with respect to the magnetic poloidal disk flux and factor out its gradient:

$$4\pi^2\mu_0\frac{\mathrm{d}p}{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}+\frac{\mu_0^2 I^{\mathrm{d}}_{\mathrm{pol}}}{r^2}\frac{\mathrm{d}I^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}+\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)=0 \tag{6.58}$$

The equation above is commonly referred to as the *Grad-Shafranov equation*, valid for all axisymmetric toroidal systems. It is remarkable that the system's axisymmetry has substantially reduced the complexity of the initial magnetohydrostatic equilibrium, which is naturally three-dimensional, whereas the Grad-Shafranov equation is one-dimensional.

Analytical solutions of the equation exist only for very simple conditions on the pressure and the poloidal disk current, since their derivatives are involved in the equation. Generally, the Grad-Shafranov equation is solved numerically; however, if the pressure and poloidal disk current profiles are measured, solving for the magnetic poloidal disk flux may yield a feasible result, but such solutions are generally analytic only if boundary conditions are given in explicit forms and if the equations describing magnetic surfaces are separable in some coordinate system.

Exploring idealised solutions is nevertheless useful from a pragmatic point of view, since they allow us to gain insight as illustrative examples, and serve as references for validating the accuracy of numerical models. The following subsection presents several examples of such analytical solutions.

6.3.1.1 Solovev's solution to the Grad-Shafranov equation

A first example of an analytical solution to the magnetic surface profile within an axisymmetric toroidal system is determined via the following hypotheses:

1. The pressure is assumed to depend linearly on the poloidal disk flux:

$$P(\Phi^{\mathrm{d}}_{\mathrm{pol}})=P^{\square}+\lambda\Phi^{\mathrm{d}}_{\mathrm{pol}} \tag{6.59}$$

2. The poloidal disk current is assumed to be constant throughout the system:

$$\frac{\mathrm{d}I^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}=0 \tag{6.60}$$

The Grad-Shafranov equation thus becomes:

$$4\pi^2\mu_0\lambda+\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)=0 \implies r\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial r}\right)+\frac{\partial^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial z^2}+4\pi^2\mu_0\lambda r^2=0 \tag{6.61}$$

Before attempting to solve the equation, we shall first consider the case $\lambda=0$, which implies constant pressure within the system, which further gives the condition:

$$\nabla p=\mathbf{J}\times\mathbf{B}=\mathbf{0} \tag{6.62}$$

Therefore, the current density is parallel to the magnetic field, reflecting on Grad-Shafranov equation:

$$\nabla\cdot\left(\frac{\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}}{r^2}\right)=0 \implies \frac{\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}}{r^2}=\nabla\times\mathbf{S} \tag{6.63}$$

for some vector field $\mathbf{S}$. Obtaining the homogeneous solution is equivalent to finding $\mathbf{S}$ such that:

$$\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}=r^2\nabla\times\mathbf{S} \tag{6.64}$$

The toroidal and poloidal components necessarily vanish:

$$\frac{\partial S^{\varrho}}{\partial\zeta}=\frac{\partial S^{\zeta}}{\partial\varrho}\quad\text{and}\quad\frac{\partial S^{\theta}}{\partial\varrho}=\frac{\partial S^{\varrho}}{\partial\theta}\implies\frac{\partial^2 S^{\varrho}}{\partial\theta\partial\zeta}=\frac{\partial^2 S^{\zeta}}{\partial\varrho\partial\theta}=\frac{\partial^2 S^{\theta}}{\partial\varrho\partial\zeta}\implies\frac{\partial}{\partial\varrho}\left(\frac{\partial S^{\zeta}}{\partial\theta}-\frac{\partial S^{\theta}}{\partial\zeta}\right)=0 \tag{6.65}$$

The only possibly non-zero component of $\nabla\times\mathbf{S}$ is thus not a flux function. This is of rather weak significance since we're trying to solve for a flux function, but it nonetheless states that $\nabla\times\mathbf{S}$ is the same for points on all magnetic surfaces with common θ and ζ coordinates. The only possible vector field $\mathbf{S}$ satisfying this requirement is a vector field that depends only on θ and ζ. Let $f(\theta, \zeta)$ denote the component of $\nabla \times \mathbf{S}$ along $\nabla\Phi^{\mathrm{r}}_{\mathrm{pol}}$. Axisymmetry allows us to eliminate the ζ dependence, so $\nabla \times \mathbf{S}$ depends only on the poloidal angle θ. Notice that we still cannot give any further conditions on $\mathbf{S}$ without advanced mathematical tools, so we shall retake such discussions for another subsection. The condition $\mathbf{J} \times \mathbf{B} = \mathbf{0}$ is commonly referred to as the *force-free* condition, which has been briefly discussed in subsection 6.1.1. The full treatment of the homogeneous solution is given in subsection 6.3.1.2.

We now proceed to find the particular solution by first introducing the substitution $\Phi^{\mathrm{d}}_{\mathrm{pol}}(r, z) = r^2\psi(r, z)$ to eliminate the extraneous radial dependencies as follows:

$$r\frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r^2\psi\right)\right] + r^2\frac{\partial \psi}{\partial z^2} + 4\pi^2\mu_0\lambda r^2 = 0 \implies \frac{\partial^2\psi}{\partial r^2} + \frac{3}{r}\frac{\partial \psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2} + 4\pi^2\mu_0\lambda = 0 \tag{6.66}$$

Note that, again, we're only searching for the particular solution. Since the highest order derivative in both the radial and the axial coordinates is of second order, there is a constant non-zero term, as well as the fact that radial and axial derivatives aren't coupled, we seek separable quadratic solutions in r and z for $\psi(r, z)$. One can nevertheless assume such functional dependence, and show it cannot be:

$$\psi(r, z) = C(r-a)^2 + C'(z-b)^2 + C'' \implies C\left[1 + 3\left(1 - \frac{a}{r}\right)\right] + C' + 2\pi^2\mu_0\lambda = 0 \tag{6.67}$$

For a system that is reflected about the horizontal plane determined by $z = 0$, the axial coordinate z can only appear in terms of pair functions, so that $b = 0$. Furthermore, the radial dependency in the above equation necessarily needs to vanish, so that $a = 0$. We thus obtain:

$$C' = -2\pi^2\mu_0\lambda - 4C \implies \psi(r, z) = Cr^2 - 2\left(\pi^2\mu_0\lambda + 2C\right)z^2 + C'' \tag{6.68}$$

The general solution for the poloidal disk flux is hence the following:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(r, z) = Cr^2\left[r^2 - 2\left(\frac{\pi^2\mu_0\lambda}{C} + 2\right)z^2 + C'\right] \tag{6.69}$$

We determine the two constants via $\Phi^{\mathrm{d}}_{\mathrm{pol}}(r_0, 0) = \Phi^{\square}_{\mathrm{pol}}$, the poloidal disk flux at the magnetic axis:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(r_0, 0) = \Phi^{\square}_{\mathrm{pol}} \quad \text{and} \quad \nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\Big|_{(r_0,0)} = \mathbf{0} \tag{6.70}$$

$$\Phi^{\square}_{\mathrm{pol}} = Cr_0^2\left(r_0^2 + C'\right) \quad \text{and} \quad 2Cr_0(2r_0^2 + C') = 0 \implies C' = -2r_0^2 \implies C = -\frac{\Phi^{\square}_{\mathrm{pol}}}{r_0^4} \tag{6.71}$$

Solovev's solution to the Grad-Shafranov equation is thus of the following form:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(r, z) = \frac{\Phi^{\square}_{\mathrm{pol}}}{r_0^4}r^2\left(2r_0^2 - r^2 + 4\left(1 - \frac{\pi^2\mu_0\lambda r_0^4}{2\Phi^{\square}_{\mathrm{pol}}}\right)z^2\right) \tag{6.72}$$

Let us discuss the implications of the particular solution. Notably, $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ vanishes in the limit $r \to 0$ for any axial coordinate. This is only expected since, in that limit, the poloidal disk surface vanishes as well, hence the net magnetic flux can only be zero. In fact, we are certain that the flux depends on the quadratic radius in that limit, since the magnetic flux is a stream function, and, in our previous digression, namely subsection 2.10.3.2, we have demonstrated such dependence for a divergence-less vector field in axisymmetric geometry. The contour lines of the particular solution are illustrated in Figure 6.1. The red line represents the so-called *separatrix*, the unique magnetic surface corresponding to zero poloidal disk flux. Magnetic surfaces with the same sign of poloidal flux as that at the magnetic axis are thus enclosed by the separatrix, and are called *closed* surfaces. Meanwhile, the magnetic surfaces in the exterior of the separatrix are called *open* surfaces, because they extend to infinity as the radial coordinate goes to zero.

6.3.1.2 Force-free solution under Solovev's hypotheses

We have briefly discussed homogeneous solutions to the Grad-Shafranov equation, but let us investigate these further. The governing equation describing force-free solutions is $\mathbf{J} \times \mathbf{B} = \mathbf{0}$. A force-free regime bears such a name because the term $\mathbf{J} \times \mathbf{B}$ is precisely the macroscopic form of the Lorentz force; hence, such regimes are indeed regimes where there is no Lorentz force acting on any particular plasma volume element. This condition, in conjunction with the Maxwell-Ampère equation, gives:

$$\nabla \times \mathbf{B} = \mu \mathbf{B} \qquad \text{for some } \mu \tag{6.73}$$

where, in general, μ might be a function of position as well as of magnetic field. Let us discuss more precisely what the function μ physically represents. Substituting the Grad-Shafranov equation, specifically given by equation (6.58), into the expression for toroidal current density given by equation (6.54):

$$\mathbf{J}_{\text{tor}} = \left(2\pi r^2 \frac{\mathrm{d}p}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} + \frac{\mu_0 I_{\text{pol}}^{\text{d}}}{2\pi} \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \right) \nabla \zeta \tag{6.74}$$

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 6.1: Magnetic surface contours according to Solovev's hypotheses, only in the upper half due to reflective symmetry about the horizontal plane. The red line is the *separatrix*, corresponding to zero poloidal disk flux and separating the *open* surfaces from *closed* ones.

The total current density is thus given by:

$$\mathbf{J} = \mathbf{J}_{\text{pol}} + \mathbf{J}_{\text{tor}} = \frac{1}{2\pi} \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \nabla \Phi_{\text{pol}}^{\text{d}} \times \nabla \zeta + \left(2\pi r^2 \frac{\mathrm{d}p}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} + \frac{\mu_0 I_{\text{pol}}^{\text{d}}}{2\pi} \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \right) \nabla \zeta \tag{6.75}$$

$$= 2\pi r^2 \frac{\mathrm{d}p}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \nabla \zeta + \frac{1}{2\pi} \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \left(\nabla \Phi_{\text{pol}}^{\text{d}} \times \nabla \zeta + \mu_0 I_{\text{pol}}^{\text{d}} \nabla \zeta \right) = 2\pi^2 \frac{\mathrm{d}p}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \nabla \zeta + \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \mathbf{B} \tag{6.76}$$

Evidently, the force-free condition mandates $\mathrm{d}p/\mathrm{d}\Phi_{\text{pol}}^{\text{d}} = 0$, thus the function μ is given by:

$$\nabla \times \mathbf{B} = \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \mathbf{B} \implies \mu = \frac{\mathrm{d}I_{\text{pol}}^{\text{d}}}{\mathrm{d}\Phi_{\text{pol}}^{\text{d}}} \tag{6.77}$$

It is only expected that μ be a flux function. Indeed, taking the divergence of equation (6.73):

$$\nabla \cdot (\nabla \times \mathbf{B}) = 0 = \nabla \cdot (\mu \mathbf{B}) = \mu (\nabla \cdot \mathbf{B}) + \mathbf{B} \cdot \nabla \mu \implies \mathbf{B} \cdot \nabla \mu = 0 \tag{6.78}$$

where evidently $\nabla \cdot \mathbf{B} = 0$ and the divergence of any curl is zero. Taking the curl of equation (6.73):

$$\nabla^2 \mathbf{B} + \mu \nabla \times \mathbf{B} + \nabla \mu \times \mathbf{B} = (\nabla^2 + \mu^2) \mathbf{B} + \nabla \mu \times \mathbf{B} = \mathbf{0} \tag{6.79}$$

which shall be accompanied by $\nabla \cdot \mathbf{B} = 0$ to form a deterministic set of equations for a fully-known dependence $I_{\text{pol}}^{\text{d}}(\Phi_{\text{pol}}^{\text{d}})$. Returning to the so far abandoned discussion regarding the homogeneous solution of the Grad-Shafranov equation, in the particular case of Solovev's solution, the poloidal disk current was assumed constant, so that $\mu = 0$. The homogeneous solution is thus given by a curl-free magnetic field; hence, it can be represented by a magnetic scalar potential. Moreover, the magnetic field itself is harmonic. In the given cylindrical coordinate system, each component is given by the appropriate expansion in cylindrical coordinates:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \mathbf{B}}{\partial r} \right) + \frac{\partial^2 \mathbf{B}}{\partial z^2} = \mathbf{0} \tag{6.80}$$

where we have eliminated the azimuthal dependence due to axisymmetry. Separating the magnetic field by component $B = R(r)Z(z)$, and further mandating regularity along the axial coordinate to obtain oscillatory axial dependencies gives:

$$\frac{\partial^2 R_k}{\partial r^2} + \frac{1}{r} \frac{\partial R_k}{\partial r} - k^2 R_k = 0 \quad \text{and} \quad \frac{\partial^2 Z_k}{\partial z^2} + k^2 Z_k = 0 \tag{6.81}$$

$$k^2 r^2 \frac{\partial^2 R_k}{\partial (k^2 r^2)} + kr \frac{\partial R_K}{\partial (kr)} - k^2 r^2 R_k = 0 \implies R_k(r) = A_k I_0(kr) + B_k K_0(kr) \tag{6.82}$$

The general solution for each component of $\mathbf{B}$ reads as follows:

$$B(r,z) = \sum_{k=0}^{\infty} \left[A_k I_0(kr) + B_k K_0(kr)\right]\left[C_k \cos{(kz)} + D_k \sin{(kz)}\right] \tag{6.83}$$

where we understand the expansion to apply to each component of the magnetic field separately. The modified Bessel functions of the second kind diverge for $r = 0$, so $B_k = 0$ for the radial component. We are thus left with:

$$B_r(r,z) = \sum_{k=0}^{\infty} I_0(kr)\left[C_k \cos{(kz)} + D_k \sin{(kz)}\right] \tag{6.84}$$

We search for physically relevant solutions of the magnetic field that vanish at infinity. Since $I_0(kr)$ diverges as $r \to \infty$, the radial component entirely vanishes, while $A_k = 0$ in expression (6.83) for the axial and toroidal components:

$$B^{(\zeta,z)}(r,z) = \sum_{k=0}^{\infty} K_0(kr)\left[C_k^{(\zeta,z)} \cos{(kz)} + D_k^{(\zeta,z)} \sin{(kz)}\right] \tag{6.85}$$

We shall, moreover, reflect further on a particular Solovev's hypothesis, notably the assumption of constant poloidal disk current throughout the plasma within all magnetic surfaces. For the particular axisymmetric system, this is possible only if the poloidal disk current were flowing at the z-axis. Moreover, the ideal magnetohydrodynamic condition $\nabla \cdot \mathbf{J} = 0$ insists that a non-zero current necessarily flows along the entirety of the z-axis; otherwise, accumulation of charge would arise. This physically represents a symmetry under translation along the z-axis, so that the axial component of the magnetic field vanishes, and the azimuthal component depends only on the radial coordinate. The only possibly non-vanishing component is therefore the azimuthal component, given by the Maxwell-Ampère equation.

Thus, the force-free homogeneous solution to the Grad-Shafranov equation according to Solovev's hypotheses is a nested family of magnetic surfaces in the shape of concentric infinite cylinders aligned and anchored at the z-axis. Notice that all of these magnetic surfaces have identically zero poloidal disk flux, since the force-free field is purely toroidal. Therefore, the homogeneous solution to the Grad-Shafranov equation under Solovev's hypotheses is identically zero.

6.3.1.3 Properties of solutions to the Grad-Shafranov equation

This first solution to the Grad-Shafranov equation, although based on rather idealised hypotheses, in particular the pressure dependence, nevertheless exhibits important properties of its general solutions, which we shall enumerate below. Note that they only apply to axisymmetric toroidal systems characterised by a unique magnetic axis.

1. Magnetic surfaces in axisymmetric toroidal systems indeed form a nested family of surfaces anchored about the magnetic axis, and are characterised by the same topology as the magnetic axis.

2. Magnetic surfaces extending to infinity are allowed. These are commonly referred to as *open* surfaces. Meanwhile, localised magnetic surfaces, those that do not extend to infinity, are named *closed* surfaces. Under Solovev's hypotheses, the closed surfaces are *deformed tori*, while open surfaces are *oblate coaxial cylindrical-like shells.*

3. A *separatrix* is the possibly unique magnetic surface distinguishing closed surfaces from open ones in axisymmetric toroidal systems, corresponding to zero magnetic flux. Thus, open surfaces share a common sign of magnetic flux with the magnetic axis, while closed surfaces have the opposite sign of magnetic flux. Under Solovev's hypotheses, the separatrix is an ellipsoidal surface.

4. The total magnetic field projects into magnetic surfaces, as is only expected by the definition of magnetic surfaces.

5. The pressure extrema in magnetohydrostatic equilibrium systems are typically located in the proximity of the magnetic axis. Under Solovev's hypotheses, it is precisely at the magnetic axis.

6. It is precisely the poloidal component of the magnetic field, and consequently the poloidal disk and ribbon fluxes, that are responsible for axisymmetric toroidal magnetic confinement. The former are produced by a toroidal current density.

7. Under Solovev's hypotheses, force-free magnetic fields, being purely toroidal in axisymmetric toroidal systems, do not contribute to magnetic confinement.

6.3.2 Outline of exact solutions to the Grad-Shafranov equation

The main point that the author wishes to convey in this subsection is how a more general solution to the Grad-Shafranov equation swiftly becomes much more complicated. We shall consider a particular family of exact analytical solutions to the Grad-Shafranov equation obtained for quadratic dependencies of the magnetohydrodynamic pressure and poloidal disk current on the magnetic poloidal disk flux, namely:

$$p(\Phi^{\mathrm{d}}_{\mathrm{pol}}) = \frac{a}{8\pi^2\mu_0}\left[\Phi^{\mathrm{d}}_{\mathrm{pol}}\right]^2 + \frac{b}{4\pi^2\mu_0}\Phi^{\mathrm{d}}_{\mathrm{pol}} \qquad \left[I^{\mathrm{d}}_{\mathrm{pol}}\right]^2 = \frac{\alpha}{\mu_0^2}\left[\Phi^{\mathrm{d}}_{\mathrm{pol}}\right]^2 + \frac{2\beta}{\mu_0^2}\Phi^{\mathrm{d}}_{\mathrm{pol}} + \left[I^{\square}_{\mathrm{pol}}\right]^2 \tag{6.86}$$

where a, b, α, and β are free parameters determined by the total plasma current, the poloidal pressure parameter, the internal inductance, and the rotational transform of the magnetic axis, as shown later on. The Grad-Shafranov equation under these hypotheses becomes:

$$\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right) + \left(\frac{\alpha}{r^2} + a\right)\Phi^{\mathrm{d}}_{\mathrm{pol}} + \left(\frac{\beta}{r^2} + b\right) = 0 \tag{6.87}$$

$$\frac{\partial^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial r^2} - \frac{1}{r}\frac{\partial\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial r} + \frac{\partial^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial z^2} + \left(\alpha + ar^2\right)\Phi^{\mathrm{d}}_{\mathrm{pol}} + \left(\beta + br^2\right) = 0 \tag{6.88}$$

As with any differential equation, we seek its homogeneous solution. Since there are no mixed r and z terms in the partial differential equation, the homogeneous solution is necessarily separable. Before advancing any further, we shall first make a proper digression about the Frobenius method and hypergeometric functions, which will prove to be useful in this particular case.

6.3.2.1 Digression: Frobenius method and hypergeometric functions

We will here consider the ordinary differential equation of the form:

$$\mathcal{L}_x(Q) = \frac{\mathrm{d}^2Q}{\mathrm{d}x^2} + p(x)\frac{\mathrm{d}Q}{\mathrm{d}x} + q(x)Q(x) = 0 \tag{6.89}$$

such that $p(x)$ and $q(x)$ are analytic in a certain neighbourhood of point x_0. Moreover, we consider $p(x)$ and $q(x)$ such that each has a pole of at most first order and at most second order at $x = x_0$, respectively. In that case, the Laurent series of each of the two coefficients reads:

$$p(x) = \frac{1}{x}\sum_{j=0}^{\infty} p_j x^j \quad \text{and} \quad q(x) = \frac{1}{x^2}\sum_{j=0}^{\infty} q_j x^j \tag{6.90}$$

We are specifically looking for an analytic solution, such that:

$$Q(x) = x^\lambda f(x) = x^\lambda \sum_{k=0}^{\infty} f_k x^k \tag{6.91}$$

where generally $\lambda \in \mathcal{C}$, and $f(x)$ is an analytic function. Without loss of generality, we impose $f(0) = 1$, as multiplying $Q(x)$ by a constant still gives a valid homogeneous solution:

$$q(x)Q(x) = x^{\lambda-2}\sum_{j=0}^{\infty} q_j x^j \sum_{k=0}^{\infty} f_k x^k = \{m = j + k\} = x^{\lambda-2}\sum_{m=0}^{\infty}\sum_{j=0}^{m} q_j f_{m-j} x^m \tag{6.92}$$

$$p(x)\frac{\mathrm{d}Q}{\mathrm{d}x} = x^{\lambda-2}\sum_{j=0}^{\infty} p_j x^j \sum_{k=0}^{\infty}(\lambda + k) f_k x^k = \{m = j + k\} = x^{\lambda-2}\sum_{m=0}^{\infty}\sum_{j=0}^{m}(\lambda + m - j) p_j f_{m-j} x^m \tag{6.93}$$

$$\frac{\mathrm{d}^2 Q}{\mathrm{d}x^2} = x^{\lambda-2}\sum_{m=0}^{\infty}(\lambda+m)(\lambda+m-1)f_m x^m \tag{6.94}$$

Implementing these series into equation (6.89), we find alongside each $x^{\lambda+m-2}$ term:

$$(\lambda+m)(\lambda+m-1)f_m + \sum_{j=0}^{m}\left[(\lambda+m-j)p_j + q_j\right]f_{m-j} = 0 \tag{6.95}$$

Regrouping the $j=0$ terms into the former term:

$$(\lambda+m)(\lambda+m-1)f_m + \left[(\lambda+m)p_0 + q_0\right]f_m + \sum_{j=1}^{m}\left[(\lambda+m-j)p_j + q_j\right]f_{m-j} = 0 \tag{6.96}$$

The assumption $f(0)=1$ provides the so-called *indicial equation* for $m=0$:

$$I(\lambda) = \lambda^2 + (p_0-1)\lambda + q_0 = 0 \implies \lambda_{1,2} = \frac{1}{2}\left(1 - p_0 \pm \sqrt{(p_0-1)^2 - 4q_0}\right) \tag{6.97}$$

where we choose the convention $\mathrm{Re}\,\lambda_1 \geq \mathrm{Re}\,\lambda_2$. Notice:

$$(\lambda_i+m)^2 + (p_0-1)(\lambda_i+m) + q_0 = I(\lambda) + m\left[2\lambda_i + (p_0-1) + m\right] = m(m \pm (\lambda_1-\lambda_2)) \tag{6.98}$$

where the positive root branch corresponds to the choice $\lambda=\lambda_1$, and negative branch to $\lambda=\lambda_2$, respectively. Thus, choosing $\lambda=\lambda_1$ gives a recursion-type relation for all $m>0$ as follows:

$$f_m = -\frac{1}{(\lambda_1-\lambda_2+m)m}\sum_{j=1}^{m}\left[(\lambda+m-j)p_j + q_j\right]f_{m-j} \tag{6.99}$$

which is well-defined for all $m>0$, and thus provides a first homogeneous solution of the second-order differential equation. The second homogeneous solution, however, depends on the general nature of the term $\lambda_1-\lambda_2$, since it may produce divergences. For our purposes, we assume $\lambda_1-\lambda_2\in\mathbb{R}$:

1. $\lambda_1-\lambda_2\notin\mathbb{Z}$ case: the other solution is obtained by directly imposing $\lambda=\lambda_2$, giving an analogous expression to the previously obtained recursion-type relation. This certainly poses no problems regarding convergence since $m\in\mathbb{N}$, so the recursion-type relation given in equation (6.99) is well-defined for all $m>0$.

2. $\lambda_1=\lambda_2$ case: The differential equation reduces to the indicial equation multiplied by uninteresting constants:

$$\mathcal{L}_x(Q) \sim (\lambda-\lambda_1)^2 \implies \mathcal{L}_x\left(\frac{\partial}{\partial\lambda}\Big[Q(x;\lambda)\Big]\right)\bigg|_{\lambda_1} = \frac{\partial}{\partial\lambda}\Big[\mathcal{L}_x(Q(x;\lambda))\Big]\bigg|_{\lambda_1} = 0 \tag{6.100}$$

$$\frac{\partial}{\partial\lambda}\sum_{m=0}^{\infty}a_m(\lambda)x^{\lambda+m} = \sum_{m=0}^{\infty}\frac{\partial a_m}{\partial\lambda}x^{\lambda+m} + Q(x;\lambda)\ln x \tag{6.101}$$

3. $\lambda_1-\lambda_2\in\mathbb{N}$ case: one first ought to verify whether directly inputting $\lambda=\lambda_2$ gives a linearly independent solution. In the affirmative case of the latter, the obtained solution is indeed the second homogeneous solution. Otherwise:

$$\mathcal{L}_x(Q) \sim (\lambda-\lambda_1)(\lambda-\lambda_2) \implies \mathcal{L}_x\left(\frac{\partial}{\partial\lambda}\Big[(\lambda-\lambda_2)y(x;\lambda)\Big]\right)\bigg|_{\lambda_2} =$$

$$= \frac{\partial}{\partial\lambda}\Big[\mathcal{L}_x\left((\lambda-\lambda_2)\,Q(x;\lambda)\right)\Big]\bigg|_{\lambda_2} \sim \frac{\partial}{\partial\lambda}\Big[(\lambda-\lambda_2)^2(\lambda-\lambda_1)\Big]\bigg|_{\lambda_2} = 0 \tag{6.102}$$

Therefore, the other solution is thus given by:

$$Q_2(x) = \frac{\partial}{\partial\lambda}\Big((\lambda-\lambda_2)Q_1(x)\Big)\bigg|_{\lambda_2} \tag{6.103}$$

We now return to our preceding considerations, namely equation (6.88). Separating:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(r,z) = R(r)Z(z) \implies \frac{\mathrm{d}^2R}{\mathrm{d}r^2} - \frac{1}{r}\frac{\mathrm{d}R}{\mathrm{d}r} + \left(ar^2 + \alpha - k^2\right)R = 0 \quad \text{and} \quad \frac{\mathrm{d}^2Z}{\mathrm{d}z^2} + k^2Z = 0 \tag{6.104}$$

such that k is an arbitrary constant. It is necessary to distinguish the solutions according to the sign of the parameter a. We shall first treat the case $a < 0$. Implementing the substitution:

$$x = r^2\sqrt{-a} \implies \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}x}{\mathrm{d}r}\frac{\mathrm{d}R}{\mathrm{d}x}\right)\frac{\mathrm{d}x}{\mathrm{d}r} - \frac{1}{r}\frac{\mathrm{d}x}{\mathrm{d}r}\frac{\mathrm{d}R}{\mathrm{d}x} + \left(ar^2 + \alpha - k^2\right)R = 0 \tag{6.105}$$

$$r\frac{\mathrm{d}}{\mathrm{d}x}\left(2r\sqrt{-a}\frac{\mathrm{d}R}{\mathrm{d}x}\right) - \frac{\mathrm{d}R}{\mathrm{d}x} + \left(-\frac{x}{2} + \frac{\alpha - k^2}{2\sqrt{-a}}\right)R = 0 \implies \frac{\mathrm{d}^2R}{\mathrm{d}x^2} + \left(-\frac{1}{4} + \frac{\alpha - k^2}{4x\sqrt{-a}}\right)R = 0 \tag{6.106}$$

The solution to R therefore exhibits certain exponential behaviour. Specifically, the coefficient suggests behaviour like $R(x) = x\exp\left[-x/2\right]Q(x)$, so:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\mathrm{d}}{\mathrm{d}x}\left(x\exp\left[-\frac{x}{2}\right]Q(x)\right)\right] = x\exp\left[-\frac{x}{2}\right]\left(\frac{\mathrm{d}^2Q}{\mathrm{d}x^2} + \frac{2-x}{x}\frac{\mathrm{d}Q}{\mathrm{d}x} + \left[\frac{1}{4} - \frac{1}{x}\right]Q\right) \tag{6.107}$$

$$x\frac{\mathrm{d}^2Q}{\mathrm{d}x^2} + (2-x)\frac{\mathrm{d}Q}{\mathrm{d}x} - \left(\frac{k^2-\alpha}{4\sqrt{-a}} + 1\right)Q = 0 \tag{6.108}$$

Recasting the obtained ordinary differential equation of second order into the following form, we observe the necessity to have previously covered the Frobenius method.

$$\mathcal{L}_x(Q) = \frac{\mathrm{d}^2Q}{\mathrm{d}x^2} + \left(\frac{2}{x} - 1\right)\frac{\mathrm{d}Q}{\mathrm{d}x} - \frac{\eta+1}{x}Q = 0 \quad \text{where} \quad \eta = \frac{k^2-\alpha}{4\sqrt{-a}} \tag{6.109}$$

The coefficients alongside $\mathrm{d}Q/\mathrm{d}x$ and Q have regular singularities at $x = 0$, notably first-order poles, so that their respective Laurent coefficients read as follows:

$$p_{(0)} = 2, \quad p_{(1)} = -1 \quad \forall n \geq 2, p_{(n)} = 0 \quad \text{and} \quad q_{(0)} = 0, \quad q_{(1)} = -\eta - 1, \quad \forall n \geq 2, q_{(n)} = 0 \tag{6.110}$$

The indicial equation gives the following solutions for λ:

$$\lambda^2 + \lambda = 0 \implies \lambda_1 = 0, \quad \lambda_2 = -1 \tag{6.111}$$

Observe $\lambda_1 - \lambda_2 \in \mathbb{N}$. Let the first solution, $Q_1(x)$, be associated with $\lambda_1 = 0$:

$$m(m-1)f_m + mp_0f_m + (m-1)p_1f_{m-1} + q_1f_{m-1} = 0 \implies f_m = \frac{m+\eta}{m(m+1)}f_{m-1} \tag{6.112}$$

$$f_m = \prod_{k=1}^{m}\frac{\eta+k}{k(k+1)} = \frac{(\eta+1)_m}{m!(2)_m} \tag{6.113}$$

where the indices denote the Pochhammer symbol, so that the confluent hypergeometric function of the first kind gives:

$$Q_1(x) = {}_1F_1(\eta+1;2;x) = \sum_{m=0}^{\infty}\frac{(\eta+1)_m}{(2)_m}\frac{x^m}{m!} \tag{6.114}$$

In the search for the other homogeneous solution associated with $\lambda_2 = -1$, one first verifies whether it gives a linearly independent second homogeneous solution:

$$m\left[m - (\lambda_1 - \lambda_2)\right]f_m + \left[(\lambda_2 + m - 1)p_1 + q_1\right]f_{m-1} = 0 \tag{6.115}$$

$$m\,(m-1)\,f_m + \left[(m-2)p_1 + q_1\right]f_{m-1} = 0 \tag{6.116}$$

f_1 is a free coefficient, that is, the recursion relation does not hold for all $m > 0$. We thus return to equation (6.96) and find the recursion relation for Laurent coefficients of $f(x)$ in terms of the general parameter λ, so that we may use expression (6.103):

$$\left[(\lambda+m)^2+(p_0-1)(\lambda+m)+q_0\right]f_m+\left[(\lambda+m-1)p_1+q_1\right]f_{m-1}=0 \tag{6.117}$$

$$f_m=\frac{\eta+\lambda+m}{(\lambda+m)(\lambda+m+1)}f_{m-1}=\frac{(\eta+\lambda+1)_m}{(\lambda+1)_m(\lambda+2)_m}f_0 \tag{6.118}$$

Notice that our negative branch solution to the indicial equation was $\lambda_2=-1$, so that the first Pochhammer symbol in the denominator exhibits divergent behaviour resolved via the $\lambda-\lambda_2$ term:

$$(\lambda-\lambda_2)Q_1(x)=(\lambda+1)\sum_{m=0}^{\infty}\frac{(\eta+\lambda+1)_m}{(\lambda+1)_m(\lambda+2)_m}x^{\lambda+m}=\sum_{m=0}^{\infty}\frac{(\eta+\lambda+1)_m}{(\lambda+2)_{m-1}(\lambda+2)_m}x^{\lambda+m} \tag{6.119}$$

$$Q_2(x)=\frac{\partial}{\partial\lambda}\Big((\lambda-\lambda_2)Q_1(x)\Big)\Big|_{\lambda_2}=\frac{\partial}{\partial\lambda}\left(\sum_{m=0}^{\infty}\frac{(\eta+\lambda+1)_m}{(\lambda+2)_{m-1}(\lambda+2)_m}x^{\lambda+m}\right)\Bigg|_{\lambda_2} \tag{6.120}$$

Computing the derivative above is very arduous. We proceed in a straightforward manner:

$$Q_2(x)=\sum_{m=0}^{\infty}\frac{(\eta)_m}{(1)_{m-1}(1)_m}\frac{\partial}{\partial\lambda}\left(x^{\lambda+m}\right)\Big|_{\lambda_2}+\frac{1}{x}\sum_{m=0}^{\infty}\frac{\partial}{\partial\lambda}\left(\frac{(\eta+\lambda+1)_m}{(\lambda+2)_{m-1}(\lambda+2)_m}\right)\Bigg|_{\lambda_2}x^m=$$

$$=\eta\ln x\sum_{m=0}^{\infty}\frac{(\eta+1)_m}{(1)_m(2)_m}x^m+\frac{1}{x}\sum_{m=0}^{\infty}\frac{\partial b_m(\lambda)}{\partial\lambda}\Big|_{\lambda_2}x^m=\eta\ln(x)_1F_1(1+\eta;2;x)+\frac{1}{x}\sum_{m=0}^{\infty}\frac{\partial b_m(\lambda)}{\partial\lambda}\Big|_{\lambda_2}x^m \tag{6.121}$$

It is much easier to compute the logarithmic derivative:

$$\frac{1}{b_m(\lambda)}\frac{\partial b_m(\lambda)}{\partial\lambda}\Big|_{\lambda_2}=\frac{\partial\ln b_m(\lambda)}{\partial\lambda}\Big|_{\lambda_2}=\frac{\partial}{\partial\lambda}\Big(\ln(\eta+\lambda+1)_m-\ln(\lambda+2)_{m-1}-\ln(\lambda+2)_m\Big)\Big|_{\lambda_2}=$$

$$=\frac{\partial}{\partial\lambda}\left(\ln\frac{\Gamma(\eta+\lambda+m+1)}{\Gamma(\eta+\lambda+1)}-\ln\frac{\Gamma(\lambda+m+1)}{\Gamma(\lambda+2)}-\ln\frac{\Gamma(\lambda+m+2)}{\Gamma(\lambda+2)}\right)\Bigg|_{\lambda_2}=$$

$$=\psi(\eta+m)-\psi(\eta)-\psi(m)-\psi(m+1)+2\psi(1) \tag{6.122}$$

The second solution thus reads:

$$Q_2(x)=\eta\ln(x)_1F_1(1+\eta;2;x)+\frac{1}{x}\left[\sum_{m=0}^{\infty}\frac{(\eta)_m}{(1)_{m-1}}\Big(\psi(\eta+m)-\psi(\eta)-\psi(m)-\psi(m+1)+2\psi(1)\Big)\right]\frac{x^m}{m!}=$$

$$=\eta\ln(x)_1F_1(1+\eta;2;x)+\frac{1}{x}+\sum_{m=1}^{\infty}\frac{(\eta)_m}{(m-1)!m!}\left[\sum_{k=0}^{m-1}\frac{1}{\eta+k}-H_{m-1}-H_m\right]x^{m-1} \tag{6.123}$$

where we have introduced the harmonic numbers, $H_m=\psi(m+1)+\gamma$, and the Euler-Mascheroni constant, $\psi(1)=-\gamma$. The last term can be expressed via ${}_1F_1(\eta;1;x)$, but still contains terms without a closed form, so we shall leave the result as is. We remind $x=r^2\sqrt{-a}$. The full solution for the homogeneous magnetic poloidal disk flux is therefore:

$$\Phi^{\mathrm{d\,(h)}}_{\mathrm{pol}}(r,z)=r^2\sqrt{-a}\exp\left[-\frac{r^2\sqrt{-a}}{2}\right]\Bigg[C_1{}_1F_1(\eta+1;2;r^2\sqrt{-a})+C_2\Bigg(\eta\ln(r^2\sqrt{-a})_1F_1(1+\eta;2;r^2\sqrt{-a})+$$

$$+\frac{1}{r^2\sqrt{-a}}+\sum_{m=1}^{\infty}\frac{(\eta)_m}{(m)!}\left[\sum_{k=0}^{m-1}\frac{1}{\eta+k}-H_{m-1}-H_m\right]\frac{\left(r^2\sqrt{-a}\right)^{m-1}}{(m-1)!}\Bigg)\Bigg]\Big[C_3\cos(kz)+C_4\sin(kz)\Big] \tag{6.124}$$

with C_1, C_2, C_3, and C_4 as arbitrary constants fixed through boundary conditions. It is important to note that, following the above considerations, $\eta\neq 0$, and this will be addressed later on. For $\eta=1$, simplifies the above computations, namely:

$$Q_1(x) = \sum_{m=0}^{\infty} \frac{x^m}{m!} = \exp(x) \quad \text{and} \quad Q_2(x) = \frac{1}{x} - \exp(x) E_1(x) - \gamma \exp(x) \tag{6.125}$$

where $E_1(x)$ denotes the principal branch of the exponential integral function, so that:

$$\Phi_{\rm pol}^{\rm d\,(h)}(r, z; \eta = 1) = r^2\sqrt{-a}\exp\left[-\frac{r^2\sqrt{-a}}{2}\right]\Bigg[C_1 \exp(r^2\sqrt{-a}) +$$

$$+ C_2\left(\frac{1}{r^2\sqrt{-a}} - \exp(r^2\sqrt{-a})E_1(r^2\sqrt{-a})\right)\Bigg]\Big[C_3\cos(kz) + C_4\sin(kz)\Big] \tag{6.126}$$

The particular solution in the $a < 0$ case may be separated into the radial and axial parts by the linearity of equation (6.88). Namely, the axial part presents no inhomogeneous terms, and as such, presents minor modifications of functional dependencies of the particular solution. For demonstrative purposes, we impose:

$$\Phi_{\rm pol}^{\rm d\,(p)}(r) = -\frac{\beta}{\alpha} + \psi(r) \implies \frac{{\rm d}^2\psi}{{\rm d}r^2} - \frac{1}{r}\frac{{\rm d}\psi}{{\rm d}r} + (ar^2 + \alpha)\psi(r) = \left(\frac{\beta}{\alpha}a - b\right) r^2 \tag{6.127}$$

Using the identical substitution as above, namely $x = r^2\sqrt{-a}$, we obtain:

$$\frac{{\rm d}^2\psi}{{\rm d}x^2} + \left(-\frac{1}{4} + \frac{\alpha}{4x\sqrt{-a}}\right)\psi(x) = \frac{1}{4}\left(\frac{b}{a} - \frac{\beta}{\alpha}\right) \tag{6.128}$$

As we have previously done in a similar fashion, we introduce:

$$\psi(x) = \frac{1}{4}\left(\frac{b}{a} - \frac{\beta}{\alpha}\right) x \exp\left[-\frac{x}{2}\right]\xi(x) \implies x\frac{{\rm d}^2\xi}{{\rm d}x^2} + (2 - x)\frac{{\rm d}\xi}{{\rm d}x} + \left[\frac{\alpha}{4\sqrt{-a}} - 1\right]\xi(x) = \exp\left[\frac{x}{2}\right] \tag{6.129}$$

Since the exponential in the right-hand side is analytic, we equally search for analytic solutions, so that:

$$\sum_{m=0}^{\infty}\left[m(m-1)\xi_m x^{m-1} + (2 - x)m\xi_m x^{m-1} - \kappa\xi_m x^m - \frac{1}{m!}\left(\frac{x}{2}\right)^m\right] = 0 \tag{6.130}$$

where we have defined $\kappa = 1 - \alpha/4\sqrt{-a} \equiv 1 + \eta(k = 0)$. The indicial equation reads $2\xi_1 - \kappa\xi_0 = 1$. We now regroup terms associated with the same power x^m, notably by separating the entire sum into terms with x^m and x^{m-1}, respectively, followed by the shift $m \to m + 1$:

$$\sum_{m=0}^{\infty}(m+1)m\xi_{m+1}x^m + 2(m+1)\xi_{m+1}x^m = \sum_{m=0}^{\infty} m\xi_m x^m - \kappa\xi_m x^m - \frac{1}{m!}\left(\frac{x}{2}\right)^m \tag{6.131}$$

$$\xi_m = \frac{\kappa + m - 1}{m(m+1)}\xi_{m-1} + \frac{1}{2^{m-1}(m+1!)} \tag{6.132}$$

The homogeneous part of the recursion relation above is easily found:

$$\xi_m^{(h)} = \xi_0\prod_{k=0}^{m}\frac{\kappa + k - 1}{k(k+1)} = \frac{(\kappa)_m}{m!(m+1)!}\xi_0 \tag{6.133}$$

To determine the inhomogeneous part, we let $\xi_m = A_m\xi_m^{(h)}$, so that:

$$A_m\xi_m^{(h)} = \frac{\kappa + m - 1}{m(m+1)}A_{m-1}\xi_{m-1}^{(h)} + \frac{1}{2^{m-1}(m-1!)} \tag{6.134}$$

$$A_m - A_{m-1} = \frac{1}{\xi_m^{(h)}2^{m-1}(m-1!)} = \frac{m!(m+1)!}{(\kappa)_m 2^{m-1}(m-1!)}\frac{1}{\xi_0} \tag{6.135}$$

The solution to the recursion relation is therefore:

$$\xi_m = \frac{(\kappa)_m}{m!(m+1)!}\left[\xi_0 + \sum_{j=1}^{m}\frac{j!}{2^{j-1}(\kappa)_j}\right] \tag{6.136}$$

$$\xi(x)=\sum_{m=0}^{\infty}\frac{\xi_0(\kappa)_m}{(2)_m}\frac{x^m}{m!}+\sum_{m=0}^{\infty}\frac{(\kappa)_m}{(2)_m}\sum_{j=1}^{m}\frac{j!}{2^{j-1}(\kappa)_j}\frac{x^m}{m!}=\xi_0{}_1F_1(\kappa;2;x)+\sum_{m=0}^{\infty}\frac{(\kappa)_m}{(2)_m}\frac{x^m}{m!}\sum_{j=1}^{m}\frac{2^{1-j}j!}{(\kappa)_j} \quad (6.137)$$

where the sum can be expressed in terms of hypergeometric functions, just not in closed form, namely:

$$\xi(x)=-\frac{1}{\kappa}+\frac{1+\kappa\xi_0}{\kappa}{}_1F_1(\kappa;2;x)+2\kappa\underbrace{\sum_{m=2}^{\infty}\frac{(\kappa+1)_{m-1}}{m!(m+1)!}\sum_{j=2}^{m}\frac{j!}{(\kappa)_j}\frac{x^j}{2^j}}_{\theta(x)} \quad (6.138)$$

One may expand the nested sum to obtain:

$$\theta(x)=\frac{1}{2\kappa}+\frac{1}{\kappa}\left[{}_2F_1(1,1;\kappa;1/2)-1-\frac{1}{2\kappa}\right]{}_1F_1(\kappa;2;x)-\frac{1}{\kappa}\sum_{m=0}^{\infty}\frac{{}_2F_1(m+2,1;m+\kappa+1;1/2)}{2^{m+1}(m+\kappa)m!}x^m \quad (6.139)$$

We shall now revert $\kappa=1+\eta(k=0)=1+\eta_0$ preventively in order to demonstrate a point:

$$\xi(x)=\left[2{}_2F_1\left(1,1;1+\eta_0;1/2\right)-2+\xi_0\right]{}_1F_1(1+\eta_0;2;x)-\sum_{m=0}^{\infty}\frac{{}_2F_1(m+2,1;m+\eta_0+2;1/2)}{2^m(m+\eta_0+1)m!}x^m \quad (6.140)$$

Notice that the first term can be absorbed into the homogeneous solution. The chosen particular solution of only radial dependence for the magnetic poloidal disk flux reads:

$$\Phi_{\text{pol}}^{\text{d (p)}}(r)=-\frac{\beta}{\alpha}+\frac{r^2\sqrt{-a}}{4}\exp\left[-\frac{r^2\sqrt{-a}}{2}\right]\left(\frac{\beta}{\alpha}-\frac{b}{a}\right)\sum_{m=0}^{\infty}\frac{{}_2F_1(m+1,1;1+\eta_0;1/2)}{2^m(m+\eta_0+1)m!}\left(r^2\sqrt{-a}\right)^m \quad (6.141)$$

So that the general solution for the $a<0$ case is given by a linear combination of the particular solution and the sum over values of k for the homogeneous solution.

We now consider the case $a>0$, and particularly, its homogeneous solution. Introducing a similar substitution, namely $x=r^2\sqrt{a}/2$, we find:

$$\frac{\mathrm{d}^2R}{\mathrm{d}x^2}+\left(1-\frac{2\eta}{x}\right)R(x)=0 \quad \text{where} \quad \eta=\frac{k^2-\alpha}{4\sqrt{a}} \quad (6.142)$$

We begin the solution procedure by introducing the substitution:

$$R(x)=x\exp\left[-\mathrm{i}x\right]w(x) \implies \exp\left[-\mathrm{i}x\right]\left[x\frac{\mathrm{d}^2w}{\mathrm{d}x^2}+2(1-\mathrm{i}x)\frac{\mathrm{d}w}{\mathrm{d}x}-2(\mathrm{i}+\eta)w(x)\right]=0 \quad (6.143)$$

We perform the substitution $z=2\mathrm{i}x$ to obtain a very familiar equation:

$$z\frac{\mathrm{d}^2w}{\mathrm{d}z^2}+(2-z)\frac{\mathrm{d}w}{\mathrm{d}x}-(1-\mathrm{i}\eta)w(x)=0 \quad (6.144)$$

whose solution is precisely the confluent hypergeometric function of the first kind, namely:

$$w(x)={}_1F_1(1+\mathrm{i}\eta;2;x) \implies R(x)=x\exp\left[-\mathrm{i}x\right]{}_1F_1(1+\mathrm{i}\eta;2;x)=F_0(\eta,x) \quad (6.145)$$

where we have introduced the so-called *Coulomb wave function* of the first kind, $F_L(\eta,x)$ for $L=0$. The second linearly independent solution is the Coulomb wave function of the second kind, $G_0(\eta,x)$, so that the general homogeneous solution to the above equation reads:

$$\Phi_{\text{pol}}^{\text{d (h)}}(r,z)=\left[C_1F_0\left(\eta,\frac{r^2\sqrt{a}}{2}\right)+C_2G_0\left(\eta,\frac{r^2\sqrt{a}}{2}\right)\right]\left[C_3\cos\left(kz\right)+C_4\sin\left(kz\right)\right] \quad (6.146)$$

The treatment of the particular solution follows an analogous fashion as in the $a<0$ case, and shall not be considered, since it presents equally challenging computations. However, what is indeed valuable to consider is how the free parameters a, b, α, and β can be determined using integral relations concerning the global plasma parameters, namely: the toroidal plasma current, $I_{\text{tor}}^{\text{pl}}$, the poloidal magnetic pressure parameter β_{pol}, the internal inductance l_{int}, and the rotational transform, or equivalently, the safety factor, at the plasma boundary or the magnetic axis, q_{b} or q_{ax}, respectively. Moreover, the knowledge of the plasma contour and the magnetic poloidal disk flux at the plasma boundary determines the four arbitrary constants, each quadruple associated with $k \in \mathbb{N}$, in the series expansion of the homogeneous solutions for the magnetic poloidal disk flux, and thus presents a somewhat methodological procedure when evaluating the magnetic field structure of a given system, so it will not be further discussed. Let us, therefore, return to the above statement regarding the free parameters.

We had previously derived how the toroidal current density relates to the Grad-Shafranov equation, namely expression (6.54), which we retake and use equation (6.87):

$$\mathbf{J}_{\mathrm{tor}} = \frac{1}{2\pi\mu_0}\left[\left(ar + \frac{\alpha}{r}\right)\Phi^{\mathrm{d}}_{\mathrm{pol}} + br + \frac{\beta}{r}\right]\hat{\zeta} \tag{6.147}$$

Integrating the above over a toroidal cross-section of the plasma Σ, we obtain:

$$2\pi\mu_0 I^{\mathrm{pl}}_{\mathrm{tor}} = a\iint_\Sigma \Phi^{\mathrm{d}}_{\mathrm{pol}} r\,\mathrm{d}r\,\mathrm{d}z + \alpha\iint_\Sigma \frac{\Phi^{\mathrm{d}}_{\mathrm{pol}}}{r}\,\mathrm{d}r\,\mathrm{d}z + b\iint_\Sigma r\,\mathrm{d}r\,\mathrm{d}z + \beta\iint_\Sigma \frac{\mathrm{d}r\,\mathrm{d}z}{r} \tag{6.148}$$

The poloidal magnetic pressure parameter is defined as follows:

$$\beta_{\mathrm{pol}} = \frac{2\mu_0\langle p\rangle}{\langle \mathbf{B}^2_{\mathrm{pol}}\rangle} = \frac{2\mu_0\iint_\Sigma pr\,\mathrm{d}r\,\mathrm{d}z}{\langle \mathbf{B}^2_{\mathrm{pol}}\rangle\iint r\,\mathrm{d}r\,\mathrm{d}z} \quad \text{along with} \quad \langle\|\mathbf{B}_{\mathrm{pol}}\|\rangle = \frac{\mu_0 I^{\mathrm{pl}}_{\mathrm{tor}}}{\oint \mathrm{d}\ell} \tag{6.149}$$

The quadratic dependence on pressure allows us to determine:

$$\frac{a}{2}\iint_\Sigma \left(\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)^2 r\,\mathrm{d}r\,\mathrm{d}z + b\iint_\Sigma \Phi^{\mathrm{d}}_{\mathrm{pol}}\, r\mathrm{d}r\,\mathrm{d}z = 2\pi^2\mu_0^2\beta_{\mathrm{p}}\frac{\iint_\sigma r\,\mathrm{d}r\,\mathrm{d}z}{\left(\oint \mathrm{d}\ell\right)^2}\left(I^{\mathrm{pl}}_{\mathrm{tor}}\right) \tag{6.150}$$

The internal inductance is defined as follows.

$$l_{\mathrm{int}} = \frac{\left(\oint \mathrm{d}\ell\right)^2}{2\pi\mu_0\left(I^{\mathrm{pl}}_{\mathrm{tor}}\right)^2}\frac{\iint_\Sigma \Phi^{\mathrm{d}}_{\mathrm{pol}}\|\mathbf{J}_{\mathrm{tor}}\|\,\mathrm{d}r\,\mathrm{d}z}{\iint_\Sigma r\,\mathrm{d}r\,\mathrm{d}z} \implies a\iint_\Sigma \left(\Phi^{\mathrm{d}}_{\mathrm{pol}}\right)^2 r\,\mathrm{d}r\,\mathrm{d}z + \alpha\iint_\Sigma \frac{(\Phi^{\mathrm{d}}_{\mathrm{pol}})^2}{r}\,\mathrm{d}r\,\mathrm{d}z + \tag{6.151}$$

$$+\, b\iint_\Sigma \Phi^{\mathrm{d}}_{\mathrm{pol}} r\,\mathrm{d}r\,\mathrm{d}z + \beta\iint_\Sigma \frac{\Phi^{\mathrm{d}}_{\mathrm{pol}}}{r}\,\mathrm{d}r\,\mathrm{d}z = 4\pi^2\mu_0^2 l_{\mathrm{int}}\left(I^{\mathrm{pl}}_{\mathrm{tor}}\right)\frac{\iint_\Sigma r\,\mathrm{d}r\,\mathrm{d}z}{\left(\oint \mathrm{d}\ell\right)^2} \tag{6.152}$$

The safety factor at any magnetic surface is determined via:

$$q(\Phi^{\mathrm{d}}_{\mathrm{pol}}) = \frac{\mu_0}{4\pi^2}F(\Phi^{\mathrm{d}}_{\mathrm{pol}})\oint_{\Phi^{\mathrm{d}}_{\mathrm{pol}}=\mathrm{const.}} \frac{\mathrm{d}\ell}{r^2\|\mathbf{B}_{\mathrm{pol}}\|} \tag{6.153}$$

such that evaluating it at either the plasma boundary or at the magnetic axis involves taking the continuous limit. The previous four equations constitute a deterministic system for the free parameters a, b, α, and β, given the four plasma quantities specified above, the plasma contour in space, and its poloidal disk flux profile at the plasma boundary.

It is ultimately worth noting that linear relations between magnetohydrodynamic pressure and the flux function with quadratic corrections hold surprisingly well in a majority of non-idealised, realistic cases. Even in weakly resistive plasmas, ideal magnetohydrostatic equilibria describe their magnetic field structure to a reasonable and acceptable accuracy. It is only in specific circumstances that one ought to consider some decaying behaviour via exponential dependencies of pressure on the flux function.

We shall consider another family of exact solutions to the Grad-Shafranov equation with somewhat similar hypotheses to Solovev's, notably the assumption of constant poloidal disk current. The functional dependence of pressure on the magnetic poloidal disk flux is assumed to be rational, a common feature in divertor-tokamak configurations. We impose:

$$I^{\mathrm{d}}_{\mathrm{pol}} = \mathrm{const.} \quad \text{and} \quad p(\Phi^{\mathrm{d}}_{\mathrm{pol}}) = \alpha + \frac{\beta}{(\Phi^{\mathrm{d}}_{\mathrm{pol}})^2} \tag{6.154}$$

which provides the Grad-Shafranov equation in the following form:

$$\nabla\cdot\left(\frac{1}{r^2}\nabla\Phi^{\mathrm{d}}_{\mathrm{pol}}\right) - \frac{8\pi^2\mu_0\beta}{(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \iff \frac{\partial^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial r^2} - \frac{1}{r}\frac{\partial\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial r} + \frac{\partial^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial z^2} - \frac{8\pi^2\mu_0\beta r^2}{(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.155}$$

The obtained equation for $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ is a homogeneous partial equation; we expect to obtain at most two linearly independent solutions whose particular combination shall be given by boundary conditions, just as for Solovev's solution. In particular, we shall employ a dimensionality-reduction technique. Having at disposal $I^{\mathrm{d}}_{\mathrm{pol}} = \text{const.}$ as well as axisymmetry, we are motivated to introduce the substitution $x = Ar^2 + \eta(z)$, and subsequently find such $\eta(z)$, if it exists, so that the partial differential equation can be reduced to an ordinary one. The coordinate transformation is given via substitutions $x = Ar^2 + \eta(z)$ and $\zeta = z$, giving:

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial \zeta}{\partial r}\frac{\partial}{\partial \zeta} = 2Ar\frac{\partial}{\partial x}, \quad \frac{\partial}{\partial z} = \frac{\partial x}{\partial z}\frac{\partial}{\partial x} + \frac{\partial \zeta}{\partial z}\frac{\partial}{\partial \zeta} = \eta'(z)\frac{\partial}{\partial x} + \frac{\partial}{\partial \zeta} \tag{6.156}$$

The second derivatives with respect to r and z require further attention:

$$\frac{\partial^2}{\partial r^2} = \frac{\partial}{\partial r}\left(2Ar\frac{\partial}{\partial x}\right) = 2A\frac{\partial}{\partial x} + 2Ar\frac{\partial^2}{\partial x^2} \tag{6.157}$$

$$\frac{\partial^2}{\partial z^2} = \frac{\partial}{\partial z}\left(\eta'(z)\frac{\partial}{\partial x} + \frac{\partial}{\partial \zeta}\right) = \eta''(z)\frac{\partial}{\partial x} + (\eta'(z))^2\frac{\partial^2}{\partial x^2} + 2\eta'(z)\frac{\partial^2}{\partial x \partial \zeta} + \frac{\partial^2}{\partial \zeta^2} \tag{6.158}$$

The Grad-Shafranov operator thus becomes:

$$(2Ar)^2\frac{\partial^2}{\partial x^2} + \eta''(z)\frac{\partial}{\partial x} + (\eta'(z))^2\frac{\partial^2}{\partial x^2} + 2\eta'(z)\frac{\partial^2}{\partial x \partial \zeta} + \frac{\partial^2}{\partial \zeta^2} \tag{6.159}$$

$$\left[4A(x-\eta(z)) + (\eta'(z))^2\right]\frac{\partial^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial x^2} + \eta''(z)\frac{\partial \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial x} + 2\eta'(z)\frac{\partial^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial x \partial \zeta} + \frac{\partial^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\partial \zeta^2} - \frac{8\pi^2\mu_0\beta(x-\eta(z))}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.160}$$

Imposing $\Phi^{\mathrm{d}}_{\mathrm{pol}}(x)$ only, the ζ derivatives vanish, leaving:

$$\left[4A(x-\eta(z)) + (\eta'(z))^2\right]\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + 2\eta''(z)\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x} - \frac{8\pi^2\mu_0\beta(x-\eta(z))}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.161}$$

We are now ready to define the function $\eta(z)$ in order to eliminate the axial dependence. Since, $\eta''(z)$, $(\eta'(z))^2$, and $\eta(z)$ terms appear, we shall define the auxiliary function via $(\eta'(z))^2 = 4a\eta(z) + b$. This ensures that only the first power of $\eta(z)$ remains in the equation, since differentiating the auxiliary definition twice with respect to z gives a constant, $\eta''(z) = a$.

$$\left[4Ax + b + 4(a-A)\eta(z)\right]\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + 2a\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x} - \frac{8\pi^2\mu_0\beta(x-\eta(z))}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.162}$$

Before proceeding, let us truly determine the explicit expression for the auxiliary function:

$$\eta'(z) = \sqrt{4a\eta(z) + b} \implies \int\left(\frac{\mathrm{d}\eta}{\sqrt{4a\eta + b}} = \mathrm{d}z\right) \implies \eta(z) = a(z-z_0)^2 - \frac{b}{4a} \tag{6.163}$$

We separate terms containing $\eta(z)$ and employ the *η-independence* argument:

$$\left[(4Ax+b)\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + 2a\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x} - \frac{8\pi^2\mu_0\beta x}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3}\right] + 4\eta(z)\left[(a-A)\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + \frac{2\pi^2\mu_0\beta}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3}\right] = 0 \tag{6.164}$$

$$\eta\text{-independent:} \quad (4Ax+b)\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + 2a\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x} - \frac{8\pi^2\mu_0\beta x}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.165}$$

$$\text{constraint:} \quad 4(a-A)\frac{\mathrm{d}^2 \Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2} + \frac{8\pi^2\mu_0\beta}{A(\Phi^{\mathrm{d}}_{\mathrm{pol}})^3} = 0 \tag{6.166}$$

We proceed to find a valid solution by solving the η-independent part and verifying that it indeed satisfies the constraint. On the contrary, if the solution doesn't obey the constraint, it suggests a separable behaviour, which can most likely be evaluated in terms of infinite series for once separated functions. Moreover, the coordinate transformation requires further definition for the, so far arbitrary, coefficient A, which we will fix using β. If a solution for $\Phi^{\mathrm{d}}_{\mathrm{pol}}$ satisfies both equations, it automatically satisfies their sum. We are thus motivated to multiply the *constraint* equation by x, and add them, which further eliminates the uncomfortable constant A and the cubic rational term:

$$(4ax+b)\frac{\mathrm{d}^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2}+2a\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x}=0 \tag{6.167}$$

Substituting $\psi=\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}/\mathrm{d}x$, we find:

$$(4ax+b)\frac{\mathrm{d}\psi}{\mathrm{d}x}+2a\psi=0 \implies \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\psi+\frac{1}{2}\ln\left(4ax+b\right)\right)=0 \implies \psi(x)=\frac{C}{\sqrt{4ax+b}} \tag{6.168}$$

where C is an integration constant. We easily find the magnetic poloidal disk flux:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(x)=\frac{C}{2}\sqrt{4ax+b}+C' \tag{6.169}$$

It is only evident that $C'=0$ due to the cubic rational term, which would simply not cancel if it contained any non-zero additive constant:

$$\frac{\mathrm{d}\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x}=\frac{Ca}{\sqrt{4ax+b}} \implies \frac{\mathrm{d}^2\Phi^{\mathrm{d}}_{\mathrm{pol}}}{\mathrm{d}x^2}=-\frac{2Ca^2}{(4ax+b)^{3/2}} \tag{6.170}$$

$$\frac{8(a-A)a^2C}{(4ax+b)^{3/2}}=\frac{64\pi^2\mu_0\beta}{AC^3(4ax+b)^{3/2}} \implies (a-A)a^2C=\frac{8\pi^2\mu_0\beta}{AC^3} \implies \beta=\frac{C^4Aa^2(a-A)}{8\pi^2\mu_0} \tag{6.171}$$

This yields an expression for the linear coefficient A in the transformation, since β is most likely measured, whereas the measured poloidal disk flux may provide the integration constant C at a given location. The full explicit general solution to the Grad-Shafranov equation is therefore given by:

$$\Phi^{\mathrm{d}}_{\mathrm{pol}}(r,z)=\frac{C}{2}\sqrt{4aAr^2+4a^2(z-z_0)^2+b\left(1-\frac{1}{4a}\right)} \tag{6.172}$$

Firstly, it is valuable to deduce the position and shape of the magnetic axis solely from the explicit expression. The magnetic surfaces are either ellipsoids or hyperboloids, depending on the sign of A. For ellipsoidal surfaces, the magnetic axis degenerates into the point $(0,z_0)$, while for hyperboloidal surfaces, the magnetic *axis* is obtained by setting $\Phi^{\mathrm{d}}_{\mathrm{pol}}(r,z)=\sqrt{b(1-1/4a)}$, giving as the solution an infinite cone described by $Ar^2+a(z-z_0)^2=0$ reflected about the $z=z_0$ plane. Indeed, in this system, the magnetic *axis* is not a simple closed curve, but a family of curves spanning the entire cone. They, in a way, represent the symmetry axes in each azimuthal plane and divide two opposite-facing pairs of cusp-mirror-like configurations of magnetic surfaces. This demonstrates that in non-linear systems, magnetic axes become non-trivial. We thus observe that it was necessary and properly executed to define magnetic axes as the zero-volume magnetic surfaces.

6.3.3 Generalisation to helical systems

Here, we revisit our considerations on magnetic field structure in systems characterised by helical symmetry discussed in subsection 5.6 in the previous chapter. We apply our findings to the usual magnetohydrostatic equilibrium equation by performing the vector product of the current density as given by equation (5.162) with the magnetic field density given by (5.148):

$$\begin{gathered}(\ell^2+k^2r^2)\Big[\mu_0\nabla p+(\mathbf{h}\times\nabla f)\times(\mathbf{h}\times\nabla\varrho)\Big]+\\ +f\nabla f+\left(\frac{\ell^2+k^2r^2}{r}\left[\frac{\partial}{\partial r}\left(\frac{r}{\ell^2+k^2r^2}\frac{\partial\varrho}{\partial r}\right)+\frac{1}{r}\frac{\partial^2\varrho}{\partial u^2}\right]-\frac{2k\ell}{\ell^2+k^2r^2}f\right)\nabla\varrho=\mathbf{0}\end{gathered} \tag{6.173}$$

The second term in the above equation (6.173) evaluates to zero due to our considerations in subsection 5.6. This term reduces to the vector product $\nabla f\times\nabla\varrho$, which we have determined to be identically zero:

$$(\mathbf{h}\times\nabla f)\times(\mathbf{h}\times\nabla\varrho)=[\mathbf{h}\cdot(\nabla f\times\nabla\varrho)]\,\mathbf{h}=\mathbf{0} \tag{6.174}$$

Equation (6.173) thus represents the generalisation of the Grad-Shafranov equation to helical systems:

$$\frac{\ell^2 + k^2 r^2}{r} \left[\frac{\partial}{\partial r} \left(\frac{r}{\ell^2 + k^2 r^2} \frac{\partial \varrho}{\partial r} \right) + \frac{1}{r} \frac{\partial^2 \varrho}{\partial u^2} \right] = \frac{2k\ell}{\ell^2 + k^2 r^2} f - f \frac{\mathrm{d}f}{\mathrm{d}\varrho} - \mu_0 (\ell^2 + k^2 r^2) \frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{6.175}$$

We demonstrate its general properties by considering the two limiting cases: $\ell = 0$, corresponding to axisymmetry, and $k = 0$, corresponding to translational symmetry. Inputting $\ell = 0$ gives:

$$k^2 r \frac{\partial}{\partial r} \left(\frac{1}{k^2 r^2} \frac{\partial \varrho}{\partial r} \right) + \frac{\partial^2 \varrho}{\partial z^2} = -f \frac{\mathrm{d}f}{\mathrm{d}\varrho} - \mu_0 k^2 r^2 \frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{6.176}$$

where $u(\ell = 0) = kz$ and the free factor k can be set to unity, giving the usual Grad-Shafranov equation for toroidal systems as given in equation (6.58). Conversely, inputting $k = 0$ gives:

$$\frac{\ell^2}{r} \frac{\partial}{\partial r} \left(\frac{r}{\ell^2} \frac{\partial \varrho}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varrho}{\partial \varphi^2} = -f \frac{\mathrm{d}f}{\mathrm{d}\varrho} - \mu_0 \ell^2 \frac{\mathrm{d}p}{\mathrm{d}\varrho} \tag{6.177}$$

where $u(k = 0) = \ell\varphi$ and the free factor ℓ can be set to unity, giving the Grad-Shafranov equation for axial systems. The function f thus clearly corresponds to the total poloidal current which governs the force-free magnetic field in the absence of toroidal current responsible for magnetic confinement.

Chapter 7

Magnetohydrodynamic instabilities

7.1 Normal modes and instability

To study the behaviour of small perturbations about a static equilibrium state, we write for all quantities $f = f^{\square} + \varepsilon \overline{f}$ for $f = \mathbf{u}$, the total flow velocity, $\mathbf{B}$, the total magnetic field, p, the scalar magnetohydrodynamic pressure, and ρ, the mass density, and linearise the magnetohydrodynamic equations about the static equilibrium quantities. We recall the magnetohydrodynamic equations:

$$\rho \frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = \mathbf{J} \times \mathbf{B} - \nabla p \implies \rho_0 \frac{\partial \overline{\mathbf{u}}}{\partial t} = -\nabla \overline{p} + (\nabla \times \overline{\mathbf{B}}) \times \frac{\mathbf{B}_0}{\mu_0} + (\nabla \times \mathbf{B}_0) \times \frac{\overline{\mathbf{B}}}{\mu_0} + \nu \nabla^2 \mathbf{u} \tag{7.1}$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B} \implies \frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times (\overline{\mathbf{u}} \times \mathbf{B}_0) + \eta \nabla^2 \overline{\mathbf{B}} \tag{7.2}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \implies \frac{\partial \overline{\rho}}{\partial t} = -\nabla \cdot (\rho_0 \overline{\mathbf{u}}) = \tag{7.3}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{p}{\rho^\gamma}\right) = 0 \implies \frac{\partial p}{\partial t} + \mathbf{u} \cdot \nabla p + \gamma p \nabla \cdot \mathbf{v} = 0 \implies \frac{\partial \overline{p}}{\partial t} = -\overline{\mathbf{u}} \cdot \nabla p_0 + \gamma p_0 \nabla \cdot \overline{\mathbf{u}} \tag{7.4}$$

In the ideal case, where $\eta, \nu \to 0$, these equations can be reduced into a single equation for the displacement vector $\boldsymbol{\xi}$ defined by $\overline{\mathbf{u}} = \dot{\boldsymbol{\xi}}$, so that integration of equations (7.1) through (7.4) gives the perturbations $\overline{\mathbf{B}}$, $\overline{p}$ and $\overline{\rho}$ in terms of $\boldsymbol{\xi}$ as follows:

$$\overline{\mathbf{B}} = \nabla \times (\boldsymbol{\xi} \times \mathbf{B}_0) \qquad \overline{p} = -\boldsymbol{\xi} \cdot \nabla p_0 - \gamma p_0 \nabla \cdot \boldsymbol{\xi} \qquad \overline{\rho} = -\nabla \cdot (\rho_0 \boldsymbol{\xi}) \tag{7.5}$$

Indeed, we employed this method precisely throughout Chapter 4 when considering homogeneous plasma in a homogeneous external magnetic field. Inserting equation (7.5) into (7.1) gives:

$$\rho_0 \ddot{\boldsymbol{\xi}} = \mathbf{F}(\boldsymbol{\xi}) = (\nabla \times \overline{\mathbf{B}}) \times \frac{\mathbf{B}_0}{\mu_0} + (\nabla \times \mathbf{B}_0) \times \frac{\overline{\mathbf{B}}}{\mu_0} + \nabla(\boldsymbol{\xi} \cdot \nabla p_0 + \gamma p_0 \nabla \cdot \boldsymbol{\xi}) \tag{7.6}$$

The previous equation, in the ideal case $\eta, \nu \to 0$, describes the time evolution of a small-amplitude perturbation applied at $t = 0$ and is fully determined by the initial values $\boldsymbol{\xi}(\mathbf{x}, 0)$, $\dot{\boldsymbol{\xi}}(\mathbf{x}, 0)$, and by appropriate boundary conditions. Moreover, it is independent of the density gradient. We shall consider the case of finite resistivity later on. Instead of solving this initial value problem for different perturbations, we shall, on the contrary, consider the equivalent problem of determining the normal modes of the system of equations given by (7.6). The linearity of the equation in terms of $\boldsymbol{\xi}$ allows us to consider a single Fourier mode, so that equation (7.6) takes the form:

$$-\rho_0 \omega^2 \boldsymbol{\xi} = \mathbf{F}(\boldsymbol{\xi}) \tag{7.7}$$

Since boundary conditions are required to solve the initial value problem, the general solution for arbitrary ω does not exist. Instead, this equation can be viewed as an eigenvalue problem for the eigenvalue ω^2 and the eigenfunctions $\boldsymbol{\xi}$. The knowledge of the spectrum, i.e., the set of admissible eigenvalues, and the corresponding eigenfunctions allows us to solve the initial value problem for a homogeneous system by expanding the initial perturbation in terms of these eigenfunctions, and reducing the eigenmode problem into a purely algebraic form.

7.1.1 Energy principle

Contrary to homogeneous systems, solving for the time evolution of a particular perturbation in an inhomogeneous system can lead to differential equations, which, in general, require numerical treatment at some point. In this chapter, the complete spectra of eigenfunctions for a given inhomogeneous system are not of our interest; we shall focus only on the unstable part of a particular spectrum. Our general solution method is as follows: we determine stability thresholds in terms of characteristic equilibrium parameters and, if these thresholds are exceeded, we mathematically characterize the properties of the unstable modes. A particularly useful tool for this objective is the energy principle. We first show through direct calculation that the linearised force operator $\mathbf{F}(\boldsymbol{\xi})$ given in equation (7.7) is self-adjoint:

$$\iiint_\Omega \boldsymbol{\eta}\cdot\mathbf{F}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} = \iiint_\Omega \boldsymbol{\xi}\cdot\mathbf{F}(\boldsymbol{\eta})\,\mathrm{d}^3\mathbf{x} \tag{7.8}$$

such that $\boldsymbol{\eta}$ is an arbitrary, *test* displacement, with both displacements satisfying $\boldsymbol{\eta}\cdot\mathbf{n} = 0 = \boldsymbol{\xi}\cdot\mathbf{n}$ on the plasma boundary $\partial\Omega$. We separate the force operator into a pressure-associated and a magnetic-associated force operators and consider their respective inner products with $\boldsymbol{\eta}$:

$$\mathbf{F}_p(\boldsymbol{\xi}) = \nabla(\boldsymbol{\xi}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\boldsymbol{\xi}) \quad \text{and} \quad \mathbf{F}_{\mathbf{B}}(\boldsymbol{\xi}) = (\nabla\times\overline{\mathbf{B}}(\boldsymbol{\xi}))\times\frac{\mathbf{B}_0}{\mu_0} + (\nabla\times\mathbf{B}_0)\times\frac{\overline{\mathbf{B}}(\boldsymbol{\xi})}{\mu_0} \tag{7.9}$$

Beginning with the pressure term, we find:

$$\begin{aligned}\iiint_\Omega \boldsymbol{\eta}\cdot\mathbf{F}_p(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} &= \iiint_\Omega \boldsymbol{\eta}\cdot\nabla(\boldsymbol{\xi}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} = \oiint_{\partial\Omega}(\boldsymbol{\xi}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\boldsymbol{\xi})\boldsymbol{\eta}\cdot\mathrm{d}\mathbf{S} + \\ -\iiint_\Omega(\nabla\cdot\boldsymbol{\eta})(\boldsymbol{\xi}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} &= -\iiint_\Omega\left[(\nabla\cdot\boldsymbol{\eta})(\boldsymbol{\xi}\cdot\nabla p_0) + \gamma p_0(\nabla\cdot\boldsymbol{\eta})(\nabla\cdot\boldsymbol{\xi})\right]\,\mathrm{d}^3\mathbf{x}\end{aligned} \tag{7.10}$$

where the latter term is symmetric. We may thus hope that treating the magnetic-associated force will cancel the first term. Moving on to the latter:

$$\iiint_\Omega \boldsymbol{\eta}\cdot\mathbf{F}_{\mathbf{B}}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} = \iiint_\Omega \boldsymbol{\eta}\cdot\left[(\nabla\times\overline{\mathbf{B}}(\boldsymbol{\xi}))\times\frac{\mathbf{B}_0}{\mu_0} + (\nabla\times\mathbf{B}_0)\times\frac{\overline{\mathbf{B}}(\boldsymbol{\xi})}{\mu_0}\right]\mathrm{d}^3\mathbf{x} \tag{7.11}$$

which consists of the first term as a current-associated term and the latter as a magnetic-associated term:

$$\iiint_\Omega(\nabla\times\overline{\mathbf{B}}(\boldsymbol{\xi}))\cdot(\mathbf{B}_0\times\boldsymbol{\eta})\,\mathrm{d}^3\mathbf{x} = \oiint_{\partial\Omega}\left[\overline{\mathbf{B}}(\boldsymbol{\xi})\times(\boldsymbol{\eta}\times\mathbf{B}_0)\right]\cdot\mathrm{d}\mathbf{S} + \iiint_\Omega\overline{\mathbf{B}}(\boldsymbol{\xi})\cdot[\nabla\times(\mathbf{B}_0\times\boldsymbol{\eta})]\,\mathrm{d}^3\mathbf{x} \tag{7.12}$$

At the plasma boundary, we have $\boldsymbol{\eta}\cdot\mathbf{n} = 0$ as well as $\mathbf{B}_0\cdot\mathbf{n} = 0$, so that the surface term evaluates to zero. We also recognise $\nabla\times(\mathbf{B}_0\times\boldsymbol{\eta}) = \overline{\mathbf{B}}(\boldsymbol{\eta})$:

$$\iiint_\Omega(\nabla\times\overline{\mathbf{B}}(\boldsymbol{\xi}))\cdot(\mathbf{B}_0\times\boldsymbol{\eta})\,\mathrm{d}^3\mathbf{x} = -\iiint_\Omega\overline{\mathbf{B}}(\boldsymbol{\xi})\cdot\overline{\mathbf{B}}(\boldsymbol{\eta})\,\mathrm{d}^3\mathbf{x} \tag{7.13}$$

which is totally symmetric in $\boldsymbol{\eta}\leftrightarrow\boldsymbol{\xi}$. We now treat the current-associated term with the remaining non-symmetric pressure-associated term via the magnetohydrostatic equilibium $\nabla p_0 = \mathbf{J}_0\times\mathbf{B}_0$:

$$\begin{aligned}\iiint_\Omega\boldsymbol{\eta}\cdot(\mathbf{J}_0\times(\nabla\times(\boldsymbol{\xi}\times\mathbf{B}_0)))\,\mathrm{d}^3\mathbf{x} &= \iiint_\Omega\mathbf{J}_0\cdot\left([(\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0 - \mathbf{B}_0(\nabla\cdot\boldsymbol{\xi})]\times\boldsymbol{\eta}\right)\,\mathrm{d}^3\mathbf{x} = \\ &= \iiint_\Omega[\mathbf{J}_0\cdot(\boldsymbol{\eta}\times\mathbf{B}_0)]\,(\nabla\cdot\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} + \iiint_\Omega\mathbf{J}_0\cdot[((\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi})\times\boldsymbol{\eta} - ((\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0)\times\boldsymbol{\eta}]\,\mathrm{d}^3\mathbf{x} = \\ &= -\iiint_\Omega(\nabla\cdot\boldsymbol{\xi})(\boldsymbol{\eta}\cdot\nabla p_0)\,\mathrm{d}^3\mathbf{x} + \iiint_\Omega\mathbf{J}_0\cdot[((\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi})\times\boldsymbol{\eta} - ((\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0)\times\boldsymbol{\eta}]\,\mathrm{d}^3\mathbf{x}\end{aligned} \tag{7.14}$$

The first integral is symmetric in conjunction with equation (7.10). The remaining integral gives:

$$\begin{aligned}&\iiint_\Omega[(\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} = \\ &= -\iiint_\Omega\boldsymbol{\xi}\cdot[(\mathbf{B}_0\cdot\nabla)(\boldsymbol{\eta}\times\mathbf{J}_0)]\,\mathrm{d}^3\mathbf{x} - \iiint_\Omega[(\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} =\end{aligned}$$

$$= \iiint_\Omega \boldsymbol{\xi}\cdot(\mathbf{J}_0\times[(\mathbf{B}_0\cdot\nabla)\boldsymbol{\eta}])\,\mathrm{d}^3\mathbf{x} - \iiint_\Omega \boldsymbol{\xi}\cdot(\boldsymbol{\eta}\times[(\mathbf{B}_0\cdot\nabla)\mathbf{J}_0])\,\mathrm{d}^3\mathbf{x} - \iiint_\Omega [(\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} =$$

$$= \iiint_\Omega [(\mathbf{B}_0\cdot\nabla)\boldsymbol{\eta}]\cdot(\boldsymbol{\xi}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} - \iiint_\Omega [(\mathbf{J}_0\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\xi}\times\boldsymbol{\eta})\,\mathrm{d}^3\mathbf{x} - \iiint_\Omega [(\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} \quad (7.15)$$

where we have tacitly used integration by parts and the usual plasma boundary conditions. Since $\mathbf{B}_0$ is a divergence-free vector field, we find:

$$[(\mathbf{J}_0\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\xi}\times\boldsymbol{\eta}) + [(\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0) + [(\boldsymbol{\eta}\cdot\nabla)\mathbf{B}_0]\cdot(\mathbf{J}_0\times\boldsymbol{\xi}) = 0 \quad (7.16)$$

Gathering everything above concludes the proof:

$$\iiint_\Omega [(\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\eta}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} = \iiint_\Omega [(\mathbf{B}_0\cdot\nabla)\boldsymbol{\eta} - (\boldsymbol{\eta}\cdot\nabla)\mathbf{B}_0]\cdot(\boldsymbol{\xi}\times\mathbf{J}_0)\,\mathrm{d}^3\mathbf{x} \quad (7.17)$$

A self-adjoint operator admits purely real eigenvalues, that is, ω^2 in equation (7.7). We easily prove this by multiplying equation (7.7) by the complex conjugate of the perturbation:

$$-\omega^2 \iiint \rho_0\boldsymbol{\xi}\boldsymbol{\xi}^*\,\mathrm{d}^3\mathrm{x} = -\omega^2\iiint \rho_0\boldsymbol{\xi}^2\,\mathrm{d}^3\mathbf{x} = \langle\boldsymbol{\xi}^*,\mathbf{F}(\boldsymbol{\xi})\rangle = \langle\boldsymbol{\xi},\mathbf{F}(\boldsymbol{\xi}^*)\rangle = -(\omega^*)^2\iiint \rho_0\boldsymbol{\xi}^2\,\mathrm{d}^3\mathbf{x} \quad (7.18)$$

Therefore, necessarily $\omega^2 = (\omega^*)^2$. Eigenmodes are thus either purely oscillating, i.e. $\omega^2 > 0 \implies \omega\in\mathbb{R}$, or purely growing or decaying, that is, $\omega^2 < 0 \implies \omega \in \mathrm{i}\mathbb{R}$. Note that, since any perturbation is real, growing eigenmodes characterised by $\mathrm{Im}\,\omega > 0$ are always accompanied by conjugate decaying modes; however, these exert an uninteresting influence. We will now set the groundwork for the energy principle. We multiply equation (7.7) by $\dot{\boldsymbol{\xi}}$ and integrate first over time, then over the plasma space:

$$\int\left[\iiint_\Omega \dot{\boldsymbol{\xi}}\cdot(\rho_0\ddot{\boldsymbol{\xi}} - \mathbf{F}(\boldsymbol{\xi}))\,\mathrm{d}^3\mathbf{x} = 0\right]\mathrm{d}t \implies \frac{1}{2}\iiint_\Omega \rho_0\dot{\boldsymbol{\xi}}^2\,\mathrm{d}^3\mathbf{x} - \frac{1}{2}\iiint_\Omega \boldsymbol{\xi}\cdot\mathbf{F}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} = H \quad (7.19)$$

and the total kinetic energy is given by the usual well-known definition:

$$K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) = \frac{1}{2}\iiint_\Omega \rho_0(\dot{\boldsymbol{\xi}}\cdot\dot{\boldsymbol{\xi}})\,\mathrm{d}^3\mathbf{x} \quad (7.20)$$

The second integral is evaluated with respect to time by simply noting that $\mathbf{F}(\cdot)$ is a linear operator. H may be interpreted as the total energy associated with the perturbation $\boldsymbol{\xi}$, which is furthermore uniquely determined by initial conditions $\boldsymbol{\xi}(t=0)$ and $\dot{\boldsymbol{\xi}}(t=0)$. We now specifically consider the quantity:

$$\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) = -\frac{1}{2}\iiint_\Omega \boldsymbol{\xi}\cdot\mathbf{F}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} \implies K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) + \delta W(\boldsymbol{\xi},\boldsymbol{\xi}) = H = \text{const.} \quad (7.21)$$

If, for all square-integrable displacements from the equilibrium $\boldsymbol{\xi}$, $\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) > 0$, then the kinetic energy of the plasma is bounded by the total available energy H, so that the associated equilibrium is stable. Now, assume that there is a displacement $\boldsymbol{\eta}$ such that $\delta W(\boldsymbol{\eta},\boldsymbol{\eta}) < 0$, therefore:

$$\exists\gamma > 0, \delta W(\boldsymbol{\eta},\boldsymbol{\eta}) = -\gamma K(\boldsymbol{\eta},\boldsymbol{\eta}) \quad (7.22)$$

Choosing initial conditions $\boldsymbol{\xi}(0) = \boldsymbol{\eta}$, $\dot{\boldsymbol{\xi}} = \sqrt{\gamma}\boldsymbol{\eta}$ gives $H = 0$, thus $K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) + \delta W(\boldsymbol{\xi},\boldsymbol{\xi}) = 0$. Cauchy-Schwarz inequality gives:

$$\left(\frac{\mathrm{d}K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t}\right)^2 = 4\left(K(\dot{\boldsymbol{\xi}},\boldsymbol{\xi})\right)^2 \le 4K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}})K(\boldsymbol{\xi},\boldsymbol{\xi}) \quad (7.23)$$

Multiplying equation (7.7) by $\boldsymbol{\xi}$ and integrating over the plasma gives:

$$\iiint_\Omega \rho_0\ddot{\boldsymbol{\xi}}\cdot\boldsymbol{\xi}\,\mathrm{d}^3\mathbf{x} = \iiint_\Omega \boldsymbol{\xi}\cdot\mathbf{F}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} \quad \text{with} \quad \frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(\boldsymbol{\xi}^2\right) = \frac{\mathrm{d}}{\mathrm{d}t}(2\boldsymbol{\xi}\cdot\dot{\boldsymbol{\xi}}) = 2\dot{\boldsymbol{\xi}}^2 + 2\boldsymbol{\xi}\cdot\ddot{\boldsymbol{\xi}} \quad (7.24)$$

$$2K(\ddot{\boldsymbol{\xi}},\boldsymbol{\xi}) = 2K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) - \frac{\mathrm{d}^2K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t^2} = 2\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) \implies \frac{\mathrm{d}^2K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t^2} = 4K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) \quad (7.25)$$

which, combined with the Cauchy-Schwarz inequality above, gives:

$$K(\boldsymbol{\xi},\boldsymbol{\xi})\frac{\mathrm{d}^2K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t^2} \geq \left(\frac{\mathrm{d}K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t}\right)^2 \tag{7.26}$$

We may introduce a new variable, $\kappa = \ln(K(\boldsymbol{\xi},\boldsymbol{\xi})/K(\boldsymbol{\eta},\boldsymbol{\eta}))$, such that $\boldsymbol{\xi}(t=0) = \boldsymbol{\eta}$, satisfying:

$$\kappa(t=0) = 0, \quad \left.\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right|_{t=0} = \left.\left[\frac{1}{K(\boldsymbol{\xi},\boldsymbol{\xi})}\frac{\mathrm{d}K(\boldsymbol{\xi},\boldsymbol{\xi})}{\mathrm{d}t}\right]\right|_{t=0} = \frac{2K(\boldsymbol{\eta},\sqrt{\gamma}\boldsymbol{\eta})}{K(\boldsymbol{\eta},\boldsymbol{\eta})} = 2\sqrt{\gamma} \tag{7.27}$$

$$[K(\boldsymbol{\eta},\boldsymbol{\eta})]^2 \exp[\kappa]\frac{\mathrm{d}}{\mathrm{d}t}\left(\exp[\kappa]\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right) \geq [K(\boldsymbol{\eta},\boldsymbol{\eta})]^2\left(\exp[\kappa]\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right)^2 \tag{7.28}$$

$$\exp[\kappa]\left(\exp[\kappa]\left(\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right)^2 + \exp[\kappa]\frac{\mathrm{d}^2\kappa}{\mathrm{d}t^2}\right) \geq \exp[2\kappa]\left(\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right)^2 \tag{7.29}$$

$$\frac{\mathrm{d}^2\kappa}{\mathrm{d}t^2} \geq 0 \implies \frac{\mathrm{d}\kappa}{\mathrm{d}t} \geq \left.\frac{\mathrm{d}\kappa}{\mathrm{d}t}\right|_{t=0} = 2\sqrt{\gamma} \implies \kappa(t) \geq 2\sqrt{\gamma}t \tag{7.30}$$

$$K(\boldsymbol{\xi},\boldsymbol{\xi}) \geq K(\boldsymbol{\eta},\boldsymbol{\eta})\exp[2\sqrt{\gamma}t] \implies K(\dot{\boldsymbol{\xi}},\dot{\boldsymbol{\xi}}) \geq |\delta W(\boldsymbol{\eta},\boldsymbol{\eta})|\exp(2\sqrt{\gamma}t) \tag{7.31}$$

Therefore, the kinetic energy and, by extension, the displacement $\boldsymbol{\xi}$ grow at least exponentially. Furthermore, note that the equality is valid for a pure normal mode with some growth rate $\sqrt{\gamma}$, while for an initial perturbation composed of two or more eigenmodes, $\sqrt{\gamma}$ represents the average initial growth rate. Naturally, the eigenmode with the greatest growth rate dominates after sufficient time. Our considerations above allow us to restate the energy principle as follows:

An equilibrium is stable if and only if for all square-integrable displacement functions $\boldsymbol{\xi}(\mathbf{x})$:

$$\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) \equiv -\frac{1}{2}\iiint_\Omega \boldsymbol{\xi}\cdot\mathbf{F}(\boldsymbol{\xi})\,\mathrm{d}^3\mathbf{x} \geq 0 \tag{7.32}$$

The energy principle is therefore a useful mathematical tool to qualitatively describe the stability of some equilibrium: by simply guessing a trial displacement $\boldsymbol{\xi}$ such that $\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) < 0$, it follows immediately that the equilibrium is unstable. The instability growth rate is also bounded from below by the value given in equation (7.22).

It is now useful to return to our proof of self-adjointness of the operator $\mathbf{F}$, and discuss the various contributions to δW, notably: the plasma contribution, δW_P, the surface contribution, δW_S, and the vacuum contribution δW_V. Recognise $\delta W(\boldsymbol{\xi},\boldsymbol{\xi}) = -\frac{1}{2}\langle\boldsymbol{\xi},\mathbf{F}(\boldsymbol{\xi})\rangle$, so that:

$$\delta W_\mathrm{P}(\boldsymbol{\xi},\boldsymbol{\xi}) = \frac{1}{2}\iiint_\Omega \Big[\gamma p_0(\nabla\cdot\boldsymbol{\xi})^2 + (\boldsymbol{\xi}\cdot\nabla p_0)(\nabla\cdot\boldsymbol{\xi})\Big]\mathrm{d}^3\mathbf{x} +$$
$$+\frac{1}{2\mu_0}\iiint_\Omega \left[\overline{\mathbf{B}}(\boldsymbol{\xi})\right]^2\mathrm{d}^3\mathbf{x} - \frac{1}{2}\iiint_\Omega \mathbf{J}_0\cdot\left([(\mathbf{B}_0\cdot\nabla)\boldsymbol{\xi} - (\boldsymbol{\xi}\cdot\nabla)\mathbf{B}_0]\times\boldsymbol{\xi}\right)\mathrm{d}^3\mathbf{x} \tag{7.33}$$

where the first term in the first integral represents the energy of a general non-magnetic sound wave, while the latter term represents pressure-driven instabilities. The second integral simply represents the potential energy stored in the magnetic field perturbation, and the last integral is the so-called *kink* term, which represents the energy released when the equilibrium current density allows the plasma to enter a lower-energy state. If one were to consider a localised region of free plasma, such that it is not bounded by confinement walls, the surface integrals in our considerations above would not evaluate to zero, and one would also have to include the vacuum region surrounding the plasma, where only the vacuum magnetic field contributions are present. We shall only state these contributions:

$$\delta W_\mathrm{S} = \frac{1}{2}\oiint_{\partial\Omega}\left[(\boldsymbol{\xi}\cdot\mathbf{n})^2\mathbf{n}\cdot\nabla\left.\left(p_0 + \frac{\mathbf{B}_0^2}{2\mu_0}\right)\right|_\text{plasma} - \mathbf{n}\cdot\nabla\left.\left(\frac{\mathbf{B}_\mathrm{V}^2}{2\mu_0}\right)\right|_\text{vacuum}\right]\mathrm{d}^3\mathbf{x} \tag{7.34}$$

$$\delta W_\mathrm{V} = \frac{1}{2\mu_0}\iiint_{\mathbb{R}^3\setminus\Omega}(\delta\mathbf{B}_\text{vac})^2\,\mathrm{d}^3\mathbf{x} \tag{7.35}$$

We can attribute certain properties to these terms depending on whether they are semi-definite positive or take on a different sign under a specific configuration. Notably, the non-magnetic sound-wave energy is always stabilising, whereas the pressure-driven instability term may introduce instability. The perturbed magnetic field energy associated with the plasma and vacuum fields, respectively, is always stabilising. The current kink term and the surface kink term may lead to instabilities depending on the configuration.

Chapter 8

Tokamaks and toroidally confined plasmas

8.1 Hamiltonian formalism applied to tokamaks

The most effective tool to represent magnetic field-lines in any magnetic confinement system is the method of Hamiltonian formalism, which successfully models both the regular and the chaotic behaviour of magnetic field-lines. We have already discussed the canonical representation of magnetic field-lines, and here, we will formally employ the variational formalism to describe the magnetic field structure in tokamaks as axisymmetric confinement systems.

Consider a magnetic field-line parametrised by $\mathbf{r}(\tau)$ in three-dimensional space. We define the action of the magnetic field-line over its segment bounded by $\mathbf{r}_1$ and $\mathbf{r}_2$ as the curvilinear integral of the magnetic vector potential $\mathbf{A}$ producing the magnetic field:

$$S[\mathbf{r}(\tau)] = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{A}(\mathbf{r}(\tau)) \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\tau}\,\mathrm{d}\tau \tag{8.1}$$

Applying the variational principle of analytical mechanics to the magnetic action defined above gives:

$$\delta S[\mathbf{r}(\tau)] = \delta \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{A}(\mathbf{r}(\tau)) \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\tau}\,\mathrm{d}\tau = 0 \implies \mathcal{L} = \mathbf{A} \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\tau} \text{ satisfies } \frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{r}} = \mathbf{0} \tag{8.2}$$

$$\frac{\partial \mathcal{L}}{\partial r^i} = \frac{\partial}{\partial r^i}(A_j \dot{r}^j) = \frac{\partial A_j}{\partial r^i}\frac{\mathrm{d}r^j}{\mathrm{d}\tau} \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}^i}\right) = \frac{\mathrm{d}A_i}{\mathrm{d}\tau} = \frac{\partial A_i}{\partial r^j}\frac{\mathrm{d}r^j}{\mathrm{d}\tau} \tag{8.3}$$

$$\frac{\partial A_i}{\partial r^j}\frac{\mathrm{d}r^j}{\mathrm{d}\tau} - \frac{\partial A_j}{\partial r^i}\frac{\mathrm{d}r^j}{\mathrm{d}\tau} = \left(\frac{\partial A_i}{\partial r^j} - \frac{\partial A_j}{\partial r^i}\right)\frac{\mathrm{d}r^j}{\mathrm{d}\tau} = \left([\nabla \times \mathbf{A}] \times \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\tau}\right)_i = 0 \tag{8.4}$$

which is physically interpreted as the fact that $\mathbf{B}$ is parallel to $\dot{\mathrm{r}}(\tau)$ as provided by the definition of a magnetic field-line. The *shortest* path in terms of magnetic vector potential between two points $\mathbf{r}_1$ and $\mathbf{r}_2$ is thus precisely the one following the magnetic field-line connecting the two points. Compare the above action with the usual classical mechanical action given by:

$$S[\mathbf{r}(\tau)] = \int_{\mathbf{r}_1}^{\mathbf{r}_2} (\mathbf{p} \cdot \mathrm{d}\mathbf{r} - \mathcal{H}\,\mathrm{d}t) \tag{8.5}$$

where $(\mathbf{r}, \mathbf{p})$ are canonical conjugate variables, $\mathcal{H}$ is the Hamiltonian of the system, and t is time. Moreover, recall Hamilton's equations:

$$\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}} \qquad \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial \mathbf{r}} \tag{8.6}$$

Now, let (u, v, w) be an orthogonal coordinate system and (A_u, A_v, A_w) the corresponding components of the vector potential $\mathbf{A}$. We choose an appropriate gauge $\mathbf{A} \to \mathbf{A} + \nabla g$ such that $A_w = -\partial g/\partial w$. The component A_w is hence eliminated. Let $\mathbf{r}(\tau) = (U(\tau), V(\tau), W(\tau))$ in the orthogonal system, so that the momenta read $P_u = A_u(U, V, W)$ and $P_v = A_v(U, V, W)$. A fundamental result in analytical mechanics is the existence of a set of canonical coordinates such that the transformed Hamiltonian is identically zero: $\mathcal{H} = \mathcal{H}(U, V, P_u, P_v) = 0$. The action given in equation (8.1) is thus expressed as:

$$S[\mathbf{r}(\tau)] = \int_{\mathbf{r}_1}^{\mathbf{r}_2} (P_u\,\mathrm{d}U + P_v\,\mathrm{d}V - \mathcal{H}\,\mathrm{d}\tau) \Longleftrightarrow S[\mathbf{r}(\tau)] = \int_{\mathbf{r}_1}^{\mathbf{r}_2} (A_u\,\mathrm{d}U + A_v\,\mathrm{d}V) = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{A} \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\tau}\,\mathrm{d}\tau \tag{8.7}$$

Behold the resemblance with equation (8.7). The corresponding field-line is, in the particular set of canonical coordinates, described by autonomous Hamilton's equations:

$$\frac{\mathrm{d}U}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial P_u}, \qquad \frac{\mathrm{d}V}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial P_v}, \qquad \frac{\mathrm{d}P_u}{\mathrm{d}\tau} = -\frac{\partial \mathcal{H}}{\partial U}, \qquad \frac{\mathrm{d}P_v}{\mathrm{d}\tau} = -\frac{\partial \mathcal{H}}{\partial V} \tag{8.8}$$

Since the Hamiltonian is identically zero in this set of coordinates, we have the freedom to choose:

$$\mathcal{H}(U, V, P_u, P_v) = P_u - A_u(U, V, W(P_v, U, V)) = 0 \tag{8.9}$$

by expressing the third coordinate W via $P_v \equiv A_v$, U and V. An equally valid choice is expressing $\mathcal{H} = P_v - A_v(U, V, W(P_u, U, V))$. The particular choice of the Hamiltonian depends on whether the variable U (or V) can be considered as a time-like independent variable that does not change its direction. Then, the field-lines can be expressed via a non-autonomous Hamiltonian system. Choose $U = \tau$ so that the U variable is the time-like variable:

$$\frac{\mathrm{d}V}{\mathrm{d}U} = \frac{\partial \mathcal{H}_u}{\partial P_v}, \qquad \frac{\mathrm{d}P_v}{\mathrm{d}U} = -\frac{\partial \mathcal{H}}{\partial V} \tag{8.10}$$

where $\mathcal{H}_u = -A_u(U, V, W(P_v, U, V))$. This two-dimensional system is known as a *one-and-a-half degrees-of-freedom* Hamiltonian system, since the time-like variable moves in only one direction (positive time). We now consider the particular form of the Hamiltonian equations for a tokamak, that is, a toroidal system. Let (ρ, φ, z) be a cylindrical coordinate system whose z-axis is directed along the symmetry axis, and φ is the azimuthal, that is, the toroidal angle. We can relate these to the quasi-toroidal coordinate system (r, θ, φ) as follows:

$$\rho = R_0 + r\cos\theta, \qquad z = r\sin\theta, \qquad r = \sqrt{(\rho - R_0)^2 + z^2}, \qquad \theta = \arctan\left(\frac{z}{\rho - R_0}\right) \tag{8.11}$$

The local quantity presented by the ratio $\varepsilon = r/R_0$ is called the *inverse aspect* ratio. Let (A_R, A_Φ, A_Z) be the components of the magnetic vector potential $\mathbf{A}$ in the cylindrical coordinate system (ρ, φ, z). We choose $W = \rho(\tau)$, so that the canonical variables and the Hamiltonian read:

$$P_Z = A_Z(\rho(\tau), \varphi(\tau), z(\tau)), \qquad P_\Phi = \rho(\tau)A_\Phi, \qquad u = \varphi, \qquad v = z \tag{8.12}$$

Let us denote $R = \rho(\tau)$, $\Phi = \varphi(\tau)$, and $Z = z(\tau)$ as spatial coordinates along a particular magnetic field line, distinguished by the capital letters just like (U, V, W). The Hamiltonian reads:

$$\mathcal{H}(Z, \Phi, P_Z, P_\Phi) = P_\Phi - R(Z, P_Z)A_\Phi(R(Z, P_Z), \Phi, Z) = 0 \tag{8.13}$$

so that the corresponding Hamilton's equations read:

$$\frac{\mathrm{d}Z}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial P_Z}, \qquad \frac{\mathrm{d}\Phi}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial P_\Phi}, \qquad \frac{\mathrm{d}P_Z}{\mathrm{d}\tau} = -\frac{\partial \mathcal{H}}{\partial Z}, \qquad \frac{\mathrm{d}P_\Phi}{\mathrm{d}\tau} = -\frac{\partial \mathcal{H}}{\partial \Phi} \tag{8.14}$$

Axisymmetry allows us to choose the toroidal angle as the time-like independent variable, so that:

$$\frac{\mathrm{d}Z}{\mathrm{d}\Phi} = \frac{\partial \mathcal{H}}{\partial P_Z}, \qquad \frac{\mathrm{d}P_Z}{\mathrm{d}\Phi} = -\frac{\partial \mathcal{H}}{\partial Z} \tag{8.15}$$

and $\mathcal{H}_\varphi = -RA_\varphi$. The relation between the canonical variables (Z, P_Z) and the geometrical coordinates (R, Φ, Z) on a magnetic field-line is given by $P_Z = A_Z(R, \Phi, Z)$. In the case of weakly diamagnetic plasmas, the Maxwell-Ampère law gives $B_\Phi = B_0R_0/R$, so that $A_Z = B_0R_0 \ln R/R_0$, and one has $P_Z = B_0R_0 \ln(R/R_0)$. The magnetic field components in the cylindrical systems are given as:

$$B_R = -\frac{\partial A_\varphi}{\partial Z} = \frac{1}{R}\frac{\partial \psi_{\mathrm{tor}}}{\partial Z}, \qquad B_\Phi = -\frac{\partial \psi_{\mathrm{tor}}}{\partial R}, \qquad B_Z = -\frac{\partial \psi_{\mathrm{tor}}}{\partial P_Z} \tag{8.16}$$

where we had identified the Hamiltonian with the reduced toroidal magnetic flux, $\psi_{\mathrm{tor}} = \Phi_{\mathrm{tor}}/2\pi$

8.1.1 Action-angle variables for magnetic field-lines

Consider an equilibrium plasma with nested toroidal magnetic surfaces. Any torus presents only two irreducible closed contours along the poloidal and the toroidal direction, which we denote as $\mathcal{C}_{\rm pol}$ and $\mathcal{C}_{\rm tor}$, respectively. We shall consider specifically the poloidal and toroidal angles as canonical coordinates of the Hamiltonian system, whose conjugate momenta are $\psi_{\rm pol}$ and $\psi_{\rm tor}$. In its action-angle form, $\mathcal{H} = \mathcal{H}(\psi_{\rm pol}, \psi_{\rm tor})$ only, and the Hamilton equations take the form:

$$\frac{\mathrm{d}\theta}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial \psi_{\rm pol}} = \omega_\theta, \qquad \frac{\mathrm{d}\varphi}{\mathrm{d}\tau} = \frac{\partial \mathcal{H}}{\partial \psi_{\rm tor}} = \omega_\varphi, \qquad \frac{\mathrm{d}\psi_{\rm pol}}{\mathrm{d}\tau} = 0 = \frac{\mathrm{d}\psi_{\rm tor}}{\mathrm{d}\tau} \tag{8.17}$$

where the Hamiltonian is $\mathcal{H}(\theta, \varphi) = \psi_{\rm tor} - RA_\varphi(R, \varphi, Z) = 0$. The radial and axial cylindrical coordinates, R and Z, are thus functions of $(\psi_{\rm pol}, \theta)$, and ω_θ and ω_φ are the *frequencies* of the system. Formally, the action-angle variables are introduced via the appropriate canonical transformation determined by the generating function:

$$F(\psi_{\rm pol}, \psi_{\rm tor}; \varphi, Z) = \int_{Z_0}^{Z} P_Z(Z'; \psi_{\rm pol})\, \mathrm{d}Z' + \int_{\varphi_0}^{\varphi} P_\varphi(\varphi'; \psi_{\rm tor})\, \mathrm{d}\varphi' \tag{8.18}$$

where Z_0 and φ_0 are arbitrary constants. The above gives the following relations:

$$\psi_{\rm pol} = \frac{1}{2\pi} \oint_{\mathcal{C}_\theta} P_Z(Z; \psi_{\rm pol})\, \mathrm{d}Z, \qquad \psi_{\rm tor} = \frac{1}{2\pi} \oint_{\mathcal{C}_\varphi} P_\varphi(\varphi; \psi_{\rm tor})\, \mathrm{d}\varphi, \qquad \theta = \frac{\partial F}{\partial \psi_{\rm tor}}, \qquad \varphi = \frac{\partial F}{\partial \psi_{\rm pol}} \tag{8.19}$$

The above integrals precisely represent the (disk) poloidal and the toroidal magnetic fluxes, so it shall tacitly be known that $\psi_{\rm pol}$ and $\psi_{\rm tor}$ represent the reduced fluxes, that is, fluxes divided by 2π. The rotational transform $\iota\!\!\!-(\psi_{\rm pol})$, that is, the safety factor $q(\psi_{\rm pol})$ can then be determined via (8.15):

$$q(\psi_{\rm pol}) = \frac{\mathrm{d}\psi_{\rm tor}}{\mathrm{d}\psi_{\rm pol}} = \frac{\mathrm{d}\varphi}{\mathrm{d}\theta} = \frac{\Delta\varphi}{2\pi} = \frac{1}{2\pi} \oint_{\mathcal{C}_\varphi} \frac{\mathrm{d}Z}{\partial \psi_{\rm pol}/\partial P_Z} \tag{8.20}$$

8.2 Single-null divertor tokamak

We now illustrate the action-angle Hamiltonian formalism for a magnetic-field-structure model equivalent to that in tokamaks. The considered topologically equivalent poloidal magnetic flux is given by:

$$\psi_{\rm pol}(R, Z) = \frac{\beta^2}{2} \left[\ln\left(\frac{R}{R_0}\right) \right]^2 - \frac{\alpha^2 Z^2}{2} \left(1 - \frac{Z^2}{2Z_0^2}\right) \tag{8.21}$$

where α and β are some parameters, while the magnetic axis radius is located at (R_0, Z_0). The magnetic surface profile is illustrated in Figure 8.1. The first term is precisely the normalised momentum P_Z/B_0R_0, as determined via relations (8.16). Here, we reinstate the preference for φ to serve as the time-like independent variable, given axisymmetry. Hamilton's equations are given as follows:

$$\frac{\mathrm{d}Z}{\mathrm{d}\varphi} = \frac{\partial \psi_{\rm pol}}{\partial P_Z} = \beta^2 P_Z, \qquad \frac{\mathrm{d}P_Z}{\mathrm{d}\varphi} = -\frac{\partial \psi_{\rm pol}}{\partial Z} = \alpha^2 Z \left(1 - \frac{Z^2}{Z_0^2}\right) \tag{8.22}$$

The fixed points of the system of equations (8.22) are determined via:

$$\frac{\mathrm{d}Z}{\mathrm{d}\varphi} \stackrel{!}{=} 0 = \frac{\partial \psi_{\rm pol}}{\partial P_Z} \qquad \frac{\mathrm{d}P_Z}{\mathrm{d}\varphi} \stackrel{!}{=} 0 = -\frac{\partial \psi_{\rm pol}}{\partial Z} \tag{8.23}$$

which correspond to a zero poloidal magnetic field via equations (8.16). This particular system contains two fixed points: one is a *hyperbolic* fixed point, which is stable in one canonical direction and unstable in the other, at $(Z_s, p_s) = (0, 0)$, and corresponds to the so-called *X-point*; one *elliptic* fixed point, which is stable in both canonical directions, at $(Z_s, p_s) = (Z_0, 0)$, corresponding to the magnetic axis, and is usually referred to as the *O-point*. The Hamiltonian flow-field is hyperbolic around the X-point, and elliptic in the neighbourhood of the O-point. The extrema of poloidal flux are found at $(R = R_0, Z = \pm Z_0)$ and $(R = R_0, Z = 0)$. Moreover, the points $(R = R_0, Z = \pm Z_0)$ correspond to minima of poloidal flux, while the point $(R = R_0, Z = 0)$ is a saddle point at which also $\psi_{\rm pol}(R = R_0, Z = 0) = 0$. Namely, the separatrix passes through the saddle point, which is therefore called the X-point. The global minimum of the poloidal flux is equal to $\Psi_{\text{pol}} = -\alpha^2 Z_0^2/4$ at the magnetic axis. We shall now explore the applications of the action-angle formalism to each type of magnetic surface.

[[Image Discarded Due To '/tikz/external/mode=list and make']]

Figure 8.1: Magnetic surface contours topologically equivalent to those in tokamaks. The separatrix corresponding to zero magnetic flux is traced in red.

1. Closed field-lines: Having chosen a gauge such that $A_R = 0$, the reduced toroidal flux can be expressed as an action variable:

$$\psi_{\text{tor}} = \frac{1}{2\pi}\iint_{S_\varphi} B_\varphi \,\mathrm{d}S = \frac{1}{2\pi}\oint_{\mathcal{C}_\theta} \mathbf{A}\cdot \mathrm{d}\boldsymbol{\ell} = \frac{1}{2\pi}\oint_{\mathcal{C}_\theta} A_Z \,\mathrm{d}Z = \frac{1}{2\pi}\oint_{\mathcal{C}_\theta} P_Z \,\mathrm{d}Z \tag{8.24}$$

where $\mathcal{C}_\theta$ is the intersection curve of any poloidal plane and the magnetic surface labelled by the poloidal flux $2\pi\psi_{\text{pol}}$. We may now express the momentum P_Z via equation (8.21), giving:

$$\psi_{\text{tor}} = \frac{1}{\pi}\int_{Z_{\min}}^{Z_{\max}} P_Z \,\mathrm{d}Z = \frac{B_0 R_0\sqrt{2}}{\pi\beta}\int_{Z_{\min}}^{Z^{\max}} \sqrt{\psi_{\text{pol}} + \frac{\alpha^2 Z^2}{2}\left(1 - \frac{Z^2}{2Z_0^2}\right)}\mathrm{d}Z =$$

$$= \left\{\frac{Z^2_{\max/\min}}{Z_0^2} = 1 \pm \sqrt{1 + \frac{\psi_{\text{pol}}}{|\Psi_{\text{pol}}|}}\right\} = \frac{B_0 R_0 \alpha}{\pi\beta Z_0\sqrt{2}}\int_{Z_{\min}}^{Z_{\max}} \sqrt{(Z^2_{\max} - Z^2)(Z^2 - Z^2_{\min})}\,\mathrm{d}Z \tag{8.25}$$

where $Z_{\min}$ and $Z_{\max}$ are the lowest and, respectively, the highest points of the closed curve $\mathcal{C}_\theta$ along the positive axial direction in the upper half-space. Note again that Ψ_{pol} denotes the magnetic poloidal disk flux at the magnetic axis. One may express the integral in terms of elliptic integrals:

$$\int_{Z_{\min}}^{Z_{\max}} \sqrt{(Z^2_{\max} - Z^2)(Z^2 - Z^2_{\min})}\,\mathrm{d}Z = \left\{Z^2 = Z^2_{\max}\cos^2\phi + Z^2_{\min}\sin^2\phi\right\} =$$

$$= \left\{\mathrm{d}Z = -\frac{(Z^2_{\max} - Z^2_{\min})}{Z}\sin\phi\cos\phi\,\mathrm{d}\phi\right\} = (Z^2_{\max} - Z^2_{\min})^2\int_0^{\pi/2} \frac{\sin^2\phi\cos^2\phi\,\mathrm{d}\phi}{\sqrt{Z^2_{\max}\cos^2\phi + Z^2_{\min}\sin^2\phi}} =$$

$$= \left\{k^2 = 1 - \frac{Z^2_{\min}}{Z^2_{\max}}\right\} = Z^3_{\max}k^4\left[\int_0^{\pi/2}\left(\frac{\sin^2\phi}{\sqrt{1 - k^2\sin^2\phi}} - \frac{\sin^4\phi}{\sqrt{1 - k^2\sin^2\phi}}\right)\mathrm{d}\phi\right] \tag{8.26}$$

We now introduce $\Delta(\phi) = \sqrt{1 - k^2\sin^2\phi}$, so that $\sin^2\phi = (1 - \Delta^2(\phi))/k^2$:

$$\int_0^{\pi/2}\frac{\sin^2\phi}{\Delta(\phi)}\mathrm{d}\phi = \frac{1}{k^2}\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\Delta(\phi)} - \frac{1}{k^2}\int_0^{\pi/2}\Delta(\phi)\,\mathrm{d}\phi = \frac{K(k) - E(k)}{k^2} \tag{8.27}$$

$$\int_0^{\pi/2}\frac{\sin^4\phi}{\Delta(\phi)}\mathrm{d}\phi = \frac{1}{k^4}\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\Delta(\phi)} - \frac{2}{k^4}\int_0^{\pi/2}\Delta(\phi)\,\mathrm{d}\phi + \frac{1}{k^4}\int_0^{\pi/2}\Delta^3(\phi)\,\mathrm{d}\phi =$$

where we recognise $K(k)$ and $E(k)$. The last integral is evaluated by first considering the product $\Delta(\phi)\sin\phi\cos\phi$ and differentiating:

$$\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\Delta(\phi)\sin\phi\cos\phi\right) = (\cos^2\phi - \sin^2\phi)\Delta(\phi) - \frac{k^2\sin^2\phi\cos^2\phi}{\Delta(\phi)} =$$

$$= \frac{(1 - 2\sin^2\phi)(1 - k^2\sin^2\phi) - k^2\sin^2\phi(1 - \sin^2\phi)}{\Delta(\phi)} = \frac{1 - 2(1 + k^2)\sin^2\phi + 3k^2\sin^2\phi}{\Delta(\phi)} \implies$$

$$\implies k^2 \frac{\mathrm{d}}{\mathrm{d}\phi} \left(\Delta(\phi)\sin\phi\cos\phi\right) = 3\Delta^3(\phi) + (2k^2-4)\Delta(\phi) + \frac{1-k^2}{\Delta(\phi)} \tag{8.28}$$

Integrating the obtained result over $(0, \pi/2)$ and noting that the left-hand side evaluates to zero since $\sin(0) = \cos(\pi/2) = 0$, we obtain:

$$\int_0^{\pi/2} \Delta^3(\phi)\,\mathrm{d}\phi = \frac{(4-2k^2)E(k) - (1-k^2)K(k)}{3} \tag{8.29}$$

This provides us with the final result:

$$\int_{Z_{\min}}^{Z_{\max}} \sqrt{(Z_{\max}^2 - Z^2)(Z^2 - Z_{\min}^2)}\,\mathrm{d}Z = \frac{Z_{\max}}{3}\left[(Z_{\max}^2 + Z_{\min}^2)E(k) - 2Z_{\min}^2 K(k)\right]$$

where, again, $k^2 = 1 - Z_{\min}^2/Z_{\max}^2$. Substituting the result into equation (8.25), we obtain:

$$\psi_{\mathrm{tor}} = \frac{2B_0R_0Z_0^2\alpha}{3\pi\beta\sqrt{2-k^2}}\left[E(k) - 2\frac{1-k^2}{2-k^2}K(k)\right] \quad \text{where} \quad k^2 = \frac{2\sqrt{1+\psi_{\mathrm{pol}}/|\Psi_{\mathrm{pol}}|}}{1+\sqrt{1+\psi_{\mathrm{pol}}/|\Psi_{\mathrm{pol}}|}} \tag{8.30}$$

Note that this marks our first concrete encounter with the bijective relation between the toroidal and the poloidal magnetic fluxes, ψ_{tor} and ψ_{pol}. The safety factor then follows trivially by differentiating the above with respect to the poloidal flux, ψ_{pol}. Moreover, we find that the total toroidal flux of the magnetic axis is zero, a trivial yet important result for the coherence of our considerations. Notably, the magnetic axis is indeed characterised by extremal poloidal disk flux, but zero toroidal flux since its volume, and consequently its cross-section, is identically zero. The toroidal magnetic flux of the separatrix is given by substituting $\psi_{\mathrm{pol}} = 0$, so that $k = 1$. Due to the divergence of $K(k)$ as k approaches unity, one has to take the limit $k \to 1^-$. The elliptic integral of the first kind has a logarithmic singularity at $k = 1$, namely:

$$\lim_{k\to 1^-} K(k) \simeq \ln\left(\frac{4}{\sqrt{1-k^2}}\right) \implies \lim_{k\to 1^-} 2(1-k^2)K(k) = 0 \tag{8.31}$$

while the elliptic integral of the second kind converges to unity as $k \to 1^-$. This gives:

$$\Psi_{\mathrm{tor}} = \frac{2}{3\pi}B_0R_0Z_0\frac{\alpha Z_0}{\beta} \quad \text{within the separatrix} \tag{8.32}$$

as the reduced magnetic toroidal flux enclosed by the separatrix. To determine the conjugate angle variable θ, let $\theta = 0$ at $Z = Z_{\max}$. Since we have chosen φ as the independent time-like variable, the generating function given in (8.18) reduces to an integral over the vertical coordinate Z' only, so that the angle θ is simply given via the relation:

$$\begin{aligned}
\theta(Z) &= \frac{\partial}{\partial\psi_{\mathrm{tor}}}\int_Z^{Z_{\max}} P_Z(Z';\psi_{\mathrm{pol}})\,\mathrm{d}Z' = \frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}\int_Z^{Z_{\max}} \frac{\partial P_Z}{\partial\psi_{\mathrm{pol}}}\,\mathrm{d}Z' = \frac{B_0^2R_0^2}{\beta^2}\frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}\int_Z^{Z_{\max}} \frac{\mathrm{d}Z'}{P_Z} = \\
&= \frac{B_0R_0}{\beta\sqrt{2}}\frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}\int_Z^{Z_{\max}} \frac{\mathrm{d}Z'}{\sqrt{\psi_{\mathrm{pol}} + \frac{\alpha^2Z^2}{2}\left(1 - \frac{Z^2}{2Z_0^2}\right)}} = \left\{\frac{Z_{\max/\min}^2}{Z_0^2} = 1 \pm \sqrt{1 + \frac{\psi_{\mathrm{pol}}}{|\Psi_{\mathrm{pol}}|}}\right\} = \\
&= B_0R_0Z_0\frac{\sqrt{2}}{\alpha\beta}\frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}\int_Z^{Z_{\max}} \frac{\mathrm{d}Z'}{\sqrt{(Z_{\max}^2 - Z'^2)(Z'^2 - Z_{\min}^2)}} = \left\{\sin^2\phi = \frac{Z_{\max}^2 - Z'^2}{Z_{\max}^2 - Z_{\min}^2}\right\} = \\
&= \frac{B_0R_0Z_0\sqrt{2}}{\alpha\beta Z_{\max}}\frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}F\left(\arcsin\left[\frac{Z_{\max}^2 - Z^2}{Z_{\max}^2 - Z_{\min}^2}\right], \sqrt{1 - \frac{Z_{\min}^2}{Z_{\max}^2}}\right)
\end{aligned} \tag{8.33}$$

where $F(\phi, k)$ is the incomplete elliptic integral of the first kind. Imposing the condition that $\theta(Z = Z_{\min}) = \pi$, we find an expression for the safety factor:

$$q(\psi_{\mathrm{pol}}) = \left(\frac{\mathrm{d}\psi_{\mathrm{pol}}}{\mathrm{d}\psi_{\mathrm{tor}}}\right)^{-1} = \frac{B_0 R_0}{\pi\alpha\beta} K\left(k\right)\sqrt{2-k^2} \tag{8.34}$$

which, by inverting relation (8.33), gives the field-line coordinates $Z(\theta)$, $R(\theta)$ as follows:

$$Z(\theta) = Z_{\max}\,\mathrm{dn}(K(k)\theta/\pi;k) \quad \text{and}$$

$$R(\theta) = R_0 \exp\left[-\frac{Z_0\sqrt{2}\alpha}{\beta}\sqrt{1+\frac{\psi_{\mathrm{pol}}}{|\Psi_{\mathrm{pol}}|}}\mathrm{sn}(K(k)\theta/\pi;k)\,\mathrm{cn}(K(k)\theta/\pi;k)\right] \tag{8.35}$$

where k retains its meaning, while $\mathrm{sn}(u;k)$, $\mathrm{cn}(u;k)$, and $\mathrm{dn}(u;k)$ denote the Jacobi elliptic sine, elliptic cosine, and the delta amplitude functions. The dependence on φ is found via the rotational transform.

2. Open field-lines: The reduced toroidal flux is defined for the open field-lines by requiring that it is a continuous function of the reduced poloidal flux at the separatrix. The contour is thus the intersection curve of the considered magnetic surface with the poloidal plane starting and ending on the $Z=0$ plane after a single full turn, that is:

$$\psi_{\mathrm{tor}} = \frac{1}{2\pi}\oint_{\mathcal{C}_{\mathrm{pol}}} P_Z(Z;\psi_{\mathrm{pol}})\,\mathrm{d}Z = \frac{B_0R_0\alpha}{\pi Z_0\beta\sqrt{2}}\int_0^{Z_{\max}}\sqrt{\frac{\psi_{\mathrm{pol}}}{|\Psi_{\mathrm{pol}}|}Z_0^4+2Z_0^2Z^2-Z^4}\,\mathrm{d}Z \tag{8.36}$$

where, now, $\psi_{\mathrm{pol}}>0$. The biquadratic form inside the integral presents two purely real and two purely imaginary roots, so it factorises as follows:

$$\int_0^{Z_{\max}}\sqrt{(Z_{\max}^2-Z^2)(Z^2+Z_{\mathrm{img}}^2)}\,\mathrm{d}Z = \frac{2\sqrt{2}kZ_0^3}{3\sqrt{2-k^2}}\left[E\left(\frac{1}{k}\right)+\frac{k^2-1}{2-k^2}K\left(\frac{1}{k}\right)\right] \tag{8.37}$$

where the integral was evaluated in a similar fashion to the previous case. Substituting:

$$\psi_{\mathrm{tor}} = \frac{2B_0R_0Z_0^2\alpha}{3\pi\beta}\frac{k}{\sqrt{2-k^2}}\left[E\left(\frac{1}{k}\right)+\frac{k^2-1}{2-k^2}K\left(\frac{1}{k}\right)\right] \quad \text{where} \quad k^2 = \frac{2\sqrt{1+\psi_{\mathrm{pol}}/|\Psi_{\mathrm{pol}}|}}{1+\sqrt{1+\psi_{\mathrm{pol}}/|\Psi_{\mathrm{pol}}|}} \tag{8.38}$$

All other relations that have been treated in the previous case can be determined by an analogous treatment. We shall, however, determine the expression of the safety factor by using the conditions $\theta(Z=Z_{\max})=0$ and $\theta(Z=0)=\pi$, so that:

$$q(\psi_{\mathrm{pol}}) = \frac{B_0R_0}{\pi\alpha\beta}K\left(\frac{1}{k}\right)\frac{\sqrt{2-k^2}}{k} \tag{8.39}$$

3. Separatrix: All details regarding the magnetic separatrix can be obtained by taking the limit $k\to1$ via either of the considerations above. Notably, in the limit $k\to1$, we have $\mathrm{dn}(u;k=1)=\mathrm{cn}(u;k=1)=\mathrm{sech}(u)$, and $\mathrm{sn}(u;k=1)=\tanh(u)$, $K(k)/\pi q(\psi_{\mathrm{pol}})\to\alpha\beta/B_0R_0$. The magnetic field-lines on the separatrix are determined via the following relations:

$$Z(\Phi) = \frac{Z_0\sqrt{2}}{\cosh\left(\alpha\beta\Phi/B_0R_0\right)}, \qquad R(\Phi) = R_0\exp\left[-\frac{\alpha Z_0\sqrt{2}}{\beta}\frac{\sinh\left(\alpha\beta\Phi/B_0R_0\right)}{\cosh^2\left(\alpha\beta\Phi/B_0R_0\right)}\right] \tag{8.40}$$

The safety factor exhibits logarithmic asymptotics near the separatrix:

$$q(\psi_{\mathrm{pol}}) = \frac{B_0R_0}{2\pi\alpha\beta}\ln\left(\frac{16Z_0^2\alpha^2}{|\psi_{\mathrm{pol}}|}\right)+\mathcal{O}(\psi_{\mathrm{pol}}) \qquad (\psi_{\mathrm{pol}}\to\pm0) \tag{8.41}$$

8.3 Equilibrium fields of toroidally confined plasmas

8.3.1 Equilibrium plasmas with circular cross-section

The most elementary model of equilibrious toroidally confined plasmas is given by the so-called *standard magnetic field* prescribed as follows:

$$\mathbf{B}(r,\theta) = B_\varphi \hat{\mathbf{e}}_\varphi + B_\theta \hat{\mathbf{e}}_\theta = \frac{B_0}{1+\varepsilon\cos\theta}\left(\hat{\mathbf{e}}_\varphi + \frac{r}{q(r)R_0}\hat{\mathbf{e}}_\theta\right) \tag{8.42}$$

where (r,θ) denote the quasi-toroidal radial and poloidal coordinates, B_0 is the magnetic field strength at the magnetic axis located at (R_0, Z_0), and $q(r)$ is the safety factor. Even though idealised, this model exhibits many characteristics of general toroidally confined plasmas, such as the radial decay of the azimuthal magnetic field, $B_\varphi \propto 1/\rho$, and nested magnetic surfaces. Here, the latter are concentric regular tori described by $r = \text{const}$.

8.3.2 Elongated equilibrium plasmas

Chapter 9

Neoclassical transport in plasmas

In this chapter, we develop a theoretical framework for neoclassical transport, which provides a more general foundation than classical transport, particularly for transport phenomena such as electrical resistivity, collision frequencies, and the diffusion of momentum and kinetic energy. The term *neoclassical* given to this theory is fully justified in the sense that the basic mechanisms at work are the same as those in the classical theory, which has been tacitly the subject of the first chapter. The neoclassical theory explains the behaviour of plasmas in inhomogeneous and curved magnetic fields with excellent accuracy. The main conceptual result of the theory is that the global geometry of magnetic field structures influences transport processes in plasmas; hence, our preceding considerations of magnetic field structures. We shall observe that, even in weakly toroidal plasmas, neoclassical effects have, in fact, a prominent influence on the general plasma particle motion.

9.1 Fokker-Planck collision theory

In this section, we lay the necessary groundwork to formally treat interparticle collisions in plasmas. We shall start by first deriving the main result of the Fokker-Planck collision theory, that is, the Fokker-Planck equation, and subsequently find analytical expressions of quantities of interest when only interparticle collisions are taken into account, in particular, electrical resistivity, diffusion of momentum, and of energy. It is worth noting that these are the demonstrations of results that were only magically revealed in the first chapter.

9.1.1 Fokker-Planck equation

Let particle species σ be the incident particle species and ρ the target particle species in a plasma. It is appropriate to introduce a change of variables $(\mathbf{r}_\sigma, \mathbf{r}_\rho) \mapsto (\mathbf{R}, \mathbf{r})$ such that $\mathbf{R}$ is the position vector of the centre of mass and $\mathbf{r}$ the position vector of a single incident particle relative to a target particle:

$$\mathbf{R} = \frac{m_\sigma \mathbf{r}_\sigma + m_\rho \mathbf{r}_\rho}{m_\sigma + m_\rho} \implies \ddot{\mathbf{R}} = \frac{m_\sigma \ddot{\mathbf{r}}_\sigma + m_\rho \ddot{\mathbf{r}}_\rho}{m_\sigma + m_\rho} = \mathbf{0} \implies \frac{\mathrm{d}\dot{\mathbf{R}}}{\mathrm{d}t} = \mathbf{0} \tag{9.1}$$

$$\mathbf{r} = \mathbf{r}_\sigma - \mathbf{r}_\rho \quad \text{and} \quad \frac{1}{\mu_{\sigma\rho}} = \frac{1}{m_\sigma} + \frac{1}{m_\rho} \tag{9.2}$$

$$\ddot{\mathbf{r}} = \ddot{\mathbf{r}}_\sigma - \ddot{\mathbf{r}}_\rho = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 m_\sigma} \frac{\mathbf{r}}{\|\mathbf{r}\|^3} - \left(-\frac{q_\sigma q_\rho}{4\pi\varepsilon_0 m_\rho} \frac{\mathbf{r}}{\|\mathbf{r}\|^3} \right) = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 \mu_{\sigma\rho}} \frac{\mathbf{r}}{\|\mathbf{r}\|^3} \tag{9.3}$$

The position vectors of the two particles can be found from the new variables:

$$\mathbf{r}_\sigma = \frac{m_\sigma \mathbf{r}_\sigma + m_\rho \mathbf{r}_\rho + m_\rho(\mathbf{r}_\sigma - \mathbf{r}_\rho)}{m_\sigma + m_\rho} = \mathbf{R} + \frac{m_\rho}{m_\sigma + m_\rho}\mathbf{r} \tag{9.4}$$

$$\mathbf{r}_\rho = \frac{m_\sigma \mathbf{r}_\sigma + m_\rho \mathbf{r}_\rho + m_\sigma(\mathbf{r}_\rho - \mathbf{r}_\sigma)}{m_\sigma + m_\rho} = \mathbf{R} - \frac{m_\sigma}{m_\sigma + m_\rho}\mathbf{r} \tag{9.5}$$

It is also appropriate to relate the deviation of the relative velocity of the incident particle in the target particle's frame of reference with that in the laboratory frame:

$$\Delta \mathbf{v}_\sigma = \mathbf{v}'_\sigma - \mathbf{v}_\sigma = \dot{\mathbf{r}}'_\sigma - \dot{\mathbf{r}}_\sigma = \dot{\mathbf{R}} + \frac{m_\rho}{m_\sigma + m_\rho}\dot{\mathbf{r}}' - \dot{\mathbf{R}} - \frac{m_\rho}{m_\sigma + m_\rho}\dot{\mathbf{r}} = \frac{m_\rho}{m_\sigma + m_\rho}\Delta \mathbf{v}_\mathrm{r} \tag{9.6}$$

As demonstrated in chapter 1, the conservation of the Laplace-Runge-Lenz vector proves particularly useful for determining the deflection angle of the incident particle:

$$\boldsymbol{\epsilon}_\sigma = -\frac{4\pi\varepsilon_0}{\mu_{\sigma\rho} q_\sigma q_\rho}\mathbf{p}_\sigma \times \boldsymbol{\mathcal{L}}_\sigma - \hat{\mathbf{r}} = -\frac{4\pi\varepsilon_0 \mu_{\sigma\rho}\mathbf{v}_\mathrm{rel}^2 b}{q_\sigma q_\rho}\,\hat{\mathbf{x}} + \hat{\mathbf{z}} \tag{9.7}$$

$$\boldsymbol{\epsilon}_\sigma = \left[\sin\theta - \frac{4\pi\varepsilon_0 \mu_{\sigma\rho}\mathbf{v}_\mathrm{rel}^2 b}{q_\sigma q_\rho}\cos\theta\right]\hat{\mathbf{x}} + \left[\frac{4\pi\varepsilon_0 \mu_{\sigma\rho}\mathbf{v}_\mathrm{rel}^2 b}{q_\sigma q_\rho}\sin\theta - \cos\theta\right]\hat{\mathbf{z}} \implies \tan\frac{\theta}{2} = \frac{q_\sigma q_\rho}{4\pi\varepsilon_0 \mu_{\sigma\rho}\mathbf{v}_\mathrm{rel}^2 b} \tag{9.8}$$

The utility of changing the reference frame becomes clear in the following. We can write the deviation of the relative velocity of the incident particle as a function of the deflection angle and perform the calculation in this frame, then change reference frames using equation (9.6) as follows:

$$\Delta \mathbf{v}_\mathrm{rel} = v_\mathrm{rel}(\cos\theta - 1)\,\hat{\mathbf{z}} + v_\mathrm{rel}\sin\theta\,\hat{\mathbf{x}} \tag{9.9}$$

The core idea of the Fokker-Planck collision theory is to define a conditional probability density, $F_\sigma(\mathbf{v}_\sigma, \Delta\mathbf{v}_\sigma)$, which represents the probability that, knowing a particle of species σ moves at velocity $\mathbf{v}_\sigma$ at instant t, it moves at velocity $\mathbf{v}_\sigma + \Delta\mathbf{v}_\sigma$ at instant $t + \Delta t$. This probability density is necessarily normalizable with respect to the velocity deviation:

$$\iiint_{\mathbb{R}^3} F_\sigma(\mathbf{v}_\sigma, \Delta\mathbf{v}_\sigma)\,\mathrm{d}^3\Delta\mathbf{v}_\sigma = 1 \tag{9.10}$$

The objective of defining such a probability density becomes evident when considering the expression of the distribution function at instant t as a function of its expression at the previous instant, $t - \Delta t$:

$$f_\sigma(\mathbf{v}_\sigma, t) = \iiint_{\mathbb{R}^3} f_\sigma(\mathbf{v}_\sigma - \Delta\mathbf{v}_\sigma, t - \Delta t)F_\sigma(\mathbf{v}_\sigma - \Delta\mathbf{v}_\sigma, \Delta\mathbf{v}_\sigma)\,\mathrm{d}^3\Delta\mathbf{v}_\sigma \tag{9.11}$$

We perform the Taylor expansion of the integrand in the previous equation:
By employing the normalization of the probability density F_σ, the following differential equation arises:
It is worth remarking on the appearance of the first and second moments of the probability function:

$$\left\langle \frac{\Delta\mathbf{v}_\sigma}{\Delta t}\right\rangle = \frac{1}{\Delta t}\iiint_{\mathbb{R}^3} \Delta\mathbf{v}_\sigma F_\sigma(\mathbf{v}_\sigma, \Delta\mathbf{v}_\sigma)\,\mathrm{d}^3\Delta\mathbf{v}_\sigma \tag{9.12}$$

$$\left\langle \frac{\Delta\mathbf{v}_\sigma\Delta\mathbf{v}_\sigma}{\Delta t}\right\rangle = \frac{1}{\Delta t}\iiint_{\mathbb{R}^3} \Delta\mathbf{v}_\sigma\Delta\mathbf{v}_\sigma F_\sigma(\mathbf{v}_\sigma, \Delta\mathbf{v}_\sigma)\,\mathrm{d}^3\Delta\mathbf{v}_\sigma \tag{9.13}$$

The first term on the right side of the previous equation represents the slowing down of incident particles, while the second represents their spread in phase space due to diffusion. Our goal is now to evaluate these terms. Considerations explored in the previous chapter indicate the dominance of small-angle deflections, hence the motivation for the following approximation:

$$\theta \simeq \frac{q_\sigma q_\rho}{2\pi\varepsilon_0 \mu_{\sigma\rho}\mathbf{v}_\mathrm{rel}^2 b} \implies \sin\theta \simeq \theta, \cos\theta \simeq 1 - \frac{\theta^2}{2} \tag{9.14}$$

$$\Delta\mathbf{v}_\mathrm{rel} \simeq -\frac{v_\mathrm{rel}\theta^2}{2}\,\hat{\mathbf{z}} + v_\mathrm{rel}\theta\,(\cos\varphi\,\hat{\mathbf{x}} + \sin\varphi\,\hat{\mathbf{y}}) \tag{9.15}$$

where φ represents the angle between the axes $\hat{\mathbf{b}}$ and $\hat{\mathbf{x}}$. It remains to determine the mean velocity deviation taken over all possible values of b and φ, that is, over $b \in [b_{\pi/2}, \lambda_D]$ and $\varphi \in [0, 2\pi)$. The cross-section associated with impact parameter b and azimuthal angle range is $b\,\mathrm{d}b\,\mathrm{d}\varphi$. In a time span Δt, the incident particle travels distance $v_\sigma\,\Delta t$, so the effective volume swept by this particle is $b\,\mathrm{d}b\,\mathrm{d}\varphi\,v_\sigma\,\Delta t$. The number of target particles having a velocity $\mathbf{v}_\rho$ is $f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho b\,\mathrm{d}b\,\mathrm{d}\varphi\,v_\sigma\,\Delta t$, so the mean relative velocity deviation of the incident particle is determined by integrating over the ranges of impact parameter and azimuthal angle to determine the mean velocity deviation:

$$\Delta\mathbf{v}_\mathrm{rel}\Big|_{\text{all } b \text{ and } \varphi} = v_\sigma\Delta t f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho\int_0^{2\pi}\mathrm{d}\varphi\int_{b_{\pi/2}}^{\lambda_D}\Delta\mathbf{v}_\sigma\,b\,\mathrm{d}b \tag{9.16}$$

Integrals with respect to axes $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ evaluate to zero due to symmetry:

$$\Delta\mathbf{v}_{\mathrm{rel}}\Big|_{\text{all } b \text{ and } \varphi} = -\mathbf{v}_{\mathrm{rel}}^2 \Delta t f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \pi \int_{b_{\pi/2}}^{\lambda_D} \theta^2 b\,\mathrm{d}b = -\frac{\Delta t}{4\pi} f_\rho(\mathbf{v}_\rho)\,\mathrm{d}\mathbf{v}_\rho \left(\frac{q_\sigma q_\rho}{\varepsilon_0 \mu_{\sigma\rho} \|\mathbf{v}_{\mathrm{rel}}\|}\right)^2 \ln\frac{\lambda_D}{b_{\pi/2}} \tag{9.17}$$

In view of equations (**??**) and (**??**), it is possible to express the mean velocity deviation undergone by the incident particle in the laboratory frame:

$$\Delta\mathbf{v}_\sigma\Big|_{\text{all } b \text{ and } \varphi} = -\Delta t \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \ln\frac{\lambda_D}{b_{\pi/2}} \frac{\mathbf{v}_\sigma - \mathbf{v}_\rho}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|^3} f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.18}$$

By taking the average over all possible velocities of the target particle, it follows:

$$\Delta\mathbf{v}_\sigma\Big|_{\text{all } b,\ \varphi \text{ and } \mathbf{v}_\rho} = -\Delta t \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \ln\frac{\lambda_D}{b_{\pi/2}} \iiint_{\mathbb{R}^3} \frac{\mathbf{v}_\sigma - \mathbf{v}_\rho}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|^3} f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.19}$$

The integral found appears difficult to evaluate; however, the following equivalences come to our aid:

$$\frac{\partial}{\partial\mathbf{v}_\sigma} = \frac{\partial}{\partial\mathbf{v}_{\mathrm{rel}}}\frac{\partial\mathbf{v}_{\mathrm{rel}}}{\partial\mathbf{v}_\sigma} = \frac{\partial}{\partial\mathbf{v}_{\mathrm{rel}}} \quad \text{and} \quad \frac{\partial\|\mathbf{v}\|}{\partial\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \tag{9.20}$$

$$\frac{\partial\|\mathbf{v}_{\mathrm{rel}}\|}{\partial\mathbf{v}_\sigma} = \frac{\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|} \quad \text{and} \quad \frac{\partial}{\partial\mathbf{v}_\sigma}\left(\frac{1}{\|\mathbf{v}_{\mathrm{rel}}\|}\right) = -\frac{\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|^3} = -\frac{\mathbf{v}_\sigma - \mathbf{v}_\rho}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|^3} \tag{9.21}$$

$$\Delta\mathbf{v}_\sigma\Big|_{\text{all } b,\ \varphi \text{ and } \mathbf{v}_\rho} = \Delta t \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \ln\frac{\lambda_D}{b_{\pi/2}} \frac{\partial}{\partial\mathbf{v}_\sigma} \iiint_{\mathbb{R}^3} \frac{f_\rho(\mathbf{v}_\rho)}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|}\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.22}$$

We have thus determined the first term associated with the slowing down of incident particles:

$$\left\langle\frac{\Delta\mathbf{v}_\sigma}{\Delta t}\right\rangle = \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \ln\frac{\lambda_D}{b_{\pi/2}} \frac{\partial}{\partial\mathbf{v}_\sigma} \iiint_{\mathbb{R}^3} \frac{f_\rho(\mathbf{v}_\rho)}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|}\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.23}$$

The procedure regarding the second term is analogous, so we start with the following:

$$\Delta\mathbf{v}_{\mathrm{rel}}\Delta\mathbf{v}_{\mathrm{rel}} = \begin{pmatrix} v_{\mathrm{rel}}\theta\cos\varphi \\ v_{\mathrm{rel}}\theta\sin\varphi \\ -v_{\mathrm{rel}}\theta^2/2 \end{pmatrix} \begin{pmatrix} v_{\mathrm{rel}}\theta\cos\varphi \\ v_{\mathrm{rel}}\theta\sin\varphi \\ -v_{\mathrm{rel}}\theta^2/2 \end{pmatrix}^\top \tag{9.24}$$

The mixed terms all vanish to zero, due to the orthogonality of trigonometric functions. The component along $\hat{\mathbf{z}}\hat{\mathbf{z}}$ is negligible, due to its quartic dependence on angle θ compared to the quadratic dependencies of the components along $\hat{\mathbf{x}}\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}\hat{\mathbf{y}}$:

$$\Delta\mathbf{v}_{\mathrm{rel}}\Delta\mathbf{v}_{\mathrm{rel}}\Big|_{\text{all } b \text{ and } \varphi} = \Delta t \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 \mu_{\sigma\rho}^2 \|\mathbf{v}_{\mathrm{rel}}\|} f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho\,(\hat{\mathbf{x}}\hat{\mathbf{x}} + \hat{\mathbf{y}}\hat{\mathbf{y}}) \tag{9.25}$$

It is worth noting the fact that the $\hat{\mathbf{z}}$ axis is precisely defined by the relative velocity, $\mathbf{v}_{\mathrm{rel}}$:

$$\hat{\mathbf{x}}\hat{\mathbf{x}} + \hat{\mathbf{y}}\hat{\mathbf{y}} = \mathbf{1} - \frac{\mathbf{v}_{\mathrm{rel}}\mathbf{v}_{\mathrm{rel}}}{\mathbf{v}_{\mathrm{rel}}^2} \implies \Delta\mathbf{v}_{\mathrm{rel}}\Delta\mathbf{v}_{\mathrm{rel}}\Big|_{\text{all } b \text{ and } \varphi} = \frac{q_\sigma^2 q_\rho^2 \Delta t f_\rho(\mathbf{v}_\rho)}{4\pi\varepsilon_0^2 \mu_{\sigma\rho}^2} \frac{\mathbf{v}_{\mathrm{rel}}^2\mathbf{1} - \mathbf{v}_{\mathrm{rel}}\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|^3}\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.26}$$

where $\mathbf{1}$ represents the unit tensor. The velocity deviation of the incident particle in the laboratory frame is found using equation (**??**):

$$\Delta\mathbf{v}_\sigma\Delta\mathbf{v}_\sigma\Big|_{\text{all } b \text{ and } \varphi} = \Delta t \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 m_\sigma^2} \frac{\mathbf{v}_{\mathrm{rel}}^2\mathbf{1} - \mathbf{v}_{\mathrm{rel}}\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|^3} f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.27}$$

$$\frac{\partial^2\|\mathbf{v}_{\mathrm{rel}}\|}{\partial\mathbf{v}_\sigma\partial\mathbf{v}_\sigma} = \frac{\partial}{\partial\mathbf{v}_\sigma}\left(\frac{\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|}\right) = \frac{\mathbf{v}_{\mathrm{rel}}^2\mathbf{1} - \mathbf{v}_{\mathrm{rel}}\mathbf{v}_{\mathrm{rel}}}{\|\mathbf{v}_{\mathrm{rel}}\|^3} \tag{9.28}$$

from which we find the second term associated with the diffusion of incident particles:

$$\left\langle\frac{\Delta\mathbf{v}_\sigma\Delta\mathbf{v}_\sigma}{\Delta t}\right\rangle = \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 m_\sigma^2} \ln\frac{\lambda_D}{b_{\pi/2}} \frac{\partial^2}{\partial\mathbf{v}_\sigma\partial\mathbf{v}_\sigma} \iiint_{\mathbb{R}^3} \|\mathbf{v}_\sigma - \mathbf{v}_\rho\| f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.29}$$

It is now appropriate to introduce the Rosenbluth potentials between species σ and ρ:

$$g_\rho(\mathbf{v}_\sigma) = \iiint_{\mathbb{R}^3} \|\mathbf{v}_\sigma - \mathbf{v}_\rho\| f_\rho(\mathbf{v}_\rho)\,\mathrm{d}^3\mathbf{v}_\rho \quad \text{and} \quad h_\rho(\mathbf{v}_\sigma) = \frac{m_\sigma}{\mu_{\sigma\rho}} \iiint_{\mathbb{R}^3} \frac{f_\rho(\mathbf{v}_\rho)}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|}\,\mathrm{d}^3\mathbf{v}_\rho \tag{9.30}$$

This allows writing the two terms in easily graspable forms:

$$\left\langle \frac{\Delta\mathbf{v}_\sigma}{\Delta t} \right\rangle = \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 m_\sigma^2} \ln \frac{\lambda_D}{b_{\pi/2}} \frac{\partial h_\rho}{\partial \mathbf{v}_\sigma} \quad \text{and} \quad \left\langle \frac{\Delta\mathbf{v}_\sigma \Delta\mathbf{v}_\sigma}{\Delta t} \right\rangle = \frac{q_\sigma^2 q_\rho^2}{4\pi\varepsilon_0^2 m_\sigma^2} \ln \frac{\lambda_D}{b_{\pi/2}} \frac{\partial^2 g_\rho}{\partial \mathbf{v}_\sigma \partial \mathbf{v}_\sigma} \tag{9.31}$$

By substituting the previous results into equation (**??**), the Fokker-Planck equation follows:

9.1.2 Dissipation of momentum

It is appropriate to apply the theory developed above to plasma dynamics models. As will be seen in the third chapter, the average velocity of particles of a species, here σ, is defined by:

$$\mathbf{u}_\sigma = \frac{1}{n_\sigma} \iiint_{\mathbb{R}^3} \mathbf{v}_\sigma f_\sigma(\mathbf{v}_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma \quad \text{where} \quad n_\sigma = \iiint_{\mathbb{R}^3} f_\sigma(\mathbf{v}_\sigma)\,\mathrm{d}^3\mathbf{v}_\sigma \tag{9.32}$$

It is evident that the time derivative of the average velocity is the first moment of equation (**??**). Integration by parts of the left side of this equation yields the following results:

$$-\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma \frac{\partial}{\partial \mathbf{v}_\sigma} \cdot \left(f_\sigma \frac{\partial h_\rho}{\partial \mathbf{v}_\sigma} \right) \mathrm{d}^3\mathbf{v}_\sigma = \iiint_{\mathbb{R}^3} f_\sigma \frac{\partial h_\rho}{\partial \mathbf{v}_\sigma}\,\mathrm{d}^3\mathbf{v}_\sigma \tag{9.33}$$

$$= -\oiint_{\partial\mathbb{R}^3} f_\sigma \frac{\partial^2 g_\sigma}{\partial \mathbf{v}_\sigma \partial \mathbf{v}_\sigma} \cdot \mathrm{d}\boldsymbol{\sigma}_{\mathbf{v}_\sigma} = \mathbf{0} \quad \text{since} \quad \lim_{\|\mathbf{v}_\sigma\| \to \infty} f_\sigma(\mathbf{v}_\sigma) = 0 \tag{9.34}$$

We thus obtain an expression for the time derivative of the macroscopic velocity of species σ:

$$n_\sigma \frac{\partial \mathbf{u}_\sigma}{\partial t} = \sum_\rho \frac{q_\sigma^2 q_\rho^2 \ln\Lambda}{4\pi\varepsilon_0^2 m_\sigma^2} \iiint_{\mathbb{R}^3} f_\sigma(\mathbf{v}_\sigma) \frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \mathbf{v}_\sigma}\,\mathrm{d}^3\mathbf{v}_\sigma \tag{9.35}$$

We define the slowing down frequency between species σ and ρ via the following relation:

$$n_\sigma \frac{\partial \mathbf{u}_\sigma}{\partial t} = -\sum_\rho n_\sigma \nu_{\sigma\to\rho} \mathbf{u}_\sigma \implies \nu_{\sigma\to\rho}\mathbf{u}_\sigma = -\frac{q_\sigma^2 q_\rho^2 \ln\Lambda}{4\pi\varepsilon_0^2 n_\sigma m_\sigma^2} \iiint_{\mathbb{R}^3} f_\sigma(\mathbf{v}_\sigma) \frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \mathbf{v}_\sigma}\,\mathrm{d}^3\mathbf{v}_\sigma \tag{9.36}$$

Consider the case of Maxwellian distributions. Since the slowing down frequency is independent of the chosen inertial reference frame, we shall move into the frame of target particles of species ρ, such that, in the most general case, the incident particle distribution function is a shifted Maxwellian about the macroscopic velocity $\mathbf{u}_\sigma$. The Rosenbluth potential $h_\rho(\mathbf{v}_\sigma)$ of the target species ρ is given by:

$$h_\rho(\mathbf{v}_\sigma) = \frac{n_\rho m_\sigma}{\mu_{\sigma\rho}} \left(\frac{m_\rho}{2\pi k_\mathrm{B} T_\rho} \right)^{3/2} \iiint_{\mathbb{R}^3} \exp\left[-\frac{m_\rho \mathbf{v}_\rho^2}{2k_\mathrm{B}T_\rho} \right] \frac{\mathrm{d}^3\mathbf{v}_\rho}{\|\mathbf{v}_\sigma - \mathbf{v}_\rho\|} = \frac{n_\rho m_\sigma}{\mu_{\sigma\rho}} \left(\frac{m_\rho}{2\pi k_\mathrm{B} T_\rho} \right)^{3/2} I(\mathbf{v}_\sigma) \tag{9.37}$$

where, implementing the substitution $\mathbf{v}_\rho = \mathbf{v}_\sigma + \overline{\mathbf{v}}_\rho$, the integral $I(\mathbf{v}_\sigma)$ reads:

$$I(\mathbf{v}_\sigma) = \iiint_{\mathbb{R}^3} \exp\left[-\frac{m_\rho(\mathbf{v}_\sigma + \overline{\mathbf{v}}_\rho)^2}{2k_\mathrm{B}T_\rho} \right] \frac{\mathrm{d}^3\overline{\mathbf{v}}_\rho}{\|\overline{\mathbf{v}}_\rho\|} = \iiint_{\mathbb{R}^3} \exp\left[-\frac{m_\rho}{2k_\mathrm{B}T_\rho} \left(\mathbf{v}_\sigma^2 + 2\mathbf{v}_\sigma \cdot \overline{\mathbf{v}}_\rho + \overline{\mathbf{v}}_\rho^2 \right) \right] \frac{\mathrm{d}^3\overline{\mathbf{v}}_\rho}{\|\overline{\mathbf{v}}_\rho\|} \tag{9.38}$$

It is only appropriate to choose a spherical coordinate system such that the polar angle $\theta = 0$ corresponds to the positive direction along $\mathbf{v}_\sigma$, giving:

$$I(\mathbf{v}_\sigma) = \int_0^{2\pi} \int_0^\infty \overline{v}_\rho \exp\left[-\frac{m_\rho}{2k_\mathrm{B}T_\rho}(v_\sigma^2 + \overline{v}_\rho^2) \right] \int_0^\pi \exp\left[-\frac{m_\rho v_\sigma \overline{v}_\rho}{k_\mathrm{B}T_\rho} \cos\theta \right] \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\overline{v}_\rho \tag{9.39}$$

Notice that the inner integral over the polar angle evaluates to:

$$\frac{k_\mathrm{B}T_\rho}{m_\rho v_\sigma \overline{v}_\rho} \int_0^\pi \mathrm{d}\left(\exp\left[-\frac{m_\rho v_\sigma \overline{v}_\rho}{k_\mathrm{B}T_\rho} \cos\theta \right] \right) = \frac{2k_\mathrm{B}T_\rho}{m_\rho v_\sigma \overline{v}_\rho} \sinh \frac{m_\rho v_\sigma \overline{v}_\rho}{k_\mathrm{B}T_\rho} \tag{9.40}$$

$$I(\mathbf{v}_\sigma) = \frac{2\pi k_\mathrm{B} T_\rho}{m_\rho v_\sigma} \int_0^\infty \left(\exp\left[-\frac{m_\rho(\overline{v}_\rho - v_\sigma)^2}{2k_\mathrm{B}T_\rho} \right] - \exp\left[-\frac{m_\rho(\overline{v}_\rho + v_\sigma)^2}{2k_\mathrm{B}T_\rho} \right] \right) \mathrm{d}\overline{v}_\rho \tag{9.41}$$

Proceed with caution. Let $\xi = (\overline{v}_\rho - v_\sigma)\sqrt{m_\rho/2k_\mathrm{B}T_\rho}$ and $\zeta = (\overline{v}_\rho + v_\sigma)\sqrt{m_\rho/2k_\mathrm{B}T_\rho}$, so that:

$$I(\mathbf{v}_\sigma) = \frac{2\pi k_\mathrm{B} T_\rho}{m_\rho v_\sigma} \sqrt{\frac{2k_\mathrm{B}T_\rho}{m_\rho}} \left(\int_{-v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}}^\infty \exp\left[-\xi^2\right] \mathrm{d}\xi - \int_{v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}}^\infty \exp\left[-\zeta^2\right] \mathrm{d}\zeta \right) = \tag{9.42}$$

$$= \frac{2\pi k_\mathrm{B} T_\rho}{m_\rho v_\sigma} \sqrt{\frac{2k_\mathrm{B}T_\rho}{m_\rho}} \int_{-v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}}^{v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}} \exp\left[-\xi^2\right] \mathrm{d}\xi = \frac{2\pi}{v_\sigma} \left(\frac{2k_\mathrm{B}T_\rho}{m_\rho} \right)^{3/2} \frac{\sqrt{\pi}}{2} \frac{2}{\sqrt{\pi}} \int_0^{v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}} \exp\left[-\xi^2\right] \mathrm{d}\xi = \tag{9.43}$$

$$= \frac{2\pi k_\mathrm{B} T_\rho}{m_\rho v_\sigma} \sqrt{\frac{2\pi k_\mathrm{B}T_\rho}{m_\rho}} \mathrm{erf} \left(v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) = \frac{1}{\|\mathbf{v}_\sigma\|} \left(\frac{2\pi k_\mathrm{B}T_\rho}{m_\rho} \right)^{3/2} \mathrm{erf} \left(\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) \tag{9.44}$$

Thus, we find the Rosenbluth potential of species ρ:

$$h_\rho(\mathbf{v}_\sigma) = \frac{n_\rho m_\sigma}{\mu_{\sigma\rho}\|\mathbf{v}_\sigma\|} \mathrm{erf} \left(\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) \tag{9.45}$$

Its velocity gradient with respect to $\mathbf{v}_\sigma$ can be found as follows:

$$\frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \mathbf{v}_\sigma} = \frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \|\mathbf{v}_\sigma\|} \frac{\partial \|\mathbf{v}_\sigma\|}{\partial \mathbf{v}_\sigma} = \frac{\mathbf{v}_\sigma}{\|\mathbf{v}_\sigma\|} \frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \|\mathbf{v}_\sigma\|} \tag{9.46}$$

$$\frac{\partial h_\rho(\mathbf{v}_\sigma)}{\partial \mathbf{v}_\sigma} = \frac{n_\rho m_\sigma \mathbf{v}_\sigma}{\mu_{\sigma\rho}\|\mathbf{v}_\sigma\|^2} \sqrt{\frac{2m_\rho}{\pi k_\mathrm{B}T_\rho}} \exp\left[-\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho} \right] - \frac{n_\rho m_\sigma \mathbf{v}_\sigma}{\mu_{\sigma\rho}\|\mathbf{v}_\sigma\|^3} \mathrm{erf} \left(\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) \tag{9.47}$$

We now only have to average the previous expression over $\mathbf{v}_\sigma$. However, this proves to be a difficult task. We instead resort to partial integration, since at infinity, the Maxwellian term ensures that the integrand evaluates to zero:

$$\mathbf{K}(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \frac{\mathbf{v}_\sigma - \mathbf{u}_\sigma}{\|\mathbf{v}_\sigma\|} \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B}T_\sigma} \right] \mathrm{erf} \left(\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) \mathrm{d}^3\mathbf{v}_\sigma \tag{9.48}$$

such that we shall at last find:

$$\nu_{\sigma\to\rho}\mathbf{u}_\sigma = -\frac{n_\rho q_\sigma^2 q_\rho^2 \ln\Lambda}{4\pi\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \frac{m_\sigma}{k_\mathrm{B}T_\sigma} \left(\frac{m_\sigma}{2\pi k_\mathrm{B}T_\sigma} \right)^{3/2} \mathbf{K}(\mathbf{u}_\sigma) \tag{9.49}$$

Notice that the components perpendicular to $\mathbf{u}_\sigma$ in $\mathbf{K}(\mathbf{u}_\sigma)$ evaluate to zero due to axial symmetry. We shall henceforth endeavour to evaluate the following:

$$\mathbf{K}(\mathbf{u}_\sigma) = 2\pi\, \hat{\mathbf{u}}_\sigma \int_0^\infty v_\sigma \exp\left[-\frac{m_\sigma(v_\sigma^2 + u_\sigma^2)}{2k_\mathrm{B}T_\sigma} \right] \mathrm{erf} \left(v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}} \right) I(\mathbf{u}_\sigma, \mathbf{v}_\sigma)\, \mathrm{d}v_\sigma \tag{9.50}$$

where the integral with respect to the polar angle reads:

$$I(\mathbf{u}_\sigma, \mathbf{v}_\sigma) = \int_0^\pi (v_\sigma \cos\theta - u_\sigma) \exp\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma} \cos\theta \right] \sin\theta\, \mathrm{d}\theta \tag{9.51}$$

whose solution is found using trigonometric substitution:

$$I(\mathbf{u}_\sigma, \mathbf{v}_\sigma) = \frac{2k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma} \cosh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma} \right] - \frac{2k_\mathrm{B}T_\sigma}{m_\sigma v_\sigma} \left(\frac{k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma^2} + 1 \right) \sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma} \right] \tag{9.52}$$

Notice that the integrand is a pair function of v_σ, and there are no issues with convergence since the integral converges absolutely. We shall henceforth separate $\mathbf{K}(\mathbf{u}_\sigma) = [K_1(\mathbf{u}_\sigma) - K_2(\mathbf{u}_\sigma)]\, \hat{\mathbf{u}}_\sigma$ to pragmatically evaluate the integral:

$$K_1(\mathbf{u}_\sigma) = \frac{4\pi k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma} \int_0^\infty v_\sigma \exp\left[-\frac{m_\sigma(v_\sigma^2 + u_\sigma^2)}{2k_\mathrm{B}T_\sigma} \right] \cosh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma} \right] \mathrm{erf} \left(\sqrt{\frac{m_\rho v_\sigma^2}{2k_\mathrm{B}T_\rho}} \right) \mathrm{d}v_\sigma \tag{9.53}$$

$$K_2(\mathbf{u}_\sigma) = \frac{4\pi k_\mathrm{B}T_\sigma}{m_\sigma}\left(\frac{k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma^2}+1\right)\int_0^\infty \exp\left[-\frac{m_\sigma(v_\sigma^2+u_\sigma^2)}{2k_\mathrm{B}T_\sigma}\right]\sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(\sqrt{\frac{m_\rho v_\sigma^2}{2k_\mathrm{B}T_\rho}}\right)\mathrm{d}v_\sigma \tag{9.54}$$

Let us extract the constant factors in the previous integrals:

$$K_1(\mathbf{u}_\sigma) = \frac{4\pi k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma}\exp\left[-\frac{m_\sigma u_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\overline{K}_1(u_\sigma) \tag{9.55}$$

$$\overline{K}_1(u_\sigma) = \int_0^\infty v_\sigma \exp\left[-\frac{m_\sigma v_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\cosh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(v_\sigma\sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}\right)\mathrm{d}v_\sigma \tag{9.56}$$

$$K_2(\mathbf{u}_\sigma) = \frac{4\pi k_\mathrm{B}T_\sigma}{m_\sigma}\left(1+\frac{k_\mathrm{B}T_\sigma}{m_\sigma u_\sigma^2}\right)\exp\left[-\frac{m_\sigma u_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\overline{K}_2(u_\sigma) \tag{9.57}$$

$$\overline{K}_2(u_\sigma) = \int_0^\infty \exp\left[-\frac{m_\sigma v_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(v_\sigma\sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}\right)\mathrm{d}v_\sigma = \tag{9.58}$$

$$= \frac{1}{2}\int_{-\infty}^\infty \exp\left[-\frac{m_\sigma v_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(v_\sigma\sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}\right)\mathrm{d}v_\sigma = \tag{9.59}$$

$$= \frac{1}{2}\int_{-\infty}^\infty \exp\left[-\frac{m_\sigma v_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\exp\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(v_\sigma\sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}\right)\mathrm{d}v_\sigma \tag{9.60}$$

Notice that the second integral, $\overline{K}_2$, generates the first upon differentiating under the integral sign with respect to the parameter inside the hyperbolic sine term:

$$\frac{\partial\overline{K}_2}{\partial u_\sigma} = \frac{m_\sigma}{k_\mathrm{B}T_\sigma}\overline{K}_1(u_\sigma) \implies \overline{K}_1(u_\sigma) = \frac{k_\mathrm{B}T_\sigma}{m_\sigma}\frac{\partial\overline{K}_2}{\partial u_\sigma} \tag{9.61}$$

Notice that defining $v_\sigma = 2u_\sigma x$ gives:

$$\overline{K}_2(u_\sigma) = u_\sigma\int_{-\infty}^\infty \exp\left[-\frac{2m_\sigma u_\sigma^2}{k_\mathrm{B}T_\sigma}(x^2-x)\right]\mathrm{erf}\left(x\sqrt{\frac{2m_\rho u_\sigma^2}{k_\mathrm{B}T_\rho}}\right)\mathrm{d}x \tag{9.62}$$

where we shall define $a = 2m_\sigma u_\sigma^2/k_\mathrm{B}T_\sigma$ and $b = \sqrt{2m_\rho u_\sigma^2/k_\mathrm{B}T_\rho}$, so that $\overline{K}_2(u_\sigma) = u_\sigma J(a,b)$ where:

$$J(a,b) = \int_{-\infty}^\infty \exp\left[-a(x^2-x)\right]\mathrm{erf}\,(bx)\,\mathrm{d}x = \exp\left[\frac{a}{4}\right]\int_{-\infty}^\infty \exp\left[-a\left(x-\frac{1}{2}\right)^2\right]\mathrm{erf}\,(bx)\,\mathrm{d}x \tag{9.63}$$

Performing the shift $t = x - \frac{1}{2}$, we obtain:

$$J(a,b) = \exp\left[\frac{a}{4}\right]\int_{-\infty}^\infty \exp\left(-at^2\right)\mathrm{erf}\left(bt+\frac{b}{2}\right)\mathrm{d}t \tag{9.64}$$

We shall define the following integral:

$$I(a,b,c) = \int_{-\infty}^\infty \exp\left(-at^2\right)\mathrm{erf}\,(bt+c)\,\mathrm{d}t \tag{9.65}$$

so that the integral $J(a,b) = \exp\,(a/4)\,I(a,b,b/2)$. We perform differentiation under the integral sign with respect to the parameter c:

$$\frac{\partial I(a,b,c)}{\partial c} = \int_{-\infty}^\infty \exp\left(-at^2\right)\frac{2}{\sqrt{\pi}}\exp\left[-(bt+c)^2\right]\mathrm{d}t = \tag{9.66}$$

$$= \frac{2}{\sqrt{\pi}}\int_{-\infty}^\infty \exp\left[-(a+b^2)t^2-2bct-c^2\right]\mathrm{d}t = \tag{9.67}$$

$$\frac{2}{\sqrt{\pi}}\int_{-\infty}^\infty \exp\left[-(a+b^2)\left(t+\frac{bc}{a+b^2}\right)^2-\frac{ac^2}{a+b^2}\right]\mathrm{d}t = \tag{9.68}$$

$$\frac{2}{\sqrt{\pi}}\exp\left[-\frac{ac^2}{a+b^2}\right]\int_{-\infty}^{\infty}\exp\left[-(a+b^2)u^2\right]\mathrm{d}u=\frac{2}{\sqrt{a+b^2}}\exp\left[-\frac{ac^2}{a+b^2}\right] \tag{9.69}$$

We shall now integrate the result with respect to the parameter:

$$I(a,b,c)=\int_0^c\frac{\partial I(a,b,c)}{\partial c}\,\mathrm{d}c=\sqrt{\frac{\pi}{a}}\,\mathrm{erf}\left(c\sqrt{\frac{a}{a+b^2}}\right) \tag{9.70}$$

where the lower bound is taken to be $c=0$ since the error function is odd, giving $I(a,b,0)=0$. Finally, setting $c=b/2$ and reverting all substitutions one by one, we obtain:

$$J(a,b)=\exp\left[\frac{m_\sigma u_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\sqrt{\frac{\pi k_\mathrm{B}T_\sigma}{2m_\sigma u_\sigma^2}}\,\mathrm{erf}\left(u_\sigma\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right) \tag{9.71}$$

$$\overline{K}_2(u_\sigma)=u_\sigma J(a,b)=\exp\left[\frac{m_\sigma u_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\sqrt{\frac{\pi k_\mathrm{B}T_\sigma}{2m_\sigma}}\,\mathrm{erf}\left(\frac{u_\sigma}{\sqrt{2k_\mathrm{B}T_\sigma/m_\sigma+2k_\mathrm{B}T_\rho/m_\rho}}\right) \tag{9.72}$$

$$\overline{K}_1(u_\sigma)=\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[\frac{m_\sigma\mathbf{u}_\sigma^2}{2k_\mathrm{B}T_\sigma}\Big/\left(1+\frac{T_\sigma m_\rho}{T_\rho m_\sigma}\right)\right]+ \tag{9.73}$$

$$+u_\sigma\sqrt{\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}}\exp\left[\frac{m_\sigma\mathbf{u}_\sigma^2}{2k_\mathrm{B}T_\sigma}\right]\mathrm{erf}\left(\frac{u_\sigma}{\sqrt{2k_\mathrm{B}(T_\sigma/m_\sigma+T_\rho/m_\rho)}}\right) \tag{9.74}$$

We may now express the integrals $K_1(\mathbf{u}_\sigma)$ and $K_2(\mathbf{u}_\sigma)$:

$$K_1(\mathbf{u}_\sigma)=\frac{1}{\|\mathbf{u}_\sigma\|\sqrt{\pi}}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[-\frac{m_\sigma\mathbf{u}_\sigma^2}{2k_\mathrm{B}T_\sigma}\Big/\left(1+\frac{m_\sigma T_\rho}{m_\rho T_\sigma}\right)\right]\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\Big/\left(1+\frac{m_\sigma T_\rho}{m_\rho T_\sigma}\right)}+ \tag{9.75}$$

$$+\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\mathrm{erf}\left(u_\sigma\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right) \tag{9.76}$$

$$K_2(\mathbf{u}_\sigma)=\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\left(1+\frac{k_\mathrm{B}T_\sigma}{m_\sigma\mathbf{u}_\sigma^2}\right)\mathrm{erf}\left(\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right) \tag{9.77}$$

The integral $\mathbf{K}(\mathbf{u}_\sigma)$ was defined as follows:

$$\mathbf{K}(\mathbf{u}_\sigma)=\frac{\mathbf{u}_\sigma}{\mathbf{u}_\sigma^2\sqrt{\pi}}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[-\frac{m_\sigma\mathbf{u}_\sigma^2}{2k_\mathrm{B}T_\sigma}\Big/\left(1+\frac{m_\sigma T_\rho}{m_\rho T_\sigma}\right)\right]\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\Big/\left(1+\frac{m_\sigma T_\rho}{m_\rho T_\sigma}\right)}+ \tag{9.78}$$

$$-\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\frac{k_\mathrm{B}T_\sigma\mathbf{u}_\sigma}{m_\sigma\|\mathbf{u}_\sigma\|^3}\,\mathrm{erf}\left(\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right) \tag{9.79}$$

Thus, the slowing down frequency between species σ and ρ turns out to be:

$$\nu_{\sigma\to\rho}=-\frac{n_\rho q_\sigma^2q_\rho^2\ln\Lambda}{2\pi^{3/2}\varepsilon_0^2\mu_{\sigma\rho}m_\sigma}\frac{1}{\mathbf{u}_\sigma^2}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}+$$

$$+\frac{n_\rho q_\sigma^2q_\rho^2\ln\Lambda}{4\pi\varepsilon_0^2\mu_{\sigma\rho}m_\sigma}\frac{1}{\|\mathbf{u}_\sigma\|^3}\,\mathrm{erf}\left(\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right) \tag{9.80}$$

Notice that, due to our definition of the slowing down frequency, if both species σ and ρ were stationary, that is $\mathbf{u}_\sigma=\mathbf{0}$, we intuitively expect that the rate of momentum loss is identically zero. Our result clearly indicates possible divergent behaviour near $\mathbf{u}_\sigma=\mathbf{0}$. Observe, however, that expanding the expression up to all non-vanishing terms of $\mathbf{u}_\sigma$ results in immediate cancellation of divergent terms, while terms of order $\mathbf{u}_\sigma^2$ and higher vanish by indeed taking the limit:

$$\lim_{\mathbf{u}_\sigma\to\mathbf{0}}\nu_{\sigma\to\rho}=\frac{n_\rho q_\sigma^2q_\rho^2\ln\Lambda}{2\pi^{3/2}\varepsilon_0^2\mu_{\sigma\rho}m_\sigma}\lim_{\mathbf{u}_\sigma\to\mathbf{0}}\left[\left(\frac{1}{\mathbf{u}_\sigma^2}-\frac{1}{\mathbf{u}_\sigma^2}\right)+\mathcal{O}(1)\right] \tag{9.81}$$

In the case of fully stationary species σ and ρ, the slowing down, or equivalently, collision frequency, is simply given by the following expression obtained by evaluating the previous limit:

$$\nu_{\sigma\to\rho} = \frac{n_\rho q_\sigma^2 q_\rho^2 \ln\Lambda}{6\sqrt{2}\pi^{3/2}\varepsilon_0^2 \mu_{\sigma\rho} m_\sigma} \left(\frac{k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{k_\mathrm{B}T_\rho}{m_\rho}\right)^{-3/2} \tag{9.82}$$

Notice that conservation of momentum, however, postulates:

$$n_\sigma m_\sigma \nu_{\sigma\to\rho} = n_\rho m_\rho \nu_{\rho\to\sigma} \tag{9.83}$$

which is indeed the case. We have thus obtained a physically relevant result.

9.1.3 Diffusion of kinetic energy

We begin our considerations of kinetic energy diffusion by first evaluating the first Rosenbluth potential for a stationary Maxwellian distribution of target particles, that is:

$$g_\rho(\mathbf{v}_\sigma) = n_\rho \left(\frac{m_\rho}{2\pi k_\mathrm{B}T_\rho}\right)^{3/2} \iiint_{\mathbb{R}^3} \|\mathbf{v}_\sigma - \mathbf{v}_\rho\| \exp\left[-\frac{m_\rho \mathbf{v}_\rho^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}^3\mathbf{v}_\rho \tag{9.84}$$

Performing the shift $\mathbf{v}_\rho = \mathbf{v}_\sigma + \overline{\mathbf{v}}_\rho$, we find:

$$g_\rho(\mathbf{v}_\sigma) = n_\rho \left(\frac{m_\rho}{2\pi k_\mathrm{B}T_\rho}\right)^{3/2} \iiint_{\mathbb{R}^3} \|\overline{\mathbf{v}}_\rho\| \exp\left[-\frac{m_\rho (\mathbf{v}_\sigma + \overline{\mathbf{v}}_\rho)^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}^3\overline{\mathbf{v}}_\rho = \tag{9.85}$$

$$= 2\pi n_\rho \left(\frac{m_\rho}{2\pi k_\mathrm{B}T_\rho}\right)^{3/2} \exp\left[-\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right] \int_0^\infty \overline{v}_\rho^3 \exp\left[-\frac{m_\rho \overline{v}_\rho^2}{2k_\mathrm{B}T_\rho}\right] \int_0^\pi \exp\left[-\frac{m_\rho v_\sigma \overline{v}_\rho}{k_\mathrm{B}T_\rho}\cos\theta\right] \sin\theta \,\mathrm{d}\theta \,\mathrm{d}\overline{v}_\rho \tag{9.86}$$

where we have evidently chosen a spherical coordinate system with $\theta = 0$ aligning with $\mathbf{v}_\sigma$. We had already encountered the integral over the polar angle:

$$g_\rho(\mathbf{v}_\sigma) = \frac{2n_\rho}{\|\mathbf{v}_\sigma\|}\sqrt{\frac{m_\rho}{2\pi k_\mathrm{B}T_\rho}} \exp\left[-\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right] \int_0^\infty \overline{v}_\rho^2 \exp\left[-\frac{m_\rho \overline{v}_\rho^2}{2k_\mathrm{B}T_\rho}\right] \sinh\left[\frac{m_\rho v_\sigma \overline{v}_\rho}{k_\mathrm{B}T_\rho}\right] \mathrm{d}\overline{v}_\rho = \tag{9.87}$$

$$= \frac{n_\rho}{\|\mathbf{v}_\sigma\|}\sqrt{\frac{m_\rho}{2\pi k_\mathrm{B}T_\rho}} \underbrace{\int_0^\infty \overline{v}_\rho^2 \left(\exp\left[-\frac{m_\rho(\overline{v}_\rho - v_\sigma)^2}{2k_\mathrm{B}T_\rho}\right] - \exp\left[-\frac{m_\rho(\overline{v}_\rho + v_\sigma)^2}{2k_\mathrm{B}T_\rho}\right]\right) \mathrm{d}\overline{v}_\rho}_{I(v_\sigma)} \tag{9.88}$$

Performing the substitutions $\overline{v}_\rho = \xi + v_\sigma$ for the first integral and $\overline{v}_\rho = \xi - v_\sigma$ for the second, we may separate the integral as follows:

$$I(v_\sigma) = 4v_\sigma \int_{v_\sigma}^\infty \xi \exp\left[-\frac{m_\rho \xi^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}\xi + \int_{-v_\sigma}^{v_\sigma} (\xi + v_\sigma)^2 \exp\left[-\frac{m_\rho \xi^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}\xi = 4v_\sigma J(v_\sigma) + K(v_\sigma) \tag{9.89}$$

Notice that the second integral can be expanded as:

$$K(v_\sigma) = \int_{-v_\sigma}^{v_\sigma} \xi^2 \exp\left[-\frac{m_\rho \xi^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}\xi + v_\sigma^2 \int_{-v_\sigma}^{v_\sigma} \exp\left[-\frac{m_\rho \xi^2}{2k_\mathrm{B}T_\rho}\right] \mathrm{d}\xi \tag{9.90}$$

The second integral generates the first upon differentiating with respect to the parameter in the exponent. Imposing $\alpha = m_\rho / 2k_\mathrm{B}T_\rho$:

$$\frac{\partial}{\partial\alpha}\left(\int_{-v_\sigma}^{v_\sigma} \exp\left[-\alpha\xi^2\right] \mathrm{d}\xi\right) = -\int_{-v_\sigma}^{v_\sigma} \xi^2 \exp\left[-\alpha\xi^2\right] \mathrm{d}\xi \tag{9.91}$$

Substituting $\zeta = \xi\sqrt{\alpha}$, we find:

$$\int_{-v_\sigma}^{v_\sigma} \exp\left[-\alpha\xi^2\right] \mathrm{d}\xi = \frac{1}{\sqrt{\alpha}} \int_{-v_\sigma\sqrt{\alpha}}^{v_\sigma\sqrt{\alpha}} \exp\left[-\zeta^2\right] \mathrm{d}\zeta = \sqrt{\frac{\pi}{\alpha}} \,\mathrm{erf}\,(v_\sigma\sqrt{\alpha}) = \sqrt{\frac{2\pi k_\mathrm{B}T_\rho}{m_\rho}} \,\mathrm{erf}\left(\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}}\right) \tag{9.92}$$

$$\int_{-v_\sigma}^{v_\sigma} \xi^2 \exp\left[-\alpha\xi^2\right] \mathrm{d}\xi = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}} \operatorname{erf}\left(v_\sigma\sqrt{\alpha}\right) - \frac{v_\sigma}{\alpha}\exp\left[-\alpha v_\sigma^2\right] \tag{9.93}$$

$$K(v_\sigma) = \left(\frac{k_\mathrm{B}T_\rho}{m_\rho} + \mathbf{v}_\sigma^2\right)\sqrt{\frac{2\pi k_\mathrm{B}T_\rho}{m_\rho}} \operatorname{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}}\right] - \frac{2k_\mathrm{B}T_\rho v_\sigma}{m_\rho}\exp\left[-\frac{m_\rho\mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right] \tag{9.94}$$

Meanwhile, $J(v_\sigma)$ is easily solved by expanding the differential, so that:

$$4v_\sigma J(v_\sigma) = \frac{4k_\mathrm{B}T_\rho v_\sigma}{m_\rho}\exp\left[-\frac{m_\rho\mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right] \tag{9.95}$$

$$I(v_\sigma) = \left(\frac{k_\mathrm{B}T_\rho}{m_\rho} + \mathbf{v}_\sigma^2\right)\sqrt{\frac{2\pi k_\mathrm{B}T_\rho}{m_\rho}} \operatorname{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}}\right] + \frac{2k_\mathrm{B}T_\rho v_\sigma}{m_\rho}\exp\left[-\frac{m_\rho\mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right] \tag{9.96}$$

We have thus obtained the Rosenbluth potential $g_\rho(\mathbf{v}_\sigma)$:

$$g_\rho(\mathbf{v}_\sigma) = n_\rho\left[\left(\frac{k_\mathrm{B}T_\rho}{m_\rho\|\mathbf{v}_\sigma\|} + \|\mathbf{v}_\sigma\|\right)\operatorname{erf}\left[\|\mathbf{v}_\sigma\|\sqrt{\frac{m_\rho}{2k_\mathrm{B}T_\rho}}\right] + \sqrt{\frac{2k_\mathrm{B}T_\rho}{\pi m_\rho}}\exp\left[-\frac{m_\rho\mathbf{v}_\sigma^2}{2k_\mathrm{B}T_\rho}\right]\right] \tag{9.97}$$

It is now only appropriate to use it to determine the diffusion of kinetic energy. We multiply the Fokker-Planck equation given in eq. (**??**) by $\mathbf{v}_\sigma^2/2$, so that:
We now wish to perform the average over the velocity coordinate of species $\mathbf{v}_\sigma$, thus effectively taking the second moment of the Fokker-Planck equation. We shall perform the following manipulations to simplify our expression. Notice that, when partially integrating, the Maxwellian nature of distribution functions ensures that all boundary terms vanish:

$$\iiint_{\mathbb{R}^3} \frac{\mathbf{v}_\sigma^2}{2}\frac{\partial}{\partial\mathbf{v}_\sigma}\cdot\left(f_\sigma\frac{\partial h_\rho}{\partial\mathbf{v}_\sigma}\right)\mathrm{d}^3\mathbf{v}_\sigma = -\iiint_{\mathbb{R}^3} \mathbf{v}_\sigma\cdot f_\sigma\frac{\partial h_\rho}{\partial\mathbf{v}_\sigma}\mathrm{d}^3\mathbf{v}_\sigma = \tag{9.98}$$

$$= -\iiint_{\mathbb{R}^3} f_\sigma\left[\frac{\partial}{\partial\mathbf{v}_\sigma}\cdot(h_\rho\mathbf{v}_\sigma) - 3h_\rho\right]\mathrm{d}^3\mathbf{v}_\sigma = \iiint_{\mathbb{R}^3}\left(3f_\sigma + \mathbf{v}_\sigma\cdot\frac{\partial f_\sigma}{\partial\mathbf{v}_\sigma}\right)h_\rho\,\mathrm{d}^3\mathbf{v}_\sigma \tag{9.99}$$

$$= \frac{1}{2}\iiint_{\mathbb{R}^3} f_\sigma\frac{\partial^2 g_\rho}{\partial\mathbf{v}_\sigma^2}\mathrm{d}^3\mathbf{v}_\sigma = \frac{1}{2}\iiint_{\mathbb{R}^3} g_\rho\frac{\partial^2 f_\sigma}{\partial\mathbf{v}_\sigma^2}\mathrm{d}^3\mathbf{v}_\sigma \tag{9.100}$$

where the second partial derivative in the last two expressions is the Laplacian in velocity coordinates, and the last equality is established using Green's 2nd identity and noticing that boundary terms vanish due to the rapid decay of both f_σ, as a Maxwellian distribution, and g_ρ, since its fastest growing term is linear in the velocity norm. Evaluating the terms related to f_σ, we find:

$$3f_\sigma + \mathbf{v}_\sigma\cdot\frac{\partial f_\sigma}{\partial\mathbf{v}_\sigma} = n_\sigma\left(\frac{m_\sigma}{2\pi k_\mathrm{B}T_\sigma}\right)^{3/2}\left[3 + \frac{m_\sigma\mathbf{u}_\sigma\cdot\mathbf{v}_\sigma}{k_\mathrm{B}T_\sigma} - \frac{m_\sigma\mathbf{v}_\sigma^2}{k_\mathrm{B}T_\sigma}\right]\exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma-\mathbf{u}_\sigma)^2}{2k_\mathrm{B}T_\sigma}\right] \tag{9.101}$$

$$\frac{\partial^2 f_\sigma}{\partial\mathbf{v}_\sigma^2} = \frac{n_\sigma}{(2\pi)^{3/2}}\left(\frac{m_\sigma}{k_\mathrm{B}T_\sigma}\right)^{5/2}\left[\frac{m_\sigma(\mathbf{v}_\sigma-\mathbf{u}_\sigma)^2}{k_\mathrm{B}T_\sigma} - 3\right]\exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma-\mathbf{u}_\sigma)^2}{2k_\mathrm{B}T_\sigma}\right] \tag{9.102}$$

Omitting the sum over target particle species, we shall express the Fokker-Planck equation as follows:

$$\frac{\partial}{\partial t}\iiint_{\mathbb{R}^3}\frac{1}{2}f_\sigma\mathbf{v}_\sigma^2\,\mathrm{d}^3\mathbf{v}_\sigma = \frac{q_\sigma^2 q_\rho^2\ln\Lambda}{4\pi\varepsilon_0^2 m_\sigma^2}\left[-\underbrace{\iiint_{\mathbb{R}^3}\left[3f_\sigma + \mathbf{v}_\sigma\cdot\frac{\partial f_\sigma}{\partial\mathbf{v}_\sigma}\right]h_\rho\,\mathrm{d}^3\mathbf{v}_\sigma}_{H(\mathbf{u}_\sigma)} + \underbrace{\frac{1}{2}\iiint_{\mathbb{R}^3} g_\rho\frac{\partial^2 f_\sigma}{\partial\mathbf{v}_\sigma^2}\mathrm{d}^3\mathbf{v}_\sigma}_{G(\mathbf{u}_\sigma)}\right] \tag{9.103}$$

$$H(\mathbf{u}_\sigma) = \frac{n_\sigma n_\rho m_\sigma}{\mu_{\sigma\rho}}\left(\frac{m_\sigma}{2\pi k_\mathrm{B}T_\sigma}\right)^{3/2}\overline{H}(\mathbf{u}_\sigma) \tag{9.104}$$

$$G(\mathbf{u}_\sigma) = \frac{n_\sigma n_\rho m_\sigma}{2k_\mathrm{B}T_\sigma}\left(\frac{m_\sigma}{2\pi k_\mathrm{B}T_\sigma}\right)^{3/2}\left[\frac{k_\mathrm{B}T_\rho}{m_\rho}\overline{G}_1(\mathbf{u}_\sigma) + \overline{G}_2(\mathbf{u}_\sigma) + \sqrt{\frac{2k_\mathrm{B}T_\rho}{\pi m_\rho}}\overline{G}_3(\mathbf{u}_\sigma)\right] \tag{9.105}$$

Henceforth, we shall evaluate the following integrals:

$$\overline{H}(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \left(3 + \frac{m_\sigma \mathbf{u}_\sigma \cdot \mathbf{v}_\sigma}{k_\mathrm{B} T_\sigma} - \frac{m_\sigma \mathbf{v}_\sigma^2}{k_\mathrm{B} T_\sigma}\right) \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B} T_\rho}}\right] \frac{\mathrm{d}^3\mathbf{v}_\sigma}{\|\mathbf{v}_\sigma\|} \tag{9.106}$$

$$\overline{G}_1(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \left(\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{k_\mathrm{B} T_\sigma} - 3\right) \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B} T_\rho}}\right] \frac{\mathrm{d}^3\mathbf{v}_\sigma}{\|\mathbf{v}_\sigma\|} \tag{9.107}$$

$$\overline{G}_2(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \left(\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{k_\mathrm{B} T_\sigma} - 3\right) \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \|\mathbf{v}_\sigma\| \,\mathrm{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B} T_\rho}}\right] \mathrm{d}^3\mathbf{v}_\sigma \tag{9.108}$$

$$\overline{G}_3(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \left(\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{k_\mathrm{B} T_\sigma} - 3\right) \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \exp\left[-\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B} T_\rho}\right] \mathrm{d}^3\mathbf{v}_\sigma \tag{9.109}$$

Notice that all of these integrals are absolutely convergent due to the Maxwellian terms. Several of them are of the same or equivalent type, so that we shall define the following integral in order to pragmatically evaluate the aforementioned:

$$I(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[\|\mathbf{v}_\sigma\| \sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}}\right] \frac{\mathrm{d}^3\mathbf{v}_\sigma}{\|\mathbf{v}_\sigma\|} = \tag{9.110}$$

$$= 2\pi \int_0^\infty \exp\left[-\frac{m_\sigma(v_\sigma^2 + u_\sigma^2)}{2k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}}\right] J(u_\sigma, v_\sigma)\, v_\sigma \,\mathrm{d}v_\sigma \tag{9.111}$$

$$J(u_\sigma, v_\sigma) = \int_0^\pi \exp\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B} T_\sigma} \cos\theta\right] \sin\theta \,\mathrm{d}\theta = \frac{2k_\mathrm{B} T_\sigma}{m_\sigma u_\sigma v_\sigma} \sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B} T_\sigma}\right] \tag{9.112}$$

$$I(\mathbf{u}_\sigma) = \frac{4\pi k_\mathrm{B} T_\sigma}{m_\sigma u_\sigma} \exp\left[-\frac{m_\sigma \mathbf{u}_\sigma^2}{2k_\mathrm{B} T_\sigma}\right] \underbrace{\int_0^\infty \exp\left[-\frac{m_\sigma v_\sigma^2}{2k_\mathrm{B} T_\sigma}\right] \sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[v_\sigma \sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}}\right] \mathrm{d}v_\sigma}_{\overline{K}_2(u_\sigma)} \tag{9.113}$$

where we recognise having already solved this integral in the previous section, whose solution is given by expression (9.72). We thus find the newly defined integral:

$$I(\mathbf{u}_\sigma) = \frac{1}{\|\mathbf{u}_\sigma\|} \left(\frac{2\pi k_\mathrm{B} T_\sigma}{m_\sigma}\right)^{3/2} \mathrm{erf}\left(\|\mathbf{u}_\sigma\| \Big/ \sqrt{\frac{2k_\mathrm{B} T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B} T_\rho}{m_\rho}}\right) \tag{9.114}$$

Three of four integrals can thus be evaluated as follows:

$$\overline{H}(\mathbf{u}_\sigma) = 3I(\mathbf{u}_\sigma) + \frac{m_\sigma}{k_\mathrm{B} T_\sigma} \frac{\partial I(\mathbf{u}_\sigma)}{\partial(\frac{m_\sigma}{2k_\mathrm{B} T_\sigma})} - \|\mathbf{u}_\sigma\| \frac{\partial I(\mathbf{u}_\sigma)}{\partial(\|\mathbf{u}_\sigma\|)} \tag{9.115}$$

$$\overline{G}_1(\mathbf{u}_\sigma) = -3I(\mathbf{u}_\sigma) - \frac{m_\sigma}{k_\mathrm{B} T_\sigma} \frac{\partial I(\mathbf{u}_\sigma)}{\partial\left(\frac{m_\sigma}{2k_\mathrm{B} T_\sigma}\right)} \tag{9.116}$$

$$\overline{G}_3(\mathbf{u}_\sigma) = \frac{\sqrt{\pi}}{2} \frac{\partial \overline{G}_1(\mathbf{u}_\sigma)}{\partial\left(\sqrt{\frac{m_\rho}{2k_\mathrm{B} T_\rho}}\right)} = \sqrt{\frac{\pi m_\rho}{2k_\mathrm{B} T_\rho}} \frac{\partial \overline{G}_1(\mathbf{u}_\sigma)}{\partial\left(\frac{m_\rho}{2k_\mathrm{B} T_\rho}\right)} \tag{9.117}$$

It is, at first glance, not exactly evident how to evaluate $\overline{G}_2(\mathbf{u}_\sigma)$. Let us introduce:

$$J(\mathbf{u}_\sigma) = \iiint_{\mathbb{R}^3} \|\mathbf{v}_\sigma\| \exp\left[-\frac{m_\sigma(\mathbf{v}_\sigma - \mathbf{u}_\sigma)^2}{2k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[\sqrt{\frac{m_\rho \mathbf{v}_\sigma^2}{2k_\mathrm{B} T_\rho}}\right] \mathrm{d}^3\mathbf{v}_\sigma = \tag{9.118}$$

$$= \frac{4\pi k_\mathrm{B} T_\sigma}{m_\sigma \|\mathbf{u}_\sigma\|} \int_0^\infty v_\sigma^2 \exp\left[-\frac{m_\sigma(v_\sigma^2 + u_\sigma^2)}{2k_\mathrm{B} T_\sigma}\right] \sinh\left[\frac{m_\sigma u_\sigma v_\sigma}{k_\mathrm{B} T_\sigma}\right] \mathrm{erf}\left[\sqrt{\frac{m_\rho v_\sigma^2}{2k_\mathrm{B} T_\rho}}\right] \mathrm{d}v_\sigma \tag{9.119}$$

Notice that we have practically already solved this integral:

$$J(\mathbf{u}_\sigma) = \frac{\pi}{2\|\mathbf{u}_\sigma\|} \left(\frac{2k_\mathrm{B} T_\sigma}{m_\sigma}\right)^3 \exp\left[-\frac{m_\sigma \mathbf{u}_\sigma^2}{2k_\mathrm{B} T_\sigma}\right] \frac{\partial^2 \overline{K}_2(\mathbf{u}_\sigma)}{\partial(\|\mathbf{u}_\sigma\|)^2} \tag{9.120}$$

$$\overline{G}_2(\mathbf{u}_\sigma) = -3J(\mathbf{u}_\sigma) - \frac{m_\sigma}{k_\mathrm{B}T_\sigma}\frac{\partial J(\mathbf{u}_\sigma)}{\partial\left(\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}\right)} \tag{9.121}$$

Reverting all substitutions one by one, we obtain:

$$\frac{m_\sigma}{k_\mathrm{B}T_\sigma}\frac{\partial I(\mathbf{u}_\sigma)}{\partial\left(\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}\right)} = \frac{4k_\mathrm{B}T_\sigma}{\sqrt{\pi}m_\sigma}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2}+$$
$$-\frac{3}{\|\mathbf{u}_\sigma\|}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\mathrm{erf}\left[\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.122}$$

$$\|\mathbf{u}_\sigma\|\frac{\partial I(\mathbf{u}_\sigma)}{\partial(\|\mathbf{u}_\sigma\|)} = \left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]\Big/\sqrt{\frac{\pi k_\mathrm{B}T_\sigma}{2m_\sigma}+\frac{\pi k_\mathrm{B}T_\rho}{2m_\rho}}+$$
$$-\frac{1}{\|\mathbf{u}_\sigma\|}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\mathrm{erf}\left[\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.123}$$

$$J(\mathbf{u}_\sigma) = \pi\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{7/2}\frac{\left(1+\frac{m_\sigma}{2k_\mathrm{B}T_\sigma}\frac{4k_\mathrm{B}T_\rho}{m_\rho}\right)}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2}}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]+$$
$$+\frac{\pi^{3/2}}{2\|\mathbf{u}_\sigma\|}\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{5/2}\left(1+\frac{m_\sigma\mathbf{u}_\sigma^2}{k_\mathrm{B}T_\sigma}\right)\mathrm{erf}\left[\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.124}$$

We may now express the desired integrals, namely:

$$H(\mathbf{u}_\sigma) = -\frac{4n_\sigma n_\rho m_\sigma}{\sqrt{\pi}\mu_{\sigma\rho}}\frac{k_\mathrm{B}T_\rho}{m_\rho}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2}+$$
$$+\frac{n_\sigma n_\rho m_\sigma}{\mu_{\sigma\rho}\|\mathbf{u}_\sigma\|}\mathrm{erf}\left[\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.125}$$

$$\frac{k_\mathrm{B}T_\rho}{m_\rho}\overline{G}_1(\mathbf{u}_\sigma) = -\frac{2k_\mathrm{B}T_\sigma}{\sqrt{\pi}m_\sigma}\frac{2k_\mathrm{B}T_\rho}{m_\rho}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\frac{\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2}} \tag{9.126}$$

$$\overline{G}_2(\mathbf{u}_\sigma) = \frac{2k_\mathrm{B}T_\sigma}{m_\sigma\|\mathbf{u}_\sigma\|}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\mathrm{erf}\left[\|\mathbf{u}_\sigma\|\Big/\sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right]+$$
$$-\frac{4\mathbf{u}_\sigma^2k_\mathrm{B}T_\sigma}{\sqrt{\pi}m_\sigma}\left(\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^2\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{7/2}+$$
$$+\frac{2k_\mathrm{B}T_\sigma}{\sqrt{\pi}m_\sigma}\frac{2k_\mathrm{B}T_\rho}{m_\rho}\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\frac{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+4\frac{2k_\mathrm{B}T_\rho}{m_\rho}}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{5/2}}\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right] \tag{9.127}$$

$$\sqrt{\frac{2k_\mathrm{B}T_\rho}{\pi m_\rho}}\overline{G}_3(\mathbf{u}_\sigma) = \frac{4\mathbf{u}_\sigma^2k_\mathrm{B}T_\sigma}{\sqrt{\pi}m_\sigma}\left(\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^2\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\frac{\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{7/2}}+$$
$$-\frac{3}{\sqrt{\pi}}\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}\left(\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^2\left(\frac{2\pi k_\mathrm{B}T_\sigma}{m_\sigma}\right)^{3/2}\frac{\exp\left[-\mathbf{u}_\sigma^2\Big/\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma}+\frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{5/2}} \tag{9.128}$$

Summing the last three integrals, we find that the contribution of the Rosenbluth g-potential is simply:

$$G(\mathbf{u}_\sigma) = \frac{n_\sigma n_\rho}{\|\mathbf{u}_\sigma\|} \operatorname{erf}\left[\|\mathbf{u}_\sigma\| \Big/ \sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.129}$$

Thus, the rate of net-energy loss is given by:

$$\frac{\partial}{\partial t}\left(\frac{1}{2}n_\sigma m_\sigma \mathbf{u}_\sigma^2 + \frac{3}{2}n_\sigma k_\mathrm{B}T_\sigma\right) = \frac{n_\sigma n_\rho q_\sigma^2 q_\rho^2 \ln\Lambda}{2\pi^{3/2}\varepsilon_0^2 \mu_{\sigma\rho}} \frac{2k_\mathrm{B}T_\rho}{m_\rho} \frac{\exp\left[-\mathbf{u}_\sigma^2 \Big/ \left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)\right]}{\left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2}} +$$

$$- \frac{n_\sigma n_\rho q_\sigma^2 q_\rho^2 \ln\Lambda}{4\pi\varepsilon_0^2 m_\rho \|\mathbf{u}_\sigma\|} \operatorname{erf}\left[\|\mathbf{u}_\sigma\| \Big/ \sqrt{\frac{2k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B}T_\rho}{m_\rho}}\right] \tag{9.130}$$

Taking the stationary limit, that is, letting $\|\mathbf{u}_\sigma\| \to 0$, the energy exchange rate is determined to be:

$$\frac{\partial}{\partial t}\left(\frac{3}{2}n_\sigma k_\mathrm{B}T_\sigma\right) = \frac{n_\sigma n_\rho q_\sigma^2 q_\rho^2 \ln\Lambda}{2\pi^{3/2}\varepsilon_0^2} \frac{2k_\mathrm{B}(T_\rho - T_\sigma)}{m_\sigma m_\rho} \Big/ \left(\frac{2k_\mathrm{B}T_\sigma}{m_\sigma} + \frac{2k_\mathrm{B}T_\rho}{m_\rho}\right)^{3/2} \tag{9.131}$$

Notice that the determined energy exchange rate is directly proportional to the temperature difference $T_\rho - T_\sigma$, thus no energy exchange occurs in the case of two mutually thermalised species, as expected. All other terms being symmetric upon exchange $\sigma \leftrightarrow \rho$, it is clear that summing the exchange rates between two species gives identically zero, thus confirming the physical relevance of the obtained result.

In order to evaluate the frequency of energy dissipation, we simply impose:

$$\frac{\langle \Delta E_\sigma \rangle}{\Delta t} = \frac{1}{2} m_\sigma \operatorname{Tr}\left(\frac{\langle \Delta \mathbf{v}_\sigma \Delta \mathbf{v}_\sigma \rangle}{\Delta t}\right) = \frac{q_\sigma^2 q_\rho^2 \ln\Lambda}{4\pi\varepsilon_0^2 m_\sigma} \frac{1}{2} \frac{\partial^2 g_\rho}{\partial \mathbf{v}_\sigma^2} \tag{9.132}$$

CONTINUE